# IPOÁ REDD+ PROJECT

Document Prepared By CARBONEXT

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|---|---|
| **Project Title** | IPOÁ REDD+ PROJECT |
| **Version** | 1.0 |
| **Date of Issue** | 20th - June - 2022 |
| **Project Location** | Novo Aripuanã municipality, State of Amazonas, Brazil |
| **Project Proponent(s)** | Carbonext Tecnologia em Soluções Ambientais Ltda<br>Carina Marcia Klein Balen<br>Sílvio Antonio Balen |
| **Prepared By** | Carbonext Tecnologia em Soluções Ambientais Ltda. |
| **Validation Body** | Undefined |
| **Project Lifetime** | 17 March 2022 – 16 March 2052; 30 - year total period |
| **GHG Accounting Period** | 17 March 2022 – 16 March 2052; 30 - year total period |
| **History of CCB Status** | Not applicable |
| **Gold Level Criteria** | Not applicable |
| **Expected Verification Schedule** | August 2022 |

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# 1 SUMMARY OF PROJECT BENEFITS

## 1.1 Unique Project Benefits

| Outcome or Impact Estimated by the End of Project Lifetime | Section Reference |
|---|---|
| 1) Provide improved quality of life for the local community based on pillars of sustainability | 2.1.12 |
| 2) Increase the number of youth and adults with internet connection, providing increased life opportunities | 2.1.11 |

## 1.2 Standardized Benefit Metrics

| Category | Metric | Estimated by the End of Project Lifetime | Section Reference |
|---|---|---|---|
| GHG emission reductions or removals | Net estimated emission removals in the project area, measured against the without-project scenario | N/A | - |
| | Net estimated emission reductions in the project area, measured against the without-project scenario | 2,809,491 $tCO_2$ | 3.2.4 |
| Forest[1] cover | For REDD[2] projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario | 4,563.75 ha | 2.2 |
| | For ARR[3] projects: Estimated number of hectares of forest cover increased in the project area measured against the without-project scenario | N/A | - |
| Improved land management | Number of hectares of existing production forest land in which IFM[4] practices are expected to occurred as a result of project activities, measured against the without-project scenario | N/A | - |
| | Number of hectares of non-forest land in which improved land management practices are expected to occur as a result of project activities, measured against the without-project scenario | N/A | - |
| Training | Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities | 105 community members | 4.1.1 |

[1] Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

[2] Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

[3] Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

[4] Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*)

| Category | Metric | Estimated by the End of Project Lifetime | Section Reference |
|---|---|---|---|
| | Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities | Around 31 females | 4.1.1 |
| Employment | Total number of people expected to be employed in project activities,[5] expressed as number of full-time employees[6] | N/A | 2.3.15 |
| | Number of women expected to be employed as a result of project activities, expressed as number of full-time employees | N/A | 2.3.15 |
| Livelihoods | Total number of people expected to have improved livelihoods[7] or income generated as a result of project activities | N/A | - |
| | Number of women expected to have improved livelihoods or income generated as a result of project activities | N/A | - |
| Health | Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario | 105 community members | 4.1.1 |
| | Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario | Around 31 women | 4.1.1 |

[5] Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.
[6] Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])
[7] Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

| Category | Metric | Estimated by the End of Project Lifetime | Section Reference |
|---|---|---|---|
| Education | Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario | Around 29 children | 4.1.1 |
| | Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario | Around 15 girls[8] | 4.1.1 |
| Water | Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario | N/A | - |
| | Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario | N/A | - |
| Well-being | Total number of community members whose well-being[9] is expected to improve as a result of project activities | 105 community members | 4.1.1 |
| | Number of women whose well-being is expected to improve as a result of project activities | Around 31 women | 4.1.1 |

[8] According to data collected in the interviews, about 50% of the children are girls

[9] Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

| Category | Metric | Estimated by the End of Project Lifetime | Section Reference |
|---|---|---|---|
| Biodiversity conservation | Expected change in the number of hectares managed significantly better by the project for biodiversity conservation,[10] measured against the without-project scenario | 4,563.75 ha | 5.1.3 |
| | Expected number of globally Critically Endangered or Endangered species[11] benefiting from reduced threats as a result of project activities,[12] measured against the without-project scenario | 11 | 5.1.1 |

[10] Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

[11] Per IUCN's Red List of Threatened Species

[12] In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

# 2 GENERAL

## 2.1 Project Goals, Design and Long-Term Viability

### 2.1.1 Summary Description of the Project (G1.2)

The primary objective of the IPOÁ REDD+ Project is to reduce the emission of 2,809,491 $tCO_2e$ (average of 93,650 $tCO_2e$/year), corresponding to 4,564 hectares of avoided deforestation over the 30-year project lifetime.

IPOÁ REDD+ Project operates within private areas totaling approximately 17,386 hectares, located in the municipality of Novo Aripuanã, which ranks 15th in the ranking of deforestation of the municipalities of the Amazon biome. According to PRODES, between 2016 and 2021, an average of 160km²/year were deforested.[13]

The IPOÁ REDD+ project is located at an approximate distance of 13km from the Juma Sustainable Development Reserve and 25km from the Acari National Park. The Juma RDS was created through State Decree AM nº 26.010 of July 3, 2006, with the objective of preserving nature, ensuring the necessary conditions and means for reproduction and improvement of the ways and quality of life and exploitation. of the natural resources of traditional populations, to value, conserve and improve the knowledge and techniques of environmental management developed by traditional populations, among others. PARNA do Acari was created on May 11, 2016, with the objective of protecting the biological diversity of part of the Acari, Camaiú, Sucunduri, Abacaxis rivers and their tributaries, their natural landscapes and associated abiotic values, in addition to guaranteeing the sustainability of the ecosystem services, contribute to the environmental stability of the region where it is located and provide the development of recreation activities in contact with nature and ecological tourism.[14][15] This project will provide a buffer zone in the surroundings of these Conservation Units, supporting the combatting of illegal deforestation.

Within the project, an estimated average of 152 ha/year of deforestation will be avoided over its lifetime. The project will generate benefits for communities and biodiversity, which are quantified and monitored

[13] PRODES, 2021. http://terrabrasilis.dpi.inpe.br/app/dashboard/deforestation/biomes/legal_amazon/increments
[14] SEMA - AM, 2022. http://meioambiente.am.gov.br/wp-content/uploads/2018/03/Decreto-26.010_2006-Cria-RDS-do-Juma.pdf (Acesso em 14/03/2022).
[15] ICMBio. https://www.gov.br/icmbio/pt-br/assuntos/biodiversidade/unidade-de-conservacao/unidades-de-biomas/amazonia/lista-de-ucs/parna-do-acari/arquivos/dcom_decreto_criacao_parna_do_acari.pdf (Acesso em 14/03/2022).

according to the CCB methodological standard. The definition of strategies is carried out in a process of participatory consultation with stakeholders.

The project aims to positively impact its participants immediately. For instance, programs to benefit the population including installation of communications capabilities in this remote region, distribution of sanitary products and high impact education actions are planned to initiate from the project's start (see Appendix 1). Later in the development, the feasibility of developing programs to promote the local bioeconomy will be studied, facilitating sale and increasing demand for products originating in the project area), together with the continuous conservation and maintenance of regional biodiversity.

The IPOÁ REDD+ project is committed to carrying out socio-environmental activities related to forest preservation and maintaining the integrity of the project proponents' farms. The model proposed by this project includes the possibility of its replication in other areas with potential to receive REDD+ projects. The central idea is to multiply the preserved areas in the surrounding regions by adopting sustainable practices, converting the region into a model for sustainable development and with the income benefits arising from the reduction of emissions. Project's expected climate, community and biodiversity benefits:

**Climate benefits**:

Avoid planned deforestation of the tropical rainforest in the region, thus avoiding the emission of 2,611,755 $tCO_2e$ over a period of 5 years of the Project's crediting period (annual average of 522,351 $tCO_2e$);

Prevent unplanned deforestation of the tropical rainforest in the region, thus avoiding the emission of 534,733 $tCO_2e$ over a period of 30 years of the Project's crediting period (annual average of 17,824$tCO_2e$);

The resulting total avoided deforestation in the region therefore avoids the emission of 3,146,488 $tCO_2e$ over a period of 30 years of the Project's crediting period (annual average of 104,883 $tCO_2e$).

**Community Benefits:**

The project has planned initiatives in the following areas:

- Health: Health trasport programme; Medical care by the NGO "Doutor das águas"; Distribution of soaps and women's sanitary products; Distribution of oral hygiene kits.
- Work safety programme: training and equipment provided.
- Communication and access: installation of internet connection in this remote rural area.
- Education programme: Refurbishment of school establishment and provision of, and continued support for, certified teacher.

- Community training: Dissemination of forest conservation practices.

**Biodiversity Benefits:**

The protection and maintenance of 4,563.75 ha of forest will favor the habitat and survival of different animal and plant species, including threatened and endangered species.

Flora and fauna inventory in order to study and continuously monitor the fauna and flora present in the Project Zone, included threatened and endangered species, along the project lifetime.

### 2.1.2 Project Scale

| Project Scale | |
|---|---|
| Project | X |
| Large project | |

### 2.1.3 Project Proponent (G1.1)

| | |
|---|---|
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#### 2.1.4 Other Entities Involved in the Project

This item does not apply to the present project.

#### 2.1.5 Physical Parameters (G1.3)

***Location of the project***

The project is located in the Brazilian Northern Region, in the state of Amazonas, in municipality of Novo Aripuanã, as the figure below. Novo Aripuanã belongs to the Mesoregion of the South Amazon and Microregion of Madeira[16].

The main way to reach the municipality is by waterway, large boats or speedboats that leave the state capital, Manaus, crossing parts of the Amazon River and Madeira River. The duration of the route is 24 hours in larger boats and 10 hours in speedboats, as there are 450 kilometers that separate the two cities, on average. The main bodies of water in the vicinity of the project area are the Sucunduru River and the Acari River.

[16] https://portalamazonia.com/amazonia-az/letra-n/novo-aripuana

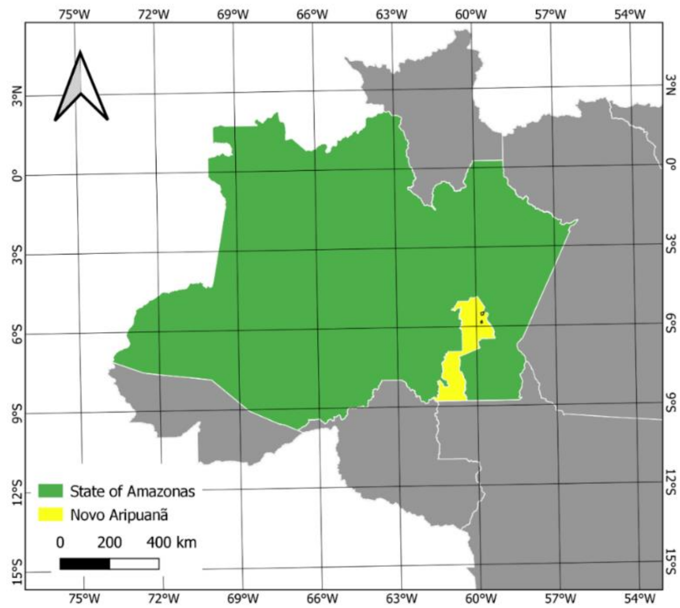


*Figure 1. Location of the municipality of Novo Aripuanã, AM, Brazil.*

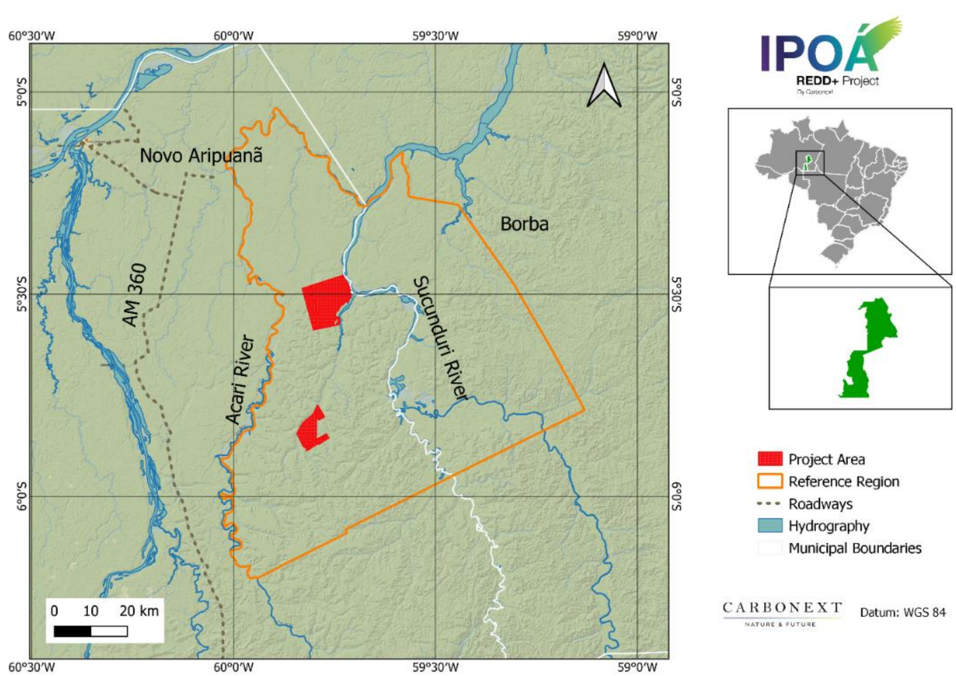


*Figure 2. Location of the Project Area.*

The IPOÁ REDD+ Project Area is 17,386.45 ha and is composed of two zones: Maanaim farm (05°31'04,33" S 59°46'00,53" W) and Campos farm (05°50'15,93" S 59°48'26,53" W). The Project Areas are divided as:

- Maanaim Farm Avoided Unplanned Deforestation (AUD) area: 9,938.49 ha.
- Maanaim Farm Avoided Planned Deforestation (APD) area: 27,06.52 ha.
- Campos Farm Avoided Unplanned Deforestation (AUD) area: 3,861.17 ha.
- Campos Farm Avoided Planned Deforestation (APD) area: 880.26 ha

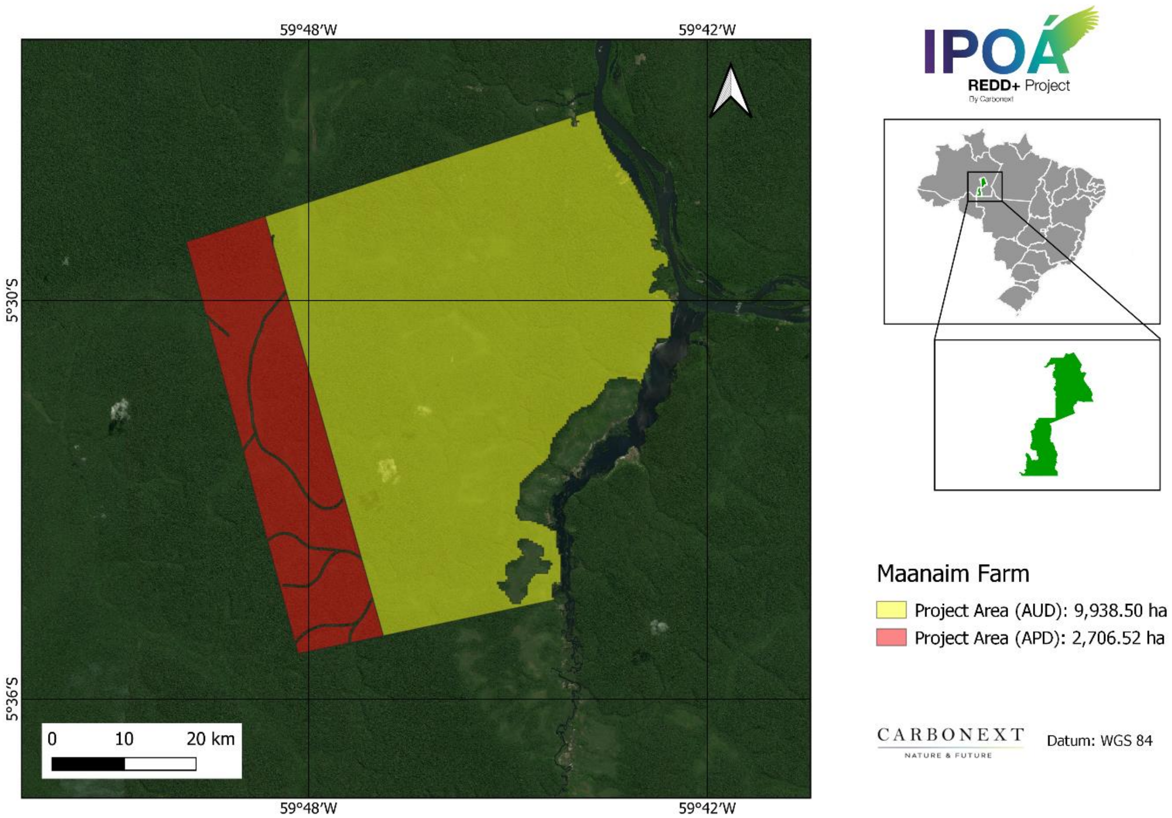


*Figure 3. Project Area Maanaim Farm.*

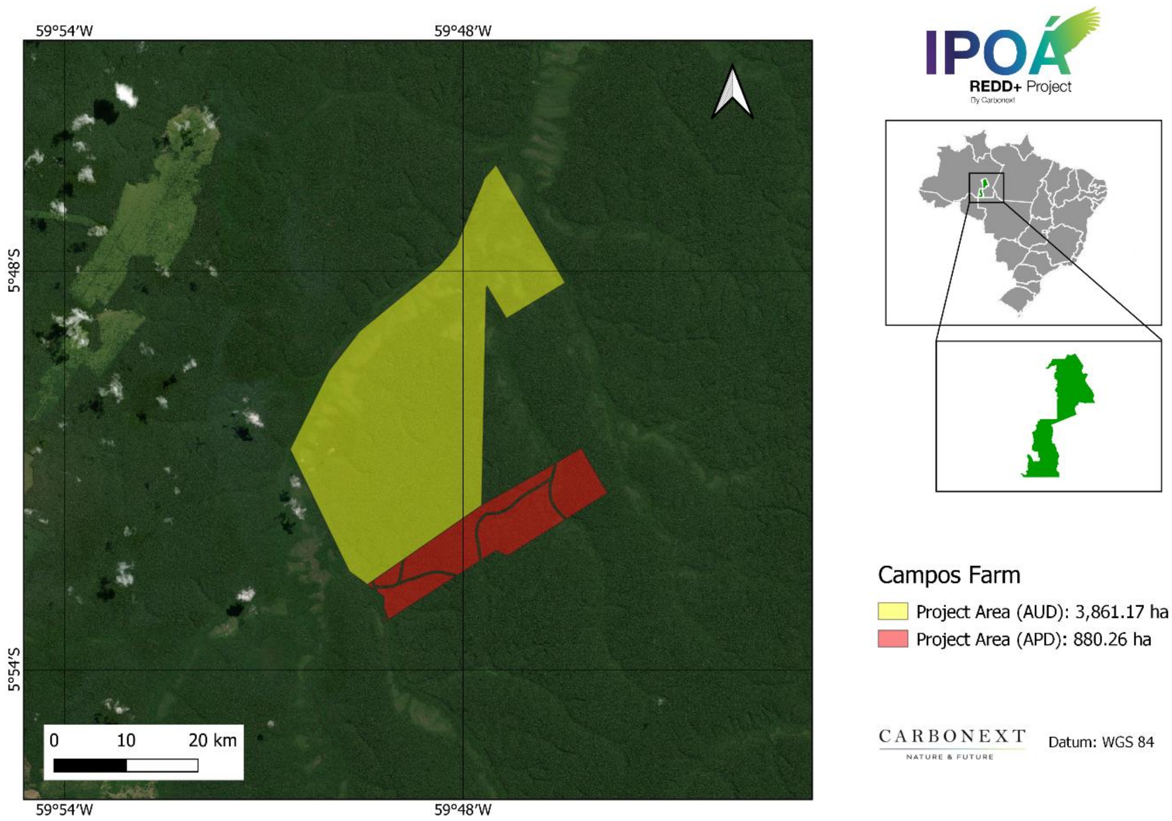


*Figure 4. Project Area Maanaim Farm.*

***Topography***

The municipality of Novo Aripuanã is located in the lowlands, whose topography is flat and slightly undulated.[17]

In the Reference Region, the highest elevation is 137.41 meters, while in the Project Area it is 102.32 meters, with the Maanaim farm area being more flat and Campos farm with greater variation.[18]

[17]http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSI%c3%87%c3%83O%20FLOR%c3%8dSTICA%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20%c3%81REA%20COM%20EXPLORA%c3%87%c3%83O%20FLORESTAL%20NO%20MUNIC%c3%8dPIO%20DE%20.pdf
[18]Mapa Índice TOPODATA (webmapit.com.br)

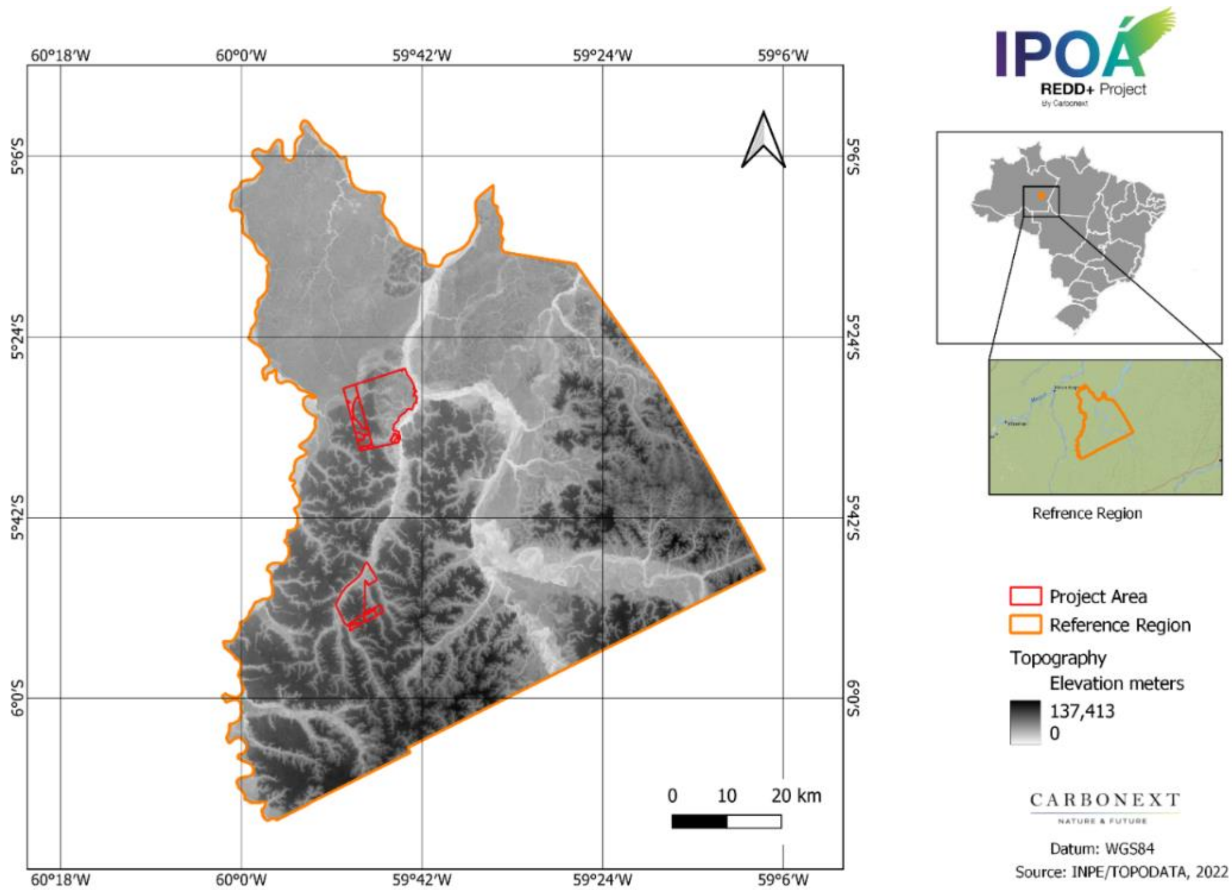


*Figure* **5***. Elevation of Reference Region.*

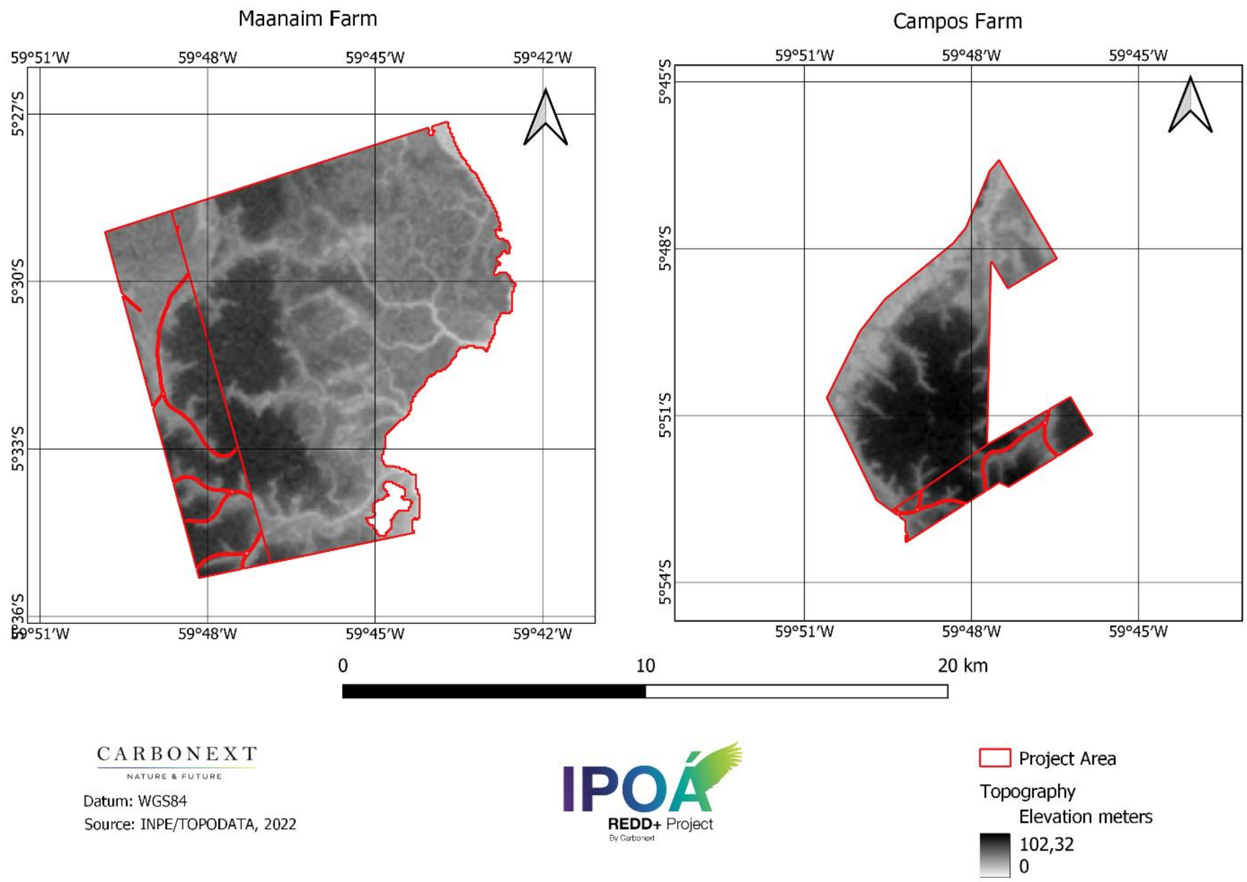


*Figure* **6***. Elevation of the Manaaim and Campos plots, Project Area.*

In the Reference Region, the maximum slope is 40º. The maximum slope observed in the Project Area is 32º, and in 81% of the area the slope does not exceed 10º.[19]

[19]Mapa Índice TOPODATA (webmapit.com.br)

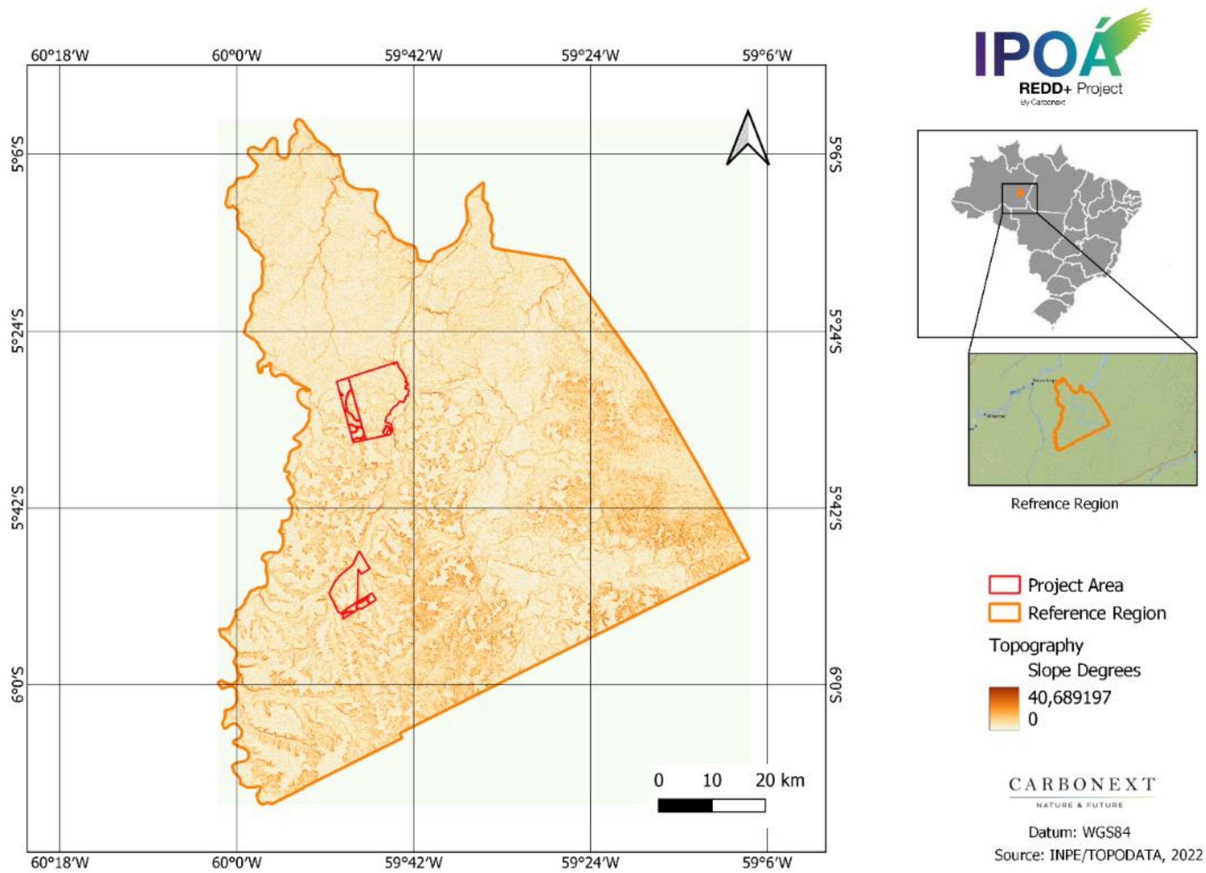


*Figure 7. Slope of Reference Region.*

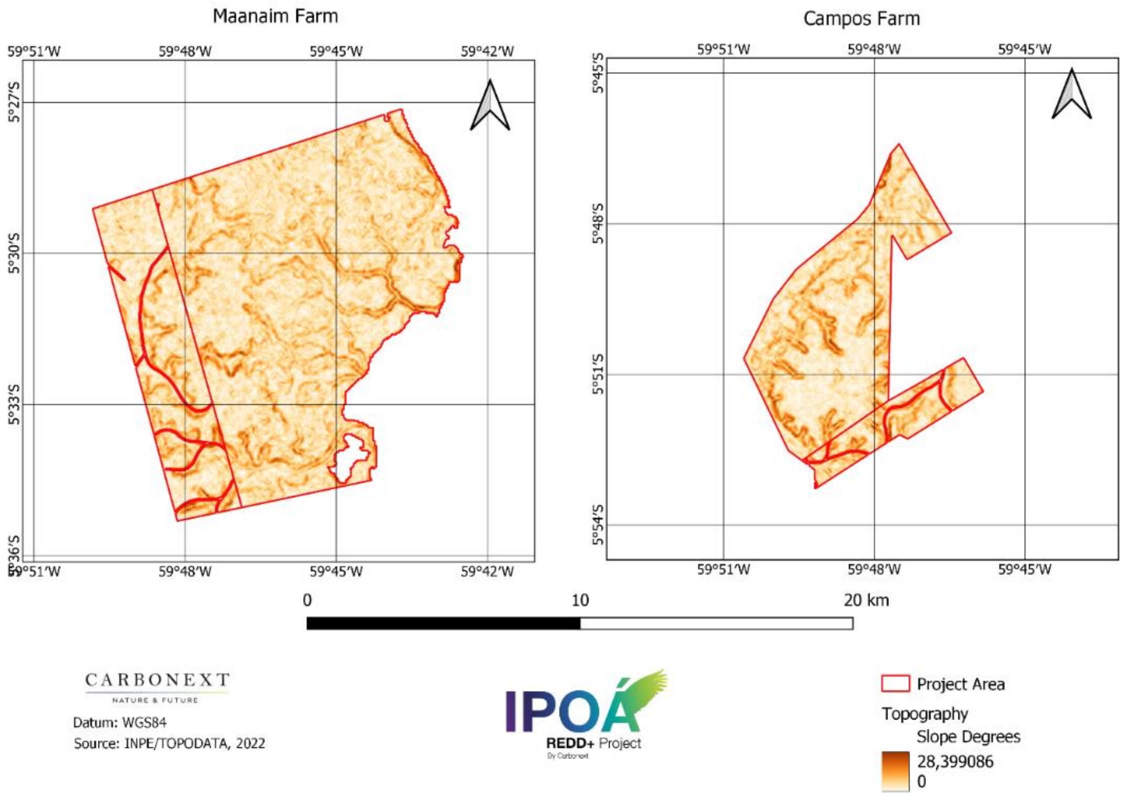


*Figure 8. Slope of the Manaaim and Campos plots, Project Area.*

***Soil***

According to the Brazilian Soil Classification System (SiBCS, 2006) 20, the Region of Reference is composed by the following soil types, as show in the figure below below:

- Haplic Tb Dystrophic Gleysol (GXbd1).
- Dystrophic Yellow Latosol (LAd1).
- Dystrophic Yellow Latosol + Orthic Quartzarenic Neosol (LAd9).
- Dystrophic Red-Yellow Latosol + Orthic Quartzarenic Neosol (LVAd10).
- Dystrophic Red-Yellow Latosol + Dystrophic Red-Yellow Ultisol (LVAd5).

[20]http://geoinfo.cnps.embrapa.br/layers/geonode%3Abrasil_solos_5m_20201104

- Dystrophic Red-Yellow Latosols + Dystrophic Tb Haplic Gleysols (LVAd9).

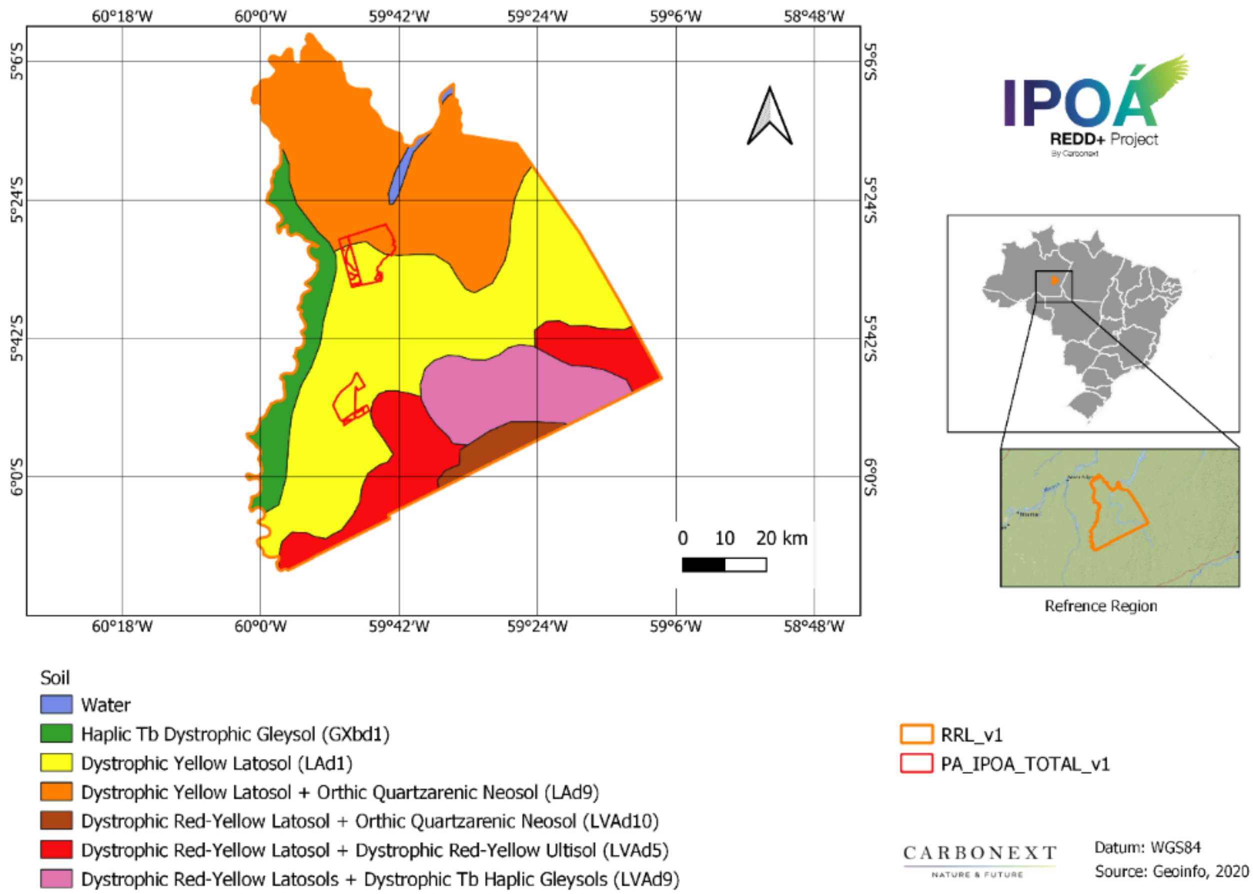


*Figure* **9**. *Soil types in Reference Region.*

The soil in the project area is 80% Dystrophic Yelow Latosol (LAd1) and 20% Dystrophic Yelow Latosol + Orthic Quartzarenic Neosol (Lad9).

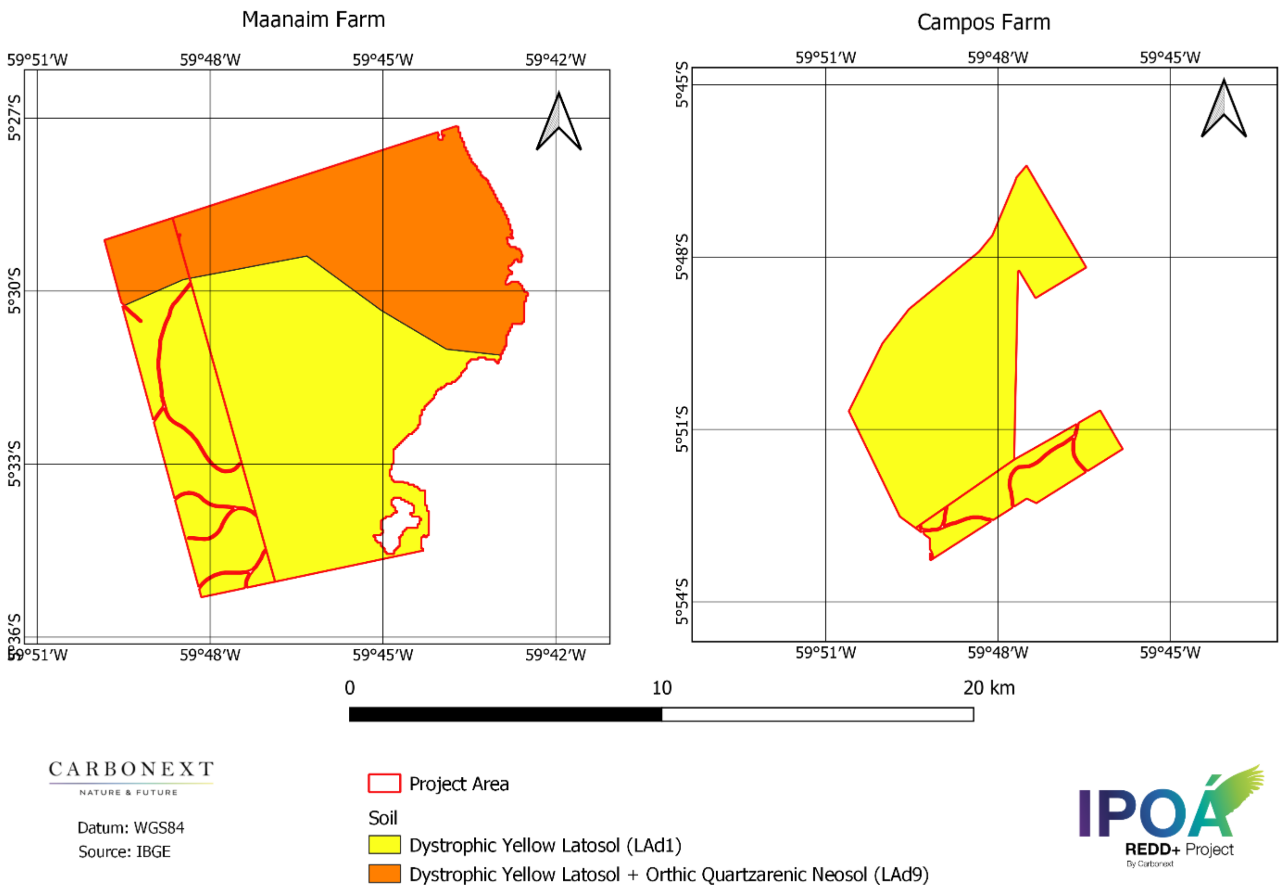


*Figure 10. Soil types in Project Area.*

Yelow Latosol are developed from clayey or sandy-clayey sedimentary materials from the Barreiras formation in the coastal region of Brazil or in the low plateaus of the Amazon region related to the Alter-do-Chão Formation, which may also occur outside these environments when they meet the color requirements defined by the SiBCS.

The yellowish color is uniform in depth, as is the clay content. The most common texture is clayey or very clayey. Another field aspect refers to the high cohesion of the structural aggregates (cohesive soils).

They have good physical conditions of moisture retention and good permeability, being intensively used for sugarcane and pasture crops, and to a lesser extent, for the cultivation of cassava, pineapple, coconut and citrus; and large areas of eucalyptus reforestation. In the Amazon, they are mainly used for pasture.

Yellow latosols can be classified in the third categorical level of the SiBCS. Those classified as dystrophic are those with low fertility. [21]

***Climate***

The climate of Amazon region is classified as equatorial, characterized by high temperatures and air humidity, besides being constantly subjected to the oscillations of the rivers[22]. Within the equatorial, according to Koopen's climatic classification the predominant type of climate in Novo Aripuanã and in the Project Area is type Af (humid or sumer humid tropical), with maximum temperatures of 37.5ºC, minimum of 16ºC and an average temperature of 26.7ºC.

The relative humidity is quite high in the municipality, where it presents minimum and maximum factors of 85 and 95%. The high rainfall of the region is one of the characteristic factors, with an average variation of 2250 to 2750 mm in the year, where rainfall rates tend to be high all year round. Rainfall generally starts in October, with the rainiest four months between January and April, and the driest between June and September.[23]

A small part of the reference region is characterized by climate Am (humid or sub-humid tropical) according to Köppen[24], a transition between the Af and Aw climatic type. It is characterized by an average temperature of the coldest month that is always above 18ºC, with a short dry season that is compensated by high precipitation totals.[25]

[21]https://www.agencia.cnptia.embrapa.br/gestor/solos_tropicais/arvore/CONT000fzyjaywi02wx5ok0q43a0r58asu5l.html
[22] http://eventoscopq.mackenzie.br/index.php/jornada/xvijornada/paper/view/2159/1295
[23]http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSI%c3%87%c3%83O%20FLOR%c3%8dSTICA%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20%c3%81REA%20COM%20EXPLORA%c3%87%c3%83O%20FLORESTAL%20NO%20MUNIC%c3%8dPIO%20DE%20.pdf
[24] https://ri.ufmt.br/bitstream/1/2160/1/TESE_2016_Jos%c3%a9%20Maur%c3%adcio%20da%20Cunha.pdf
[25] embrapa.br

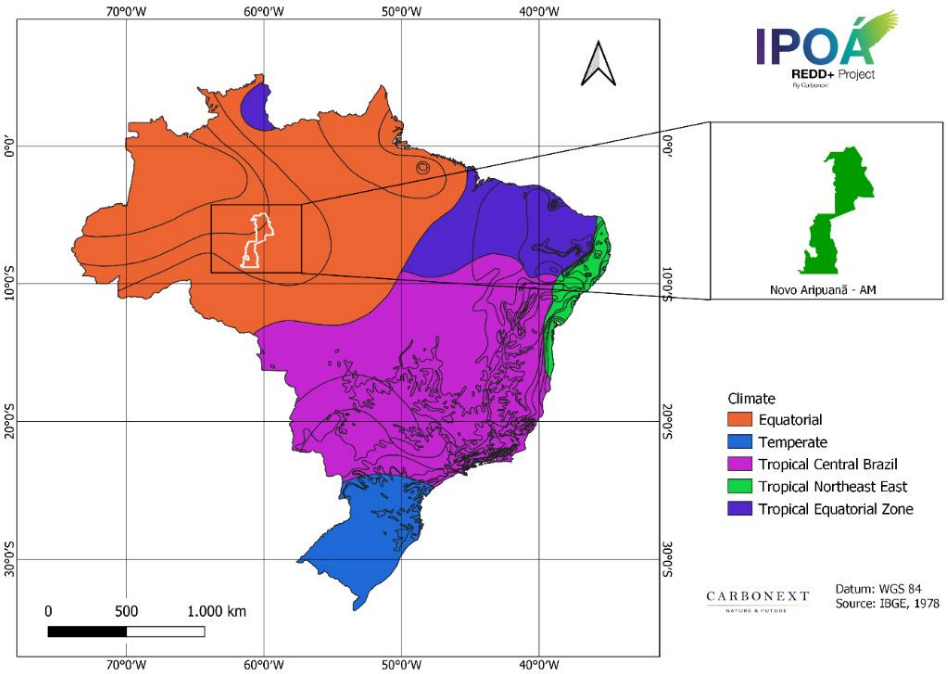


*Figure 11- Amazonian equatorial climate in Novo Aripuanã.*

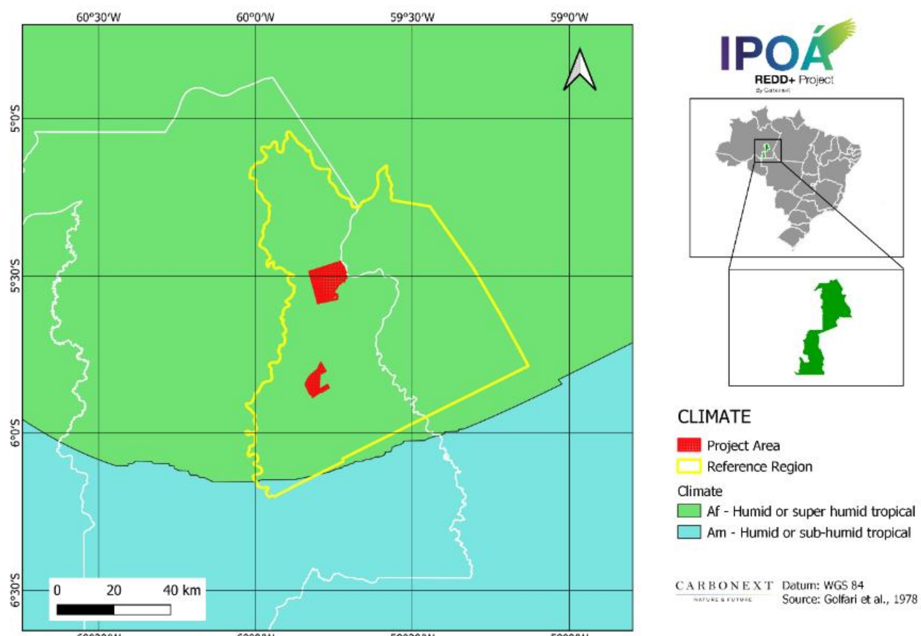


*Figure 12- Climate types according to Koopen's climate classification.*

***Hydrology***

The river system of the municipality of Novo Aripuanã is represented by the Madeira River Basin and its tributary the Aripuanã River[26]. The flood period is from December to May, with a maximum quota in the month of April. The floodplains is full from June to November, with minimum waters in the month of September in the vicinity of Novo Aripuanã, suffering with the influence of the floods of the Amazon River - that occur in the period of May and July.[27]

The Project Area and entire Reference Region are located in Baixo Madeira Hydrografic Microregion[28].

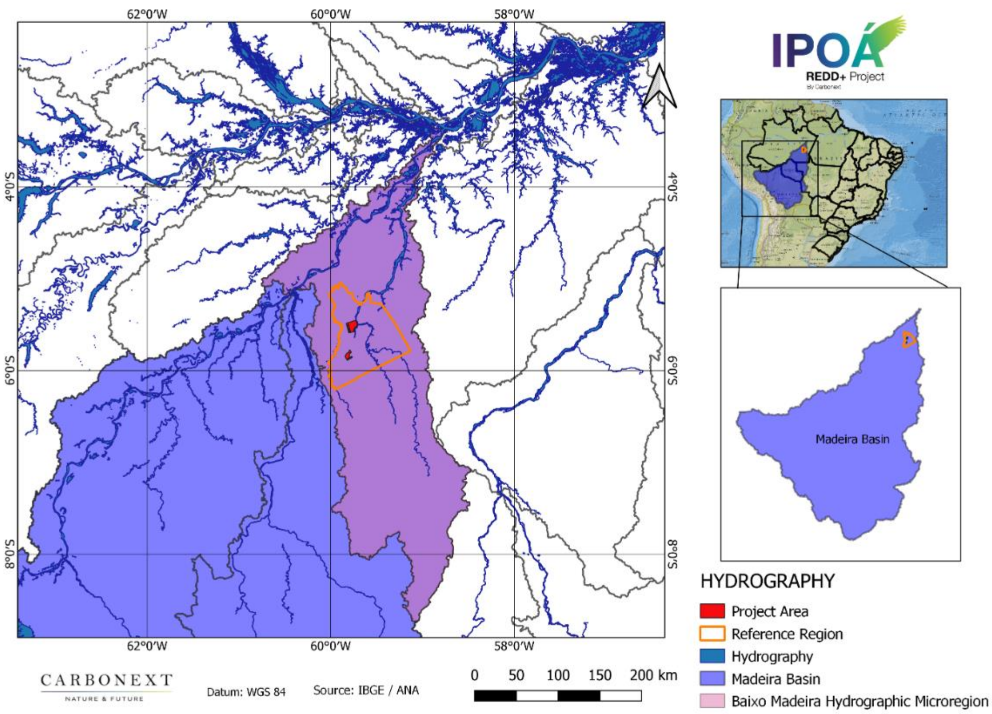


*Figure 13. Baixo Madeira Hydrografic Microregion.*

[26]http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSI%c3%87%c3%83O%20FLOR%c3%8dSTICA%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20%c3%81REA%20COM%20EXPLORA%c3%87%c3%83O%20FLORESTAL%20NO%20MUNIC%c3%8dPIO%20DE%20.pdf

[27]http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSI%c3%87%c3%83O%20FLOR%c3%8dSTICA%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20%c3%81REA%20COM%20EXPLORA%c3%87%c3%83O%20FLORESTAL%20NO%20MUNIC%c3%8dPIO%20DE%20.pdf.

[28]https://geoftp.ibge.gov.br/informacoes_ambientais/estudos_ambientais/bacias_e_divisoes_hidrograficas_do_brasil/2021/Divisao_Hidrografica_Nacional_DHN250/mapas/mapa_das_divisoes_hidrograficas_do_brasil_2021.pdf

Water bodies close to the project area are important for river transport. Access to Maanaim farm is via the Sucunduri River and to reach Campos farm, the path is carried out through the Capinarana Stream.

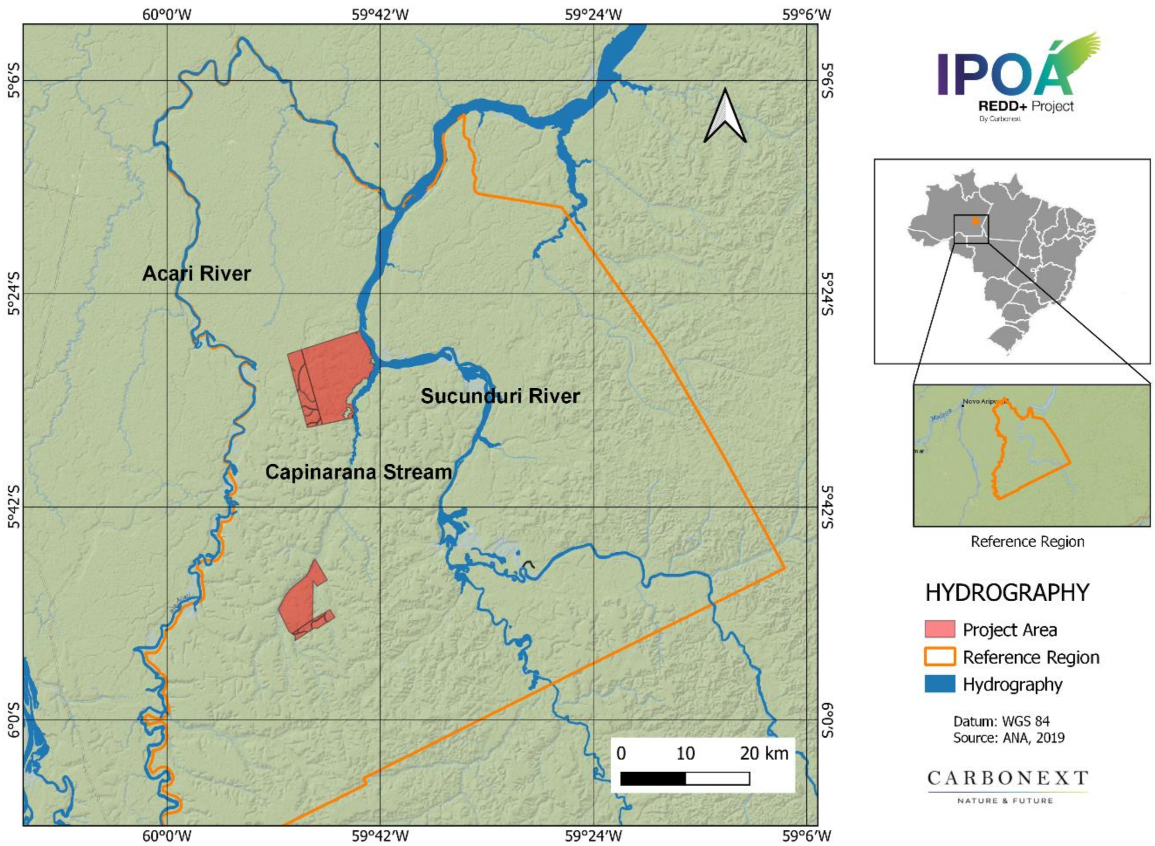


*Figure 14. Hydrology in Project Area.*

***Vegetation***

The vegetation present in the municipality of Novo Aripunã (AM) is characterized as Dense Tropical Rainforest, with emergent species and a group of palm trees, which compete for light in the upper tree stratum[29]. The Dense Tropical Rainforest is characterized by the presence of phanerophytes, woody lianas, and epiphytes in abundance, and is subdivided into five formations that vary according to topography.

In the reference region, the prevalence of vegetation is Submontane Dense Tropical Rainforest (48%) followed by Lowland Dense Tropical Rainforest (44%).

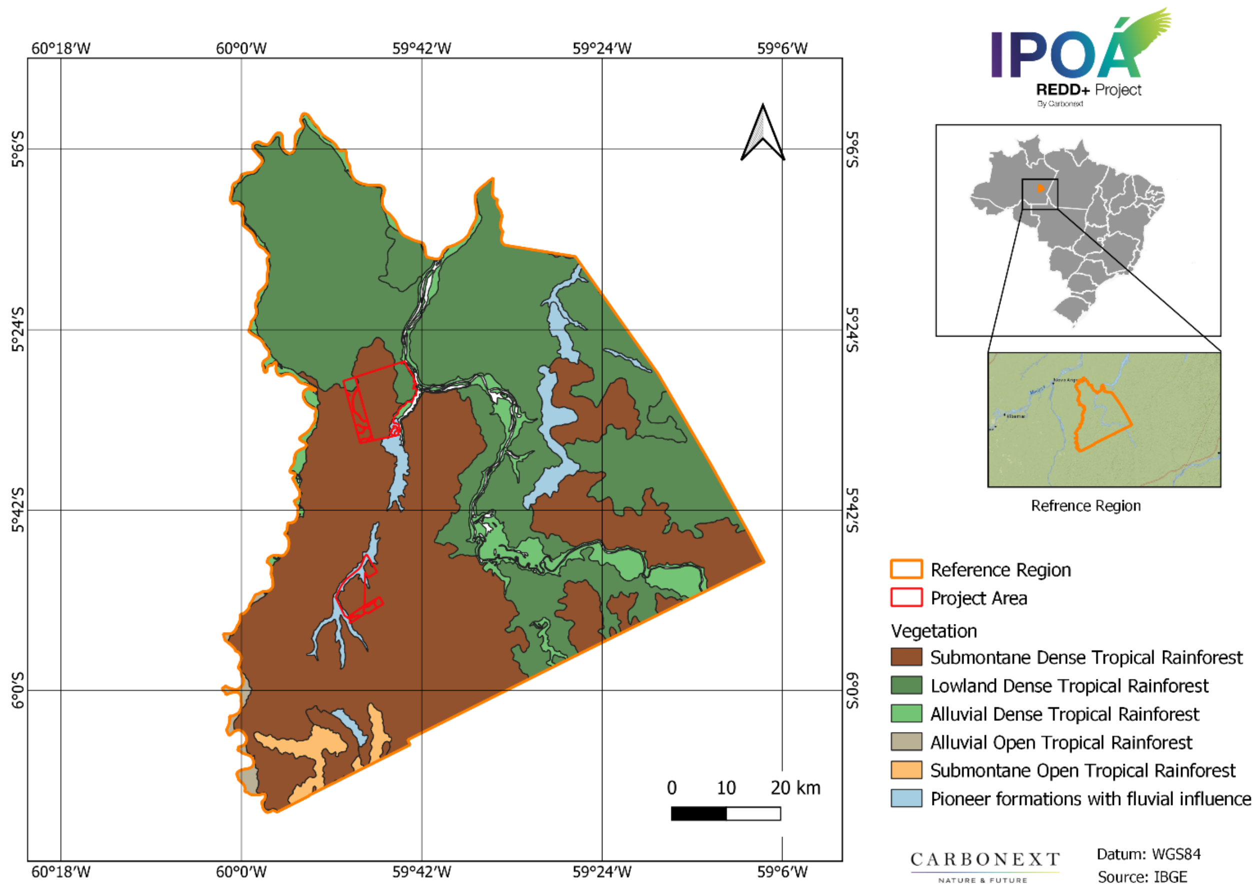


*Figure 15. Vegetational typology of Reference Region.*

[29]http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSI%c3%87%c3%83O%20FLOR%c3%8dSTICA%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20%c3%81REA%20COM%20EXPLORA%c3%87%c3%83O%20FLORESTAL%20NO%20MUNIC%c3%8dPIO%20DE%20.pdf

Three types of Dense Tropical Rainforest are observed for the Project Area regions are a) Alluvial; b) Lowland and c) Submontane. Subformations can be defined with open or uniform canopy.[30]

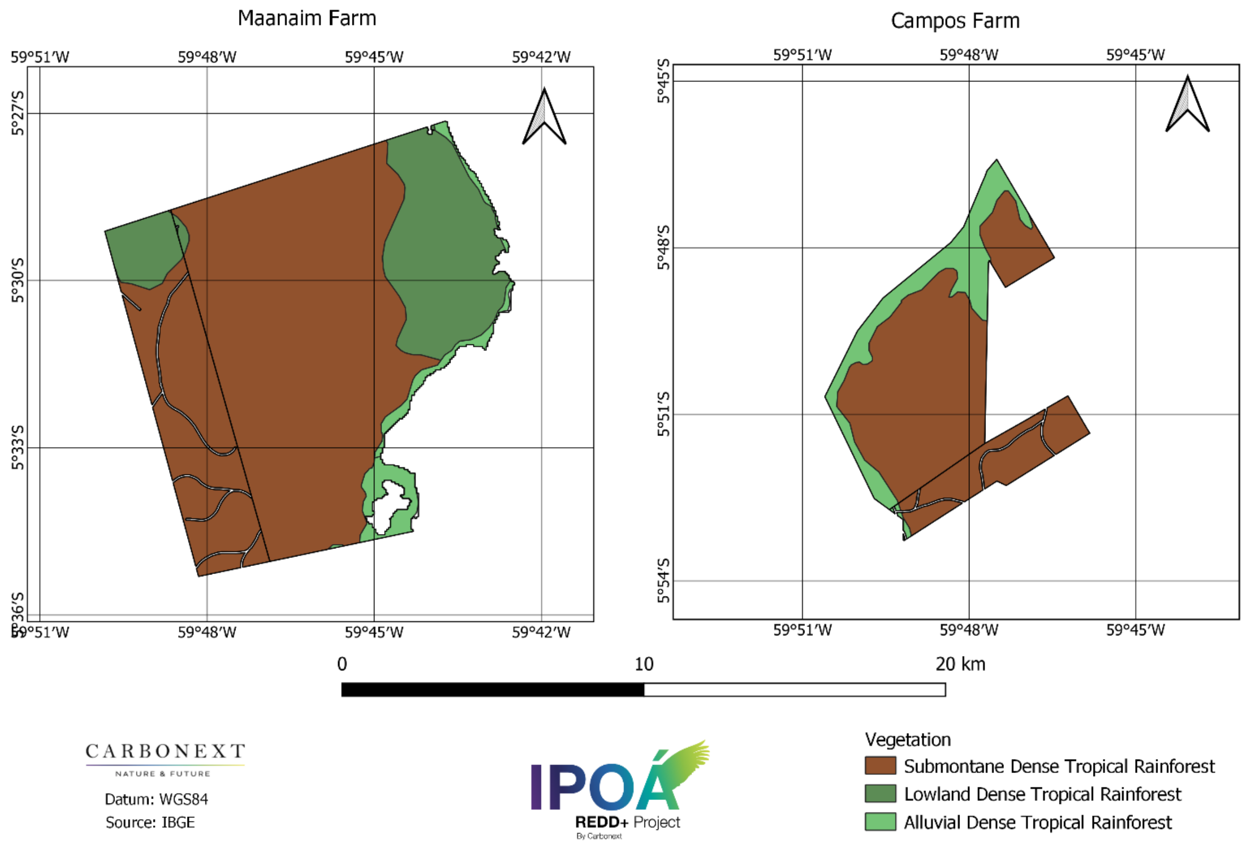


*Figure 16. Vegetational typology of the Manaaim (A) and Campos (B) plots, Project Area.*

a) Alluvial: it does not vary topographically. It is a riverine formation or riparian forest that occurs along watercourses. It consists of fast growing phanerophytic individuals, generally with smooth bark, conical trunk, and sometimes with tabular roots. It is a formation with many palm trees in the dominated

[30] http://redd.mma.gov.br/images/gttredd/daltonvaleriano_definicaofloresta.pdf

stratum and submata. It presents many woody and herbaceous lianas, besides a large number of epiphytes and few parasites.[31]

b) Lowlands: it is situated between 4° N latitude and 16° S latitude, from 5 m to 100 m above the sea; from 16° S latitude to 24° S latitude from 5 m to 50 m; from 24° S latitude to 32° S latitude from 5 m to 30 m. It is a formation that in general occupies the coastal plains, capped by pliopleistocene tablelands of the Barreiras Group. [32]

c) Submontane: occurs between 4° N and 16° S latitude from 100 m to 600 m; from 16° S to 24° S latitude from 50 m to 500 m; from 24° S to 32° S latitude from 30 m to 400 m. Its main features are high phanerophytes, some exceeding 50 m in Amazonia and rarely 30 m in other parts of the country. The submata is composed of naturally regenerating seedlings, few nannophanerophytes and chamaephytes, in addition to the presence of small palm trees and herbaceous lianas in larger numbers.[33] The main species that characterize the emerging Submontane Dense Ombrophylous Forests are: Dinizia excelsa Ducke (angelimpedra), Bertholletia excelsa (chestnut tree) and Cedrelinga cateniformis (Ducke) Ducke (cedrorana). The uniform stratum is characterized by Manilkara spp. (maçarandubas), Protium spp. (breus) and Pouteria spp. (abius)[34].


*Figure 17. Dense Tropical Rainforest schematic profile. 1 - Altomontane; 2 - Montane; 3 - Submontane; 4- Lowland; 5- Alluvial. Source: Veloso, Rangel Filho and Lima (1991).*

[31] https://ambientes.ambientebrasil.com.br/natural/regioes_fitoecologicas/regioes_fitoecologicas_-_floresta_ombrofila_densa.html
[32] https://ambientes.ambientebrasil.com.br/natural/regioes_fitoecologicas/regioes_fitoecologicas_-_floresta_ombrofila_densa.html
[33] https://ambientes.ambientebrasil.com.br/natural/regioes_fitoecologicas/regioes_fitoecologicas_-_floresta_ombrofila_densa.html
[34] http://scielo.iec.gov.br/pdf/bmpegcn/v4n1/v4n1a02.pdf

### 2.1.6 Social Parameters (G1.3)

***The Madeira Microregion and deforestation factors***

The microregion Madeira is located in the mesoregion of South Amazon. It has an area of 221,036.579 km², which represents 14.07% of the total area of the State (AM). The region is home to 165,663 inhabitants (IBGE, 2010) with a population density of 0.75 hab./km², and is composed of 5 municipalities: Novo Aripuanã, Apuí, Borba, Humaitá, Manicoré.

Located in the southeastern portion of Amazonas, this region has different transportation logistics than the rest of the state and is strategically important in containing the advance of deforestation. The region is endowed with federal and state highways that constitute the main means of transportation, although the region also presents an extensive water network, only the Madeira River is widely navigable. The channel of the Madeira River, in the southeastern region of Amazonas, is cut by the Transamazon Highway (BR-230), which passes through of these municipalities[35].

The occupation of the Madeira Microregion was more intense in the 1960s with the construction of the BR-230 highway. The central axis was the occupation of the Amazon with the increase of the exploitation of natural and agricultural resources through the construction of roads aiming at exportation (MAHAR 1978; FEARNSIDE, 1986). In recent years, the Madeira micro-region has suffered from deforestation for the expansion of farming and cattle raising activities, mainly due to the advance of the Arc of deforestation towards the Amazon[36].

In 2014, Novo Aripuanã was in the top positions in the deforestation ranking in southeastern Amazonas, along with Manicoré, and Apuí[37]. Novo Aripuanã saw a jump in deforestation in mid-2019[38]. The margins of the BR-230 and AM-174, which connect Apuí to Novo Aripuanã, are formed almost entirely by a landscape of deforested areas and pastures. The AM-174 é one of the largest drivers of deforestation in southern Amazonas [39]. As soon as the soil becomes unproductive, new areas of pasture open up. The dense forest,

[35] https://1library.org/article/microrregi%C3%A3o-madeira-no-estado-do-amazonas.y816jjwz
[36] https://repositorio.unb.br/bitstream/10482/15216/1/2013_KatiaVianaCavalcante.pdf
[37] https://www.revistaterceiramargem.com/index.php/terceiramargem/article/view/318/225
[38] https://www.redalyc.org/journal/5999/599963212009/599963212009.pdf
[39] https://www.sciencedirect.com/science/article/abs/pii/S0921800918304130

which was once the basis for extractivism, is being quickly substituted with the practice of burning, the formation of pastures, and the production of cattle[40].

According to the Mapbiomas database, in 2020 about 3% of the total area of the municipality of Novo Aripuanã has already been converted to pasture, which is the predominant land use after deforestation in the region[41].

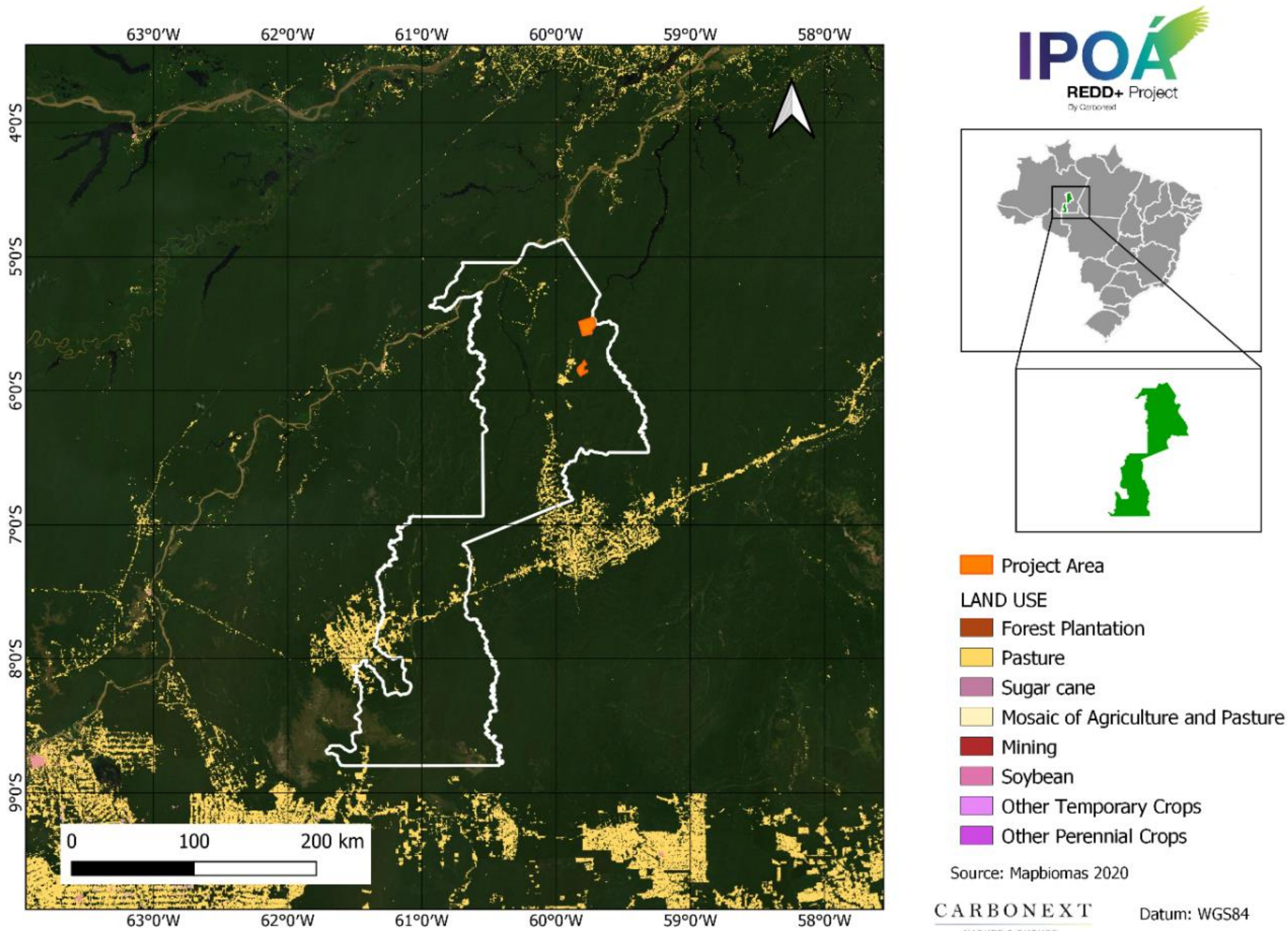


*Figure 18. Land use in the municipality of novo Aripuanã, AM, Brazil.*

[40] https://www.redalyc.org/journal/5999/599963212009/599963212009.pdf
[41] https://mapbiomas.org/download

***Municipality of Novo Aripuanã***

The Novo Aripuanã municipality is made up of territory separated from Borba and Manicoré, so its history is closely linked to those municipalities. Its territory was inhabited by the indigenous ethnic groups Toras, Barés, Muras, Urupás, Araras Indians, among others (IBGE,2012).

In 2021, its population was 26,443 people with population density reached 0.52 per km² (IBGE,2010;2021). Most of the population in this municipality is recognized as pardo/mixed (80%), followed by white (12,4%) and black (5,5%) races. About 66% of the population lives in the urban area, while the remaining 34% is in the urban area. Regarding to the distribution of inhabitants by gender, 53% are men and 47% are women (IBGE,2010).

*Figure 19. Municipality of Novo Aripuanã, AM, Brazil[42].*

***Healthy***

The municipality of Novo Aripuanã has seven health care facilities, all of which are public facilities that are part of SUS (Brazilian Unique Health System), with approximately 3.26 units for 10,000 inhabitants. Hospitalizations happen mainly due to problems in the genitourinary system (IBGE,2009;2010).

[42] https://ibge.gov.br/cidades-e-estados/am/novo-aripuana.html

The basic sanitation service is provided only for water supply and is carried out exclusively by a state government entity. Its situation is classified as +inadequate for the more than 6,000 households served (IBGE,2017).

In 2020, the infant mortality rate in the municipality reached 8.13 deaths per thousand live births, a considerable drop aiming at the previous year's rate that reached 35.17 (IBGE,2020).

***Education***

In terms of quality of education, the municipality has the largest number of schools focused on elementary and middle school with 95% concentrated in this segment and the remaining 5% for high school (IBGE,2020). In educational level, most of the population falls into no education or incomplete elementary school (71,8%). Only about 0,2% of the population has completed higher education (IBGE,2010).

The Basic Education Development Index (IDEB) for elementary education in the city was 3.9, in which the city ranks 35th in the state and performs below the state average of 4.5 (IBGE, 2019).

***Land use and economic activities***

Agriculture, livestock, services, and industry are the main economic activities in Novo Aripuanã, where the municipality has a Municipal Human Development Index (HDI) of 0.554. The economy is concentrated in agricultural and livestock establishments where they are managed by individual producers (71,4%) with collective title deed (93,4%) and the service sector activities are mainly linked to public administration, agriculture, and education (IBGE,2010;2017). The per capita GDP of the municipality is R$ 8,263.52 (IBGE,2019).

Land use is divided into four segments: farming, pasture, and forest areas. Farming areas are subdivided into two groups of uses: temporary (44,9%) and permanent (55,1%). While pasture areas are subdivided into three conditions: natural (43,7%), good condition (52,6%) or poor condition (3,7%). The most significant agricultural species for the region are cassava, watermelon, and banana, while in the forestry segment the highlight is açaí. Of the livestock activities, cattle and chicken raising are the most significant (IBGE,2017).

The forest areas present in the municipality are classified as natural areas (38%) or areas designated for permanent preservation or legal reserve (62%). At the same time, there are 3 local establishments with areas of agroforestry systems with forest species for crops and grazing by animals (IBGE,2017).

***Relevant historical conditions***

The history of Novo Aripuanã dates back to the 17th century, where the first recorded expeditions aiming to explore the Madeira River were in early 1617, with an accentuation from 1637 onwards. Among these expeditions was that of Pedro Teixeira.

The village of Borba was founded in 1728, being the first to be established in the Upper Madeira region. With the creation of the village, the economy of the region began to be boosted, with the cultivation of tobacco and coffee, and these products began to be exported to Belém. During all this time, the territory that currently makes up the municipality of Novo Aripuanã belonged to the village of Borba.

### 2.1.7 Project Zone Map (G1.4-7, G1.13, CM1.2, B1.2)

The IPOÁ Project Zone was delimited by the Project Area and a supplementary area where additional project activities will be undertaken. Thus, the Project Zone includes the communities of Vila Batista, Nova Esperança and Vale da Benção, identified in Section *2.1.9 Stakeholder Descriptions*, whose population will benefit from the project activities.

The high conservation value (HCV) areas (identified in Sections *4.1.3 High Conservation Values* and 5*.1.2 High Conservation Values*) area are delimited by the Project Area, with the exception of the HCV 5 and HCV 6 areas which are located in the communities listed above and indicated on the map.

No negative offsite biodiversity impacts (Section *5.3 Offsite Biodiversity Impacts*) or negative climate impacts resulting from the implementation of the project's activities are expected. There are also no direct impacts to other stakeholders from project activities (Section *4.3 Other Stakeholder Impacts*).

The areas of the Project Zone that are outside the property of the proponent are not in private areas registered in the Land Management System (SIGEF) maintained by the National Institute for Colonization and Agrarian Reform (INCRA), the Brazilian federal agency whose objective is to maintain the national registry of rural properties in the country. The areas are also not located in Federal Conservation Units or Indigenous Territories, as described in Section *2.5.1 Statutory and Customary Property Rights*.

The project zone map is seen below, in compliance with CCB Standards requirements.

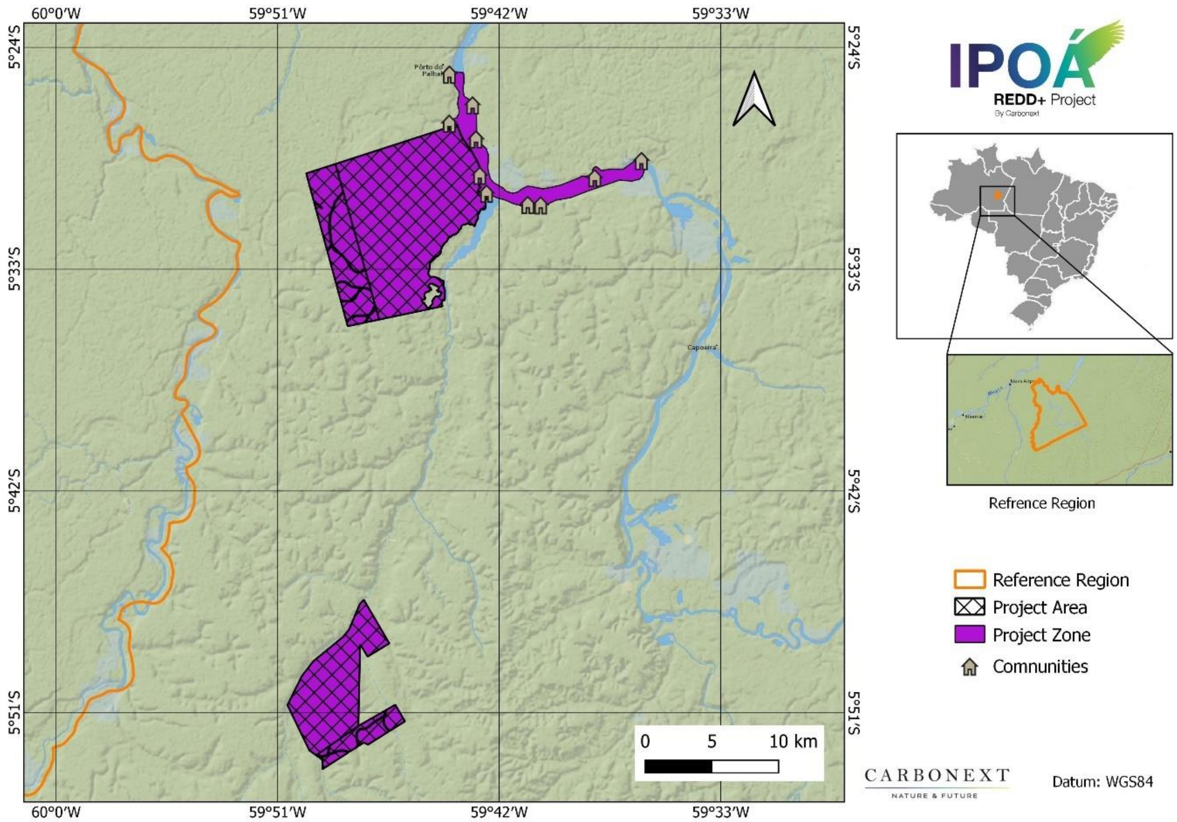


Figure 20. Project Zone Map.

### 2.1.8 Stakeholder Identification (G1.5)

Once the project area and its reference region had been identified, we sought to understand more about which stakeholders would be located in or around the project area, or even within the municipality where it is located.

The identification of the project's stakeholders was based on secondary information and, later, on interviews conducted in the field on February and March, 2022. The first step to identify the stakeholders was through the help of remote sensing software, with the projection of the project area in conjunction with cartographic databases, in order to verify possible overlaps between the project area and other destined or non-destined areas and the shared limits between areas. Then, following the recommendation for non-permanence risk analysis (AFOLU Non-Permanence Risk Tool V4.0), the surrounding region was established at 20 km from the project area boundary, where it was possible to identify existing infrastructure that could indicate human occupation in the surrounding area.

From this, it was possible to identify some focal groups that might be affected directly (being ascertained) and indirectly (positively or negatively) by the project. The general objective of this identification was to generate a bridge between organizations, groups and individuals with visions and interests; and to cover the most relevant aspects of the project, we invited them to participate in the decision making.

By satellite images and local visits was identified a community called Vila Batista (1km from PA), which were visited in order to understand the profile of the beneficiary families and start the process of collecting information and community needs for the elaboration of the activity plans. Also were identify the communities of Nova Esperança (1 km from PA) and Vale da Benção (8 km from PA) were identified and visited to identify the needs of the communities through interviews. The geographical limits of the Project Zone were defined according to the locations of the communities Vila Batista, Nova Esperança and Vale da Benção, as shown in *2.1.7 Project Zone Map.*

The identification of the stakeholders was done according to the characteristics of each stakeholder and focus groups according to the role and geographic distribution, as well as the facilities to define a time and place for the consultation, according to table below. In the next section, the stakeholders identified in the process of mapping for both the reference area and the project area, as well as the information derived from interviews with the social professional in the area, are described.

*Table 1 - Stakeholder Identification.*

| Type | Description |
|---|---|
| I | People within the project area directly or indirectly affected by the carbon project and their representatives or advocates. |
| II | Other affected stakeholders outside the project area. |
| III | Companies/institutions operating in the project area that may be affected. |
| IV | Local policy makers and local authority representatives. |
| V | Any regional authorities. |
| VI | National government institutions or National Focal Point. |

### 2.1.9 Stakeholder Descriptions (G1.6, G1.13)

***Institutional Consultation***

The stakeholders were defined as introduced in *2.1.8 Stakeholder Identification*. Thus, the public consultation with the institutions and associations of interest was carried out between March and April 2022. To this end, e-mails with a summary of the project were sent to the institutions listed below:

*Table 2. Description of Institutional Stakeholders – State and Municipal Bodies.*

| Stakeholder | Description and Relevance to the Project | Median distance to the PAs (km) |
|---|---|---|
| Novo Aripuanã City Hall | Responsible for public management, promoting sustainable, social and economic development. | 80 |
| Municipal Secretariat of Social Assistance of Novo Aripuanã | Implements the municipality's social assistance policy, aimed at meeting the social interests and aspirations of the population in situations of social risk. | 80 |
| Municipal Secretariat of Environment of Novo Aripuanã | Responsible for carrying out activities related to environmental management, promoting environmental education, regulation, control, regularization, protection, conservation and recovery of natural resources | 80 |
| Municipal Secretary of Hab. Int. Social and Land Regularization of Novo Aripuanã | Responsible for the formulation, planning, coordination, and articulation of social interest housing and urban development policies, with the objective of promoting universal access to housing. | 80 |
| Municipal Secretariat of Education of Novo Aripuanã | Promote, implement, and maintain the education for the municipal population according to the Law of Directives and Bases of National Education (Lei nº 9.394/1996). | 80 |
| Municipal Secretariat of Agriculture and Supply in Novo Aripuanã | Promote rural development based on the search for alternatives to priority problems and on local potentialities | 80 |
| Municipal Secretariat of Health in Novo Aripuanã | Health unit that provides health care services in the locality of the Novo Aripuanã - AM district of the city. | 96 |

| Stakeholder | Description and Relevance to the Project | Median distance to the PAs (km) |
|---|---|---|
| Borba City Hall | Responsible for public management, promoting sustainable, social and economic development. | 120 |
| Borba City Councilor | | 120 |
| Sucunduri Lodge | Popular hotel and tourism business in the project region. | 20 |
| IDESAM - Institute for Conservation and Sustainable Development of the Amazon | Non-profit organization. Works to promote the valuation and sustainable management of natural resources in the Amazon, seeking alternatives for environmental conservation, social development and the mitigation of climate change. | 268 |
| IDAM - Institute of Agricultural Development and Sustainable Forestry of the State of Amazonas | Responsible for providing Technical Assistance and Extension services to farmers and educational producers in the State through participatory processes that ensure sustainability, citizenship and improvement in the quality of life. | 260 |
| ICMBio - CR1 | Management of the National Flona Aripuanã | 80 |
| ICMBio: Humaitá Integrated Management Center | Management of the Acari National Park management | 426 |
| ICMBio: Integrated Management Center | Management of the Pau-Rosa National Forest | 426 |
| FAS - Amazonas Sustainable Foundation | Acting in the Juma and Rio Madeira RDS | 264 |
| ISA - Instituto Sociambeintal | Responsible for the Setemã and Coatá-Laranjal Indigenous Lands | 257 |
| ODESPI - Organization of Development and Economic Sustainability for Indigenous Peoples | Associations for the defense of social rights: acting in causes of a social nature, such as the defense of human rights, defense of the environment, defense of ethnic minorities, etc. | 266 |

| Stakeholder | Description and Relevance to the Project | Median distance to the PAs (km) |
|---|---|---|
| General Council of the Sateré-Mawé Tribe - CGTSM | Organizational and support groups of the referred indigenous communities | 457 |
| Union of Sateré Mawé and Munduruku Indigenous Peoples | | 457 |
| Association of Sateré Mawé Indigenous Women | | 464 |
| Mura Indigenous Organization of the Municipality of Novo Aripuanã and Borba | | 494 |
| Association of Residents and Friends of the Juma Reserve | Organizational and support groups of the RDS do Juma | 80 |
| SEMA/AM - Secretary of Environment of the State of Amazonas | Manager of the Juma RDS and manager of the Madeira River RDS | 263 |
| DEMUC - Department of Climate Change and Conservation Unit Management | Responsible for UC Management in the State of AM | 263 |
| Federal University of Amazonas (UFAM) | Educational institution. Carries out research in the field of forestry engineering and sustainability. | 261 |
| Universi Amazonas State University (UEA) | | 262 |

***Community Public Consultation***

The public consultations with the communities directly involved (Project Zone) were carried out face-to-face, through the exhibition of promotional material (data-show and banner) and lectures. As mentioned above, the communities involved are described below:

- Vila Batista Community
- Nova Esperança Community
- Vale da Benção Community

Widely publicized informational meetings will be held with communities and other stakeholders to explain project activities and respond to stakeholder questions and concerns related to the project. During the course of the project, other stakeholders may be raised and consulted in the future as well. More details about community meetings and public consultation are described in section *2.3.7 Steakholder Consultations.*

### 2.1.10 Sectoral Scope and Project Type

The project corresponds to the VCS Scope VM0007

Sectoral Scope: 14 - Agriculture, Forestry, Land Use

Project Category: Avoided Unplanned Deforestation (AUD project activity)

This is a grouped project.

### 2.1.11 Project Activities and Theory of Change (G1.8)

The Theory of Change is referred to as the causal model or an evaluation tool that maps out the logical sequence of how a given service will achieve what it sets out to do. In practice, it involves describing the various steps from the idea of the goal to its execution, establishing the actions or strategies that will lead to the expected results, explaining how it can and will happen[43].

Using the available information about the previous social and environmental conditions in the project area and surrounding areas collected through the social and environmental diagnosis carried out in loco, it was possible to identify the main demands of the communities. Subsequently, these demands were analyzed by the technical team and the project activities, from which the execution would imply benefits, were outlined considering climatic, community and biodiversity factors.

In a participative process with the communities, a workshop was held to prioritize the activities [44], where in a clear and inclusive way the residents themselves draw their main demands and selected the central

[43] SBIA_Part_1.pdf (verra.org).
[44] A prioritization matrix is a tool that provides a way to rank a diverse set of items in an order of importance.

demands for the creation of short-, medium- and long-term solutions. From the technical and community perspective, the main strategies and activities needed to meet these demands were mapped out within the context of the IPOÁ Project (section *6 Appendices - Appendix 1*).

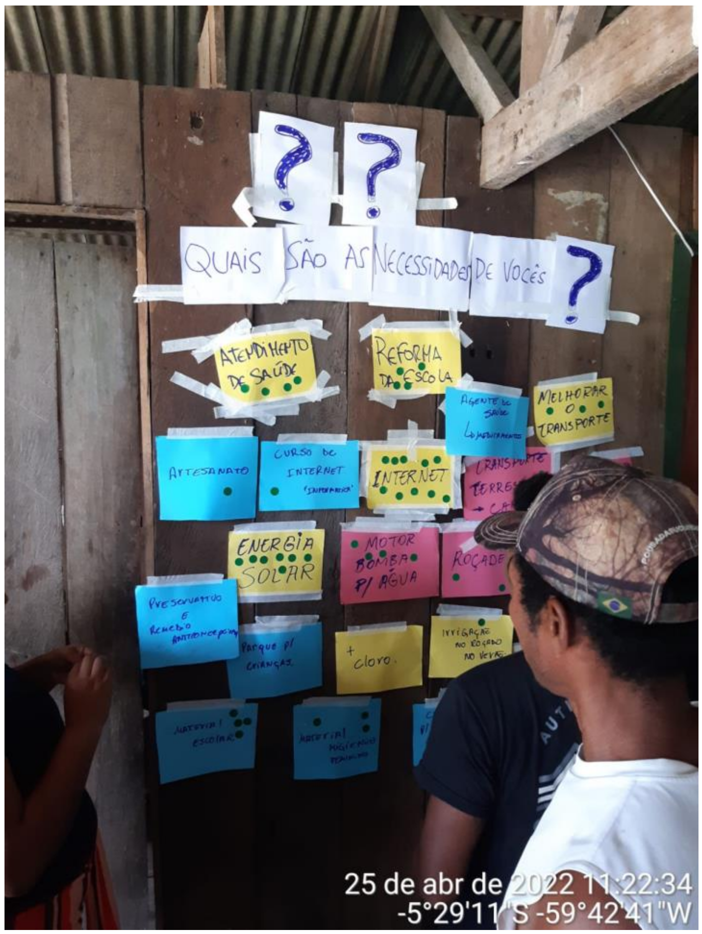


*Figure 21 - Prioritization of activities.*

Further investigations can be conducted to improve the current diagnosis and to help manage the achievement of results, considering a continuous improvement management methodology so that more robust activities, pertinent to the reality and objectives of the project, can be designed and delivered to the area using the methodology of the theory of change. Thus, the activities, consequential benefits and expected results in a sequential period have been designed and can be found in the next table, where the project's Theory of Change.

***Area Management***

The effort to manage the areas, aims to create and improve the technical, administrative, and institutional capacities that can promote new agreements and implement best practices related to forest protection and social welfare.

***Socio-Environmental Monitoring***

The objective of the monitoring strategy is to generate a balance between control and enforcement activities, with continuous diagnosis of the current state of the ecosystem and the advance or stabilization of deforestation. In addition, to monitor the socio-environmental indicators for the strengthening and continuous monitoring of the communities involved in the project.

Tools will be included to monitor all project activities and accountability, providing useful knowledge to implement actions in the medium and long term.

***Communication***

Creating effective communication and dissemination channels within communities and other stakeholders, as well as providing access to information about the concepts and components of REDD+ that will be vital to communities. This would help in creating a collective environment as well as develop trust and skills to design project strategies.

***Strategic actions***

The objective of this strategic line is to generate a balance between stakeholder priorities and potential public or private funding in project development in the region. By implementing other actions during the course of the project, which are not included in the strategic lines presented above, it is expected that there will be opportunities to include actions that will be directly related to conservation and the community that we have not considered in the current situation of the project. Such actions, in the medium and long term, will help to solve problems and offer new solutions for conservation and communities.

This strategic line should integrate all the objective lines of this project through meetings, new projects, and new financial sources, with the main parameter or indicator being the strategic alliances created with entities.

*Table 3. Theory of Change.*

| **Strategic Axis** | **Activity description** | **Expected climate, community, and/or biodiversity** | | | **Relevance to project's objectives** |
|---|---|---|---|---|---|
| | | **Outputs (short term)** | **Outcomes (medium term)** | **Impacts (long term)** | |
| **Area Management** | Manage the natural resource economy during the implementation of project activities.<br><br>Provide and encourage participation in events, projects and training sessions or workshops with the multiple stakeholders operating within the Reference Region and its surrounding area. | Increase the number of resources for implementation of activities.<br><br>Create and generate strategic alliances with entities interested in forest conservation.<br><br>Identify conflicts and overlaps of land use within the project area and collect socially based information.<br><br>Develop management guidelines for conservation of the project area. | Strengthen the area administration, with technical and administrative capacity building institutional capacities to promote new agreements and implement better practices related to the forest protection.<br><br>Creation of project strategies for conflict resolution on land use change and deforestation. | Generate actions that help resolve socio-environmental conflicts and promote the protection and maintenance of the ecosystem.<br><br>Self-management and sustainability for the implementation of project activities. | *Climate:* Reduce threats to the forest through actions to avoid deforestation.<br><br>*Community*: Improve the well-being of communities associated with the project through management of ecosystems and their associated services.<br><br>*Biodiversity*: Encourage harmonious integration between conservation and human development. |

| Strategic Axis | Activity description | Expected climate, community, and/or biodiversity | | | Relevance to project's objectives |
|---|---|---|---|---|---|
| | | **Outputs (short term)** | **Outcomes (medium term)** | **Impacts (long term)** | |
|  **Socio-Environmental Monitoring** | Generate a balance between control and inspection activities, with diagnosis of the current state of the diagnosis of the current state of the ecosystem, to assist in decision making.<br><br>Create monitoring procedures of socio-environmental indicators for the strengthening and continuous follow-up of the communities involved in the project. | Develop tools and plans for monitoring the project area and activities, as well as their implementation and execution of the plans and activities. | Adapt new tools, technologies, and the background knowledge used to implement an appropriate mitigation and adaptation plan for mid- and long-term actions. | Increasing capacity for local territorial control, governance and natural resource management. | *Climate*: The diagnosis of the current state of the ecosystem and, through actions that allow an orderly and planned management against deforestation, the maintenance of the vegetation cover and biodiversity present in the region.<br><br>*Community:* Community monitoring as a tool for planning social actions that can strengthen communities.<br><br>*Biodiversity*: Monitor and inform about the presence or absence of fauna and flora species that are considered of high conservation value in the ecosystem. |

| Strategic Axis | Activity description | Expected climate, community, and/or biodiversity | | | Relevance to project's objectives |
|---|---|---|---|---|---|
| | | **Outputs (short term)** | **Outcomes (medium term)** | **Impacts (long term)** | |
|  **Communication** | Telecommunication and technology Connectivity and Accessibility. | Installation, maintenance and facilitation of internet access.<br><br>Create communication channels between project implementers and the communities involved in the project.<br><br>Design and Implementation strategies to disseminate information about the project and sensitize the community on the conservation and management of the forest. | Encourage effective and voluntary participation of communities in the implementation of monitoring and other project activities.<br><br>Ensure that communities have transparent access to timely project-related information on platforms or in the media. | Increase transparency and access to information for all stakeholders.<br><br>Increase capacity for integration of all sectors under a common goal, conservation. | *Climate*: Aggregate knowledge about the importance of the ecosystem and prevent changes in land use.<br><br>*Community*: Broaden the understanding of the communities associated with the project through access to information about REDD+ concepts and activities; Creation of a fixed communication channel for the transmission of information, problem solving and reporting.<br><br>*Biodiversity*: Improve community surveillance schemes and monitoring of high conservation value species. |

| Strategic Axis | Activity description | Expected climate, community, and/or biodiversity | | | Relevance to project's objectives |
|---|---|---|---|---|---|
| | | **Outputs (short term)** | **Outcomes (medium term)** | **Impacts (long term)** | |
|  **Strategic actions** | Build technical, administrative, and institutional to help promote stakeholder empowerment, including the formulation and adoption of administrative measures. | Generate a balance of priorities and funding to develop activities that will help manage the project area and improve the well-being of communities.<br><br>Follow closely the process of design, implementation and strengthening of laws decrees and agreements established with public and private national institutions. | Integrate wherever possible the parties involved in the REDD+ project through meetings and new projects. And seek partnerships with agencies and experts for the improvement of actions in the REDD+ project | Create actions that will help to resolve socio-environmental resolve socio-environmental conflicts and promote the protection and production of the ecosystem.<br><br>Self-management and sustainability are essential for the implementation of project activities. | *Climate*: New projects, actions and alliances aimed at reducing deforestation and institutional barriers.<br><br>*Community*: Whenever possible, involve small businesses in building solutions seeking gender equality and creating local solutions, jobs through the implementation of project activities.<br><br>*Biodiversity*: Promote the protection and conservation of ecosystems and their services, diversifying sources of income in the territory. |

### 2.1.12 Sustainable Development

The State of Amazonas has a State Commission for Ecological-Economic Zoning (ZEE)[45], created by State Decree 23.477/2003, modified by Decree 24.048/2004. The first ZEE initiative of the State of Amazonas, called Macrozoneamento Ecológico-Econômico do Estado do Amazonas (Ecological-Economic Macro-Zoning of the State of Amazonas), was instituted by the state law no. 3.417, of July 31, 2009, [46]. The creation of the Legal Amazon Ecological-Economic Macro-Zoning - MacroZEE was instituted by Decree no. 4.297, of July 10, 2002, and altered by Decree no. 7.378, of December 1, 2010, [47].

The MacroZEE[48] dialogs and maintains a two-way relationship with the main initiatives that are already transforming the Amazon and have strong political and social legitimacy, generally referenced in the Sustainable Amazon Plan (PAS), such as, for example, the Action Plan for Prevention and Control of Deforestation in the Amazon (PPCDAm), the Regional Development Policies (PNDR) and Defense Policies (PND), the National Plan on Climate Change[49] (PNMC), the Territories of Citizenship Program, the Regional Development Plans, such as the Marajó, BR-163 and Xingu Plans, the Legal Amazon Land Regularization Program (Terra Legal), the Public Forest Management Law (federal law nº 11.284/2006), the Community and Family Forest Management Program (federal decree nº 6.874/2009) and the Regional Development Plan for the Amazon (PRDA).

Because of the increase in deforestation in the Legal Amazon, the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm) came into effect, initiating mitigation actions and ongoing actions to reduce deforestation. The IPOÁ REDD+ project is in line with the main premises of the PPCDAm[50]. In order to achieve this goal, it will be necessary to join governmental initiatives with independent actions, such as those proposed under this project, but it is not mandatory for the IPOÁ project to report to the Brazilian government on the actions taken on the national SDGs.

At the international level, the IPOÁ REDD+ Project is in line with the logic of the environmental priorities set by the Brazilian Federal Administration at COP 14 in December 2008, where a 70% deforestation by the year 2018 was declared, and then further targets of zero illegal deforestation by 2030, and offsetting

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[45]O Zoneamento Ecologico Economico na Amazonia Legal.pdf (zee.ma.gov.br)
[46] Lei nº 3.417 de 31/07/2009 (normasbrasil.com.br).
[47] Decreto n° 7 .378 _2_de 01/12/2010 (abce.org.br).
[48] Referências (mma.gov.br).
[49] Federal Government Site: https://www.gov.br/agricultura/pt-br/acesso-a-informacao/acoes-e-programas/ppa/plano-plurianual-ppa-2016-2019-1/relatorio_avaliacao_programa_2050-mudanca_do_clima.pdf.
[50] PPCDAm (mma.gov.br). Acessado 12/05/2022.

greenhouse gas emissions from illegal sources of vegetation removal were declared. The latter are elements of the Brazilian NDC, which the country intends to adopt more firmly in the framework of the Paris Climate Accords (COP-21)[51]. Recently Brazil made official at COP 26, in November 2021, that the country's goal is to eliminate deforestation by 2028, increase to 50% the goal of reducing greenhouse gas emissions by 2030, and reduce by 30% its methane emissions in the atmosphere[52]. At this COP, the Glascow Climate Pact was signed, which reinforced the parties' commitments to keep warming below the 1.5°C limit [53]. Another important decision taken was the authorization of the global carbon market, which means that countries will be able to exchange carbon credits among themselves, favoring Brazil because of the Amazon Forest (IDESAM, 2021). To achieve these goals, an essential component will be to ally governmental initiatives with independent actions such as the one proposed under this project. Going into more detail, the IPOÁ project is aligned with Brazil's commitments to the UN Sustainable Development Goals (SDGs).

The SDGs - Sustainable Development Goals are a campaign of the UN, the United Nations Organization, to promote positive changes in the world of the future. The SDGs seek to promote urgent measures to ensure the 17 goals proposed in this UN sustainable agenda by conserving natural resources, combating climate change, and adopting more sustainable production and consumption practices. The SDGs are central to Brazil's 2030 Agenda and poverty can be the main mission for all internal or external initiatives, programs, projects and activities, including REDD+ projects.

In 2015, the conclusion of the negotiations on the 2030 Agenda culminated in an ambitious result for Brazil, which was the proposal of 17 Sustainable Development Goals; 231 Sustainable Development Global Indicators to monitor and measure progress in the implementation of the SDGs; and 169 targets, representing the central axis of the 2030 Agenda, which will guide actions in the three dimensions of sustainable development[54]. Brazil has highlighted this as a great opportunity to eradicate poverty that may be within the mandate of this new Agenda.

In 2017, the Brazilian government's first "Voluntary National Reviews" focusing on institutional frameworks were put in place to incorporate the Sustainable Development Goals into public policy. In terms of conservation and sustainable use of natural resources, environmental sustainability is not only based on reducing damage to the environment, but also on the efficient management of ecosystem services that

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[51] Source MMA: http://redd.mma.gov.br/pt/redd-e-a-indc-brasileira.
[52] Source FGV: https://portal.fgv.br/artigos/quem-e-brasil-cop-26. Access 12/05/2022.
[53] Source IDESAM (2021): https://idesam.org/opiniao/cop26-entre-erros-e-acertos/.
[54] ODS – Metas Nacionais dos Objetivos de Desenvolvimento Sustentável (portalods.com.br). Access 04/05/2022.

favor human development through increasing economic opportunities as well as social and ecological resilience.

After taking into account the national sustainable development targets and indicators, the IPOÁ REDD+ Project has aligned its activities in such a way that, at the local level, progress is being made towards achieving the targets of each of the following SDGs of the 2030 Agenda[55]: Health and well-being, safe working environment, gender equality, climate action, life in terrestrial ecosystems, among others. These SDGs and the indicators that the project seeks to target with its activities are presented in the table below.

[55] Pacto Global.

*Table 4. Goals and indicators of sustainable development considered by the project.*

| Sustainable development goal (SDG) | SDG Indicator | Indicator type | Unit | Value/ adjusted unit |
|---|---|---|---|---|
| | 3.8 Achieve universal health coverage, including financial risk protection, access to quality essential health services, and access to safe, effective, quality, and affordable essential medicines and vaccines for all. | Health | N° of healthcare users served | Proportion of people served with improved logistics |
| | 4.1 Ensure that all girls and boys complete free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes.<br><br>4.a Build and improve physical facilities for education that are appropriate for children and sensitive to disability and gender, and that provide safe and non-violent, inclusive and effective learning environments for all. | Social | Primary and secondary school completion rate.<br><br>N° of items that the school has access to (electricity, internet, etc) | In development |
| | 5.5 Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in political, economic, and public life.<br><br>5.b Increase the use of basic technologies, in particular information and communication technologies, to promote women's empowerment.<br><br>5.6 Ensure universal access to sexual and reproductive health and reproductive rights, as agreed in accordance with the Program of Action of the International Conference on Population and Development and the | Equality | Proportion of women participating in meetings and decision-making.<br><br>Proportion of people who own a cell phone, by gender.<br><br>Proportion of women aged 15-49 who make informed decisions about | Increase proportion of women participating in meetings and decision making.<br><br>Ensure that all genders have access to technologies, such as the Internet. |

| Sustainable development goal (SDG) | SDG Indicator | Indicator type | Unit | Value/ adjusted unit |
|---|---|---|---|---|
| | Beijing Platform for Action and the outcome documents of its review conferences | | their sexual relations, contraceptive use, and reproductive health care | Track the number of women who use contraception. |
|   | 8.8 Protect labor rights and promote safe and secure work environments for all workers, including migrant workers, in particular migrant women, and people in precarious employment. | Economic and Health | Accounting for work incidents and accidents. | Decrease the work accident rate |
|   | 13.2 Integrate climate change measures into national policies, strategies and planning | Environment | Tons of carbono dioxide | Reduce the number of tons of carbon dioxide emissions |
|   | 15.1 Ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in accordance with obligations under international agreements.<br><br>15.2 Promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests, and substantially increase afforestation and reforestation globally. | Environment | % Area of forest under sustainable management.<br><br>Total conserved area (ha) | Forest area as a proportion of the total Project IPOÁ area: expected to be 17,386 hectares; Red List Index of species; Number of biological and botanical activities implemented. |

| Sustainable development goal (SDG) | SDG Indicator | Indicator type | Unit | Value/ adjusted unit |
|---|---|---|---|---|
| | 15.5 Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity, and, by 2020, protect and prevent the extinction of endangered species. | | Number of red list species be nefited. | |
|   | 17.8 Fully operationalize the Technology Bank and the Science, Technology and Innovation Capacity Building Facility for Least Developed Countries by 2017, and increase the use of enabling technologies, in particular information and communication Technologies. | Economic and social | Proportion of individuals using the Internet | Guarantee the right to use technology of the communities |

In addition to the international laws and regulations already mentioned, the present project is also in accordance with: Convention on the International Trade in Endangered Species of Wild Flora and Fauna, 1973; Cartagena Protocol on Biosafety to the Convention on Biological Diversity, 2000; International Tropical Timber Agreement, 1994; United Nations Convention on Biological Diversity, 1992; United Nations Framework Convention on Climate Change, 1992; United Nations Declaration of the Rights of indigenous Peoples, 2007; The Kyoto Protocol to the Convention on Climate Change, 1997.

The present project is developed in accordance with the Cancun Safeguards for REDD+ activities, therefore following all its clauses, which relate to both biodiversity and forest conservation, as well as indigenous and traditional communities' inclusion, interests, and rights, as follows below. These clauses have also been adopted by Brazil, as reported by the Environment Ministry (MMA)[56]:

> *"When undertaking [REDD+] activities the following safeguards should be promoted and supported: (a) That actions complement or are consistent with the objectives of national forest programmers and relevant international conventions and agreements; (b)Transparent and effective national forest governance structures, taking into account national legislation and sovereignty; (c)Respect for the knowledge and rights of indigenous peoples and members of local communities, by taking into account relevant international obligations, national circumstances and laws, and noting that the United Nations General Assembly has adopted the United Nations Declaration on the Rights of Indigenous Peoples; (d) The full and effective participation of relevant stakeholders, in particular indigenous peoples and local communities, in the [REDD+] actions ; (e) That actions are consistent with the conservation of natural forests and biological diversity, ensuring that the [REDD+] actions are not used for the conversion of natural forests, but are instead used to incentivize the protection and conservation of natural forests and their ecosystem services, and to enhance other social and environmental benefits; (f) Actions to address the risks of reversals; (g) Actions to reduce displacement of emissions."*[57]

[56] Source: http://redd.mma.gov.br/images/publicacoes/salvaguardas_1sumario.pdf
[57] Source: UNFCCC Decision 1/CP.16, Annex I, paragraph 2, available at: https://www.un-redd.org/glossary/cancun-safeguards

### 2.1.13 Implementation Schedule (G1.9)

During the project lifetime, from the development to implementation, some important dates and milestones will take place, such as introductory meeting dates, start and end dates of each project activity, start and end dates of the GHG accounting period, monitoring program, verification program, etc. The table below presents some of these milestones, as others are still being defined.

*Table 5. Implementation Schedule*

| Milestone | Date | Strategic lines | Milestone(s) in the project's development and implementation |
|---|---|---|---|
| 1 | December, 2021 | Area Management | Closing the contract for the start of the REDD+ project. |
| 2 | March, 2022 | Socio-environmental Monitoring | Social mapping of the surrounding communities. |
| 3 | March, 2022 | Socio-environmental Monitoring | Knowledge visit to the Project Zone to identify families and present the project. Social research and first contacts through the technical manager. |
| 4 | March, 2022 | Strategic actions | Presentation of the project to the community leaders of: Vila Batista, Nova Esperança and Vale da Benção. |
| 5 | April, 2022 | Area Management | Presentation of the project to the residents, application of socioeconomic questionnaires. |
| 6 | April, 2022 | Socio-environmental Monitoring | Delivery of oral hygiene kits, pads, and soap. |
| 7 | June, 2022 | Environmental monitoring | Maps monitoring the degradation and deforestation history of the region to build the project's baseline. |
| 8 | July, 2022 | Area Management | Visit to Fazenda Campos, visit by the proponent to talk to community leaders. |

| | | | |
|---|---|---|---|
| 9 | July, 2022 | Comunicação | Creation of a communication channel with the community of Vila Batista. |
| 10 | Throughout 2022 | Area Management | Assistance in education-related bureaucratic issues with government agencies. |
| 11 | Until 2024 | Environmental monitoring | Fauna, and flora inventory. |
| 12 | At least once before every verification | Environmental monitoring | Update, monitoring, and mapping fauna parameters. |
| 13 | At least once before every verification | Fire prevention | Fire monitoring via satellite. |
| 14 | One year prior to every verification | Area management | Monitoring of deforestation and quantification of emission reductions |
| 15 | At least one time prior to Every verification | Area management | Monitoring of community impact and biodiversity indicators |
| 16 | At least every one years from 2022 onwards | Area management | Verifications |

### 2.1.14 Project Start Date

***AUD component:***

The activities that led to the generation of GHG emission reductions began in March 17, 2022.

The project proponent (landowner) acquired the Campos property on August 21, 2019, and the Maanaim property on October 13, 2019, with the objective of carbon credit generation and forest management, as shown in the purchase and sale agreements.

In December 2021, the landowner and Carbonext (Project Developer) signed an agreement for the development and implementation of the REDD+ project.

After the recognition and analysis step, Carbonext started the implementation of project activities in the field on March 17, 2022. This event is considered as the Project Start Date.

On this occasion, an experienced social technician from Carbonext, accompanied by a local consultant, visited the Project Area, carrying out public consultations with the communities in the Project Zone, presenting the principles and benefits generated by the REDD+ project. The presentation lecture given to local inhabitants is considered the first activity to generate reductions in the project's GHG emissions, as it made communities aware of preserving the forest and encouraged them to monitor possible illegal deforestation.

***APD component:***

Regarding APD activities, May 3rd, 2022, was the date that the project owner submitted a request to the official organ IPAAM, to obtain permission to deforest a legal proportion of the Maanaim and Campos properties in the Project Area (PA). The documents required and submitted to IPAAM for filing the request are available for auditing. This planned deforestation activity was necessary because, from that moment on, maintaining forests without carbon credits within the PA was no longer financially viable for the project owner, who initially intended to extract the timber to sell and evaluate a possible permanent agricultural activity after deforestation. It is clarified that the approved avoided planned deforestation request is the most advanced documentation that the project owner was able to obtain without in fact deforesting the property. If a deforestation permit had been obtained, the project owner would then have been obliged to carry out the planned deforestation, or face complications in the IPAAM organ.

### 2.1.15 Benefits Assessment and Crediting Period (G1.9)

The crediting period is from March 17, 2022, to March 16, 2052, therefore comprising a duration of 30 years.

### 2.1.16 Differences in Assessment/Project Crediting Periods (G1.9)

There is no difference between the GHG emission accounting period and the community and biodiversity benefits assessment periods.

### 2.1.17 Estimated GHG Emission Reductions or Removals

The estimated total net GHG emission reductions or removals by the Project Activity over 30 years is 3,122,412 $tCO_2e$. This represents an average of 104,080.42 $tCO_2e$ per year, summing all types of avoided deforestation (AUD and APD).

*Table 6. Total ex ante estimated actual net carbon stock changes and emissions of non-$CO_2$ gases in the project area.*

| Year | Estimated GHG emission reductions or removals ($tCO_2e$) |
|---|---|
| 2022 | 413,723 |
| 2023 | 717,058 |
| 2024 | 702,100 |
| 2025 | 690,885 |
| 2026 | 168,062 |
| 2027 | 16,584 |
| 2028 | 16,544 |
| 2029 | 16,504 |
| 2030 | 16,464 |
| 2031 | 16,424 |
| 2032 | 16,384 |
| 2033 | 16,344 |
| 2034 | 16,304 |
| 2035 | 16,265 |
| 2036 | 16,225 |
| 2037 | 16,186 |
| 2038 | 16,147 |
| 2039 | 16,107 |
| 2040 | 16,068 |
| 2041 | 16,029 |
| 2042 | 15,990 |

| Year | Estimated GHG emission reductions or removals ($tCO_2e$) |
|---|---|
| 2043 | 15,951 |
| 2044 | 15,913 |
| 2045 | 15,874 |
| 2046 | 22,018 |
| 2047 | 21,964 |
| 2048 | 15,729 |
| 2049 | 15,690 |
| 2050 | 15,652 |
| 2051 | 15,614 |
| 2052 | 15,609 |
| Total estimated ERs | 3,122,412 |
| Total number of crediting years | 30 |
| Average annual ERs | 104,080.42 |

### 2.1.18 Risks to the Project (G1.10)

Some natural and human-induced risks are expected to occur to the climate, community, and biodiversity benefits during the project lifetime. They each have corresponding measures taken to mitigate the risk, as presented in the table below. This item is to be revised along the project lifetime, at each monitoring period, in accordance with renewed assessments of each factor.

Further risks to the project were analysed using the guidelines of the Non-Permanence Risk Tool, version 4.0, and the results are presented in the IPOÁ Non-Permanence Risk Report. The Adaptive Management Plan (section *2.3.8 Continued Consultation and Adaptive Management*) identifies, assesses, and creates a mitigation plan to address them.

*Table 7 - Potential risks to the IPOÁ REDD+ Project.*

| Identify Risk | Potential impact of risk on climate, community and/or biodiversity benefits | Actions needed and designed to mitigate the risk |
|---|---|---|
| Disputes over access/use rights (or overlapping rights) | Disputes can compromise the continuity of the project and consequently the benefits to the climate, biodiversity and communities. | Asset control and judicial monitoring of possible claims. |
| Changes in policies and laws governing external actors | Changes can compromise the continuity of the project and consequently the benefits to the climate, biodiversity and communities. | Legal monitoring and adaptation to new rules |
| Increase of fires | Carbon emissions, biodiversity loss and risks for the communities. | In the Amazon biome, anthropogenic forest burning is common, as landowners use fire to "clean" their land[58]. However, after analysis using geographic information systems, we did not identify such a risk in the Project Area.<br>Thus, we consider this a risk because fire can occur due to natural factors or even anthropic action, even though it is a very low risk. And to maintain the low probability of the risk occurring, we propose the constant monitoring by satellite images and the awareness of residents in the communities about the risks of fires and the procedure of communication to Carbonext or responsible agencies if this occurs. |
| Deforestation/Degradation | Deforestation or degradation of forest cover. | Remote satellite monitoring (Carbonext's MonitoraCarbon System) to identify and act on deforestation; on-site visits and community environmental education to combat degradation; measures for filing a report of occurrences and/or recovery of the area if necessary |

[58] Queimadas e desmatamento estão relacionados na Amazônia; entenda | Natureza | G1 (globo.com)

### 2.1.19 Benefit Permanence (G1.11)

The community activities are designed to transform local economies into activities based on the low carbon economy that become perennial and financially attractive in the long term.

The project's priority will ensure the substitution of pastureland use, the main cause of deforestation in the region, with alternative activities with low environmental impact, in order to maintain the benefits to the climate, communities and biodiversity.

Such activities are all designed to be managed in the future by the communities themselves, without the need for external interventions. This item may be revised considering the manifestations of stakeholders in the public consultation in progress.

### 2.1.20 Financial Sustainability (G1.12)

There are no economic activities being carried out by the proponent in the Project Area. The project implementation annual costs are estimated at R$202,136.90 throughout the project.

As of 2023, costs are expected to be covered by the revenues obtained from the sales of carbon credits. The Project Proponent have sufficient cash funds to keep the project until breakeven. This financial analysis may change due to needs identified in the ongoing public consultation.

### 2.1.21 Grouped Projects

This is a grouped project, and it was designed as an Avoided Unplanned Deforestation and Avoided Planned Deforestation, applying VM0007 methodology, Version 1.6, dated 08 September 2020.

In this sense, this project activity is designed to include more than one “project activity instance”, such as the accession of new communities or landowners to the project across its lifetime. Thus, this grouped project is designed to allow the expansion of the project activity, subsequent to project validation.

**1) Eligibility Criteria for Grouped Projects (G1.14)**

For this grouped project, the following set of eligibility criteria for the inclusion of new project activity instances has been defined, which is applicable for VM0007 REDD+ AUD activities, and the geographic area demarcated by the Reference Region:

This grouped project has one clearly defined geographic area within which project activity instances may be developed, which is defined using geodesic polygons, corresponding to the current Reference Region. The determination of baseline scenario and demonstration of additionality were based upon the initial project activity instances (2 farms, Maanaim and Campos), that are presented in this CCB-PD for validation. For inclusion of new geographic areas, it will be demonstrated that such areas are subject to the same baseline scenario and rationale for the demonstration of additionality as the geographic area that does include initial project activity instances.

A single baseline scenario is determined for the entire designated geographic area (Reference Region), in accordance with VM0007 methodology. The additionality of the initial project activity instances was demonstrated for each designated geographic area, in accordance with the methodology applied to the project. All factors relevant to the determination of the baseline scenario or demonstration of additionality (i.e., common practice; laws, statutes, regulatory frameworks or policies relevant to demonstration of regulatory surplus; historical deforestation rates) were assessed across the grouped project geographic area and respective Reference Region.

The project proponent has defined the capacity limit for this project activity in terms of the Reference Region. If exceptionally a new instance is located outside the Reference Region, it will be guaranteed that all limit premises will respect the same conditions of similarity of historical deforestation rates as applied in the Reference Region to the initial project instances.

**2) Scalability Limits for the Grouped Projects (G1.15)**

The project scalability is delimited by the geographic area, in this project defined as the reference region. The geographical and jurisdictional limits for this expansion are predicted to be within the municipalities of Novo Aripuanã, however this may change at the monitoring events. If exceptionally a new instance is located outside the Reference Region, it will be guaranteed that all limit premises will respect the same conditions of similarity of historical deforestation rates as applied in the Reference Region to the initial project instances. Also, the eligibility criteria described in the section above shall be met. For the addition of new instances to the project, the interested institutions must present a financial plan as well as an implementation schedule to guarantee the permanence of the project and the fulfillment of its objectives in accordance with the CCB standard.

**3) Risk Mitigation Approach for Grouped Projects (G1.15)**

The risks associated with the non-continuity of benefits are reduced since the project activities proposed for the new instances will be designed and led in partnership with the owners of these new instances, who are directly involved in the management and administration of the areas. Carbonext's technical team, which has the technical capacity to determine the project strategies and activities, in addition to offering technical support to the landowners for the correct execution of the activities and monitoring will develop this plan in case the project expansion occurs. Thus, Carbonext's continuous technical and scientific monitoring generates a support framework for monitoring, reporting and verification issues, which ensures the quality of the work and a continuous process of adjustment and improvement.

## 2.2 Without-project Land Use Scenario and Additionality

### 2.2.1 Land Use Scenarios without the Project (G2.1)

The present project activity has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction. Instead, this project only foresees the reduction of emissions that would occur in the Project Area.

The baseline scenario is the same as the conditions existing prior to the project initiation. The description of scenarios is detailed in section *3.1.4 Baseline Scenario*.

### 2.2.2 Most-Likely Scenario Justification (G2.1)

The most-likely scenario to occur is planned and unplanned deforestation followed by pasture with cattle ranching.

Forest land is expected to be converted to non-forest land in the baseline scenario, under two mechanisms, or components of the baseline: Avoided Unplanned Deforestation (AUD) and Avoided

Planned Deforestation (APD). The landowners cannot afford efforts and costs to keep long term vigilance of the project boundaries to avoid unplanned land use change from uncontrolled trespassing and deforestation, without carbon credit income.

A study conducted by Soares (2022) in the neighboring municipality of Apuí identified that the municipality of Novo Aripuanã is a new pole of expansion of the agricultural frontier, following the dynamics of the search for cheap land, investment in deforestation and agricultural production.[59]

According with Map Biomas[60], pasture is the most common alternative land use in the municipality of Novo Aripuanã. In 2010, pastures covered only 1% of the municipal territory. However, in 2020 this percentage rose to 3%, totaling 132,085 hectares. This value represents 99,41% of non-natural land uses in Novo Aripuanã.

The increasing trend was also observed in the number of cattle identified in the municipality of Novo Aripuanã by the agricultural census. In 2010, the number of heads was 13,989, while in 2020 this amount increased to 42.100 [61], an increase of 300%.

The percentage of the total area of this land use class increased from 0.57% in 2010 to 1.67%, approaching the project area. More information is provided in Section 3.1.4.

### 2.2.3 Community and Biodiversity Additionality (G2.2)

As mentioned earlier in the item *2.2.2 Most-Likely Scenario Justification*, the most likely land use scenario in the absence of the project is the expansion of pasture resulting in forest deforestation for the creation of cattle ranching. According to the Forest Code (Law 12.651/2012), 80% of private properties in the Legal Amazon are defined as Legal Reserve, as described in:

> *"Article 16. The forests and other types of native vegetation, excepting those located in Areas of Permanent Preservation, as well as those not subject to the politics of restricted use or subject to specific legislation, are susceptible to suppression, as long as a portion of vegetation is preserved, as Legal Reserve, at a minimum:*
>
> *I – eighty percent (80%), in rural estates located in forest zones located in the Legal Amazon.*"

[59] https://www.teses.usp.br/teses/disponiveis/91/91131/tde-11022022-160839/publico/Pedro_Gandolfo_Soares_versao_revisada.pdf
[60] https://mapbiomas.org/
[61] https://sidra.ibge.gov.br/tabela/3939

Thus, the landowner has the right to convert forest to other diverse land uses in the proportion of 20% of his property. In the case of Manaaim and Campos ranches, in the absence of the IPOÁ REDD+ Project, the owner had demonstrated the intention to use this part of the property for cattle ranching, removing the characteristics of the local landscape and negatively impacting local biodiversity.

The removal of the forest for the implementation of ranching activities besides resulting in GHG emissions and destruction of the vegetation present, removes the habitat of several faunal populations. In addition to the space destined for cattle cultivation, in a future with ranching activities on the farms it would still be necessary to remove the forest in the direction of the Sucunduri River for the flow of the beef cattle, resulting in greater impacts to local biodiversity.

The creation of the project REDD + IPOÁ serves as an aid to the protection of biodiversity of the JUMA RDS, a neighboring conservation unit located 13km from the PA, serving as a refuge for wildlife as well as a temporary or permanent habitat for animal populations (such as birds and bats), including endemic and threatened ones, that may share the same environment. The presence of the REDD+ IPOÁ project also allows the survey and monitoring of flora and fauna species, including those at different levels of threat and endemism, enabling a better understanding of the region's scenario for conservationist decisions.

As for the social and environmental aspects, in the absence of the project, the communities of Vale da Benção, Nova Esperança and Vila Batista would continue without infrastructure support for communication, and the IPOÁ REDD+ Project is dynamizing efforts to bring internet to the region. Furthermore, planned socio-environmental actions regarding aspects of health improvement, work safety, connectivity and accessibility, educational program and training (better described in section 4.4.1 Community Monitoring Plan) will only be possible due to the presence of the IPOÁ REDD+ Project. Without the project's activities, it is expected that communities will continue to have precariousness in meeting their basic needs.

Considering that the project region is at risk of deforestation due to the cited motives above, the IPOÁ REDD+ Project climate, community and biodiversity benefits of conservation would not occur without the project's activities. The benefits include the following:

- Protection and conservation of 4,563.75 ha.
- Creation of jobs position, directly or indirectly.
- Increase of life quality and access to internet, health care and education in the communities.
- Monitoring of fauna and flora species, including endangered ones.

- Technical capacity building in different topics.

#### 2.2.4 Benefits to be used as Offsets (G2.2)

This item is not applicable, no community and biodiversity benefits will be used as offsets.

### 2.3 Stakeholder Engagement

#### 2.3.1 Stakeholder Access to Project Documents (G3.1)

Project documents, such as project description documentation and monitoring reports, will be available through the VERRA website[62] and will be made public and interested parties will be invited to review them and submit their comments. In addition, project stakeholders will be invited to participate in all public consultation periods and will receive a project summary.

All communities will have access to project information and documents pertinent to them in the local language. These documents will be given to them in an accessible and inclusive language so that they can be understood, and we will make them public in regular meetings with the communities, showing the project's progress and other questions.

Finally, we will provide a short and very illustrative copy of the Project Description that will be presented after the Validation period and registration in VERRA; of the Monitoring Reports after each verification period; and of the validation and verification processes by providing timely information on the auditor's visit, the location of the visits, the date, and how this process will occur (details in section *2.3.5 Information to Stakeholders on Validation and Verification Process*). Thus, facilitating communication between the parties.

#### 2.3.2 Dissemination of Summary Project Documents (G3.1)

The project summary documentation, including the minimum points for the overall project profile described in G1.1-9[63], has been disseminated by email to all project stakeholders and will be actively

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[62] https://registry.verra.org/
[63] https://verra.org/wp-content/uploads/2016/05/SBIA_Part_1.pdf

disseminated to the communities with distribution of the summary copy to each of them in the local language (Portuguese). The summaries of the Monitoring Reports will be disseminated in the same way after each verification period.

In addition, the summaries of the project documents will be publicly available through the VERRA website.

### 2.3.3 Informational Meetings with Stakeholders (G3.1)

Between March 14 and 21, 2022, a technical social analyst from Carbonext conducted an initial diagnosis in the vicinity of the IPOÁ project area, traveling along the Acari and Suncunduri Rivers and the Arara and Capinarana Igarapés. This trip was important to make the first contacts with the communities and the detailed description of each family present in this journey.

On March 16, 2022, in the community of Nova Esperança, an informative meeting about the IPOÁ REDD+ project was held with the residents. The social analyst who was responsible for the presentation of the project used clear communication to notify about the main points of the project and clarified the doubts raised by the inhabitants. To illustrate the project area, a banner (Figure 24) was used with a short description of the project and the contact email of Carbonext, in addition, the WhatsApp contact of the social analyst was presented to the community leader who has a cell phone. On this occasion the project was also presented to the leaders of Vila Batista and Vale da Benção, where both signed the consent form of the project.

*Figure 22 - Visit to the leaders of the communities of Vale da Benção and Vila Batista.*


*Figure 23 - Presentation of the project in the Nova Esperança community.*

Between April 23 and 25, 2022 we conducted a new visit to the project communities, and on this occasion socio-economic questionnaires were applied to families in all communities, articulation and public consultation was carried out with the residents present in the three communities involved in the IPOÁ REDD+ project: Vila Batista, Nova Esperança and Vale da Benção. During the visit, 44 people were interviewed in 6 family nuclei distributed in the three communities.

The informative meeting held on April 25th in the community of Vila Batista involved 24 participants from the three communities. This meeting was fundamental for the residents to understand more about the project, for those who were not present at the previous month's visit to get to know the project, answer the questions that the residents had, and understand and write down the demands they have. In none of the meetings so far, we have had negative answers or opinions about the project's development in the region.

The means of dissemination of the April meeting was done with the articulation executed the day before by the social analyst and the technical project analyst of Carbonext, who visited all the houses that would correspond to one of the three communities involved (Project Zone) and that were identified in March 2022, informing one by one about the meeting, in addition to the distribution of flyers for all residents presents.


*Figure 24 - Invitation flyer for community consultation meeting.*

### 2.3.4 Community Costs, Risks, and Benefits (G3.2)

The risks that were reported to the communities involved in the project were taken from section *2.1.18 Risks to the Project*, this section indicates that the risks cited were evidenced from analyses performed using the guidelines in the Non-Permanence Risk Tool, version 4.0.

During the community consultations, information about the project was provided to the communities of Vila Batista, Vale da Benção and Nova Esperança, where visual material was used to demonstrate the demarcation information and the Project Area. The explanation was delivered to the communities by a social specialist, being communicated in an accessible way, where the community members could understand the purpose of the project, and participate in identification of its potential costs, risks, and benefits.

Despite the premises that the Project Developer envisages, for example, no financial cost for the communities, and that the project activities aim to promote ecosystem services, the identification of such risks and benefits in a transparent and participatory aspect is essential to the project design.

*Table 8 - Potential risks to the IPOÁ REDD+ Project.*

| **Identify Risk** | **Potential impact of risk on climate, community and/or biodiversity benefits** | **Actions needed and designed to mitigate the risk** |
|---|---|---|
| Disputes over land rights or access/use rights | 1. Loss of potential property rights for community members.<br>2. Displacement of community by potential invaders. | Asset control and judicial monitoring of possible claims. |
| Changes in policies and laws governing external actors | Various changes in community circumstances resulting from legal changes: land tenure; resource use rights; access rights; carbon credit rights; obligations to conserve; financial penalties for legal violations, etc. | Legal monitoring and adaptation to new rules carried out by project developer legal team. |
| Natural risks including fire; plagues; ecosystem imbalances, etc. | Various impacts resulting from: fauna depletion; loss of sustenance from hunting and gathering; alterations in weather patterns and phenomena. | In the Amazon biome, anthropogenic forest burning is common, as landowners use fire to "clean" their land[64]. However, after analysis using geographic information systems, we did not identify such a risk in the Project Area. |

[64] Queimadas e desmatamento estão relacionados na Amazônia; entenda | Natureza | G1 (globo.com)

| Identify Risk | Potential impact of risk on climate, community and/or biodiversity benefits | Actions needed and designed to mitigate the risk |
|---|---|---|
| | | Thus, we consider this a risk because fire can occur due to natural factors or even anthropic action, even though it is a very low risk. And to maintain the low probability of the risk occurring, we propose the constant monitoring by satellite images and the awareness of residents in the communities about the risks of fires and the procedure of communication to Carbonext or responsible agencies if this occurs. |
| Deforestation/Degradation | Deforestation or degradation of forest cover. | Automatic satellite monitoring (Carbonext's MonitoraCarbon System) to identify and act on deforestation; on-site visits and community environmental education to combat degradation; measures for filing a report of occurrences and/or recovery of the area if necessary |

Regarding the table above, it should be noted that the approach to land rights situation or access/use rights is designed to actively facilitate possible community claims, which have reasonable claims to legitimacy, and but are not fully recognised, legally. Any claims to land rights which are fully legally recognised will be honoured with the corresponding proportion of carbon credit sales proceeds.

### 2.3.5 Information to Stakeholders on Validation and Verification Process (G3.3)

The means of communication with the community stakeholders in the pre-validation and verification visits was project agent visit supported by presentation including visual cues. At this time, the audit process was explained to the community, including the return audit visits for initial validation and verification, followed by annual verification visits.

The communication methods above were designed to be accessible intelligible to the target audience, delivered in Portuguese in person, in relatable language.

### 2.3.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

As of the project agent's validation visit to this remote region, all audit field visits will be communicated along with invitations to stakeholders to comment on the project documents in the online public consultation period. This is achieved, initially, through Whatsapp message contact with certain community members who have this capability, who are requested to pass the message on. After the project's telecommunication initiatives are operational, the message will be broadcast to all community households.

The site visits are informed with at least a month's notice, so that stakeholders can plan to be available. As the project developer is in the process of building the relationship with to the community, this section will progress along the subsequent Monitoring Reports.

### 2.3.7 Stakeholder Consultations (G3.4)

***Consultations***

The communities involved in the project were invited to participate fully and effectively in the decision-making process and in the implementation of the project throughout its lifetime, with regular meetings held to disseminate information.

In the meetings and public consultations held in the communities of Nova Esperança and Vila Batista the main information about the IPOÁ REDD+ project was transmitted, and the residents were encouraged to comment, suggest and participate in the development of the project. So far, there were no negative comments about the existence of the project and its development, in general, all comments from residents were positive for the project development.

With the first meeting, in March in the community of Nova Esperança, we had the consent of all the communities involved, through the signatures of the leaders of the three communities: Carlinhos from Vila Batista, Magno from Nova Esperança, and Bráulio from Vale da Benção. Although the communities do not have any kind of association or documents that recognize these leaders, in the second visit to the communities, questionnaires were applied and interviews were carried out with all the residents present, and all of them recognized these figures as representatives of the communities.

In the second visit there was also the articulation with the communities involved in the project, to invite the largest number of residents present to attend the meeting and be aware of the project. In this meeting, held in April in the community of Vila Batista, the residents were encouraged to ask questions, suggest improvements, and participate in the decision-making process in the choice of activities that will be implemented in the region. The meeting had (according to the meeting's attendance list) 24 residents present.

***Project Designer***

The strategy was to incorporate the communities in the decision-making processes of the IPOA REDD+ Project, and this occurred during meetings and public consultations held in the communities of Vila Batista and Nova Esperança, where the communities involved were informed about the project and had full and effective participation in the process.

Based on the data collected through the social and environmental diagnosis carried out in loco about the social and environmental conditions in the project area, the main demands of the communities were identified. Subsequently, these demands were analyzed by the technical team and the project activities were outlined considering climatic, community and biodiversity factors.

In a participatory process with the communities, an activity prioritization workshop was held (see section *2.1.11 - Project Activities and Theory of Change*), where in a clear and inclusive way the residents themselves draw their main demands and select the central demands for the creation of short-, medium- and long-term solutions, thus contributing to the design of the project activities. From the technical perspective and from the perspective of the communities themselves, the main strategies and activities needed to meet these demands were mapped out within the context of the IPOÁ Project.

With the design of the communities' main activities of interest, the project team started a long process of structuring and planning these activities raised by the communities, in order to execute and benefit from them. And this process of incorporating the communities in the decision-making processes will be part of the IPOÁ REDD+ project during its thirty-year duration, so this section will progress in relation to the subsequent monitoring reports, and new methodologies and tools may be used to guarantee the effective participation of the communities.

### 2.3.8 Continued Consultation and Adaptive Management (G3.4)

The project understands as fundamental the existence of continuous communication channels between the project proponents, the stakeholders, and the communities involved. To guarantee that communication and consultations are permanent throughout the existence of the project, a communication plan was created for the IPOÁ REDD+ Project, but for this plan to be applied, it was essential for the communities to have a fixed means of communication, which was implemented in 2022. Below, we have the table with the project's communication plan.

*Table 9 - Communication plan.*

| Communication Action | Focus | Half | Frequency | Public | Evidence |
|---|---|---|---|---|---|
| **Informative Meeting about the Project** | Inform residents and stakeholders about the project. And consult with the communities about the implementation of the project | Presential | Annually | Community residents and stakeholders who want to participate | Attendance list<br><br>List of suggestions |
| **Consultative Meeting** | Consult with the communities about the project implementation | Presential | Annually | Community residents | Attendance list<br><br>List of suggestions |
| **Request for Information** | Soliciting community leaders for information about community and forest situations | Telephone | Monthly | To community leaders | Questionnaires |
| **Notifications** | Notify the community about technical visits, auditor visits, etc. | Telephone, and occasionally in person | Whenever necessary | To community leaders | - |
| **Submission of Project Summary** | Inform identified stakeholders by e-mail about public consultation periods and verifications | E-mail | Annually | Community residents and stakeholders who want to participate | Evidence of shipments and receipts |

Depending on the type of concerns and demands raised by stakeholders and communities, these will be included in the adaptive management process and will be taken into consideration in the development of project activities and operation. As we are in the middle of a community outreach process, this section will advance on subsequent monitoring reports.

#### 2.3.9 Stakeholder Consultation Channels (G3.5)

In the community visits of the IPOÁ REDD+ project, consultations were conducted directly with communities and other stakeholders or their legitimate representatives. They take place in a location easily accessible to the community and the project was presented in full to the participants by social experts. For stakeholders, the project was presented by email, with the project summary sent to them with the appropriate level of information, and they were invited to comment on the project.


*Figure 25 - IPOÁ REDD+ Project Consultation in April 2022.*

A banner containing information about the project was available during the consultation, containing a map with the project area and the main information about the project, such as location, name of the farms, expected benefits, and project contact. During the meeting, the community had an open space to ask questions and give suggestions about the project.

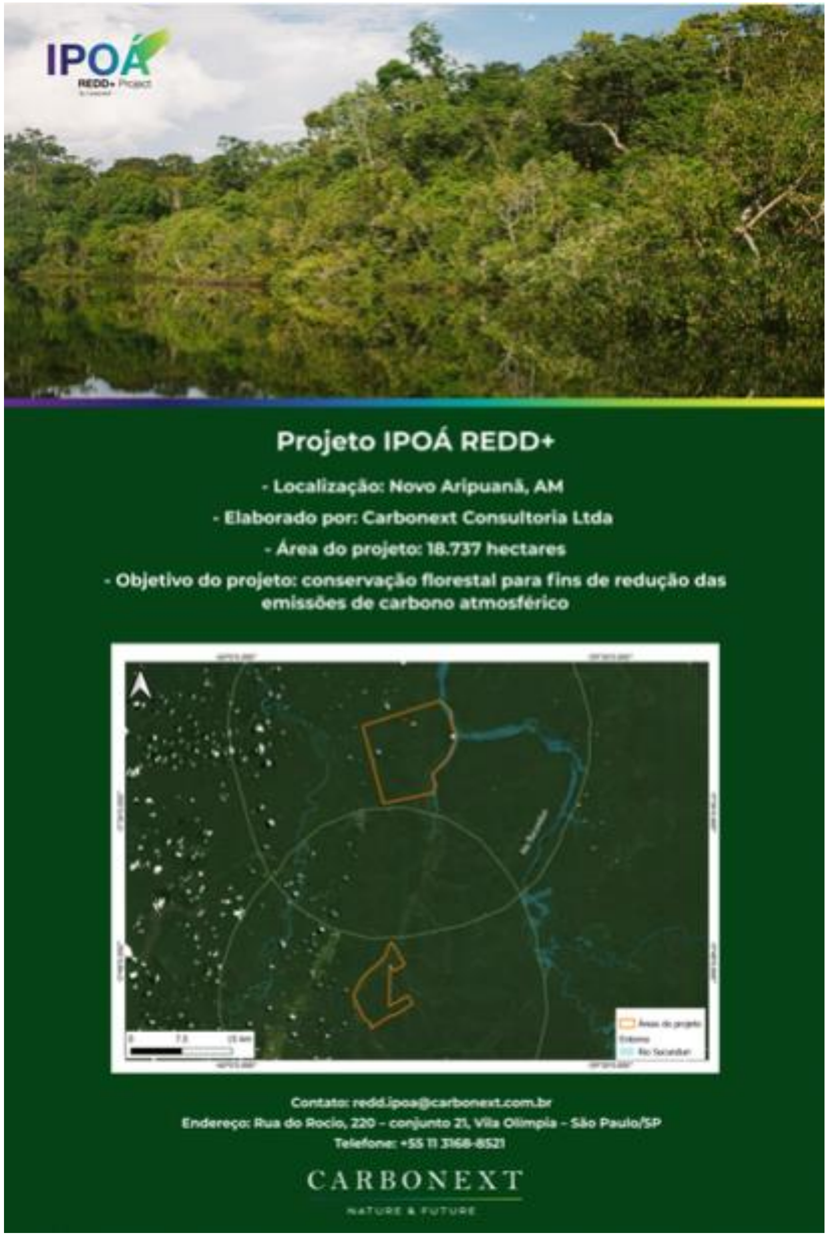


*Figure 26 - Banner IPOÁ REDD+ Project.*

#### 2.3.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

To ensure the effective participation of communities in decision making, as well as the implementation of necessary actions, it is essential to identify all communities and other stakeholders in the project.

From this, they are invited to participate in the decision-making process, as well as to be informed about any activity related to the implementation of the project.

The field visits were important to understand the local context and to propose activities that might be of interest to the community. Participatory workshops were held with the goal of understanding the residents' main wishes, dreams, desires, and needs.

It is essential to adopt viable communication channels that are effectively used by the communities and other stakeholders, allowing a greater flow of information and transparency about the actions and any relevant activity to be informed.

Furthermore, with communication facilitated, any activity to be developed in the project will respect the communities' decision as to the time and place to be executed.

Cultural and gender information are considered by the proponent. Also encouraging the participation of minority cultural groups and broad female participation.

### 2.3.11 Anti-Discrimination Assurance (G3.7)

Carbonext has a Code of Ethics and Organizational Conduct in which prohibits any type of abuse or harassment, whether moral, sexual or discriminatory. It does not allow:

- Investments of a sexual or moral nature.
- Exposure of inappropriate material or any other inappropriate conduct.
- Verbal or written offenses.
- Belittling treatment or threats.
- Jokes, nicknames or any offensive reference to sex, race, age, colour and religion.

Owners and others involved the project design are audited within administrative and judicial bodies, through certificates and declarations, in order to verify the involvement with inappropriate conduct that could implicate any risk to the project.

### 2.3.12 Feedback and Grievance Redress Procedure (G3.8)

***Communication Channel***

The project proponent understands that stakeholders want and need to be involved in project design, implementation, monitoring and evaluation throughout the project lifetime. Therefore, communication channel was established for stakeholders to continually express their concerns and to solve eventual conflicts and grievances that arise during project planning, implementation, and monitoring. The main communication channel is the project's own email (redd.ipoa@carbonext.com.br), which is managed by the Carbonext team.

It is expected that this communication channel will be a mechanism to ensure that the project proponent and all other entities involved in project design and implementation are not involved in or complicit in any form of discrimination or harassment with respect to the project. All denouncements will be available to stakeholders and auditors. In case of conflicts, the project proponent and stakeholders will be free to propose and take any appropriate corrective action. This is the formalized grievance redress procedure that will be used throughout the project lifetime to address disputes with stakeholders, which may arise during project planning and implementation.

The process for receiving, hearing, responding to and attempting to resolve Grievances will be performed within a reasonable time period. It is expected that each grievance is responded in a delay of 7 days, proposing and/or taking corrective actions. This Feedback and Grievance Redress Procedure has three stages:

1. *The Project Proponent shall attempt to amicably resolve all grievances and provide a written response to the grievances in a manner that is culturally appropriate. (Action must be taken in 7 days);*
2. *Any grievances that are not resolved by amicable negotiations shall be referred to mediation by a neutral third party. (Action must be taken in 30 days);*
3. *Any grievances that are not resolved through mediation shall be referred either to a) arbitration, to the extent allowed by the laws of the relevant jurisdiction or b) competent courts in the relevant jurisdiction, without prejudice to a party's ability to submit the grievance to a competent supranational adjudicatory body, if any. (The time to accomplish this stage is dependent on local jurisdiction delays.)*

#### 2.3.13 Accessibility of the Feedback and Grievance Redress Procedure (G3.8)

Stakeholders and community have a direct communication channel with the project team, in which they can give feedbacks or report grievances. Meetings with the communities are held periodically, it the purpose of gathering feedbacks and guarantee the project activities' quality. They are also invited to contact the field workers and monitors whenever they have something to report.

Feedbacks and grievance received during the public comment period will be publicly available in the project description.

### 2.3.14 Worker Training (G3.9)

During the community consultation, it was raised information on the communities' main needs and how the project can assist in the improvement of their life quality and socioeconomic condition. A relevant set of training activities are being organized and planned to take place periodically to build useful skills and knowledge (*section 6 - Appendix 1*). Also, with the purpose of decrease GHG emission and increase the community participation on the project implementation, workshops and specific trainings on sustainable farming and sustainable forest management will be offered. The trainings will be held by specialized personnel, whenever possible, by local labour, and the participants will receive certificates attesting their participation.

### 2.3.15 Community Employment Opportunities (G3.10)

Currently the project is studying the job opportunities that may be created, however none have been identified yet. If job opportunities are available in the project, the positions will be offered without any discrimination of gender, age, religion, marital status or ethnicity. The selection will be made according to the person that best qualifies for the requirements of the position. Women and elderly will be encouraged to apply for the position. Once hired, the worker(s) will go through a training process and a trial period, that lasts 3 months. All the workers hired will be registered.

### 2.3.16 Relevant Laws and Regulations Related to Worker's Rights (G3.11)]

The IPOÁ Project operates in full compliance with the labor laws of Brazil and strives to set an example of best practice regarding labor terms, conditions, and practices.

Decree-Law No. 5,452, May 1, 1943, is approval of the Consolidation of Labor Laws (CLT), which the then Brazilian president Getúlio Vargas and a group of lawyers and legislators drafted to ensure a series of titles and regulations in the relationship between employers and employees. Since then, the CLT has undergone a series of modifications and improvements.

The main and biggest changes, occurred with the Labor Reform under Law No. 13,467 of July 13, 2017, in order to adapt the legislation to the new labor relations.

The project proponent complies with all laws and regulations covering workers' rights and has no claims against them.

### 2.3.17 Occupational Safety Assessment (G3.12)

The Occupational Safety Assessment of the IPOÁ REDD+ Project was developed based on the stages of hazard identification and risk analysis, as well as the proposition of prevention, mitigation and contingency measures.

***Hazard Identification***

The prevention and mitigation of hazards related to field activities must consider from planning to completion. Thus, the possible scenarios and hazards were identified considering transportation, internal displacement, conditions and physical and biological agents to which the worker may be exposed. For better management, the hazards were categorized as (A) accidental, (B) biological, (P) physical, (C) chemical and (E) ergonomic.

***Risk Assessment***

To analyze each identified risk, the risk matrix of probability versus severity was used, as shown in Figure 27. Five levels of probability and five levels of severity were defined, whose application results in the final risk classification, as shown in Figure 28. The risk degree will define the priority for taking prevention and mitigation actions.

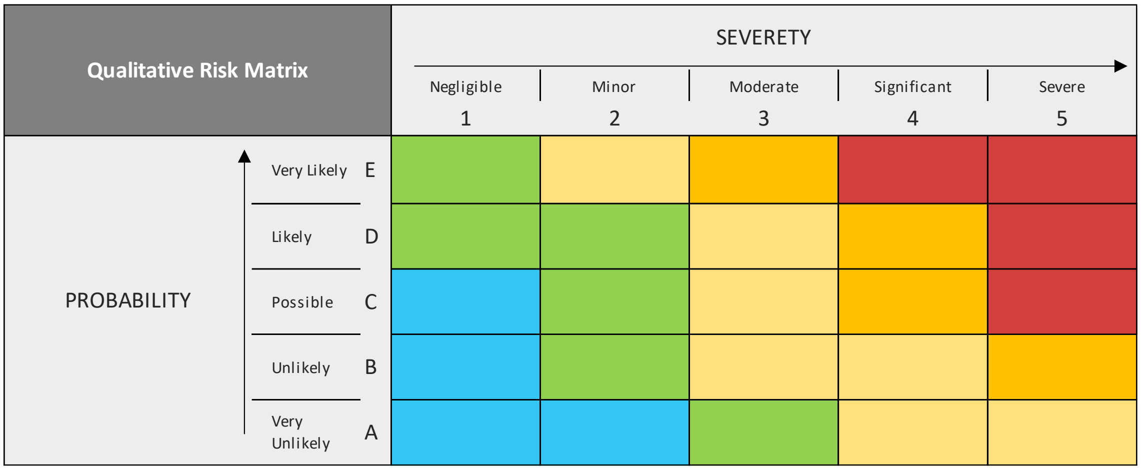


*Figure 27. Risk matrix: (probability x severity).*

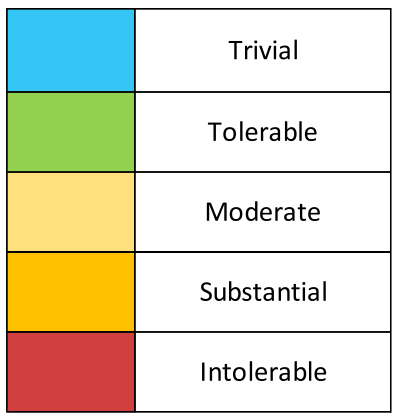


*Figure 28. Risk classification.*

Severity values were analyzed for each risk considering the scale and criteria described in Figure 29.

| Risk Estimation: severity of consequence | |
|---|---|
| Index | Definition |
| 1 | Mild injuries without the need for medical attention, discomfort or discomfort. |
| 2 | Reversible serious injury or illness. |
| 3 | Irreversible critical injury or illness that may limit functional capacity. |
| 4 | Disabling or fatal injury or illness. |
| 5 | Multiple deaths or disabilities (>10). |

*Figure 29. Severity of consequence.*

Probability values were analyzed for each risk considering the scale and criteria described in Figure 30.

| Qualitative Estimation: categories of exposure | | |
|---|---|---|
| Index | Category | Description |
| A | Very low exposure | < 5% |
| B | Low exposure | >5% a <30% |
| C | Moderate exposure | >30% e <70% |
| D | Hight exposure | >70%<95% |
| E | Very hight exposure | >95% |

*Figure 30. Categories of exposure.*

The main prevention and mitigation actions of the identified risks are presented below, considering their causes and priority category.

Prevention, mitigation and emergency procedures

The proponent developed the occupational health and safety manual, which presents risk prevention and mitigation actions, as well as procedures for emergency situations.

The manual also has contacts for activation in cases of emergency, a training schedule, a checklist for planning activities and work safety equipment. The manual is available for the audit.

## 2.4 Management Capacity

### 2.4.1 Project Governance Structures (G4.1)

Carbonext is responsible for all the activities necessary to develop the Project Description in the standards and methodology compatible with the project; calculation of the generation of carbon credits; assistance for the project validation and verification under VCS/CCB and registration process. The team responsible for the mentioned activities is divided into four principal interconnected areas (Figure 31):

i. Legal Department: Legal team specialized in legal matters of the properties, proponents, land tenure, and others. This analysis is made through certificates from the federal and state court of justice, presence of land conflicts, overlaps with public areas and possible risks of expropriation.
ii. REDD+ Project Development Analysts: Technical team specialized in the development of REDD+ projects, being the main responsible for the project design, selection of activities, determination of specific objectives and to provide technical support to the owner and to monitor those activities are being carried out.
iii. GIS (Geographic Information System): Team responsible for determining and reviewing the project baseline, deforestation dynamics, project leakage and monitoring via satellite images the occurrence of deforestation in the project area.
iv. Social Analysts: Team responsible for socioeconomic analysis, engagement and articulation with communities and other stakeholders, development of social activities and indicators, in addition to monitoring socio-economic indicators.

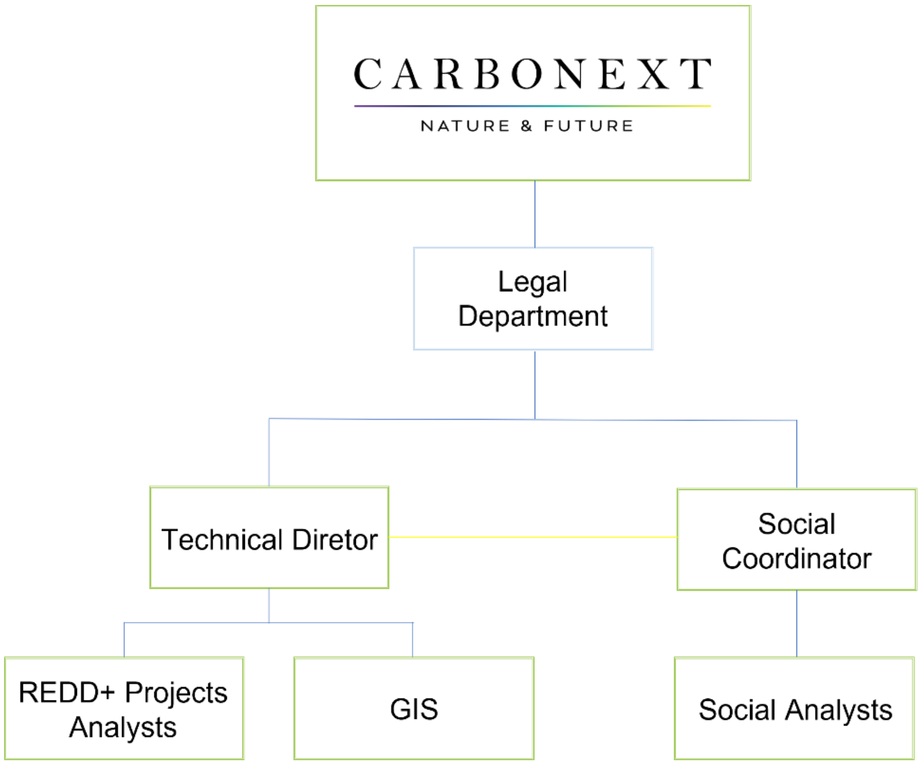


*Figure 31. Carbonext Internal Governance Struture in Project Management.*

Both will carry out the monitoring of the project areas in order to avoid invasions, changes in the biome or any other activity that may lead to deforestation and biodiversity loss. Mutual contact between the proponents will be maintained continuously during the project period. The communities and other stakeholders are essential for the development of the activities and are expected to have a participatory process to build the actions according to the communities' characteristics and needs.

Regarding the actions proposed by the project, Carbonext and other proponents will manage and execute them according to the activities and the planning/schedule.

### 2.4.2 Required Technical Skills (G4.2)

Carbonext's technical team has experienced professionals in various areas of expertise needed for project planning and monitoring, such as forest engineers, social workers, biologists, environmental managers, oceanographers, geoprocessing analysts, and other related fields.

Carbonext will perform the monitoring of carbon stocks from deforestation rates and greenhouse gas emissions produced or avoided during the REDD+ project, so it has a specialized team that has previous experience with other REDD+ projects. In addition, it has a team of social experts, who have the necessary skills to involve the community and stakeholders, so the team will be able to ensure community participation, designing activities and the monitoring of socio-economic indicators.

For the fauna and flora inventory, a third-party team will be hired to collect data in the field and generate the diagnostic report. Preference will be given to local teams that have knowledge of local conditions, but this activity will only take place after the first verification of the project and emission of carbon credits.

### 2.4.3 Management Team Experience (G4.2)

Carbonext was founded in 2010 with the aim of preserving the Amazon rainforest by designing REDD+ carbon credit projects. Since then, we have designed three REDD+ projects to the VCS standard and have conserved over 315,000 hectares to date, with new projects underway. In addition, Carbonext has a multidisciplinary technical team consisting of forest engineers, biologists, environmental managers, ecologists, geographers, social workers and others who constitute a qualified team of researchers and experts capable of conducting project activities. For more information about Carbonext, please visit our website: https://carbonext.com.br/.

Below are summaries of the expertise of the technical team dedicated to the validation and monitoring of the IPOÁ project.

| Name | Janaina Dallan |
|---|---|
| Position | Carbonext CEO |

| | |
|---|---|
| Education | Degree in Forestry Engineering from Luiz de Queiroz College of Agriculture (ESALQ) and a Master's in business of the environment |
| Resume summary | She has been working with the carbon market since 2001 on projects around the world and, as CEO of Carbonext, she is directly involved in coordinating the development and implementation of carbon credit projects. Throughout her career, she has managed UNFCCC CDM (Clean Development Mechanisms) projects through several international companies such as Golder Associates, Global Energy Partners, Orbeo / Societé Generale, One Carbon and Ecofys. She is currently a member of the United Nations Framework Convention on Climate Change (UNFCCC), in which she has been a member of the Registration and Issuance Team (RIT) since 2013, she is president of the Brazil Alliance for Nature-Based Solutions and is a member of the experts for the Taskforce for Scaling Voluntary Carbon Markets (TSVCM). |
| Contact Information | janaina@carbonext.com.br |
| LinkedIn | https://www.linkedin.com/in/janainadallan/ |

| Name | Luiz Fernando de Moura |
|---|---|
| Position | Technical Director |
| Education | Forestry Engineer by Luiz de Queiroz College of Agriculture (ESALQ) the M.Sc. and Ph.D. in Wood technology by the Université Laval (Quebec, Canada). |

| | |
|---|---|
| Resume summary | He is responsible to coordinate the technical group at Carbonext, working with projects for the Carbon Markets including Forestry projects.<br>Dr. de Moura had participation in the preparation of “Energia Verde Carbonization Project - Mitigation of Methane Emissions in the Charcoal Production of Grupo Queiroz Galvão, Maranhão, Brazil”, registered on March 21, 2011.<br><br>Dr. de Moura has also participation as project designer in Florestal Santa Maria REDD+ Projet, Fortaleza Ituxi REDD+ project, UNITOR REDD+ project and Evergreen REDD+ project. |
| Contact Information | luiz.moura@carbonext.com.br |
| LinkedIn | https://www.linkedin.com/in/luiz-fernando-de-moura-6077089/ |

| Name | Franciele Salvador |
|---|---|
| Position | Legal Manager e Compliance Officer |
| Education | Lawyer by the Cenecista College of Joinville |
| Resume summary | Responsible for managing Carbonext's legal group, working as a strategic support in projects for the Carbon Markets.<br><br>Previous experience working in large companies, as well as specializations in Business Law, Compliance, Contract: Business Vision and Practice. |
| Contact Information | franciele.salvador@carbonext.com.br |
| LinkedIn | https://www.linkedin.com/in/franciele-salvador/ |

| | |
|---|---|
| Name | David Swallow |
| Position | Technical REDD+ Coordinator |
| Education | Degree in Anthropology from the University of Edinburgh and an MSc in Ecology, Evolution and Conservation from Imperial College London. |
| Resume summary | Experience with forest carbon, certified supply chains and impact investing, and ten years of experience developing carbon projects and social aspects of REDD+ in the Brazilian market. |
| Contact Information | david@carbonext.com.br |
| LinkedIn | https://www.linkedin.com/in/david-swallow-6b960b14/ |

| | |
|---|---|
| Name | Ághata Comparin Artusi |
| Position | REDD+ Analyst |
| Education | Bachelor in Biological Sciences (URI) and Msc in Biodiversity Conservation and Sustainable Development (IPÊ/ESCAS) |
| Resume summary | Experience in forestry, socio-environmental and ecological research development. Worked with brazilian biodiversity monitoring, forest inventories and currently has a conservationist focus on REDD+ Projects and carbon stock. |
| Contact Information | aghata.artusi@carbonext.com.br |
| LinkedIn | www.linkedin.com/in/aghata-artusi/ |

| | |
|---|---|
| Name | Débora Santos de Oliveira Brum |
| Position | REDD+ Analyst |
| Education | Environmental Manager from the University of São Paulo (USP) and Master in Corporate Sustainability from the Federal University of São Carlos (UFSCAR). |
| Resume summary | Experience with Geographic Information Systems (GIS) in national and international urban cadastre projects and with Greenhouse Gas (GHG) emissions inventories. |
| Contact Information | Debora.oliveira@carbonext.com.br |
| LinkedIn | linkedin.com/in/débora-santos-de-oliveira |

| | |
|---|---|
| Name | Vinicio Rossi Sugui |
| Position | REDD+ Analyst |
| Education | Environmental Engineer (FOC) and M.Sc. in Urban and Industrial Environment (UFPR). MBA in ESG and Impact student. |
| Resume summary | Twelve years' experience in environmental projects and programs; environmental management; sustainability; air pollutant emissions; climate changes; environmental construction control; environmental licensing processes. |
| Contact Information | vinicio.sugui@carbonext.com.br |
| LinkedIn | www.linkedin.com/in/vinicio-sugui |

| Name | Rui Almeida |
| --- | --- |
| Position | Social Project REDD+ Coordinator |
| Education | Graduated in Education, attending post-graduation course in project management, attending master's course in Conflict Management. |
| Resume summary | Experience in projects in the Amazon with traditional communities, quilombolas and indigenous peoples. Experience with the implementation of a financial fund in quilombola communities. Sustainable development, social dialogue, and land and conflict mediation.Training in Nonviolent Communication and advanced knowledge of social dialogue tools, digital technologies, communication methodologies and development of support material for communication projects. |
| Contact Information | rui.almeida@carbonext.com.br |
| LinkedIn | https://www.linkedin.com/in/ruicardosoalmeida/ |

| Name | Thiago Manoel Sozinho dos Santos |
| --- | --- |
| Position | Social Analyst for REDD+ |
| Education | Degree in Forestry Engineering from Amazonia Federal Rural University and a Master’s in forestry economy in Paraná Federal University |
| Resume summary | He has ten years working on the follow themes: protected areas, environment, carbon market, community relations, research, geotechnologies, customer success, project management, BI and Data Science. |
| Contact Information | thiago.sozinho@carbonext.com.br |
| LinkedIn | https://www.linkedin.com/in/thiago-sozinho/ |

| | |
|---|---|
| Name | Matheus Trez |
| Position | GIS and Remote Sensing Coordinator |
| Education | Environmental Engineer from the Piracicaba Engineering School |
| Resume summary | Responsible for coordinating team of analysts in the development of REDD+ projects and the prospection and feasibility analysis of new carbon projects in the Amazon. Experience in the development of REDD+ projects with participation in the elaboration of the VCS Santa Maria REDD+ Forest Project, UNITOR REDD+ project and Evergreen REDD+ project. |
| Contact Information | mateus.trez@carbonext.com.br |
| LinkedIn | https://www.linkedin.com/in/mateustrez/ |

### 2.4.4 Project Management Partnerships/Team Development (G4.2)

As mentioned in section *2.4.2 Required Technical Skills*, for the fauna and flora inventory, a company or independent individuals will be hired to conduct the data survey in the field, giving preference to local teams that know the local reality. These activities will take place after the first verification report, so the contracted team has not yet been defined.

With regard to other capacities needed for project formulation or implementation, no technical or administrative gaps have been identified so far that need to be filled by another entity. However, if during the verification work the need to engage the services of independent professionals or firms to perform the work is identified, the project team will take the necessary steps to find, hire, and then manage any subcontractors. In addition, the project team will determine the time, resources, and level of participation of any outside entities providing services to the project in accordance with internal processes.

### 2.4.5 Financial Health of Implementing Organization(s) (G4.3)

Carbonext has a financial department that is responsible for taking care of the company's financial health, so that it can run its operation on a regular basis in addition to being able to execute projects without running financial risks. To this end, the company has an annual budgeting process that is periodically reviewed in order to give managers visibility of what the main expenses will be for the period, including expenses related to the execution of projects. In this way, a monetary amount is set aside for each project that will be executed, in order to avoid any cash flow risk.

Thus, the project proponents have the necessary funds to maintain the project activities until the start of the GHG revenue, as presented in the Non-Permanence Risk Report[65.]

### 2.4.6 Avoidance of Corruption and Other Unethical Behavior (G4.3)

In Brazil, the two main legal instruments that aim to prevent acts of corruption and unethical attitudes are (i) Law 12,846/2013 (Anti-Corruption Law); and (i) Law 8,429/1992 (Administrative Misconduct Law).

Law no. 12,846/2013, known as the Anti-Corruption Law, establishes rules regarding the administrative liability and civil liability of legal entities in the case of practice of acts against the public administration, national and/or foreign. Under Law 8,429/1992, administrative improbity is an immorality qualified by the dishonesty and acts of bad faith by the public agent.

Both practices, in addition to the legal risks, could represent enormous reputational risk to the project and to all that are involved. Due to this risks, Carbonext Legal Due Diligence aims to verify and, if possible, mitigate any involvement of the landowner (partner) with such practices.

The clearance certificates examined by Carbonext's legal team (applicable for all the project proponents/partners and real estate assets) during the Legal Due Diligence process cover a large spectrum of jurisdictional and administrative matters, such as, for example, consultation before State Courts of Justice and Superior Courts, State and Federal Public Prosecutors, Environmental Agencies, Protest Registry Offices, Real Estate Registry Offices, etc., significantly reducing the chances of both debts and/or acts of corruption (regarding the landowner partner or the real estate assets) go

[65] Evidence presented to the auditors

unnoticed. Eventual discovery of relevant debts and/or acts of corruption involving the landowner (partner), is one of the many reasons for the termination of the intended partnership, as provided for in the contract.

Carbonext has an Anti-corruption Policy and a code of ethics and organizational conduct in place to ensure non-involvement with corruption[66]. So that employees, customers, partners and other stakeholders can, with complete security and anonymity, report situations that are not in accordance with our values or with our Code of Ethics and Organizational Conduct and / or Compliance Policy, we created a channel for complaints to record, investigate and promote decisions involving the complaints received, to ensure compliance and continuous improvement of processes and internal controls. To register your denunciation, send an e-mail to compliance@carbonext.com.br

### 2.4.7 Commercially Sensitive Information (*Rules* 3.5.13 – 3.5.14)

Some information required by the VCS and CCB standards is considered confidential or commercially sensitive and cannot be made public by the project proponent. This information will be supplied to the audit team, but not available for the public. The documents are:

- Agreements and contracts between the parties involved.
- Financial evidence of the proponents' funds for the project.
- Property Right Documents.

## 2.5 Legal Status and Property Rights

### 2.5.1 Statutory and Customary Property Rights (G5.1)

The project area is privately owned by the proponents, who enjoy all rights of possession, access and use, as demonstrated in the documentation available for audit.

The project zone encompasses, in addition to the Project Area, the communities of Vila Batista, Vila Nova Esperança and Vale da Benção, which are not in private areas registered in the Land

[66] Anti-corruption policy and code of ethics and organizational conduct in force: https://carbonext.com.br/governanca-corporativa/

Management System (SIGEF) maintained by the National Institute for Colonization and Agrarian Reform (INCRA), the Brazilian federal agency whose objective is to maintain the national registry of rural properties in the country. The communities are also not located in Federal Conservation Units or Indigenous Territories, characterized as areas of public domain, as shown in Figure 32.

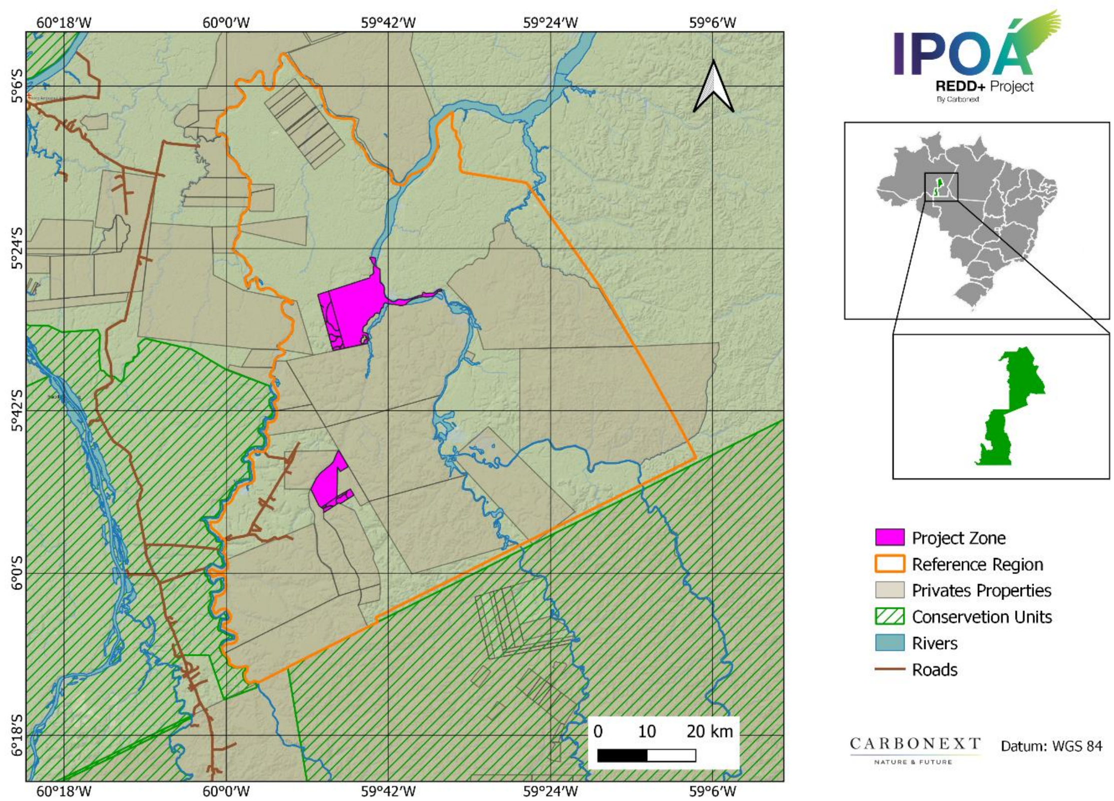


Figure 32. Properties and access rights.

It is relevant to cite the various Brazilian squatters rights laws – *usocapião* in Portuguese – which can be categorized as:

- "Ordinary", governed by Law Nº10.406[67];

[67] Source: http://www.planalto.gov.br/ccivil_03/leis/2002/l10406compilada.htm

- “Extraordinary” covered by Art. 1.238 of the Civil Code[68];
- “Special rural” described under Art 191 of the Federal Constitution[69] and;
- Extrajudicial measures, possible to be carried out in a notary office, and the framework for which is provided by Art. 1.071 of Civil Law nº 13.105/15[70];

To cite just some of the laws which affect these three modalities. The clauses of these laws state that if a person possesses an area, in some cases limited to 50 hectares, as if it were their own, for an uninterrupted period of between five and 15 years, depending on the context, they are entitled to acquire the property document.

This is the main legal framework within which legitimate property rights claims may be made within this project’s context. However, at the time of writing the present PD, no legal action relating to the ownership or possession of the properties has been identified.

### 2.5.2 Recognition of Property Rights (G5.1)

All the project lands are private property owned by the project proponents. and property rights are recognized, respected and supported. All the properties have documents that ensure rights over the land. Documentation will be available to the auditors.

As illustrated by the specific case discussed in section *2.5.6 Ongoing Disputes*, the measures designed and implemented by the project to help to secure full community property rights are: Carbonext will repeat the questionnaire and monitor the land title situation at each monitoring period. Carbonext commits to guaranteeing the rights of all those with legitimate and legally approved land claims to have those rights upheld and be entitled to corresponding project benefits.

[68] Source: http://www.planalto.gov.br/ccivil_03/leis/2002/l10406compilada.htm
[69] Source: http://www.planalto.gov.br/ccivil_03/constituicao/constituicao.htm
[70] Source: http://www.planalto.gov.br/ccivil_03/_ato2015-2018/2015/lei/l13105.htm

### 2.5.3 Free, Prior and Informed Consent (G5.2)

The project will not encroach on or affect private property, community property, or government property, where the project area consists of private land, and no outside property will be affected by the project, as demonstrated in Section *2.5.1 Statutory and Customary Property Rights*.

The entire project area and the community of Vila Batista is on the proponent's private property. The communities of Nova Esperança and Vale da Benção are not in private areas registered in the Land Management System (SIGEF) maintained by the National Institute for Colonization and Agrarian Reform (INCRA), the Brazilian federal agency whose objective is to maintain the national registry of rural properties in the country.

As property rights are not being affected and project activities may not lead to the removal or relocation of habitation or activities important to their culture and livelihood, the Free, Prior and Informed Consent (FPIC) is not required, according to CCB standards requirements.

However, for transparency purposes, in the stakeholder consultation with the local leaders of the nearby communities, consent for the development of the project was obtained in the field and the signature or fingerprint was collected. By which the free, prior, and informed consent of those whose property rights are affected by the project was obtained through a transparent and consensual process. That is, that given by a fully empowered agent, voluntarily and free of defects in consent, as required by Brazilian law for legal transactions in general.

As for the consent between the parties proposing the project, the contract between the proponents was widely and freely negotiated, being accepted and signed by those who had the legal powers to do so, as demonstrated by the partnership agreement signed and the corporate documents presented and attached to the project.

### 2.5.4 Property Rights Protection (G5.3)

The implementation and development of the IPOÁ project should not lead to the involuntary removal or relocation of any stakeholder. Indeed, the due rights to property and resulting carbon credit rights are actively facilitated as part of Carbonext’s policy, detailed in section *2.5.2 Recognition of Property Rights*.

The conservation agreements freely signed by all partners are the result of previous familiarization workshops and the commitment of both parties to identify and define the activities that will be developed in the area. This ensures that they have not been forced to relocate activities important to their culture and livelihoods. Equally, the mitigation measures implemented by the project proponent must be kept in constant communication with partners and all stakeholders around the project area and the project expansion area.

### 2.5.5 Illegal Activity Identification (G5.4)

In the baseline scenario, the scenario without the project faces the threats of illegal deforestation followed by cattle ranching, which is a strong activity in the region. In order to reduce these activities, the project will train the workers and community members to monitor the area. Also, the project has remote monitoring by satellite images from different sensors (Sentinel and Landsat) and radar images (Sentinel), with a maximum frequency of 7 days. This allows the project proponents to identify any illegal trespassing and take action.

### 2.5.6 Ongoing Disputes (G5.5)

There are no legally formalized ongoing disputes in the project area, neither current nor unresolved disputes over land rights.

During two pre-validation visits to the communities of the PZ, in March and May 2022, the issue of land rights was addressed in the community questionnaires. A community member, alleged to have a document that supports a potential claim to approximately 800 hectares within the Maanaim property area, however he was unable to produce any evidence to support this. The situation is being monitored in order to grant and facilitate claims to land rights, if they do exist, which will be updated at least at each monitoring event. Carbonext's general policy regarding such disputes and resulting changes to carbon credit rights is expouhnded in section *2.5.2 Recognition of Property Rights*.

The project intends to include the resident, known as in the project activities, aiming at the benefit to the community, in order to reduce the risks related to possible disputes.

In relation to public entities, whether state, municipal or federal, of direct or indirect administration, the certificates analyzed indicate the legitimacy of the titling, from its ground zero, ruling out the possibility of irregular “dominion” over public lands (grilagem), a fact that would make the development of the project impossible, because regardless of the domain duration, it could not be regularized (public lands cannot be usucapidated), except with the effective participation of the public entity that owns the area, through some legislative measure (law, decree , ordinance, etc.) that expressly legalizes this domain.

Under Brazilian legislation, we can say that the certificates from environmental, administrative, jurisdictional, and certifying bodies indicate that both possession and property belong to the proponent, being tame, peaceful, stabilized, fully legal and extensively documented and proven. In other words, the owner has full right to use, enjoy and dispose of the properties that make up the project, having legitimacy to allow the project to be developed.

### 2.5.7 National and Local Laws (G5.6)

There is no specific governing legislation in Brazil for carbon credit projects n or even for the carbon credits generated. Therefore, projects must be aligned with the laws and principles that govern the legal sphere.

In this context, there is a hierarchy between norms in the Brazilian legislation, which starts with the Federal Constitution. All branches of law applicable to the carbon-generating project and the applicable rules have their origin, foundation, and validity in the Federal Constitution.

When we talk about Environmental Law and the Environment, Article 225 of the Constitution is the basis for all norms, principles, objectives, and policies. The National Land Policy begins with article 184. Furthermore, rights such as possession, property, and free enterprise are also based on the Federal Constitution.

Thus, considering the premises contained in the brief explanation above, there are several rules applicable, directly, and indirectly, to the project generated from carbon credits, each one regulating a specific aspect.

- **Law 11, 284/2006** - Provides for the management of public forests for sustainable production, institutes the Brazilian Forest Service - SFB. Among other topics, this law deals with the management of public forests for sustainable production; Direct management; Forest concessions; Management and inspection bodies; Principles and concepts[71].

- **Law 12,651/2012 (Forest Code)** - The Forest Code establishes general rules on the protection of vegetation, Permanent Preservation areas and Legal Reserve areas, forest exploitation, the supply of forest raw materials, control of the origin of forest products and the control and prevention of forest fires. In addition, it provides economic and financial instruments to achieve its objectives. Creates the CAR which was later regulated by MMA Normative Instruction No. 2 of May 5, 2014[72].

- **Law 9,433/1997 (The National Policy on Water Resources)** - The National Policy on Water Resources is established based on the assumption that water is a limited natural resource, with economic value and public domain. Likewise, the management of water resources must always provide for the multiple use of water, and the management of water resources must be decentralized and count on the participation of the Public Power, users and communities[73].

- **Law 12,305/2010 (National Solid Waste Policy)** - The National Solid Waste Policy brings together the set of principles, objectives, instruments, guidelines, goals and actions adopted by the Federal Government, alone or in cooperation with states, the Federal District, municipalities or individuals, with a view to the integrated management and environmentally appropriate management of solid waste. This is a milestone in Brazilian environmental legislation, as it is the first federal standard created with a focus on the problem of solid waste. Thus, the afore mentioned law deals with relevant issues related to social, environmental, and economic interests in practically all activities. It includes as instruments the environmental, sanitary and agricultural monitoring and inspection, technical and financial cooperation between the public and private sectors for the development of research on new products, methods, processes and technologies of management, recycling, reuse, waste treatment and final disposal environmentally sound tailings; scientific and technological research; environmental education, among others[74].

- **Law 12,187/09 (National Policy on Climate Change - PNMC)** - The PNMC's objectives are to make economic and social development compatible with the protection of the climate system, the

[71] http://www.planalto.gov.br/ccivil_03/_Ato20042006/2006/Lei/L11284.htm
[72] http://www.planalto.gov.br/ccivil_03/_ato2011-2014/2012/lei/l12651.htm
[73] http://www.planalto.gov .br/ccivil_03/leis/l9433.htm
[74] http://www.planalto.gov.br/ccivil_03/_ato2007-2010/2010/lei/l12305.htm

reduction of anthropogenic greenhouse gas emissions in relation to their different sources , the strengthening of human removals by sinks of greenhouse gases in the national territory, the implementation of measures to promote adaptation to climate change by the 3 (three) spheres of the Federation, with the participation and collaboration of economic and social agents stakeholders or beneficiaries, in particular those especially vulnerable to its adverse effects; the preservation, conservation and recovery of environmental resources, with particular attention to the great natural biomes considered National Heritage; the consolidation and expansion of legally protected areas and the encouragement of reforestation and the recomposition of vegetation cover in degraded areas; and encouraging the development of the Brazilian Emissions Reduction Market – MBRE [75].

- **Decree no. 9,578/2018 -** Provides for the National Fund on Climate Change (FNMC), dealt with in Law no. 12,114, of December 9, 2009, and the National Policy on Climate Change, dealt with in Law n. 12,187, of December 29, 2009. Among the topics covered by the decree is the application of FNMC resources to projects to reduce carbon emissions from deforestation and forest degradation, with priority for natural areas threatened with destruction and relevant to conservation strategies biodiversity (art. 7, V)[76].

- **Resolution no. 001/1986 of CONAMA** - Deals with Environmental Licensing. It is an administrative procedure through which the Public Administration establishes conditions and limits for the exercise of certain activities[77].

- **Law 9,985/2000 (SNUC Law)** - This Law establishes the National System of Nature Conservation Units - SNUC, establishes criteria and norms for the creation, implementation and management of conservation units and presents a series of important concepts for a proper understanding of conservation units[78].

- **Law 6,938/1981 (National Environmental Policy)** - The National Environmental Policy has the objective of preserving, improving and recovering the environmental quality conducive to life, aiming to ensure conditions for socio-economic development, the interests of national security and the protection of human dignity[79].

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[75] http://www.planalto.gov.br/ccivil_03/_ato2007-2010/2009/lei/l12187.htm
[76] http://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/decreto/d9578.htm
[77] http://www.ibama.gov.br/sophia/cnia/legislacao/MMA/RE0001-230186.PDF
[78] http://www.planalto.gov.br/ccivil_03/leis/l9985.htm
[79] https://www.planalto.gov.br/ccivil_03/Leis/L6938.htm

- **Law 10,406/2002 (Civil Code)** - Deals with various rights and obligations, including possession, property and legal business[80].

- **Law 4,504/1964 (Land Statute) -** The Land Statute regulates the rights and obligations concerning rural real estate, for the purposes of implementing the Agrarian Reform and promoting the Agricultural Policy. It lists provisions regarding the rights of ownership and property in addition to their use[81].

- **Law 6,015/1973 (Public Records Law)** - The law deals with public records and, especially, in its chapter V it refers to the registration of rural properties, through which the ownership of rural property is demonstrated[82].

- **Land Squatting Laws:** Law Nº10.406, Art 1.238 of the Civil Code, Art 191 of the Federal Constitution among other laws which govern Land Squatting are fully discussed under section 2.5.1[83].

- **Consolidation of labor laws (CLT)**: Regulates provisions relating to labor legislation and institutes the Permanent Program for Consolidation, Simplification and bureaucratization of Infralegal Labour Standards and the National Labor Award, and amends Decree No. 9,580 of November 22, 2018[84].

- **Regulatory norms (NR)** such as: NR-5 - Internal Commission For Accident Prevention NR-12 - Safety At Work In Machinery And Equipment; NR-24 - Sanitary And Comfort Conditions In The Workplace; NR-26-Safety Signs[85].

**State Laws:**

80 http://www.planalto.gov.br/ccivil_03/leis/2002/l10406compilada.htm
81 http://www.planalto.gov.br/ccivil_03/leis/l4504.htm
82 http:/ /www.planalto.gov.br/ccivil_03/leis/l6015compilada.htm
83 https://www.planalto.gov.br/ccivil_03/LEIS/2002/L10406compilada.htm
84 https://www.planalto.gov.br/ccivil_03/decreto-lei/Del5452.htm; https://www.planalto.gov.br/ccivil_03/_Ato2015-2018/2018/Decreto/D9579.htm#art109
85 https://www.gov.br/trabalho-e-previdencia/pt-br/composicao/orgaos-especificos/secretaria-de-trabalho/inspecao/seguranca-e-saude-no-trabalho/ctpp-nrs/normas-regulamentadoras-nrs

- **Law 1,532/82:** Disciplines the State Policy for the Prevention and Control of Pollution, Improvement and Recovery of the Environment and Protection of Natural Resources[86].

- **Law 2,416/96:** Provides for the requirements for granting a license for the exploration, processing and industrialization of forest products and by-products for timber purposes[87].

- **Law 3,135/07:** Establishes the State Policy on Climate Change, Environmental Conservation and Sustainable Development of Amazonas[88].

- **Law 3,785/12:** Provides for environmental licensing in the State of Amazonas, revokes Law n. 3,219, of December 28, 2007[89].

- **Law 4,406/16**: Establishes the State Policy for Regularization, provides for the Rural Environmental Registry - CAR, the Rural Environmental Registry System - Sicar-AM, the Environmental Regularization Program - PRA, in the State of Amazonas[90].

- **Decree no. 10,028/87:** State System for Licensing Activities with Potential Impact on the Environment[91].

- **Decree no. 15,780/94**: Amends Art. 57 of State Decree No. 10,028, of February 4, 1987, which regulated Law No. 1,532, of July 6, 1982, which provides for the State Activity Licensing System with Potential Impact on the Environment[92].

- **Decree no. 15,842/94**: Amends Art. 44 of State Decree No. 10,028, of February 4, 1987, which regulated Law No. 1,532 of July 6, 1982, which provides for the State Licensing System for Activities with Potential of Impact on the Environment[93].

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[86] http://www.ipaam.am.gov.br/wp-content/uploads/2021/01/LDE-1.532-82-Pol%C3%ADtica-Estadual-da-Preven%C3%A7%C3%A3o-e-Controle-da-Polui%C3%A7%C3%A3o.pdf
[87] https://www.mpam.mp.br/images/stories/lei_2416_96.pdf
[88] https://legisla.imprensaoficial.am.gov.br/diario_am/12/2007/6/4939?q=3.135
[89] https://legisla.imprensaoficial.am.gov.br/diario_am/12/2012/7/2494?q=3.785).
[90] https://legisla.imprensaoficial.am.gov.br/diario_am/12/2016/12/1433?q=4.406
[91]https://www.mpam.mp.br/attachments/article/4836/Decreto%2010.028-87.pdf#:~:text=DECRETO%20N%C2%BA%2010.028%2C%20DE%2004%20DE%20FEVEREIRO%20DE,e%20aplica%C3%A7%C3%A3o%20de%20penalidades%20e%20d%C3%A1%20outras%20provid%C3%AAncias
[92] https://www.mpam.mp.br/images/stories/decreto_15780_94.pdf
[93] https://www.mpam.mp.br/images/stories/decreto_15842_94.pdf

As such, the Project IPOÁ REDD+ is in compliance with all identified applicable laws, and there are no cases of non-compliance laws, statutes and other regulations were identified.

### 2.5.8 Approvals (G5.7)

Brazil does not currently have a specific authority for the management and approval of projects that generate carbon credits, so there was no need for any such approval. Furthermore, there are no communities officially classed as indigenous or traditional in the Project Zone, eliminating the need for approval at a public hearing or by the community management council.

Approvals for the project were obtained internally within the structure of the proponents, being approved by the representatives of both parties, in addition to having been approved regarding the legal and technical feasibility.

Although there is no specific need for approval by any administrative body, it is worth mentioning that the project was built and will be conducted strictly within the legal dictates and, consequently, within all the regulations and principles of administrative entities in Brazil, such as IBAMA, FUNAI, INCRA, ICMBIO, etc.

### 2.5.9 Project Ownership (G5.8)

Both farms (Maanaim and Campos) are owned by Mrs. Carina Marcia Klein Balen and Mr. Sílvio Antônio Balen. Documentation evidencing ownership is available for audit.

The project's development and management is a partnership between Carbonext Tecnologia em Soluções Ambientais and the owners of the Project Area.

## 2.6 Management of Double Counting Risk (G5.9)

The IPOÁ project has not received any environmental or social credit, including any form of marketable climate (GHG or renewable energy related certificates) in the past, nor does it intend to in the future.

For information on strategies to eliminate double counting in the project crediting period, see Section *2.5.15 Double Counting*.

### 2.7 Emissions Trading Programs and Other Binding Limits

Not applicable to the Project, because the emissions reduction as a result of this project will not be used for compliance under any other trading program or mechanism. The current VCS and CCB project is entirely independent of any other carbon project or payment of ecosystem service scheme being developed.

### 2.8 Other Forms of Environmental Credit

The project will not claim other Forms of Environmental Credit under another program.

### 2.9 Participation under Other GHG Programs

The IPOÁ Project has not been registered under any emissions trading programs. The project will not claim credit for the same GHG emission reduction under another program.

### 2.10 Projects Rejected by Other GHG Programs

The IPOÁ Project has not applied for or been rejected by any other GHG programs.

### 2.11 Double Counting (G5.9)

The project neither has nor intends to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program. The VCS Program has a central project database, which lists each approved project. The VCS Project Database is the central storehouse of information on all projects validated to VCS criteria and all Verified Carbon Units issued under the program. Every VCU can be tracked from issuance to retirement in the database, allowing buyers to ensure every credit is real, additional, permanent, independently verified, uniquely numbered and fully traceable online. This project has not been registered under any other credited activity, and no VCUs have been assigned to the project area so far. Thus, any possibility of double counting of credits is eliminated.

# 3 CLIMATE

## 3.1 Application of Methodology

### 3.1.1 Title and Reference of Methodology

Appro*ved VCS Methodology VM0007 "REDD+ Methodology Framework (REDD+ MF)", Version 1.6, 08 September 2020. At: https://verra.org/wp-content/uploads/2020/09/VM0007-REDDMF_v1.6.pdf (last visited 19 April 2022).*

***Carbon pool modules:***

- VMD0001 "Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools" (CP-AB), v1.1
- VMD0005 "Estimation of carbon stocks in the long-term wood products pool" (CP-W), v1.1

***Baseline module:***

- VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL), v1.3
- VMD0007 "Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation" (BL-UP), v3.3

***Leakage modules:***

- VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation/forest degradation and avoided planned wetland degradation (LK-ASP), v1.3
- VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU), v1.2

- VMD0011 "Estimation of emissions from market-effects" (LK-ME), v1.1

***Miscellaneous Modules:***

- VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E–BPB), v1.2
- VMD0016 Methods for stratification of the project area (X-STR), v1.2
- VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC), v2.2

***Tools:***

- Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD), Version 01, 19 Oct 2007 At: https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-02-v1.pdf (last visited 19 April 2022)
- CDM – Executive Board "Tool for testing significance of GHG emissions in A/R CDM project activities (Version 01)" EB 31. At: https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-04-v1.pdf (last visited 19 April 2022).
- AFOLU "Non-Permanence Risk Tool" VCS Version 4, Procedural Document, 19 September 2019, v4.0. At: http://www.v-c-s.org/program-documents (last visited 19 April 2022).

### 3.1.2 Applicability of Methodology

The VM0007 REDD+ Methodology Framework is applicable to both Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD) project types, both of which apply to this project, and which fall under the AFOLU project category "REDD" as defined in the VCS AFOLU Guidance document. This methodology is a compilation of modules and tools that together define the project activity and necessary methodological steps. By choosing the appropriate modules, a project-specific methodology can be constructed.

***All Project Activities:***

No land areas are registered under the CDM or under any other GHG program (both voluntary and compliance-oriented). Therefore, no area must be excluded from the project area.

***All REDD+ Activity Types:***

This REDD+ activity is applicable due to the following conditions:

- Land in the project area has qualified as forest for at least the 10 years prior to the project start date.
- Baseline deforestation in the project area fall within the categories of Unplanned deforestation (VCS category AUDD) and Planned deforestation/degradation (VCS category APD).
- Leakage avoidance activities do not include: i) Agricultural lands that are flooded to increase production (e.g., rice paddy); ii) Intensifying livestock production through use of feed-lots and/or manure lagoons.

***Avoiding Unplanned Deforestation:***

This avoiding unplanned deforestation activities are applicable, as the baseline agents of deforestation: i) clear the land for tree harvesting and ranching; ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) are either residents in the reference region for deforestation or immigrants.

In the baseline scenario of this avoiding unplanned deforestation project activity, post deforestation land use does not constitute reforestation (i.e., pasture is the final land use).

***Avoiding Planned Deforestation/Degradation:***

This avoiding planned deforestation activity is applicable, considering that conversion of forest lands to a deforested condition is legally permitted until the 20% limit relative to the total property area, as described in *1.4 Compliance with Laws, Statutes and Other Regulatory Frameworks*.

The selected modules, along with their VCS site descriptions and applicability conditions, are detailed below, showing their correspondence to the project-specific conditions:

***Carbon pool modules:***

**VMD0001 "Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools" (CP-AB), v1.1:** "This module allows for ex ante estimation of carbon stocks in above- and belowground tree and non-tree woody biomass in the baseline case (for both pre- and post-deforestation stocks) and project case and for ex post estimation of change in carbon stocks in above- and belowground tree biomass in the project case". This module is applicable to all forest types and age classes. Inclusion of the aboveground tree biomass pool as part of the project boundary is mandatory as per the framework module REDD-MF.

**VMD0005 "Estimation of carbon stocks in the long-term wood products pool" (CP-W), v1.1:** "This module allows for ex-ante estimation of carbon stocks in the long-term wood products pool in the baseline case. Carbon stocks treated here are those stocks remaining in wood products after 100 years; the bulk of emissions associated with timber harvest, processing and waste, and eventual product retirement occur within this timeframe, and this module employs the simplifying assumption that the proportion remaining after 100 years is effectively 'permanent'". This module is applicable to all cases where wood is harvested for conversion to wood products for commercial markets, for all forest types and age classes. This module is applicable in the baseline, as the wood products pool is included as part of the project boundary as per applicability criteria in the framework module REDD-MF, specifically:

- Timber harvest occurs prior to or in the process of deforestation, and timber is destined for commercial markets;
- The wood products pool is determined to be significant (using T-SIG).

***Baseline module:***

**VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL), v1.3:** "This module allows for estimating carbon stock changes and GHG emissions related to planned deforestation, planned forest degradation and planned wetland degradation in the baseline scenario.

The module was revised on 8 September 2020 by Restore America's Estuaries and Silvestrum Climate Associates to expand its applicability to activities on tidal wetlands."

The module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land. Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries Modules BL-DFW and LK-DFW must be used to determine potential leakage.

**VMD0007 "Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation" (BL-UP), v3.3:** "This module allows for estimating carbon stock changes and GHG emissions related to unplanned deforestation and unplanned wetland degradation in the baseline scenario. Forest degradation is not considered.

This module was assessed and approved as part of VM0007.

The module was revised on 31 July 2012 by The Field Museum to improve uncertainty estimates in the baseline scenario and allow for new modeling approaches (version 3.0). The revision assessment reports are available on the VMD0017 (X-UNC) page.

The module was revised on 8 September 2011 by The Field Museum to add population as an option for estimating the baseline deforestation rate (version 2.0).

The module was revised on 8 September 2020 by Restore America's Estuaries and Silvestrum Climate Associates to expand its applicability to activities on tidal wetlands."

The applicability conditions of the module state that "The module is applicable for estimating baseline emissions from unplanned deforestation (conversion of forest land to non-forest land in the baseline case). The following conditions must be met to apply this module. The forest landscape configuration can be mosaic, transition, or frontier.

The module must be applied to all project activities where the baseline agents of deforestation:

(i) clear the land for settlements, crop production (agriculturalist), ranching or aquaculture, where such clearing for crop production, ranching or aquaculture does not amount to large scale industrial agri/aquaculture activities;"

(ii) "have no documented and uncontested legal right to deforest the land for these purposes; and"

(iii) "are either resident in the region (reference region—cf. Section 1 below) or immigrants."

The applicability conditions of this module are met in that:

i. the landscape configuration is transition, as discussed in section 3.1.3;
ii. the baseline agents of deforestation clear the land for ranching (see section 3.1.4 Baseline scenario);
iii. their practices are predominantly illegal and not within their legal rights, as argued in section 3.1.5 Additionality, under Scenario 3.

***Leakage modules:***

**VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation/forest degradation and avoided planned wetland degradation (LK-ASP), v1.3:**

This module allows for estimating GHG emissions caused by the activity shifting leakage of planned deforestation carbon projects.

The module is applicable for estimating the leakage emissions due to activity shifting from forest lands that are legally authorized and documented to be converted to non-forest land, including activity shifting to forested wetland that is drained or degraded because of project implementation. The module is also applicable for estimating the leakage emissions due to activity shifting from non-forested wetlands that are legally authorized and documented to be converted and degraded. Under these situations, displacement of baseline activities can be controlled and measured directly by monitoring the baseline deforestation or wetland degradation agents or class of agents.

This tool must be used for projects in areas where planned deforestation happens on forested wetlands, regardless of the absence of wetland within the project boundaries.

The module is mandatory if Module BL-PL has been used to define the baseline, and the applicability conditions in Module BL-PL must be complied with in full.

**VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU), v1.2:**

This Module provides methods for estimating emissions from displacement of unplanned deforestation (leakage due to activity shifting). This Module is applicable for estimating carbon stock changes and greenhouse gas emissions related to the displacement of activities that cause deforestation of lands outside the Project Area due to the avoided unplanned deforestation in the Project Area. Activities subject to potential displacement are: conversion of forest land to grazing lands, crop lands, and other land uses. The module is mandatory if BL-UP has been used to define the baseline and the applicability criteria in BL-UP must be complied with in full.

**VMD0011 "Estimation of emissions from market-effects" (LK-ME), v1.1:**

"This module allows for estimating GHG emissions caused by the market-effects leakage related to extraction of wood for timber, fuelwood or charcoal in the baseline for carbon projects".

This module is applicable for calculating market-effects leakage from REDD+ projects that are anticipated to reduce levels of wood harvest substantially and permanently. When REDD +project activities result in reductions in wood harvest, it is likely that production could shift to other areas of the country to compensate for the reduction.

As referenced in the Framework (REDD-MF) the module is mandatory where:

- The process of deforestation involves timber harvesting for commercial markets.
- The baseline is calculated using BL-DFW AND fuel wood or charcoal is harvested for commercial markets.

In all other circumstances the module shall not be used.

When REDD+ project activities result in reductions in wood harvest, it is likely that production could shift to other areas of the country to compensate for the reduction. The module is mandatory where the process of deforestation involves timber harvesting for commercial markets (Commercial markets are here defined as sale of products to end users and public and private companies with sales conducted distant (>50 km) from the project area).

***Miscellaneous Modules:***

**VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E–BPB), v1.2:**

This module provides a step-wise approach for estimating greenhouse emissions from biomass and peat burning. This module is applicable to REDD+ project activities with emissions from biomass burning

In the baseline scenario, fire is used to clear the land, and emissions of $CO_2$, $N_2O$ and $CH_4$ result. Where used in the baseline, accounting must occur under both the baseline and with project scenarios in both the project area and in the leakage belt. Where fires occur ex-post in areas that coincide with areas deforested or degraded in the baseline case, the module shall be used to account greenhouse gas emissions.

**VMD0016 Methods for stratification of the project area (X-STR), v1.2:**

This module provides guidance on stratifying the project area into discrete, relatively homogeneous units to improve accuracy and precision of carbon stock and carbon stock change estimates.

Any module referencing strata i shall be used in combination with this module. Strata are only used for pre-deforestation forest classes and are the same in baseline and project cases. Post-deforestation (conversion) land-uses are not stratified, instead using average post-deforestation stock values (e.g., “Simple Conservative” or “Historical Area-weighted” approaches per BL-UP).

**VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC), v2.2:**

This module allows for estimating uncertainty in baseline estimations and in estimations of with-project sequestration, emissions and leakage. This module is mandatory when using methodology REDD+ MF. It is applicable for estimating the uncertainty of estimates of emissions and removals of $CO_2$-e generated from REDD+ and WRC project activities. The module focuses on the following sources of uncertainty:

- Determination of rates of deforestation and degradation.

- Uncertainty associated with estimation of stocks in carbon pools and changes in carbon stocks.
- Uncertainty associated with estimation of peat emissions.
- Uncertainty in assessment of project emissions.

Where an uncertainty value is not known or cannot be simply calculated, a project must justify that it is using an indisputably conservative number and an uncertainty of 0% may be used for this component.

Guidance on uncertainty – a precision target of a 95% confidence interval half-width equal to or less than 15% of the recorded value must be targeted. This is especially important in terms of project planning for measurement of carbon stocks; sufficient measurement plots should be included to achieve this precision level across the measured stocks.

***Tools:***

**VT0001 "Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities" (T-ADD), v3.0.**

This tool is applicable when AFOLU activities on the land within the proposed project boundary do not lead to violation of any applicable law even if the law is not enforced. The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario.

Tool for testing significance of GHG emissions in A/R CDM project activities

This tool facilitates the determination of which GHG emissions by sources, possible decreases in carbon pools, and leakage emissions are insignificant for a particular CDM A/R project activity.

The tool shall be used in the application of an A/R CDM approved methodology to an A/R CDM project activity:

a. To determine which decreases in carbon pools and increases in emissions of the greenhouse gases measured in $CO_2$ equivalents that result from the implementation of the A/R project activity, are insignificant and can be neglected.

b. To ensure that it is valid to neglect decreases in carbon pools and increases in GHG emissions by sources stated as being insignificant in the applicability conditions of an A/R CDM methodology.

The above modules and tools are applicable because they meet the applicability conditions of the modules as set out below:

*Table 10: Applicability Conditions and Justifications for the REDD+ Methodology Framework Module.*

| Applicability Condition | Justification |
|---|---|
| Land in the project area has qualified as forest at least 10 years before the project start date. | The project area complies with this condition, with complete forest cover demonstrated for the years 2012 and 2022 |
| The project area can include forested wetlands (such as bottomland forests, floodplain forests, mangrove forests) as long as they do not grow on peat. Peat shall be defined as organic soils with at least 65% organic matter and a minimum thickness of 50 cm. If the project area includes forested wetlands growing on peat (e.g. peat swamp forests), this methodology is not applicable. | As demonstrated in Section *2.1.5 Physical Parameters*, no organic soils (or peatlands) exist within the project area. |
| Project proponents must be able to show control over the project area and ownership of carbon rights for the project area at the time of verification. | As demonstrated in Section *2.5.9 Project Ownership*, the project proponents have clear, uncontested title to both property rights and the carbon rights. |

| | |
|---|---|
| Baseline deforestation and baseline forest degradation in the project area fall within one or more of the following categories: Unplanned deforestation (VCS category AUDD); Planned deforestation (VCS category APD); Degradation through extraction of wood for fuel (fuelwood and charcoal production) (VCS category AUDD). | Baseline deforestation in the project area falls within APD and AUD:<br><br>- APD category, as the agent of deforestation is the project proponent.<br>- AUD category: deforestation agents are cattle ranchers, timber harvesters and infrastructure. |
| Baselines shall be renewed every 6 years from the project start date. | The baseline will be renewed in October 2028. |
| All land areas registered under the CDM or under any other carbon trading scheme (both voluntary and compliance-orientated) must be transparently reported and excluded from the project area. The exclusion of land in the project area from any other carbon trading scheme shall be monitored over time and reported in the monitoring reports. | The IPOÁ Project REDD+ is not registered in any carbon trading scheme or program. |
| If land is not being converted to an alternative use but will be allowed to naturally regrow (i.e. temporarily unstocked), this framework shall not be used. | Forest clearing in the baseline is followed by establishment of pasture, which prevents forest regrowth. |
| Leakage avoidance activities shall not include: Agricultural lands that are flooded to increase production (e.g. paddy rice); Intensifying livestock production through use of “feed-lots” and/or manure lagoons. | Leakage avoidance activities do not include flooding agricultural land or creating feed-lots or manure lagoons. |

*Table 11: VMD0006 Estimation of Baseline Carbon Stock Changes and Greenhouse Gas Emissions from Planned Deforestation and Planned Degradation"*

| Applicability Condition | Justification |
|---|---|
| The module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land. | The proponents have requested to IPAAM permission to clear up to 20% of the Maanaim and Camps proprieties, according to the legal limit prescribed in the Brazilian Forest Code. The landowner has a plan to deforest in the next few years, which will not be executed in case of success in obtaining VCUs. |
| Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries modules BL-DFW and LK-DFW shall be used to determine potential leakage. | No fuelwood collection has been conducted within the project area. This statement can also be confirmed through the public consultation carried out with the community, as none of the interviewees mentioned collecting wood for fuelwood or any other activity. |

*Table 12: "VMD0007 Estimation of Baseline Carbon Stock Changes and Greenhouse Gas Emissions from Unplanned Deforestation"*

| Applicability Condition | Justification |
|---|---|
| Baseline agents of deforestation shall: (i) clear the land for settlements, crop production (agriculturalist) or ranching, where such clearing for crop production or ranching does not amount to large scale industrial agriculture activities; (ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) are either resident in the reference region or immigrants. Under any other condition this framework shall not be used. | The baseline agents of deforestation clear the land for ranching as final land use. These land squatters have no legal right to use or deforest the land. These agents of deforestation are from nearby communities or immigrants looking for land to illegally harvest wood and convert the land for pasture. |

| | |
|---|---|
| Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries modules BL-DFW and LK-DFW shall be used to determine potential leakage. | No fuelwood collection has been conducted within the project area. This statement can also be confirmed through the public consultation carried out with the community, and none of the interviewees mentioned collecting wood for charcoal or any other activity. |

*Table 13: VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation/forest degradation and avoided planned wetland degradation (LK-ASP), v1.3*

| Applicability Condition | Justification |
|---|---|
| The module is applicable for estimating the leakage emissions due to activity shifting from forest lands that are legally authorized and documented to be converted to non-forest land, including activity shifting to forested wetland that is drained or degraded as a consequence of project implementation. | The proponents have the official request and the plan to deforest 20% of the Maanaim and Campos proprieties. |
| This tool must be used for projects in areas where planned deforestation happens on forested wetlands, regardless of the absence of wetland within the project boundaries. | There are no forested wetlands within the project area. |
| The module is mandatory if Module BL-PL has been used to define the baseline, and the applicability conditions in Module BL-PL must be complied with in full. | Module BL-PL was applied and all the conditions were complied with, as mentioned in Table 11. |

*Table 14: VT0001 “Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities” (T-ADD), v3.0.*

| Applicability Condition | Justification |
|---|---|
| AFOLU activities the same or similar to the proposed project activity on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project shall not lead to violation of any applicable law even if the law is not enforced | As stated in the section 1.14, the project is in compliance with all relevant laws, statutes, and regulatory frameworks. |
| The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario. Project proponent(s) proposing new baseline methodologies shall ensure consistency between the determination of a baseline scenario and the determination of additionality of a project activity. | The approach to define the baseline scenario using this tool was described in the section 3.4. |

### 3.1.3 Project Boundary

The VM0007 Methodology includes the six carbon pools listed in the table below, indicating whether they were included or excluded within the proposed AUD and APD project activity, as well as their respective justifications.

*Table 15. Carbon sinks.*

| Source | | Gas | Included or excluded | Justification/Explanation |
|---|---|---|---|---|
| Baseline | Biomass burning | $CO_2$ | Excluded | Excluded as recommended by the applied methodology. Counted as carbon stock change. |

| Source | | Gas | Included or excluded | Justification/Explanation |
|---|---|---|---|---|
| | | $CH_4$ | Included | Included as non-$CO_2$ emissions from biomass burning in the baseline scenario, according to the methodology. |
| | | $N_2O$ | Included | Included as non-$CO_2$ emissions from biomass burning in the baseline scenario, according to the methodology. |
| | | Other | Excluded | No other GHG gases were considered in this project activity. |
| | Livestock emissions | $CO_2$ | Excluded | Not a significant source |
| | | $CH_4$ | Excluded | Excluded for simplification. This is conservative. |
| | | $N_2O$ | Excluded | Excluded for simplification. This is conservative. |
| | | Other | Excluded | No other GHG gases were considered in this baseline activity. |
| Project | Biomass burning | $CO_2$ | Excluded | No biomass burning increase is predicted to occur in the project scenario compared to the baseline case. Therefore, considered insignificant. |
| | | $CH_4$ | Included | Included as non-$CO_2$ emissions from biomass burning in the project scenario, according to the methodology. |
| | | $N_2O$ | Included | Included as non-$CO_2$ emissions from biomass burning in the project scenario, according to the methodology. |
| | | Other | Excluded | No other GHG gases were considered in this project activity. |
| | Livestock emissions | $CO_2$ | Excluded | Not a significant source. |
| | | $CH_4$ | Excluded | No livestock increase is predicted to occur in the project scenario compared to the baseline case. Therefore, considered insignificant. |
| | | $N_2O$ | Excluded | No livestock increase is predicted to occur in the project scenario compared to the baseline case. Therefore, considered insignificant. |
| | | Other | Excluded | No other GHG gases were considered in this project activity. |

All project installations and management activities, based on the description provided in Section *2.1.11 Project Activities and Theory of Change*, will take place in the Project Zone, located as shown below. Thus, it is considered that the Project Boundaries are the same as the Project Zone.

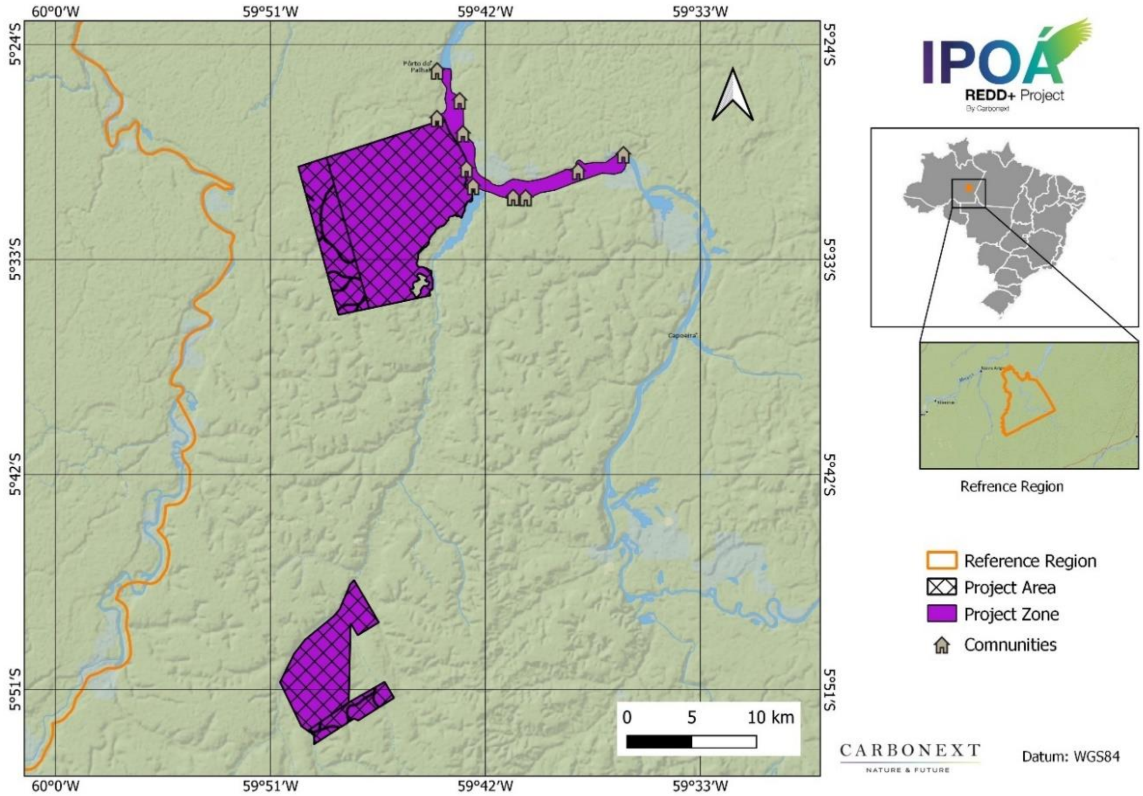


Figure 33 – Project Zone Map.

#### 3.1.4 Baseline Scenario

Forest land is expected to be converted to non-forest land in the baseline scenario, under two mechanisms, or components of the baseline: Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD). The landowners cannot afford efforts and costs to keep long term vigilance of the project boundaries to avoid unplanned land use change from uncontrolled trespassing and deforestation, without carbon credit income.

The AUD baseline component involves a predicted 968,03 hectares of deforestation over the first project lifetime, as further described below.

The APD component is regarding a 3.595,72 ha subsection (20%) of the Maanaim and Campos property, which has documented forest clearing plans. Under the latter, 20% of the property would be clear-cut in the baseline scenario, with wood to be sold on the timber and fuelwood markets, followed by transformation of deforested areas into pasturelands.

Given the context above, the project falls within the AFOLU REDD+ categories of both Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD).

The stepwise approach of the T-ADD Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activitiesis fully presented in section *3.1.5 Addiotionality*, below for AUD and APD.

The baseline will be revised every 6 years in accordance with the latest VCS rules.

***Avoided Unplanned Deforestation (AUD)***

Regarding the AUD component of the baseline, studies indicate that the municipality of Apuí, neighboring Novo Aripuanã, was the hub of livestock expansion in the southern region of the State of Amazonas. This same phenomenon has been observed in the municipality of Novo Aripuanã.

Apuí municipality has the second largest cattle herd size in Amazonas State, second only to Boca do Acre, and cattle ranching represented 51% of the municipal GDP at the time of the most recent government agricultural census (IBGE 2017[94]; Carrero et al. 2015[95]). This cattle production is strongly linked with illegal deforestation levels and deforestation. Idesam (2011[96]) note, in connection with the local cattle farmer syndicate representing ~80% of Apuí's cattle farmers, that if their Legal Reserves had just 40% of their required areas, it would be sufficient to render cattle farming inviable. Further

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[94] Source IBGE Censo Agropecuário IBGE | Cidades@ | Brasil | Pesquisa | Censo Agropecuário | Características dos estabelecimentos
[95] Carrero et al (2015): A Cadeia Produtiva de Carne Bovina no Amazonas
[96]Idesam (2011) REDD+ Estudo de Oportunidades para a Região Sul do Amazonas (accessed 13/08/21): https://idesam.org/publicacao/REDD_Estudo_de_Oportunidades_Sul_Amazonas.pdf

studies show the correlation between growth of cattle farming and the curve of deforestation in the municipality is significant (Carrero et al. 2015).

A study conducted by Soares (2022) identified that the municipality of Novo Aripuanã is a new pole of expansion of the agricultural frontier, following the dynamics of the search for cheap land, investment in deforestation and agricultural production.[97]

According with Map Biomas[98], pasture is the most common alternative land use in the municipality of Novo Aripuanã.

*Table 16 - Percentage of land uses in Novo Aripuanã municipality in 2020.*

| Classes | Area (ha) | % of Total Area |
|---|---|---|
| Forest Formation | 3,798,689.79 | 92.25% |
| Pasture | 132,085.10 | 3.21% |
| Grassland (Non-Forest Natural Formation) | 102,279.13 | 2.48% |
| Water | 76,887.18 | 1.87% |
| Savanna Formation | 6,023.94 | 0.15% |
| Wetland | 1,111.07 | 0.03% |
| Urban Area | 446.94 | 0.01% |
| Temporary Crops | 338.99 | 0.01% |

[97] https://www.teses.usp.br/teses/disponiveis/91/91131/tde-11022022-160839/publico/Pedro_Gandolfo_Soares_versao_revisada.pdf
[98] https://mapbiomas.org/

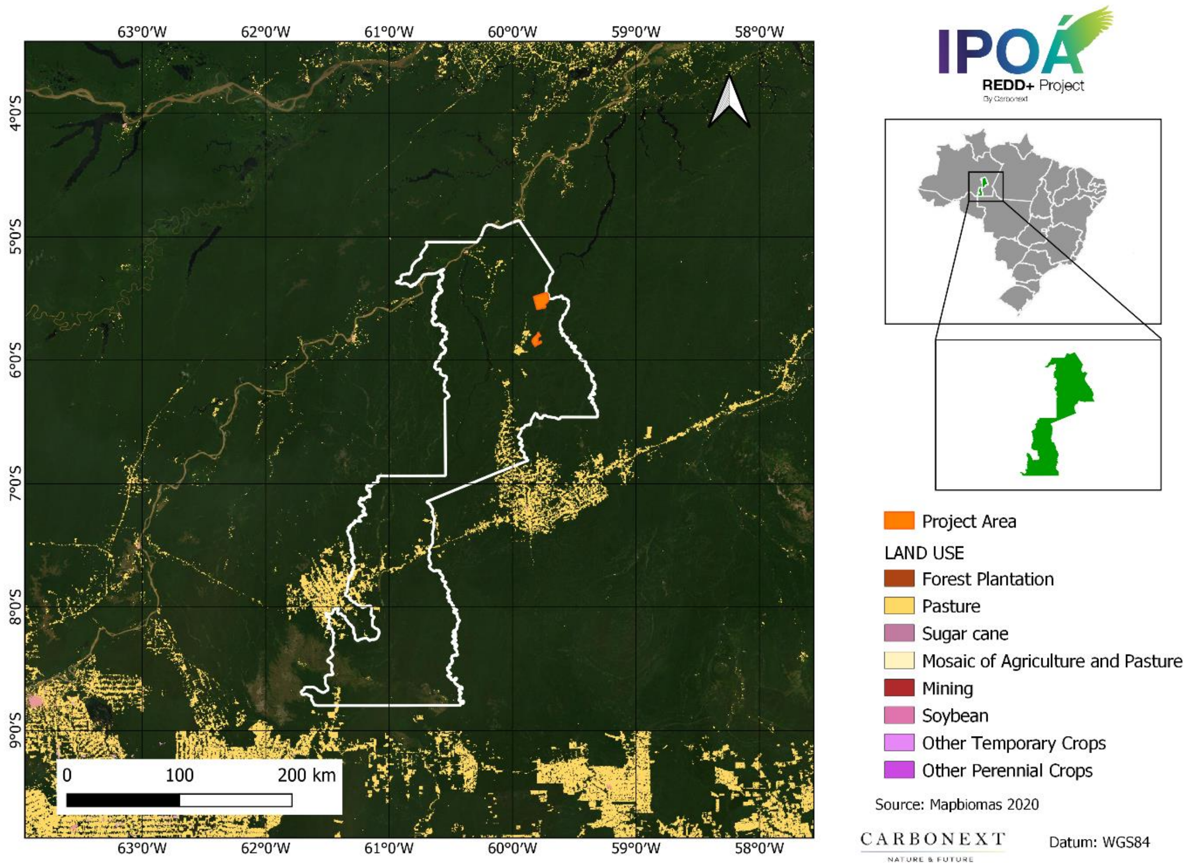


*Figure 34 – General Land Use Scenario in Novo Aripuanã municipality in 2020.*

In 2010, pastures covered only 1% of the municipal territory. However, in 2020 this percentage rose to 3%, totaling 132,085 hectares[99]. This value represents 99,41% of non-natural land uses in Novo Aripuanã.

[99] https://mapbiomas.org/

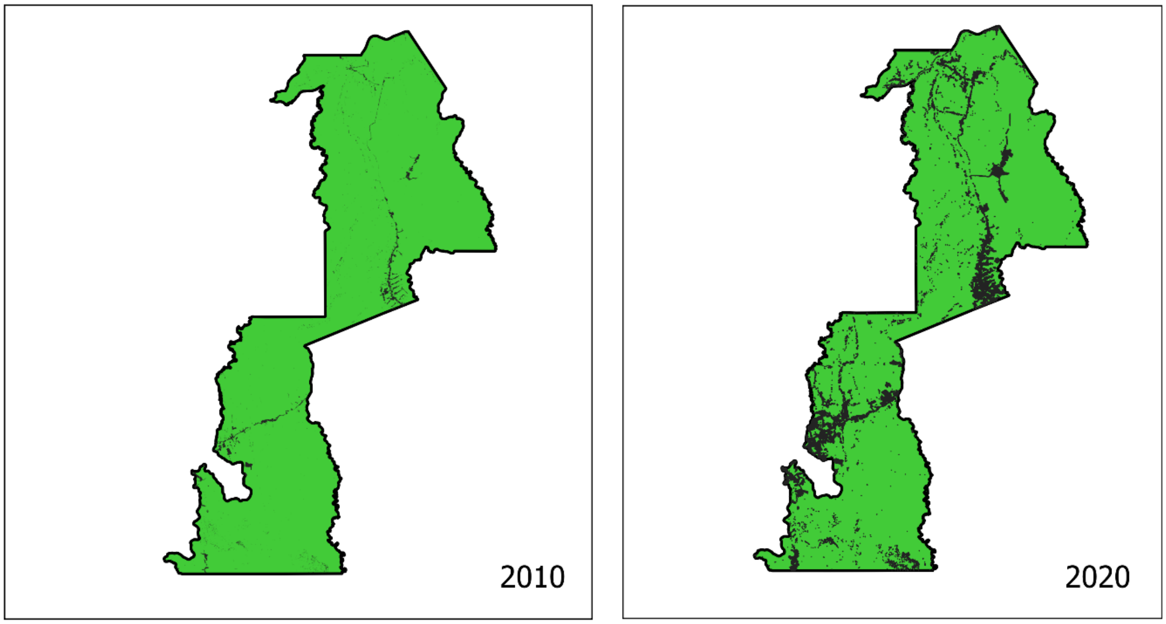


*Figure 35. Pasture in Novo Aripuanã municipality.*

The increasing trend was also observed in the number of cattle identified in the municipality of Novo Aripuanã by the agricultural census. In 2010, the number of heads was 13,989, while in 2020 this amount increased to 42,100 [100], an increase of 300%. As seen in the figure below, the increase in the herd and Novo Aripuanã peaked in 2018.

[100] https://sidra.ibge.gov.br/tabela/3939

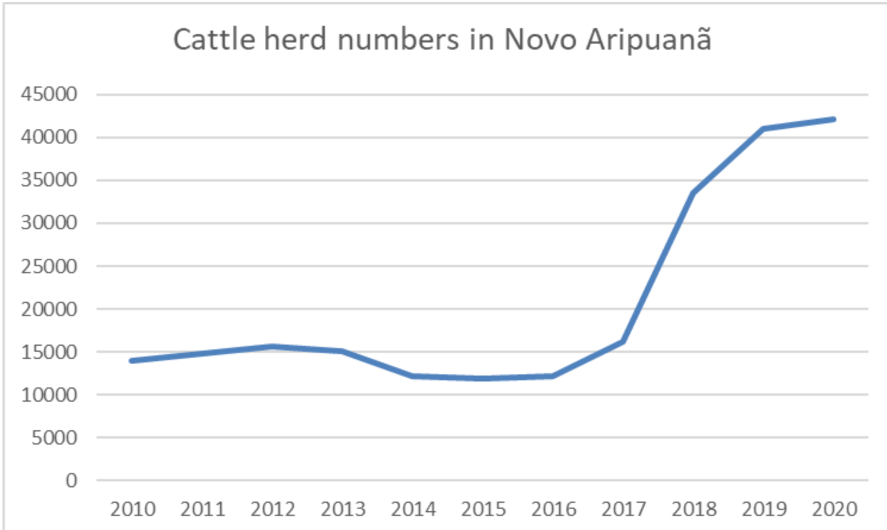


*Figure 36. Increase in cattle herd in Novo Aripuanã.*

As the main deforestation agent in the region, cattle ranching (pasture) is expected to increase in the project region. The percentage of the total area of this land use class increased from 0.57% in 2010 to 1.67%, approaching the project area.

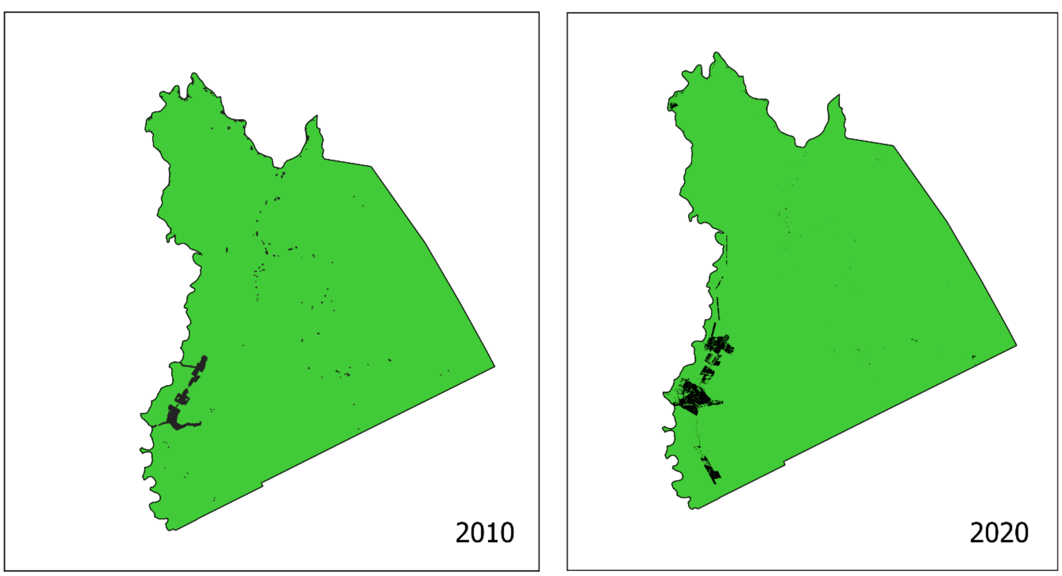


*Figure 37. Pasture in Reference region.*

Therefore low-technology beef cattle ranching is identified as the most probable baseline scenario, being the agent of deforestation and the post-deforestation land-use. The alternative, often overlapping, possible baseline land-uses are presented in Section *3.1.5 Additionality*.

Furthermore, there are several socio-political factors which strongly suggest deforestation rates in the Amazon can be expected to increase even more compared to the current ones. Among the main factors for increased deforestation after 2020, the following can be cited:

Trends in Brazilian legislation: several measures have contributed to slackening of environmental legislation in recent years, mainly under the Bolsonaro administration, with an emphasis on the following aspects:

- **Greater ease of obtaining "forest clearing authorizations":** An example can be observed in the state of Goiás, which reported a 1,100% increase in the number of permits for deforestation in 2020[101]. The new environmental licensing in Goiás, plus technologies that facilitate the inspection work of the Secretariat of Environment and Sustainable Development (Semad), in addition to efforts in the analysis of applications, are responsible for increasing the numbers of deforestation permits in the State. According to data from Semad's Environmental Licensing Superintendency, there was a 673% increase of area deforested: 6.5 $km^2$ in 2019, to 43.8 $km^2$ in 2020. Thus, as occurred in the State of Goiás, the facilitation of the issuance of authorization for the removal of native vegetation can occur at any time in the Amazon Biome. Indeed, attempts at further facilitation have been sought recently (in 2020), as indicated under the next topic.
- **Granting of tacit (or automatic) environmental licensing, in case of delay of the environmental agency:** The controversial automatic release of environmental permits and permits by completion of term, that is, after a period stipulated for the government agency to respond (120 days), was voted for on 29/04/2020, by a virtual plenary. Provisional Measure 915 originally referred to the so-called "Economic Freedom Law" edited by the government, but ended up bringing, within the texts, changes that directly affect the rite of environmental licensing throughout the country. The change could lead to the automatic authorization of forest suppression in the Amazon and Atlantic Forest enforced by delay, and without analysis of the environmental agency. This means that,

[101] https://www.meioambiente.go.gov.br/noticias/2089-emiss%C3%A3o-de-licen%C3%A7as-para-supress%C3%A3o-de-vegeta%C3%A7%C3%A3o-tem-aumento-de-1-100-in-a-year-in-goi%C3%A1s.html

once the 120-day period is expired, the request would be automatically granted with a "tacit license"[102]. Fortunately, environmentalists have revised the Provisional Measure 915, to prevent deforestation licensing from being subject to expiration of term[103].

- **Loosening legislation for timber exports:** As reported by Reuters, over 2019 Brazil exported "thousands of cargoes of wood from an Amazonian port without authorization from the federal environmental agency, increasing the risk that they have been extracted from illegally deforested land". The rule change, which scraps IBAMA's authorization for most timber exports, came after five cargoes of wood arrived in US and European ports without these mandatory documents. Foreign authorities contacted Brazil to ask about the missing authorizations, with the head of Ibama in Pará then retroactively granting the authorizations. The problem, however, is much more widespread than just these five shipments. In Pará state, more than half of the roughly 3,000 officially registered shipments in the past year, containing an estimated 54,000 $m^3$ of wood that left one port, was not officially authorized. Companies had requested authorizations from IBAMA for those shipments but exported them before the agency had time to respond. Furthermore, many shipments were exported without seeking IBAMA's approval. Shipments went to the US, the Netherlands, France, Germany, Belgium, and possibly other countries. Before the rules changed, IBAMA was required to authorize all wood exports before they leave port. Even so, most shipments continued to require the proper paperwork to be given the green light, however only a few would be randomly selected for physical inspection [104]. Arbitrarily, the president of IBAMA ensured that all future unauthorized exports of wood, previously classified as illegal, became legal: he took advantage of the lack of attention of the press to the topic during Carnival, at the end of February 2020, to quietly revoke a 2011 IBAMA policy that required an authorization from the agency before forest products could receive export licenses. From that date on, such permits would be required only for endangered tree species or in other special circumstances. Given the repeal, the floodgates were opened for large shipments of illegal timber from the Brazilian Amazon to go abroad[105]. It was also revealed that in February 2020, loggers from Pará asked IBAMA to effect that

[102] https://www.correiobraziliense.com.br/app/noticia/brasil/2020/04/29/interna-brasil,849652/camara-pode-aprovar-hoje-licenciamento-ambiental-automatico.shtml
[103] https://epbr.com.br/ambientalistas-alteram-mp-915-para-prevent-licensing-environmental-by-course-of-time/
[104] https://www.businesslive.co.za/bd/world/americas/2020-03-04-brazil-may-be-exporting-illegally-deforested-wood/
[105] https://brasil.mongabay.com/2020/04/ao-afrouxar-leis-de-exportacao-brasil-permite-saida-de-madeira-ilegal-da-amazonia/

rule change: the companies wanted to sell wood abroad presenting only the Document of Forest Origin (DFO, "*DOF – Documento de Origem Florestal*" in Portuguese), made by the companies themselves and that originally only served to allow the transport of the goods to the port. This change was immediately accepted by the president of Ibama[106].

- **Legislation favoring landgrabbers:** An analysis conducted by IPAM (Environmental Research Institute of the Amazon) showed that 35% of deforestation in the Amazon between August 2018 and July 2019 was recorded in non-designated areas without information. Regarding land regularization, environmental NGOs warn about two ongoing projects. While, Bill 510/21 was being presented in the Senate, the vote on Bill 2633/20[107] was being considered in the House of Representatives. Both derive from the original text of Provisional Measure 910, known as "*MP da Grilagem*" (Landgrabbers' Provisional Measure), which proposes to change the law to favor large occupants of recently invaded public lands. Bill 510/21 once again changes the deadline for public land invasions to be legalized (from 2011 to 2014) and allows large areas (up to 2500 hectares) to be processed without the need for inspection. Indeed, given that land-grabs of undesignated public lands is responsible for more than 1/3 of the deforestation in the country, it is to be expected that amnesty for landgrabbers and illegal deforesters would be an incentive to intensify this practice in the coming years. Bill 2633/20 has a loophole that would allow legalization, via bidding, of public areas invaded after the deadline for occupation provided for by law (i.e., 2014). Of the 49.8 million hectares of forests under state and federal responsibility, but not yet allocated to any category of use, 11.6 million hectares, or 23%, were irregularly declared as rural properties for private use, under the National System of Rural Environmental Registration (CAR). If the entire area registered to date as private property were legalized, 2.2 to 5.5 million hectares could be deforested in the coming years, according to the deforestation limits defined by the Forest Code and taking into account that deforestation is often beyond permitted limits. In recent years, grabbing of uncategorized public forests has increased: in 2019, this was the land category displaying the most forest felled in the Amazon, according to data from the deforestation alert system of INPE (National Institute of Space Research), Deter. The trend continued in 2020. The following are some of the conditions defined by Provisional Measure 910, for appropriation

[106] https://g1.globo.com/natureza/noticia/2020/11/17/documentos-mostram-que-ibama-facilitou-exportacao-de-madeira-extraida-ilegalmente.ghtml
[107] https://ipam.org.br/35-do-desmatamento-na-amazonia-e-grilagem-indica-analise-do-ipam/

of public lands by individuals: i) the area must be registered in the Rural Environmental Registry (CAR, "*Cadastro Ambiental Rural*"): as it is known, any information can be inputted into the "CAR" system up until the present moment without any veracity checking, and ii) the claimant must be carrying out agricultural activities in the area (i.e., should preferably have deforested the area)[108]. The Provisional Measure defines that, for areas meeting the requirements above and having up to 15 fiscal modules (i.e. up to 1,650 hectares), the title is to be granted without the need for inspection. Before the Provisional Measure, the exemption from inspection was granted to areas with up to four fiscal modules (i.e. with a maximum of 440 hectares). The exemption from inspection may allow large illegally deforested areas to be taken over by individuals. This is because the Provisional Measure prohibits the regularization only of areas that have been subject to fines or environmental embargoes, and not all environmental violations are known and fined by the government[109]. Given that Boa Fe farm is surrounded by public lands and that cases of land-grabbing have been proven in the Reference Region, an abnormal increase in deforestation in that region is expected in the coming years. This is primarily because Brazilian legislation increasingly gives every indication that it is highly welcoming to acts of land-grabbing, by granting amnesty to landgrabbers and agents of illegal deforestation.

- **Increased demand for beef in the international market:** China's demand for beef is still a reflection of swine flu, which has decimated between 40% and 60% of the country's pig stock (about one third of the world's pork production). In addition to this conjuncture, China also contributed to the growth of imports, since it was the only major world economy to record economic growth in 2020, even amid the coronavirus pandemic, and a more long-term factor, which is the gradual increase in income of the Chinese population, which results in progressively higher consumption of more expensive protein sources, such as beef. Analysts estimate that the price of beef should remain under pressure for the next few years, due to the livestock cycle: the low supply of oxen is not something that can quickly be solved, because cattle production is a multi-year cycle, given that production beginning today will deliver animals in two, three, or four years[110]. In 2020, Brazil broke its beef export record, with more than 2 million tons sold (8% more than in 2019). For 2021,

[108]https://ipam.org.br/cientistas-mapeiam-grilagem-em-florestas-publicas-na-amazonia/#:~:text=O%20impact%20da%20grilagem%20se,main%20g%C3%A1s%20%20effect%20estufa
[109] https://amazonia.org.br/como-a-mp-da-grilagem-pode-mudar-o-mapa-de-regioes-da-amazonia/
[110] https://www.bbc.com/portuguese/brasil-55664305

the projection indicates an increase of 5% on the 2020 values[111], indicating a strong increasing export trend for the coming years. Beef exports have continued to increase, growing by almost 7% in 2020 and close to 8% in 2021, increasing by more than 15% in the 2020/2021[112] period. Chinese importers increased their purchases of Brazilian beef by more than 150% in 2020[113].

It is expected that unplanned deforestation is likely to occur in the Project Area in the absence of the REDD+ Project. In accordance with the predominant characteristics of the region, the baseline scenario is of cattle farming conducting illegal levels of deforestation.

In the absence of the REDD+ project, it is assumed that the PA would undergo deforestation – to be supplanted by cattle farms – at an average annual rate of 0.24% per year, resulting in a total of 968.03 hectares deforested by 2052.

The rate of deforestation adopted for calculation of the REDD+ Project benefits was obtained from historical measurements of deforestation utilizing Mapbiomas sources, in accordance with the VM0007 methodology requirements.

Above and belowground carbon pools were determined by means of a literature survey regarding the Project region. Considering that the baseline process of deforestation involves timber harvesting for commercial markets, the content of carbon fixed into long-term wood products was also considered in calculation of net deforestation emissions. It is assumed that the Project Activity preserves soil organic carbon and litter pools to a greater extent than BAU activities. However, for conservativeness purposes, the project proponents decided not to include the soil and litter carbon pools in the REDD+ Project benefits.

Fossil fuel emissions were not accounted for in the Reference Area (baseline case) or for the Project Activity. It is assumed that the Project Activity also reduces emissions from fossil fuel burning, in

[111] https://revistagloborural.globo.com/Noticias/Criacao/Boi/noticia/2021/01/apos-recorde-brasil-projeta-alta-de-5-nas-exportacoes-de-carne-bovina-em-2021.html#:~:text=Segundo%20Abrafrigo%2C%20pa%C3%ADs%20alcan%C3%A7ou%20marca,em%20rela%C3%A7%C3%A3o%20ao%20ano%20anterior&text=As%20exports%C3%A7%C3%B5es%20de%20carne%20bovine,by%20fortes%20shipments%20%C3%A0%20China
[112] https://www.avisite.com.br/index.php?page=noticias&id=21284
[113] https://www2.safras.com.br/eng/2020/09/23/meat-exports-in-brazil-will-be-an-important-differential-in-2021/

comparison with BAU activities. However, this factor was also not accounted for conservativeness purposes and difficulties in monitoring during the project period.

In summary, regarding IPOÁ's AUD component, SCENARIO 3: cattle farming is the baseline scenario of the present REDD+ project, as further justified by the stepwise application of the VCS VT0001 v3.0 Tool for Additionality, which is fully presented in section *3.1.5 Additionality*, below.

***Avoided Planned Deforestation (APD)***

IPOÁ's APD component avoids deforestation of 3.586,77 ha of planned deforestation from 2022 to 2026.

This component concerns forest lands which have been legally authorized and documented to be converted to non-forest land by the project proponents, plans which and will be abandoned thanks to the present REDD+ Project. The proponents have requested to IPAAM permission to clear up to 20% of the Maanaim and Campos propriety, according to the legal limit prescribed in the Brazilian Forest Code. In addition, the landowner has a plan to deforest in the next few years, which will not be executed in case of success in obtaining VCUs.

The figure below shows the suppression schedule in the properties Maanaim and Campos.

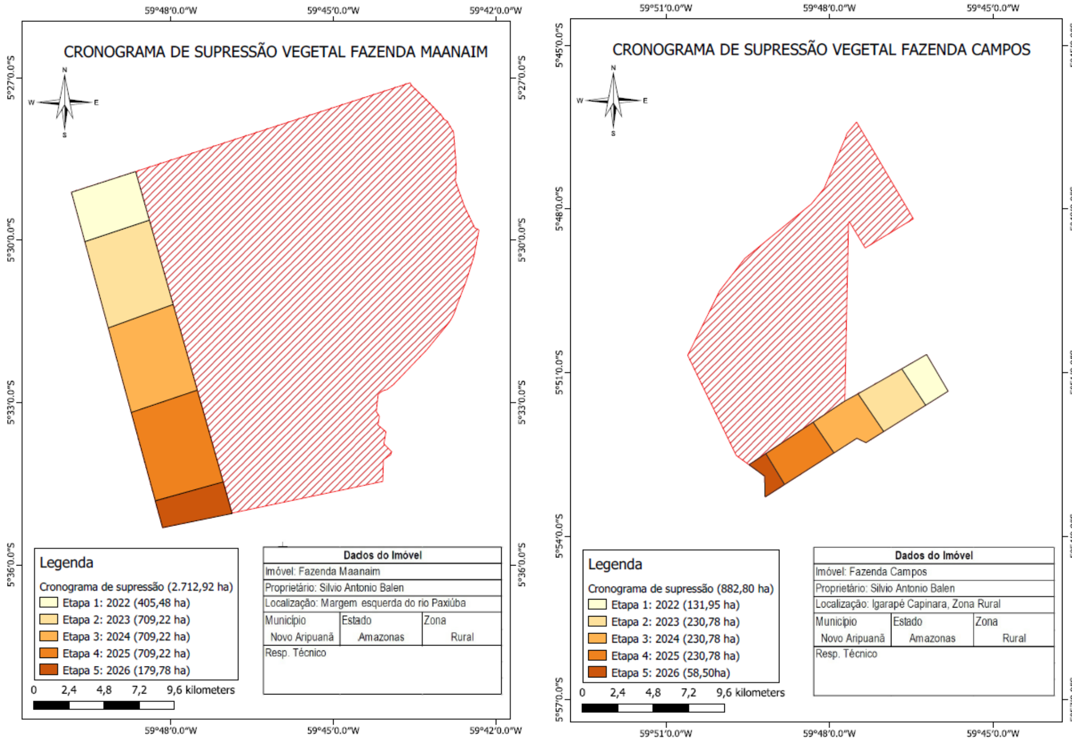


*Figure 38. Suppression schedule in properties Maanaim and Campos.*

The following annual activities were planned within the deforestation schedule:

- May to October – clear-cut area and harvest timber wood
- November to December – remove any remaining vegetation by burning
- December to January – aerial grass seeding
- July – purchase livestock and begin grazing activities

And regarding the APD Deforestation Agent: The agent of deforestation regarding the Avoided Planned Deforestation component of the project is the proponent landowner.

In summary, regarding IPOÁ's APD component, SCENARIO 3: cattle farming is the baseline scenario of the present REDD+ project, as further justified by the stepwise application of the VCS VT0001 v3.0 Tool for Additionality, which is fully presented in section *3.1.5 Additionality*, below.

### 3.1.5 Additionality

According to the procedure defined in the VM0007 methodology, the latest version of the tool referenced in T-ADD was used to identify credible alternative land use scenarios and evaluate both the alternatives and the proposed project scenarios, and to demonstrate the additionality of the project:

"Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD)" (Version 01)"[114]

The tool is applicable to this project, given the following applicability conditions of the tool:

- Conservation of forests within the proposed project boundary performed with or without being registered as the VCS REDD+ project activity[115] shall not lead to violation of any applicable law even if the law is not enforced.
- This tool is not applicable to small - scale afforestation and reforestation project activities.

Regarding the former point, as described in section 2.5 (Legal Status and Property Rights) the project is in compliance with all relevant local, regional, and national laws, statutes, and regulatory frameworks, therefore satisfying this applicability condition.

And regarding the second applicability condition, the project does not involve afforestation and reforestation activities, as pointed out in sections *2.1.11 Project Activities and Theory of Change* and *2.1.13 Implementation Schedule*, therefore fulfilling this applicability condition.

The additionality analysis, applying the T-ADD tool (Version 1), is presented below for AUD and APD activities.

[114] Source, CDM (accessed 13/06/22): https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-02-v1.pdf
[115] This sentence was adapted from the original version in the AR-AM Tool-02-v1, to fit the context of the present VCS REDD project. The original is: "Forestation of the land within the proposed project boundary performed with or without being registered as the A/R CDM project…"

**AUD ACTIVITIES ANALYSIS**

**STEP 0. Preliminary screening based on the starting date of the VCS AFOLU project[116] activity**

According to the step, "If project participants claim that the afforestation or reforestation VCS AFOLU project activity has a starting date after 31 December 1999 but before the date of its registration, then the project participants shall:

I. Provide evidence that the starting date of the VCS AFOLU project activity was after 31 December 1999, and

II. Provide evidence that the incentive from the planned sale of CERs was seriously considered in the decision to proceed with the project activity. This evidence shall be based on (preferably official, legal and/or other corporate) documentation that was available to third parties at, or prior to, the start of the project activity."

Regarding point I. above, the project start date (AUD activities) is defined as 17th March 2022. The project properties already had the explicit contractual intentions of conserving forests and REDD+ projects prior to 3/17/22, however, after 31 December 1999, as evidenced by:

- The purchase and sale agreements that shows the acquirence the Campos property on August 21, 2019, and the Maanaim property on October 13, 2019, with the objective of carbon credits generation and forest management.
- The contract between Carbonext and the landowner.

After the recognition and analysis stage, Carbonext started the implementation of project activities in the field on March 17, 2022. This event is considered as the Project Start Date.

On this occasion, an experienced social technician from Carbonext, accompanied by a local consultant, visited the Project Area, carrying out public consultations with the communities in the Project Zone, presenting the principles and benefits generated by the REDD+ project. The presentation lecture given to local inhabitants is considered the first activity to generate reductions in the project's

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[116] The AR-AM Tool-02-v1.PDF is the updated additionality tool recommended by VM0007, however it is intended for use on A/R CDM project activities. Therefore the term "A/R CDM project", or similar, was altered to "VCS AFOLU project" in all instances (accessed 13/08/21) https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf

GHG emissions, as it made communities aware of preserving the forest and encouraged them to monitor possible illegal deforestation

Regarding point II, the sale of carbon credits, in this case VERs, was seriously considered in the decision to proceed with the project activity. This is proven by the contract between the landowner and Carbonext being signed (on December 2021) prior to the project start date (17th March 2022). Evidence to this effect was duly provided at audit.

**STEP 1. Identification of alternative land use scenarios to the proposed AFOLU project activity**

**Sub-step 1a. Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity**

a) The identified credible non-excluding alternative land use scenarios to the proposed VCS AFOLU project activity include:

SCENARIO 1: Continuation of the pre-project land use:

Under the pre-project scenario, there are no activities being carried out in the Project Area or Project Zone. Thus, there are no costs or revenues for this scenario.

Scenario 1 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project:

Project activities performed without being registered as the VCS AFOLU project. As there are no economic activities besides REDD+ being implemented, the scenario involves costs but no income. These annual costs are estimated at R$202,136.90 throughout the project.
The forest conservation and maintenance activities of Scenario 2 are therefore not financially attractive without carbon credit income. Notwithstanding this fact, Scenario 2 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 3: Cattle farming:

As argued in the previous PD section (3.1.4) pasture is 99,41% of non-natural land uses in Novo Aripuanã. The trends identified and the proximity to the municipality of Apuí are making Novo Aripuanã the new hub of cattle farming expansion in the south of the State of Amazonas, with a 300% increase in the bovine herd in a decade (2010 - 2020).

Therefore, this is a credible scenario.

b) All identified land use scenarios above may be deemed realistic and credible, as they currently exist and are technically feasible in the project region. For all land use scenarios, credibility is justified by current BAU practices attested by the literature and local observations.

c) Outcome of Sub-step 1a: The credible alternative land-use scenarios that could have occurred on the land within the project boundary are SCENARIOS 1, 2 and 3, above.

**Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations**

a) The following procedure was applied:

i) <u>Demonstration that all land use scenarios identified in the sub-step 1a are in compliance with all mandatory applicable legal and regulatory requirements</u>:

SCENARIOS 1 and 2: Maintenance of forest on the area is what is specified as legal within the Law: Documentation of legal possession of the property is the only relevant aspect of law regarding No Economic Activity components of the scenario. Thus, this scenario is in compliance with legal and regulatory requirements.

ii) <u>Demonstration that applicable mandatory legal requirements are systematically not enforced and that non-compliance with requirements is widespread</u>:

SCENARIO 3: <u>Cattle farming</u>:

In the AUD case the activity does not meet the mandatory legal requirements. As widely observed, the southern region of the State of Amazonas is the epicenter of illegal deforestation. Novo Aripuanã is among the municipalities with a high incidence of environmental crimes, local political support for predatory activities, and the absence of federal inspection agencies[117].

[117]https://oglobo.globo.com/brasil/nova-fronteira-do-desmatamento-sul-do-amazonas-tem-capitais-das-queimadas-garimpo-ausencia-do-estado-25314282.

Studies show that in the region, livestock is responsible for 75% of illegal deforestation on public lands in the Amazon, a practice driven by the lack of inspection, land grabbing and the advance of pasture areas[118].

In Novo Aripuanã, deforestation for the opening of pastures takes place closer to the headquarters of the municipality and to the Acari Settlement Project and also in some isolated points that configure land grabbing and land tenure in remote areas. and difficult to access, especially on the banks of the Acari River[119].

According to an IMAZON researcher, efforts should be made to combat illegal deforestation in municipalities such as Novo Aripuanã, which are in the region driven by the agricultural expansion known as AMACRO[120].

Finally, Carrero et al. (2015) note that 71% of slaughterhouses in the region are illegal, arguing in favor of the illegality of the beef chain of production being generally.

Therefore this scenario is not in compliance applicable mandatory legal requirements are systematically not enforced and that this scenario is widespread.

b) Outcome of Sub-step 1b: Is has been demonstrated that SCENARIOS 1, 2 and 3 are plausible alternative land use scenarios to this VCS AFOLU project activity.

**Sub-step 1c. Selection of the baseline scenario:**

SCENARIO 3: Cattle farming is selected as the baseline scenario for the IPOÁ REDD+ project, as it is the predominant land-use in the project region (see sub-step 1a) and although the cattle faming in region is likely to be illegal (see sub-step 1b), this is not enforced and their illegality is ultimately a factor in favour of this activity.

SCENARIO 3: Cattle farming is the baseline scenario of the present REDD+ project.

Given that the scenarios above have been identified, in accordance with the CDM AR-AM Tool, it is proceeded to Step 2 (Barrier analysis).

[118] https://noticias.uol.com.br/meio-ambiente/ultimas-noticias/redacao/2021/10/27/amazonia-87-do-desmate-em-terras-publicas-ocorreu-em-areas-nao-destinadas.htm
[119] Idesam (2011) REDD+ Estudo de Oportunidades para a Região Sul do Amazonas (accessed 13/08/21): https://idesam.org/publicacao/REDD_Estudo_de_Oportunidades_Sul_Amazonas.pdf
[120] https://imazon.org.br/imprensa/desmatamento-na-amazonia-cresce-quase-70-e-atinge-pior-fevereiro-em-10-anos/

**STEP 2. Barrier analysis**

This step serves to identify barriers and to assess which of the land use scenarios identified in the sub-step 1b are not prevented by these barriers.

**Sub-step 2a. Identification of barriers that would prevent the implementation of at least one alternative land use scenarios**

According to T-ADD "Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities". The barriers which the previously identified land-use scenarios may potentially face are the following:

Investment barriers:

- Debt funding is not available for the land-use scenarios.
- No access to international capital markets.
- Lack of access to credit.

Institutional barriers:

- Risk related to changes in government policies or laws.
- Lack of enforcement of forest or land-use-related legislation.

Barriers due to social conditions and land-use practices:

- Widespread illegal practices (e.g. illegal grazing, non-timber product extraction and tree felling).

Barriers relating to land tenure, ownership, inheritance, and property rights:

- Lack of suitable land tenure legislation and regulation to support the security of tenure.
- Possibilities of large price risk due to the fluctuations in the prices of products related to the project activity over the project period in the absence of efficient markets.

**Sub-step 2b.** Elimination of land use scenarios that are prevented by the identified barriers

**This step aims to determine which land use scenarios identified in sub-step 1b are prevented by at least one of the barriers listed in subs-step 2a.**

Investment barriers:

Scenarios 1 and 2: Investment barriers exist in terms of Scenarios 1 and 2, as there are very few options for conservation of forest on private properties, above and beyond legal requirements, without generating any financial returns. The notable exception to this are Private Natural Heritage Reserves ("RPPNs" in Portuguese) which in the State of Amazonas are on average much smaller than the project area (0.2%, on average, of the Project Area as described in Step 4 below) and they generate further costs and no particular incentives or subsidies, as shown in Decree 5746/2006.[121].

Scenario 3: However, regarding Scenario 3 (Cattle farming), having illegal levels of deforestation generally does not prevent this land-use scenario from obtaining funding. There are numerous funding resources and credit lines for cattle raising activities in Brazil[122]. Moreover, the implementation of cattle raising activities can be funded by initial capital obtained with timber sales after deforestation, and subsequently benefit from speculation of the value of deforested land.

Institutional barriers: This barrier hinders legally compliant alternative land use scenario (specifically, Scenario 1 - 2), as it represents unfair competition from non-compliant producers (Lack of enforcement of forest or land-use-related legislation), causing uncertainty to investors (Risk related to changes in government policies or laws).

Barriers due to social conditions and land-use practices: This barrier hinders any compliant alternative land-use scenarios (Scenarios 1 – 2), due to widespread illegal practices, which lead to unfair competition from a great portion of regional producers and insecurity related to land tenure (trespassing by squatters). On the other hand, this factor favors cattle raising and pastureland (Scenario 3) which, as previously described, often depends on illegal practices to be financially viable.

Barriers relating to land tenure, ownership, inheritance, and property rights: This barrier hinders the legally compliant land-use scenarios (Scenarios 1 – 2) and represents an incentive to squatters and land-grabbers, which are in many cases associated with cattle farming[123] Scenario 3.

**Sub-step 2c**. Determination of baseline scenario (if allowed by the barrier analysis)

[121] Decree 5746/2006: http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2006/decreto/d5746.htm
[122]Sources:https://www.bcb.gov.br/estabilidadefinanceira/creditorural;https://www.bndes.gov.br/wps/portal/site/home/onde-atuamos/agropecuaria
[123]Sources:https://antigo.mma.gov.br/estruturas/225/_arquivos/9___a_grilagem_de_terras_pblicas_na_amaznia_brasileira_225.pdf ;

As put into context above, Scenario 3 (cattle farming) would not be hindered or prevented by any of the barriers identified in Sub-step 2a. Thus, Scenario 3 is confirmed as the most plausible alternative land-use scenario, in the absence of the project activity.

In this sense, according to the tool:

"Forestation" (or the REDD+ project activity, in this case) "without being registered as a VCS AFOLU project activity" is not included in the list of land use scenarios that are not prevented by any barrier"

And the list contains only one land use scenario, therefore it is necessary to continue with Step 4: Common practice test.

**STEP 4. Common practice test**

According to T-ADD "CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities", the previous steps are complemented with an analysis of the extent to which similar activities have already diffused in the geographical area of the proposed VCS AFOLU project activity. This test is a credibility check to demonstrate additionality that complements the barrier analysis (Step 2). Other registered VCS AFOLU project activities, as the Juma Sustainable Development Reserve Project, shall not be included in this analysis.

This analysis took into account similar activities in the project region, being the municipalities of Novo Aripuanã, Amazonas, and the type of land area analyzed were private natural heritage reserves, a land category known as RPPN in Brazil. This is because they are private conservation reserves, and in this sense are comparable to the IPOÁ area.

It was found through the official organ ICMbio, that in the state of Amazonas there are only 14 such areas, with an average of 62.8 ha. The nearest RPPN is located 260 km from the IPOÁ REDD+ project and is of 52.06 ha in area.

Therefore, we can cite at least two essential differences: the private conservation areas are incomparably smaller (0.2%) in size to the IPOÁ REDD+ project, and they are very distant (260 km) from the project area, and not located in the same jurisdictional area (municipality).

**In this regard, the proposed VCS REDD+ project activity is not the baseline scenario and, hence, it is additional.**

**APD ACTIVITIES ANALISIS**

**STEP 0. Preliminary screening based on the starting date of the VCS AFOLU project[124] activity**

According to the step, "If project participants claim that the afforestation or reforestation VCS AFOLU project activity has a starting date after 31 December 1999 but before the date of its registration, then the project participants shall:

I. Provide evidence that the starting date of the VCS AFOLU project activity was after 31 December 1999, and

II. Provide evidence that the incentive from the planned sale of CERs was seriously considered in the decision to proceed with the project activity. This evidence shall be based on (preferably official, legal and/or other corporate) documentation that was available to third parties at, or prior to, the start of the project activity."

Regarding point I. above, the project start date (APD activities) is defined as 3th May 2022, as it was the date that the project owner submitted a request to the official organ IPAAM, to obtain permission to deforest a legal proportion of the Aroeira property in the Project Area (PA).

The justification is that maintaining the PA forested was no longer judged to be financially viable without carbon credits by the project owner, as of that date.

Regarding point II, the sale of carbon credits, in this case VERs, was seriously considered in the decision to proceed with the project activity. This is proven by the contract between the land owner and Carbonext being signed (on December 2021) prior to the project start date (3th May 2022). Evidence to this effect was duly provided at audit.

[124] The AR-AM Tool-02-v1.PDF is the updated additionality tool recommended by VM0007, however it is intended for use on A/R CDM project activities. Therefore the term "A/R CDM project", or similar, was altered to "VCS AFOLU project" in all instances (accessed 13/08/21) https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf

**STEP 1. Identification of alternative land use scenarios to the proposed AFOLU project activity**

**Sub-step 1a. Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity**

a) The identified credible non-excluding alternative land use scenarios to the proposed VCS AFOLU project activity include:

SCENARIO 1: Continuation of the pre-project land use:

Under the pre-project scenario, there are no activities being carried out in the Project Area or Project Zone. Thus, there are no costs or revenues for this scenario.

Scenario 1 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project:

Project activities performed without being registered as the VCS AFOLU project. As there are no economic activities besides REDD+ being implemented, the scenario involves costs but no income. These annual costs are estimated at R$202,136.90 throughout the project.
The forest conservation and maintenance activities of Scenario 2 are therefore not financially attractive without carbon credit income. Notwithstanding this fact, Scenario 2 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 3: Cattle farming:

As argued in the previous PD section (3.1.4) pasture is 99,41% of non-natural land uses in Novo Aripuanã. The trends identified and the proximity to the municipality of Apuí are making Novo Aripuanã the new hub of cattle farming expansion in the south of the State of Amazonas, with a 300% increase in the bovine herd in a decade (2010 - 2020).

Therefore, this is a credible scenario.

b) All identified land use scenarios above may be deemed realistic and credible, as they currently exist and are technically feasible in the project region. For all land use scenarios, credibility is justified by current BAU practices attested by the literature and local observations.

c) Outcome of Sub-step 1a: The credible alternative land-use scenarios that could have occurred on the land within the project boundary are SCENARIOS 1, 2 and 3, above.

**Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations**

a) The following procedure was applied:

i) Demonstration that all land use scenarios identified in the sub-step 1a are in compliance with all mandatory applicable legal and regulatory requirements:

SCENARIOS 1 and 2: Maintenance of forest on the area is what is specified as legal within the Law: Documentation of legal possession of the property is the only relevant aspect of law regarding No Economic Activity components of the scenario. Thus, this scenario is in compliance with legal and regulatory requirements.

ii) Demonstration that applicable mandatory legal requirements are systematically not enforced and that non-compliance with requirements is widespread:

SCENARIO 3: Cattle farming: Deforestation and cattle farming on the area is specified as legal within the Law in APD case: deforestation of up to 20% of a property is permitted in accordance with the Brazilian Forest Code, which stipulates that a property should officially allocate 80% of its total area as LR (Legal Reserve) for conservation. The remaining 20% of land may be deforested by license. Thus, this scenario is in compliance with legal and regulatory requirements.

b) Outcome of Sub-step 1b: Is has been demonstrated that SCENARIOS 1, 2 and 3 are plausible alternative land use scenarios to this VCS AFOLU project activity.

**Sub-step 1c. Selection of the baseline scenario:**

SCENARIO 3: Cattle farming is selected as the baseline scenario for the IPOÁ REDD+ project, as it is the predominant land-use in the project region (see sub-step 1a) and deforestation of up to 20% of the property for cattle farming is a legal and planned activity (see sub-step 1b).

SCENARIO 3: Small-scale beef cattle farming is the baseline scenario of the present REDD+ project.

Given that the scenarios above have been identified, in accordance with the CDM AR-AM Tool, it is proceeded to Step 2 (Barrier analysis).

**STEP 2. Barrier analysis**

This step serves to identify barriers and to assess which of the land use scenarios identified in the sub-step 1b are not prevented by these barriers.

**Sub-step 2a. Identification of barriers that would prevent the implementation of at least one alternative land use scenarios**

According to T-ADD "Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities". The barriers which the previously identified land-use scenarios may potentially face are the following:

Investment barriers:

- Debt funding is not available for the for the land-use scenarios.
- No access to international capital markets.
- Lack of access to credit.

Institutional barriers:

- Risk related to changes in government policies or laws.
- Lack of enforcement of forest or land-use-related legislation.

Barriers due to social conditions and land-use practices:

- Widespread illegal practices (e.g. illegal grazing, non-timber product extraction and tree felling).

Barriers relating to land tenure, ownership, inheritance, and property rights:

- Lack of suitable land tenure legislation and regulation to support the security of tenure.
- Possibilities of large price risk due to the fluctuations in the prices of products related to the project activity over the project period in the absence of efficient markets.

**Sub-step 2b.** Elimination of land use scenarios that are prevented by the identified barriers

**This step aims to determine which land use scenarios identified in sub-step 1b are prevented by at least one of the barriers listed in subs-step 2a.**

Investment barriers:

Scenarios 1 and 2: Investment barriers exist in terms of Scenarios 1 and 2, as there are very few options for conservation of forest on private properties, above and beyond legal requirements, without generating any financial returns. The notable exception to this are Private Natural Heritage Reserves ("RPPNs" in Portuguese) which in the State of Amazonas are on average much smaller than the project area (0.2%, on average, of the Project Area as described in Step 4 below) and they generate further costs and no particular incentives or subsidies, as shown in Decree 5746/2006[125].

Scenario 3: However, regarding Scenario 3 (Cattle farming), access to finance is facilitated, even more so when conducted legally. There are numerous funding resources and credit lines for cattle raising activities in Brazil[126]. Moreover, the implementation of cattle raising activities can be funded by initial capital obtained with timber sales after deforestation, and subsequently benefit from speculation of the value of deforested land.

Institutional barriers:

Scenarios 1 and 2: This barrier disproportionally hinders legally compliant alternative land-use scenario that do not have revenue streams, as it promotes unfair competition from non-compliant producers (Lack of enforcement of forest or land-use-related legislation), causing uncertainty to investors (Risk related to changes in government policies or laws).

Scenario 3: This barrier does not hinder scenario 3, since there is no indication that changes in legislation will negatively impact the implementation of legal cattle ranching. On the contrary, in recent years Brazilian environmental policy has become more flexible with the legalization of activities potentially impacting the environment, such as the case of Provisional Measure 1040 of March 29, 2021, which provides automatic environmental licensing for companies and Decree 10,084, of 2019, which revoked the agroecological zoning of sugarcane. The lack of enforcement benefits illegal activities, but does not prevent the execution of legal cattle farming, which are a consolidated market in Brazil.

Barriers due to social conditions and land-use practices: This barrier hinders compliant alternative land-use scenarios without revenues (Scenarios 1 – 2), due to widespread illegal practices, which lead to unfair competition from a great portion of regional producers and insecurity related to land tenure (trespassing by squatters). On the

[125] Decree 5746/2006: http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2006/decreto/d5746.htm
[126]Sources:https://www.bcb.gov.br/estabilidadefinanceira/creditorural; https://www.bndes.gov.br/wps/portal/site/home/onde-atuamos/agropecuaria

other hand, this factor does not prevent cattle raising and pastureland (Scenario 3) which, as previously described, which are a consolidated market in Brazil, increasingly supported by the country's environmental policy.

Barriers relating to land tenure, ownership, inheritance, and property rights: This barrier hinders the legally compliant land-use scenarios without revenues (Scenarios 1 – 2) and represents an incentive to squatters and land-grabbers. On the other hand, this factor does not prevent cattle raising and pastureland (Scenario 3) which, as previously described, which are a consolidated market in Brazil, increasingly supported by the country's environmental policy.

**Sub-step 2c**. Determination of baseline scenario (if allowed by the barrier analysis)

As put into context above, Scenario 3 (cattle farming) would not be hindered or prevented by any of the barriers identified in Sub-step 2a. Thus, Scenario 3 is confirmed as the most plausible alternative land-use scenario, in the absence of the project activity.

In this sense, according to the tool:

"Forestation" (or the REDD+ project activity, in this case) "without being registered as a VCS AFOLU project activity" is not included in the list of land-use scenarios that are not prevented by any barrier"

And the list contains only one land use scenario, therefore it is necessary to continue with Step 4: Common practice test.

**STEP 4. Common practice test**

According to T-ADD "CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities", the previous steps are complemented with an analysis of the extent to which similar activities have already diffused in the geographical area of the proposed VCS AFOLU project activity. This test is a credibility check to demonstrate additionality that complements the barrier analysis (Step 2). Other registered VCS AFOLU project activities, as the Juma Sustainable Development Reserve Project, shall not be included in this analysis.

This analysis took into account similar activities in the project region, being the municipalitie of Novo Aripuanã, Amazonas, and the type of land area analyzed were private natural heritage reserves, a land category known as RPPN in Brazil. This is because they are private conservation reserves, and in this sense are comparable to the IPOÁ area.

It was found through the official organ ICMbio, that in the state of Amazonas there are only 14 such areas, with an average of 62.8 ha. The nearest RPPN is located 260 km from the IPOÁ REDD+ project and is of 52.06 ha in area.

Therefore, we can cite at least two essential differences: the private conservation areas are incomparably smaller (0.2%) in size to the IPOÁ REDD+ project, and they are very distant (260 km) from the project area, and not located in the same jurisdictional area (municipality).

In this regard, the proposed VCS REDD+ project activity is not the baseline scenario and, hence, it is additional.

### 3.1.6 Methodology Deviations

No methodology deviation was adopted in application of methodology.

## 3.2 Quantification of GHG Emission Reductions and Removals

### 3.2.1 Baseline Emissions

The net baseline carbon stock changes and greenhouse gas emissions of the project were calculated through two different modules of the VM0007 Methodology. In order to estimate the baseline emissions relating to APD, being forest lands on which the proponents have applied to IPAAM for permission to clear forests (20% of the Maanaim propriety and 20% of the Campos propriety according to the legal limit prescribed in the Brazilian Forest Code), the VMD0006 module (BL-PL) “Estimation of Baseline Carbon Stock Changes and Greenhouse Gas Emissions from Planned Deforestation and Planned Degradation” was applied. Regarding the estimation of baseline emissions from unplanned deforestation, AUD scenario, where agents of deforestation clear the land for ranching as final land use, the VMD0007 module (BL-UP) was applied: “Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation”.

The baseline net GHG emissions for planned deforestation (APD) were determined as:

$$\Delta C_{BSL,planned} = \sum_{t=1}^{t*} \sum_{i=1}^{M} (\Delta C_{BSL,i,t} + GHG_{BSL-E,i,t})$$

Where:

$\Delta C_{BSL,planned}$ Net greenhouse gas emissions in the baseline from planned deforestation up to year *t**; t $CO_2$-e

$\Delta C_{BSL,i,t}$ Net carbon stock changes in all pools in the baseline stratum *i* in year *t*; t $CO_2$-e

$GHG_{BSL\text{-}E,i,t}$ Greenhouse gas emissions as a result of deforestation activities within the project boundary in the baseline stratum *i* in year *t*; t $CO_2$-e $yr^{-1}$

*i* 1, 2, 3, … *M* strata

*t* 1, 2, 3, … *t** years elapsed since the projected start of the project activity

Where:

$\Delta C_{BSL,\ planned}$ Net greenhouse gas emissions in the baseline from planned deforestation up to year t*; t CO2-e

$\Delta C_{BSL,\ i,t}$ Net carbon stock changes in all pools in the baseline stratum *i* in year *t*; t $CO_2$-e

$GHG_{BSL-E,i,t}$ Greenhouse gas emissions as a result of deforestation activities within the project boundary in the baseline stratum *i* in year *t*; t $CO_2$-e $yr^{-1}$

$i$ 1, 2, 3 ... *M* strata

$t$ 1, 2, 3 ... t years elapsed since the start of the REDD+ project activity

The baseline net GHG emissions for unplanned deforestation (AUD) were determined by applying the module VMD0007 according to four parts, such as:

Part 1 DEFINITION OF BOUNDARIES

Part 2 ESTIMATION OF ANNUAL AREAS OF UNPLANNED DEFORESTATION

Part 3 LOCATION AND QUANTIFICATION OF THREAT OF UNPLANNED DEFORESTATION

Part 4 ESTIMATION OF CARBON STOCK CHANGES AND GREENHOUSE GAS EMISSIONS

As explained, the X-UNC106 Estimation of Uncertainty for REDD+ Project Activities module focuses on the following sources of uncertainty applicable to this project: i) Determination of rates of deforestation and degradation; ii) Uncertainty associated with estimation of stocks in carbon pools and changes in carbon stocks.

It is assumed that there is zero uncertainty in baseline rate of deforestation (item “i” above), as numbers are equal to a long-term average (BL-UP; which is the case in this project, where deforestation rate was taken as the average of the reference period) and are based on actual deforestation plans (BL-PL; which is the case in this project, as a planned deforestation plan is available).

Regarding initial and post-deforestation carbon stocks (item “ii” above), initial forest biomass data were obtained from peer-reviewed literature applying to the project’s geographic region, in accordance biome, and climate regime as in the project area. All values were obtained from the National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015). Regarding the total biomass data for the project estimations, the numbers were as follows:

- Dense Tropical Submontane Rainforest (code “Ds”): 420.66 t/ha (total Ab +Bb)

- Dense Tropical Lowland Rainforest (code Dd): 421,87 t/ha (total Ab +Bb)

- Dense Tropical Alluvial Rainforest (code Da): 478,92 t/ha (total Ab +Bb)

For the post-deforestation carbon stock (i.e., pasture carbon stock; item “ii” above), the same value adopted by the authority responsible for the National GHG Inventory was used in calculations; thus, it can be considered that an official national value has been taken and uncertainty can be regarded as zero in this case.

The calculation of uncertainty was done at a 95% confidence level, considering standard deviation values and number of samples provided by the bibliography. The results were Ds: 7.33%, Da: 9.02% and Db: 5.84% of uncertainty. Given that initial carbon stocks are the only source of uncertainty in the estimate of the total net GHG emission reductions, and uncertainty was estimated to be under 15% for all forest types, and no deduction resulted for uncertainty, as the allowable uncertainty in this methodology REDD+ MF is +/- 15% at a 95% confidence level.

***Location analysis***

As deforestation location analysis was not required due to being a transition deforestation configuration, the following conservative approach is defined as mandatory in accordance with the methodology: "Future deforestation is assumed to happen first in the strata with the lowest carbon stocks (in all relevant carbon pools)". Specifically, the stratum with the lower carbon stock of the three classes, where future deforestation is predicted to occur first, is the submontane dense tropical rainforest.

*Table 17 - Deforestation risk strata of the IPOÁ REDD+ Project*

| Stratum | | |
|---|---|---|
| **Dense Tropical Submontane Rainforest (ha)** | **Dense Tropical Lowland Rainforest (ha)** | **Dense Tropical Alluvial Rainforest (ha)** |
| 10,084.0 | 2,244.40 | 1,471.20 |
| Low to high biomass → | | |

***Estimation of the annual areas of unplanned baseline deforestation in the RRD***

*Table 18 - Projection of annual baseline deforestation for the Project Area (Location Analysis), over the project lifetime, in the Dense Tropical Submontane Rainforest stratum.*

| Stratum | | | |
|---|---|---|---|
| **Dense Tropical Submontane Rainforest** | | | |
| **Annual baseline deforestation for the Project Area** | | | |
| **Year** | **Conservative approach** | **Total yearly area (ha/year)** | **Total cumulative area (ha)** |
| 2022 | 19.5 | 19.5 | 19.5 |
| 2023 | 24.4 | 24.4 | 43.9 |
| 2024 | 24.4 | 24.4 | 68.3 |
| 2025 | 24.3 | 24.3 | 92.6 |
| 2026 | 24.3 | 24.3 | 116.9 |
| 2027 | 24.2 | 24.2 | 141.1 |
| 2028 | 24.2 | 24.2 | 165.3 |
| 2029 | 24.1 | 24.1 | 189.4 |
| 2030 | 24.0 | 24.0 | 213.4 |
| 2031 | 24.0 | 24.0 | 237.4 |
| 2032 | 23.9 | 23.9 | 261.3 |
| 2033 | 23.9 | 23.9 | 285.2 |
| 2034 | 23.8 | 23.8 | 309.0 |
| 2035 | 23.7 | 23.7 | 332.7 |
| 2036 | 23.7 | 23.7 | 356.4 |
| 2037 | 23.6 | 23.6 | 380.0 |

| Stratum | | | |
|---|---|---|---|
| Dense Tropical Submontane Rainforest | | | |
| Annual baseline deforestation for the Project Area | | | |
| Year | Conservative approach | Total yearly area (ha/year) | Total cumulative area (ha) |
| 2038 | 23.6 | 23.6 | 403.6 |
| 2039 | 23.5 | 23.5 | 427.1 |
| 2040 | 23.5 | 23.5 | 450.6 |
| 2041 | 23.4 | 23.4 | 474.0 |
| 2042 | 23.3 | 23.3 | 497.3 |
| 2043 | 23.3 | 23.3 | 520.6 |
| 2044 | 23.2 | 23.2 | 543.8 |
| 2045 | 23.2 | 23.2 | 567.0 |
| 2046 | 32.1 | 32.1 | 599.2 |
| 2047 | 32.1 | 32.1 | 631.2 |
| 2048 | 23.0 | 23.0 | 654.2 |
| 2049 | 22.9 | 22.9 | 677.1 |
| 2050 | 22.9 | 22.9 | 699.9 |
| 2051 | 22.8 | 22.8 | 722.7 |
| 2052 | 22.8 | 22.8 | 745.5 |

*Table 19. Projection of annual baseline deforestation for the Project Area (Location Analysis), over the project lifetime, in the Dense Tropical Lowland Rainforest stratum.*

| Stratum | | | |
|---|---|---|---|
| Dense Tropical Lowland Rainforest | | | |
| Annual baseline deforestation for the Project Area | | | |
| Year | Conservative approach | Total yearly area (ha/year) | Total cumulative area (ha) |
| 2022 | 0.0 | 0.0 | 0.0 |
| 2023 | 0.0 | 0.0 | 0.0 |
| 2024 | 0.0 | 0.0 | 0.0 |
| 2025 | 0.0 | 0.0 | 0.0 |
| 2026 | 0.0 | 0.0 | 0.0 |
| 2027 | 0.0 | 0.0 | 0.0 |
| 2028 | 0.0 | 0.0 | 0.0 |
| 2029 | 0.0 | 0.0 | 0.0 |
| 2030 | 0.0 | 0.0 | 0.0 |
| 2031 | 0.0 | 0.0 | 0.0 |
| 2032 | 0.0 | 0.0 | 0.0 |
| 2033 | 0.0 | 0.0 | 0.0 |
| 2034 | 0.0 | 0.0 | 0.0 |
| 2035 | 0.0 | 0.0 | 0.0 |
| 2036 | 0.0 | 0.0 | 0.0 |
| 2037 | 0.0 | 0.0 | 0.0 |

| Stratum | | | |
|---|---|---|---|
| Dense Tropical Lowland Rainforest | | | |
| Annual baseline deforestation for the Project Area | | | |
| Year | Conservative approach | Total yearly area (ha/year) | Total cumulative area (ha) |
| 2038 | 0.0 | 0.0 | 0.0 |
| 2039 | 0.0 | 0.0 | 0.0 |
| 2040 | 0.0 | 0.0 | 0.0 |
| 2041 | 0.0 | 0.0 | 0.0 |
| 2042 | 0.0 | 0.0 | 0.0 |
| 2043 | 0.0 | 0.0 | 0.0 |
| 2044 | 0.0 | 0.0 | 0.0 |
| 2045 | 0.0 | 0.0 | 0.0 |
| 2046 | 0.0 | 0.0 | 0.0 |
| 2047 | 0.0 | 0.0 | 0.0 |
| 2048 | 0.0 | 0.0 | 0.0 |
| 2049 | 0.0 | 0.0 | 0.0 |
| 2050 | 0.0 | 0.0 | 0.0 |
| 2051 | 0.0 | 0.0 | 0.0 |
| 2052 | 0.0 | 0.0 | 0.0 |

*Table 20. Projection of annual baseline deforestation for the Project Area (Location Analysis), over the project lifetime, in the Dense Tropical Alluvial Rainforest stratum.*

| Stratum | | | |
|---|---|---|---|
| Dense Tropical Alluvial Rainforest | | | |
| Annual baseline deforestation for the Project Area | | | |
| Year | Conservation approach | Total yearly area (ha/year) | Total cumulative area (ha) |
| 2022 | 0.0 | 0.0 | 0.0 |
| 2023 | 0.0 | 0.0 | 0.0 |
| 2024 | 0.0 | 0.0 | 0.0 |
| 2025 | 0.0 | 0.0 | 0.0 |
| 2026 | 0.0 | 0.0 | 0.0 |
| 2027 | 0.0 | 0.0 | 0.0 |
| 2028 | 0.0 | 0.0 | 0.0 |
| 2029 | 0.0 | 0.0 | 0.0 |
| 2030 | 0.0 | 0.0 | 0.0 |
| 2031 | 0.0 | 0.0 | 0.0 |
| 2032 | 0.0 | 0.0 | 0.0 |
| 2033 | 0.0 | 0.0 | 0.0 |
| 2034 | 0.0 | 0.0 | 0.0 |
| 2035 | 0.0 | 0.0 | 0.0 |
| 2036 | 0.0 | 0.0 | 0.0 |
| 2037 | 0.0 | 0.0 | 0.0 |

| Stratum | | | |
|---|---|---|---|
| Dense Tropical Alluvial Rainforest | | | |
| Annual baseline deforestation for the Project Area | | | |
| Year | Conservation approach | Total yearly area (ha/year) | Total cumulative area (ha) |
| 2038 | 0.0 | 0.0 | 0.0 |
| 2039 | 0.0 | 0.0 | 0.0 |
| 2040 | 0.0 | 0.0 | 0.0 |
| 2041 | 0.0 | 0.0 | 0.0 |
| 2042 | 0.0 | 0.0 | 0.0 |
| 2043 | 0.0 | 0.0 | 0.0 |
| 2044 | 0.0 | 0.0 | 0.0 |
| 2045 | 0.0 | 0.0 | 0.0 |
| 2046 | 0.0 | 0.0 | 0.0 |
| 2047 | 0.0 | 0.0 | 0.0 |
| 2048 | 0.0 | 0.0 | 0.0 |
| 2049 | 0.0 | 0.0 | 0.0 |
| 2050 | 0.0 | 0.0 | 0.0 |
| 2051 | 0.0 | 0.0 | 0.0 |
| 2052 | 0.0 | 0.0 | 0.0 |

The resulting average annual rate of annual deforestation rate in the submontane dense stratum is 0.24%.

Regarding baseline deforestation areas from planned deforestation, the baseline areas for planned deforestation were as follows, as evidenced during the project audit.

*Table 21 - Annual baseline deforestation for the Project Area from planned deforestation in Dense Tropical Submontane Rainforest.*

| Stratum: Dense Tropical Submontane Rainforest (ha) | | |
|---|---|---|
| Gross baseline emissions from planned deforestation | | |
| Year | ha/year | ha (cumulative) |
| 2022 | 537.43 | 537.43 |
| 2023 | 940.00 | 1,477.43 |
| 2024 | 940.00 | 2,417.43 |
| 2025 | 940.00 | 3,357.43 |
| 2026 | 238.29 | 3,595.72 |

*Table 22. Annual baseline deforestation for the Project Area from planned deforestation in Dense Tropical Lowland Rainforest.*

| Stratum: Dense Tropical Lowland Rainforest (ha) | | |
|---|---|---|
| **Gross baseline emissions from planned deforestation** | | |
| **Year** | **ha/year** | **ha (cumulative)** |
| 2022 | 0.0 | 0.0 |
| 2023 | 0.0 | 0.0 |
| 2024 | 0.0 | 0.0 |
| 2025 | 0.0 | 0.0 |
| 2026 | 0.0 | 0.0 |

*Table 23. Annual baseline deforestation for the Project Area from planned deforestation in Dense Tropical Alluvial Rainforest.*

| Stratum: Dense Tropical Alluvial Rainforest (ha) | | |
|---|---|---|
| **Gross baseline emissions from planned deforestation** | | |
| **Year** | **ha/year** | **ha (cumulative)** |
| 2022 | 0.0 | 0.0 |
| 2023 | 0.0 | 0.0 |
| 2024 | 0.0 | 0.0 |
| 2025 | 0.0 | 0.0 |
| 2026 | 0.0 | 0.0 |

***Biomass***

As previously described in section 2.1.5, three forest classes with varying biomass levels are present in Project Area: Dense tropical submontane rainforest (Ds) (covering 73% of the total area); Dense Tropical Lowland Rainforest (Dd) (16%); and Dense Tropical Lowland Alluvial Rainforest (Da) (11%) biomass values for all forest types' were obtained from literature: Ministério da Ciência, Tecnologia e Inovação, (2015),with above-ground values ranging from 265 to 302 t/ha, shown below.

In order to calculate the carbon pools (above and belowground) for each stratum, the total biomass of the total area of each stratum was multiplied by a CF (Carbon Fraction) in Dry Matter equal to 0.4722 (conversion from dry mass to tC), and then by 44/12 (conversion factor from tC to $tCO_2$[127])

*Table 24. The total biomass of the total area of each stratum.*

| Stratum | AUD Area (ha) | Proportion (%) | Aboveground biomass (total) (t/ha) | Belowground biomass (t/ha) | Total aboveground per stratum (t) | Total belowground per stratum (t) |
|---|---|---|---|---|---|---|
| Dense Tropical Submontane Rainforest | 10,084 | 73% | 265 | 156 | 2,672,419 | 1,569,516 |
| Dense Tropical Lowland Rainforest | 2,244 | 16% | 266 | 156 | 596,512 | 350,333 |
| Dense Tropical Alluvial Rainforest | 1,471 | 11% | 302 | 177 | 443,890 | 260,697 |
| **Total** | **13,800** | **100%** | | | **3,712,822** | **2,180,546** |

| | |
|---|---|
| **Under Deforestation Risk Analysis** | **13,800** |
| | **CF** |
| | **0.47** |

| | **Carbon Pools** | |
|---|---|---|
| **Stratum** | **Aboveground, per stratum ($tCO_2$)** | **Belowground, per stratum ($tCO_2$)** |
| Dense Tropical Submontane Rainforest | 4,605,469 | 2,704,799 |
| Dense Tropical Lowland Rainforest | 1,027,990 | 603,740 |
| Dense Tropical Alluvial Rainforest | 764,970 | 449,268 |
| **Total** | 6,398,429 | 3,757,808 |
| | **$C_{AB\_tree}$** | **$C_{BB\_tree}$** |
| Dense Tropical Submontane Rainforest | 457 | 268 |
| Dense Tropical Lowland Rainforest | 458 | 269 |
| Dense Tropical Allvial Rainforest | 520 | 305 |

[127] Default value 0.47 tC t-1 d.m. (3_CP-AB, pg. 12). https://verra.org/wp-content/uploads/2017/11/VMD0001v1.1.pdf

***Baseline emissions from unplanned deforestation (AUD)***

In order to estimate emissions from unplanned deforestation that would occur in the Project Area in the absence of project (i.e. in the baseline case), the annual estimated area of deforestation (see "Projection of annual baseline deforestation for the Project Area") multiplied by the sum of aboveground and belowground carbon stocks in forest for each biomass stratum (see the table in the biomass section above). The result of this procedure is shown in the tables below.

*Table 25 - Summary of gross baseline emissions from unplanned deforestation that would occur within the Project Area in Dense Tropical Submontane Rainforest.*

| Stratum | | | | |
|---|---|---|---|---|
| Dense Tropical Submontane Rainforest | | | | |
| Gross baseline emissions from deforestation | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ |
| 2022 | 19.5 | 19.5 | 14,109.18 | 14,109.18 |
| 2023 | 24.4 | 43.9 | 17,723.83 | 31,833.00 |
| 2024 | 24.4 | 68.3 | 17,680.77 | 49,513.78 |
| 2025 | 24.3 | 92.6 | 17,637.82 | 67,151.60 |
| 2026 | 24.3 | 116.9 | 17,594.98 | 84,746.58 |
| 2027 | 24.2 | 141.1 | 17,552.24 | 102,298.81 |
| 2028 | 24.2 | 165.3 | 17,509.60 | 119,808.41 |
| 2029 | 24.1 | 189.4 | 17,467.06 | 137,275.47 |
| 2030 | 24.0 | 213.4 | 17,424.63 | 154,700.10 |
| 2031 | 24.0 | 237.4 | 17,382.30 | 172,082.41 |
| 2032 | 23.9 | 261.3 | 17,340.08 | 189,422.49 |
| 2033 | 23.9 | 285.2 | 17,297.96 | 206,720.44 |
| 2034 | 23.8 | 309.0 | 17,255.94 | 223,976.38 |
| 2035 | 23.7 | 332.7 | 17,214.02 | 241,190.40 |
| 2036 | 23.7 | 356.4 | 17,172.20 | 258,362.60 |
| 2037 | 23.6 | 380.0 | 17,130.49 | 275,493.09 |
| 2038 | 23.6 | 403.6 | 17,088.87 | 292,581.96 |
| 2039 | 23.5 | 427.1 | 17,047.36 | 309,629.32 |
| 2040 | 23.5 | 450.6 | 17,005.95 | 326,635.27 |
| 2041 | 23.4 | 474.0 | 16,964.64 | 343,599.91 |
| 2042 | 23.3 | 497.3 | 16,923.43 | 360,523.34 |
| 2043 | 23.3 | 520.6 | 16,882.32 | 377,405.66 |
| 2044 | 23.2 | 543.8 | 16,841.31 | 394,246.97 |
| 2045 | 23.2 | 567.0 | 16,800.40 | 411,047.37 |
| 2046 | 32.1 | 599.2 | 23,302.93 | 434,350.29 |
| 2047 | 32.1 | 631.2 | 23,246.32 | 457,596.62 |
| 2048 | 23.0 | 654.2 | 16,646.51 | 474,243.12 |

| Stratum | | | | |
|---|---|---|---|---|
| **Dense Tropical Submontane Rainforest** | | | | |
| **Gross baseline emissions from deforestation** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$** |
| 2049 | 22.9 | 677.1 | 16,606.07 | 490,849.19 |
| 2050 | 22.9 | 699.9 | 16,565.73 | 507,414.92 |
| 2051 | 22.8 | 722.7 | 16,525.49 | 523,940.41 |
| 2052 | 22.8 | 745.5 | 16,519.62 | 540,460.03 |

*Table 26. Summary of gross baseline emissions from unplanned deforestation that would occur within the Project Area in Dense Tropical Lowland Rainforest.*

| Stratum | | | | |
|---|---|---|---|---|
| **Dense Tropical Lowland Rainforest** | | | | |
| **Annual baseline deforestation for the Project Area** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum | | | | |
|---|---|---|---|---|
| Dense Tropical Lowland Rainforest | | | | |
| Annual baseline deforestation for the Project Area | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$** |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 27. Summary of gross baseline emissions from unplanned deforestation that would occur within the Project Area in Dense Tropical Alluvial Rainforest.*

| Stratum | | | | |
|---|---|---|---|---|
| Dense Tropical Alluvial Rainforest | | | | |
| Annual baseline deforestation for the Project Area | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

***Baseline emissions from planned deforestation (APD)***

In order to estimate emissions from planned deforestation that would occur in the Project Area in the absence of project (i.e. in the baseline case), the annual estimated area of deforestation (see "Projection of annual baseline deforestation for the Project Area") multiplied by the sum of aboveground and belowground carbon stocks in forest for each biomass stratum (see the table in the biomass section above). The result of this procedure is shown in the tables below.

*Table 28 - Summary of gross baseline emissions from planned deforestation that would occur within the Project Area in Dense Tropical Submontane Rainforest.*

| Stratum: Submontane dense | | | | |
|---|---|---|---|---|
| Gross baseline emissions from deforestation | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 537.43 | 537.43 | 400,391.85 | 400,391.85 |
| 2023 | 940.00 | 1,477.43 | 700,311.36 | 1,100,703.21 |
| 2024 | 940.00 | 2,417.43 | 685,394.10 | 1,786,097.31 |
| 2025 | 940.00 | 3,357.43 | 674,220.10 | 2,460,317.40 |
| 2026 | 238.29 | 3,595.72 | 151,437.66 | 2,611,755.06 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 29 - Summary of gross baseline emissions from planned deforestation that would occur within the Project Area in Dense Tropical Lowland Rainforest.*

| **Stratum: Submontane Lowland** | | | | |
|---|---|---|---|---|
| **Gross baseline emissions from deforestation** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 30 - Summary of gross baseline emissions from planned deforestation that would occur within the Project Area in Dense Tropical Lowland Rainforest.*

| **Stratum: Submontane Alluvial** | | | | |
|---|---|---|---|---|
| **Gross baseline emissions from deforestation** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |

| | | | | |
|---|---|---|---|---|
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

***Emissions from biomass burning in the baseline***

Based on the IPCC 2006 Inventory Guidelines, estimating greenhouse gas emissions from biomass burning was determined as follows. The same methodology was applied for both AUD and APD components of the project:

$$E_{BiomassBurn,i,t} = \sum_{g=1}^{G}(\left(\left(A_{burn,i,t} * B_{i,t} * COMF_i * G_{g,i}\right) * 10^{-3}\right) * GWP_g)$$

Where:

$E_{BiomassBurn,\ i,t}$ Greenhouse gas emissions due to biomass burning as part of deforestation activities in stratum *i* in year *t*; $tCO_2$-e of each GHG ($CO_2$, $CH_4$, $N_2O$)

$A_{burn,i,t}$ Area burnt for stratum *i* at time *t*; ha

$B_{i,t}$ Average aboveground biomass stock before burning stratum *i*, time *t*; tonnes d.m. ha-1

$COMF_i$ Combustion factor for stratum *i*; dimensionless (default value derived from Table 2.6 of IPCC, 2006)[128]

$G_{g,i}$ Emission factor for stratum i for gas g; kg t-1 dry matter burnt (default values derived from Table 2.5 of IPCC, 2006)[129]

$GWP_g$ Global warming potential for gas g; t $CO_2$/t gas g (default values from IPCC: $CH_4$ = 28; $N_2O$ = 265)

$g$ 1, 2, 3 ... *G* greenhouse gases

$i$ 1, 2, 3 ... *M* strata

$t$ 1, 2, 3 ... *t* years elapsed since the start of the REDD+ project activity

The table below shows the parameters used to calculate biomass burning for the baseline scenario, as well as results accounted for $CH_4$ and $N_2O$ emissions generated as a consequence of incomplete biomass burning of non-commercial wood after logging, over a 30-year period.

Biomass burning only occurs for the AUD component, therefore the summaries below are regarding the unplanned deforestation, only.

[128] E-BPB refers to Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, “Primary tropical moist forest” https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf (last visited in 20/06/2022)

[129] E-BPB refers to Table 2.5; page 2.47 “Tropical forest” of Volume 4, Chapter 2, of the IPCC 2006 Inventory Guidelines: For $CH_4$: 6.8 - 2 = 4.8 g kg-1 dry matter burnt (conservative); For $N_2O$: 0.20 g kg-1 dry matter burnt (unique value proposed): https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf (last visited in 20/06/2022)

*Table 31 - Summary of biomass burning emissions ($CH_4$) in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense tropical submontane rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 19.46 | 19.46 | 326.09 | 326.09 |
| 2023 | 24.45 | 43.91 | 409.64 | 735.73 |
| 2024 | 24.39 | 68.30 | 408.64 | 1,144.37 |
| 2025 | 24.33 | 92.63 | 407.65 | 1,552.02 |
| 2026 | 24.27 | 116.90 | 406.66 | 1,958.68 |
| 2027 | 24.21 | 141.11 | 405.67 | 2,364.35 |
| 2028 | 24.15 | 165.27 | 404.68 | 2,769.03 |
| 2029 | 24.09 | 189.36 | 403.70 | 3,172.73 |
| 2030 | 24.04 | 213.40 | 402.72 | 3,575.45 |
| 2031 | 23.98 | 237.38 | 401.74 | 3,977.20 |
| 2032 | 23.92 | 261.30 | 400.77 | 4,377.96 |
| 2033 | 23.86 | 285.16 | 399.79 | 4,777.76 |
| 2034 | 23.80 | 308.96 | 398.82 | 5,176.58 |
| 2035 | 23.75 | 332.71 | 397.85 | 5,574.43 |
| 2036 | 23.69 | 356.39 | 396.89 | 5,971.32 |
| 2037 | 23.63 | 380.02 | 395.92 | 6,367.24 |
| 2038 | 23.57 | 403.60 | 394.96 | 6,762.20 |
| 2039 | 23.52 | 427.11 | 394.00 | 7,156.20 |
| 2040 | 23.46 | 450.57 | 393.04 | 7,549.25 |
| 2041 | 23.40 | 473.97 | 392.09 | 7,941.34 |
| 2042 | 23.34 | 497.32 | 391.14 | 8,332.48 |
| 2043 | 23.29 | 520.60 | 390.19 | 8,722.66 |
| 2044 | 23.23 | 543.84 | 389.24 | 9,111.90 |
| 2045 | 23.17 | 567.01 | 388.29 | 9,500.20 |
| 2046 | 32.14 | 599.16 | 538.58 | 10,038.78 |
| 2047 | 32.07 | 631.22 | 537.27 | 10,576.05 |
| 2048 | 22.96 | 654.19 | 384.74 | 10,960.79 |
| 2049 | 22.91 | 677.09 | 383.80 | 11,344.59 |
| 2050 | 22.85 | 699.94 | 382.87 | 11,727.46 |
| 2051 | 22.80 | 722.74 | 381.94 | 12,109.40 |
| 2052 | 22.79 | 745.53 | 381.80 | 12,491.20 |

*Table 32 - Summary of biomass burning emissions ($CH_4$) in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case– AUD component.*

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 33 - Summary of biomass burning emissions ($CH_4$) in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case– AUD component.*

| Stratum: Dense tropical alluvial rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |

| | | | | |
|---|---|---|---|---|
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 34 - Summary of biomass burning emissions ($CH_4$) in the sum of strata, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 19.46 | 19.46 | 326.09 | 326.09 |
| 2023 | 24.45 | 43.91 | 409.64 | 735.73 |
| 2024 | 24.39 | 68.30 | 408.64 | 1,144.37 |
| 2025 | 24.33 | 92.63 | 407.65 | 1,552.02 |
| 2026 | 24.27 | 116.90 | 406.66 | 1,958.68 |
| 2027 | 24.21 | 141.11 | 405.67 | 2,364.35 |
| 2028 | 24.15 | 165.27 | 404.68 | 2,769.03 |
| 2029 | 24.09 | 189.36 | 403.70 | 3,172.73 |
| 2030 | 24.04 | 213.40 | 402.72 | 3,575.45 |
| 2031 | 23.98 | 237.38 | 401.74 | 3,977.20 |
| 2032 | 23.92 | 261.30 | 400.77 | 4,377.96 |
| 2033 | 23.86 | 285.16 | 399.79 | 4,777.76 |
| 2034 | 23.80 | 308.96 | 398.82 | 5,176.58 |
| 2035 | 23.75 | 332.71 | 397.85 | 5,574.43 |

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2036 | 23.69 | 356.39 | 396.89 | 5,971.32 |
| 2037 | 23.63 | 380.02 | 395.92 | 6,367.24 |
| 2038 | 23.57 | 403.60 | 394.96 | 6,762.20 |
| 2039 | 23.52 | 427.11 | 394.00 | 7,156.20 |
| 2040 | 23.46 | 450.57 | 393.04 | 7,549.25 |
| 2041 | 23.40 | 473.97 | 392.09 | 7,941.34 |
| 2042 | 23.34 | 497.32 | 391.14 | 8,332.48 |
| 2043 | 23.29 | 520.60 | 390.19 | 8,722.66 |
| 2044 | 23.23 | 543.84 | 389.24 | 9,111.90 |
| 2045 | 23.17 | 567.01 | 388.29 | 9,500.20 |
| 2046 | 32.14 | 599.16 | 538.58 | 10,038.78 |
| 2047 | 32.07 | 631.22 | 537.27 | 10,576.05 |
| 2048 | 22.96 | 654.19 | 384.74 | 10,960.79 |
| 2049 | 22.91 | 677.09 | 383.80 | 11,344.59 |
| 2050 | 22.85 | 699.94 | 382.87 | 11,727.46 |
| 2051 | 22.80 | 722.74 | 381.94 | 12,109.40 |
| 2052 | 22.79 | 745.53 | 381.80 | 12,491.20 |

*Table 35 - Summary of biomass burning emissions ($N_2O$) in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense tropical submontane rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 19.46 | 19.46 | 128.59 | 128.59 |
| 2023 | 24.45 | 43.91 | 161.54 | 290.13 |
| 2024 | 24.39 | 68.30 | 161.15 | 451.28 |
| 2025 | 24.33 | 92.63 | 160.75 | 612.03 |
| 2026 | 24.27 | 116.90 | 160.36 | 772.39 |
| 2027 | 24.21 | 141.11 | 159.97 | 932.37 |
| 2028 | 24.15 | 165.27 | 159.59 | 1,091.95 |
| 2029 | 24.09 | 189.36 | 159.20 | 1,251.15 |
| 2030 | 24.04 | 213.40 | 158.81 | 1,409.96 |
| 2031 | 23.98 | 237.38 | 158.43 | 1,568.39 |
| 2032 | 23.92 | 261.30 | 158.04 | 1,726.43 |
| 2033 | 23.86 | 285.16 | 157.66 | 1,884.09 |
| 2034 | 23.80 | 308.96 | 157.27 | 2,041.36 |
| 2035 | 23.75 | 332.71 | 156.89 | 2,198.25 |
| 2036 | 23.69 | 356.39 | 156.51 | 2,354.76 |
| 2037 | 23.63 | 380.02 | 156.13 | 2,510.89 |
| 2038 | 23.57 | 403.60 | 155.75 | 2,666.64 |
| 2039 | 23.52 | 427.11 | 155.37 | 2,822.02 |

| Stratum: Dense tropical submontane rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2040 | 23.46 | 450.57 | 155.00 | 2,977.01 |
| 2041 | 23.40 | 473.97 | 154.62 | 3,131.63 |
| 2042 | 23.34 | 497.32 | 154.24 | 3,285.87 |
| 2043 | 23.29 | 520.60 | 153.87 | 3,439.74 |
| 2044 | 23.23 | 543.84 | 153.49 | 3,593.24 |
| 2045 | 23.17 | 567.01 | 153.12 | 3,746.36 |
| 2046 | 32.14 | 599.16 | 212.39 | 3,958.74 |
| 2047 | 32.07 | 631.22 | 211.87 | 4,170.61 |
| 2048 | 22.96 | 654.19 | 151.72 | 4,322.33 |
| 2049 | 22.91 | 677.09 | 151.35 | 4,473.68 |
| 2050 | 22.85 | 699.94 | 150.98 | 4,624.67 |
| 2051 | 22.80 | 722.74 | 150.62 | 4,775.28 |
| 2052 | 22.79 | 745.53 | 150.56 | 4,925.85 |

*Table 36 - Summary of biomass burning emissions ($N_2O$) in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 37 - Summary of biomass burning emissions ($N_2O$) in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense tropical alluvial rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical alluvial rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 38 - Summary of biomass burning emissions ($N_2O$) in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 19.46 | 19.46 | 128.59 | 128.59 |
| 2023 | 24.45 | 43.91 | 161.54 | 290.13 |
| 2024 | 24.39 | 68.30 | 161.15 | 451.28 |
| 2025 | 24.33 | 92.63 | 160.75 | 612.03 |
| 2026 | 24.27 | 116.90 | 160.36 | 772.39 |
| 2027 | 24.21 | 141.11 | 159.97 | 932.37 |
| 2028 | 24.15 | 165.27 | 159.59 | 1,091.95 |
| 2029 | 24.09 | 189.36 | 159.20 | 1,251.15 |
| 2030 | 24.04 | 213.40 | 158.81 | 1,409.96 |
| 2031 | 23.98 | 237.38 | 158.43 | 1,568.39 |
| 2032 | 23.92 | 261.30 | 158.04 | 1,726.43 |
| 2033 | 23.86 | 285.16 | 157.66 | 1,884.09 |
| 2034 | 23.80 | 308.96 | 157.27 | 2,041.36 |
| 2035 | 23.75 | 332.71 | 156.89 | 2,198.25 |
| 2036 | 23.69 | 356.39 | 156.51 | 2,354.76 |
| 2037 | 23.63 | 380.02 | 156.13 | 2,510.89 |
| 2038 | 23.57 | 403.60 | 155.75 | 2,666.64 |
| 2039 | 23.52 | 427.11 | 155.37 | 2,822.02 |
| 2040 | 23.46 | 450.57 | 155.00 | 2,977.01 |
| 2041 | 23.40 | 473.97 | 154.62 | 3,131.63 |
| 2042 | 23.34 | 497.32 | 154.24 | 3,285.87 |
| 2043 | 23.29 | 520.60 | 153.87 | 3,439.74 |
| 2044 | 23.23 | 543.84 | 153.49 | 3,593.24 |
| 2045 | 23.17 | 567.01 | 153.12 | 3,746.36 |
| 2046 | 32.14 | 599.16 | 212.39 | 3,958.74 |
| 2047 | 32.07 | 631.22 | 211.87 | 4,170.61 |
| 2048 | 22.96 | 654.19 | 151.72 | 4,322.33 |
| 2049 | 22.91 | 677.09 | 151.35 | 4,473.68 |
| 2050 | 22.85 | 699.94 | 150.98 | 4,624.67 |
| 2051 | 22.80 | 722.74 | 150.62 | 4,775.28 |

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2052 | 22.79 | 745.53 | 150.56 | 4,925.85 |

*Table 39 - Summary of biomass burning emissions ($CH_4$ and $N_2O$) in the sum of strata, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Sum of Strata | | | | | |
|---|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ and $N_2O$ | | | | | |
| Year | ha/year | ha (cumulative) | $CH_4$ $tCO_2$/ yr | $N_2O$ $tCO_2$/yr | Total $tCO_2$/year |
| 2022 | 19.46 | 19.46 | 326.09 | 128.59 | 454.69 |
| 2023 | 24.45 | 43.91 | 409.64 | 161.54 | 571.17 |
| 2024 | 24.39 | 68.30 | 408.64 | 161.15 | 569.79 |
| 2025 | 24.33 | 92.63 | 407.65 | 160.75 | 568.40 |
| 2026 | 24.27 | 116.90 | 406.66 | 160.36 | 567.02 |
| 2027 | 24.21 | 141.11 | 405.67 | 159.97 | 565.64 |
| 2028 | 24.15 | 165.27 | 404.68 | 159.59 | 564.27 |
| 2029 | 24.09 | 189.36 | 403.70 | 159.20 | 562.90 |
| 2030 | 24.04 | 213.40 | 402.72 | 158.81 | 561.53 |
| 2031 | 23.98 | 237.38 | 401.74 | 158.43 | 560.17 |
| 2032 | 23.92 | 261.30 | 400.77 | 158.04 | 558.81 |
| 2033 | 23.86 | 285.16 | 399.79 | 157.66 | 557.45 |
| 2034 | 23.80 | 308.96 | 398.82 | 157.27 | 556.10 |
| 2035 | 23.75 | 332.71 | 397.85 | 156.89 | 554.74 |
| 2036 | 23.69 | 356.39 | 396.89 | 156.51 | 553.40 |
| 2037 | 23.63 | 380.02 | 395.92 | 156.13 | 552.05 |
| 2038 | 23.57 | 403.60 | 394.96 | 155.75 | 550.71 |
| 2039 | 23.52 | 427.11 | 394.00 | 155.37 | 549.37 |
| 2040 | 23.46 | 450.57 | 393.04 | 155.00 | 548.04 |
| 2041 | 23.40 | 473.97 | 392.09 | 154.62 | 546.71 |
| 2042 | 23.34 | 497.32 | 391.14 | 154.24 | 545.38 |
| 2043 | 23.29 | 520.60 | 390.19 | 153.87 | 544.06 |
| 2044 | 23.23 | 543.84 | 389.24 | 153.49 | 542.73 |
| 2045 | 23.17 | 567.01 | 388.29 | 153.12 | 541.42 |
| 2046 | 32.14 | 599.16 | 538.58 | 212.39 | 750.97 |
| 2047 | 32.07 | 631.22 | 537.27 | 211.87 | 749.14 |
| 2048 | 22.96 | 654.19 | 384.74 | 151.72 | 536.46 |
| 2049 | 22.91 | 677.09 | 383.80 | 151.35 | 535.15 |
| 2050 | 22.85 | 699.94 | 382.87 | 150.98 | 533.85 |
| 2051 | 22.80 | 722.74 | 381.94 | 150.62 | 532.56 |
| 2052 | 22.79 | 745.53 | 381.80 | 150.56 | 532.37 |

*Table 40 - Summary of biomass burning emissions ($CH_4$) in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| **Stratum: Dense tropical submontane rainforest** | | | | |
|---|---|---|---|---|
| **Biomass Burning Emissions: $CH_4$** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | 9,004.57 |
| 2022 | 537.43 | 537.43 | 9,004.57 | 24,754.15 |
| 2023 | 940.00 | 1,477.43 | 15,749.58 | 40,503.74 |
| 2024 | 940.00 | 2,417.43 | 15,749.58 | 56,253.32 |
| 2025 | 940.00 | 3,357.43 | 15,749.58 | 60,245.84 |
| 2026 | 238.29 | 3,595.72 | 3,992.52 | 9,004.57 |
| 2027 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2028 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2029 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2030 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2031 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2032 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2033 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2034 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2035 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2036 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2037 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2038 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2039 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2040 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2041 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2042 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2043 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2044 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2045 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2046 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2047 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2048 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2049 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2050 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2051 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2052 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |

*Table 41 - Summary of biomass burning emissions ($CH_4$) in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case– APD component.*

| **Stratum: Dense tropical lowland rainforest** | | | | |
|---|---|---|---|---|
| **Biomass Burning Emissions: $CH_4$** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 42 - Summary of biomass burning emissions ($CH_4$) in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case– APD component.*

| Stratum: Dense tropical alluvial rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical alluvial rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 43 - Summary of biomass burning emissions ($CH_4$) in the sum of strata, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 537.43 | 537.43 | 9,004.57 | 24,754.15 |
| 2023 | 940.00 | 1,477.43 | 15,749.58 | 40,503.74 |
| 2024 | 940.00 | 2,417.43 | 15,749.58 | 56,253.32 |
| 2025 | 940.00 | 3,357.43 | 15,749.58 | 60,245.84 |
| 2026 | 238.29 | 3,595.72 | 3,992.52 | 9,004.57 |
| 2027 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2028 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2029 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2030 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2031 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2032 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2033 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2034 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2035 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2036 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2037 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2038 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2039 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2040 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2041 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2042 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2043 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2044 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2045 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2046 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2047 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2048 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2049 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2050 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2051 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |
| 2052 | 0.00 | 3,595.72 | 0.00 | 9,004.57 |

*Table 44 - Summary of biomass burning emissions ($N_2O$) in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Dense tropical submontane rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 537.43 | 537.43 | 3,550.91 | 3,550.91 |
| 2023 | 940.00 | 1,477.43 | 6,210.77 | 9,761.68 |
| 2024 | 940.00 | 2,417.43 | 6,210.77 | 15,972.46 |
| 2025 | 940.00 | 3,357.43 | 6,210.77 | 22,183.23 |
| 2026 | 238.29 | 3,595.72 | 1,574.43 | 23,757.66 |
| 2027 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2028 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2029 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2030 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2031 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2032 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2033 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2034 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2035 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |

| Stratum: Dense tropical submontane rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2036 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2037 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2038 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2039 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2040 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2041 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2042 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2043 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2044 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2045 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2046 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2047 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2048 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2049 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2050 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2051 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2052 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |

*Table 45 - Summary of biomass burning emissions ($N_2O$) in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 46 - Summary of biomass burning emissions ($N_2O$) in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Dense tropical alluvial rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical alluvial rainforest | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 47 - Summary of biomass burning emissions ($N_2O$) in the sum of strata, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 537.43 | 537.43 | 3,550.91 | 3,550.91 |
| 2023 | 940.00 | 1,477.43 | 6,210.77 | 9,761.68 |
| 2024 | 940.00 | 2,417.43 | 6,210.77 | 15,972.46 |
| 2025 | 940.00 | 3,357.43 | 6,210.77 | 22,183.23 |
| 2026 | 238.29 | 3,595.72 | 1,574.43 | 23,757.66 |
| 2027 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2028 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2029 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2030 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2031 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2032 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2033 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2034 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2035 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2036 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2037 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2038 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2039 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2040 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2041 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2042 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2043 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2044 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2045 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2046 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2047 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2048 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2049 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Biomass Burning Emissions: $N_2O$ | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2050 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2051 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |
| 2052 | 0,00 | 3,595.72 | 0,00 | 23,757.66 |

*Table 48 - Summary of biomass burning emissions ($CH_4$ and $N_2O$) in the sum of strata, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Sum of Strata | | | | | |
|---|---|---|---|---|---|
| Biomass Burning Emissions: $CH_4$ and $N_2O$ | | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$CH_4$ $tCO_2$/ yr** | **$N_2O$ $tCO_2$/yr** | **Total $tCO_2$/year** |
| 2022 | 537.43 | 537.43 | 9,004.57 | 3,550.91 | 12,555.48 |
| 2023 | 940.00 | 1,477.43 | 15,749.58 | 6,210.77 | 21,960.35 |
| 2024 | 940.00 | 2,417.43 | 15,749.58 | 6,210.77 | 21,960.35 |
| 2025 | 940.00 | 3,357.43 | 15,749.58 | 6,210.77 | 21,960.35 |
| 2026 | 238.29 | 3,595.72 | 3,992.52 | 1,574.43 | 5,566.95 |
| 2027 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2028 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2029 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2030 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2031 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2032 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2033 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2034 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2035 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2036 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2037 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2038 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2039 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2040 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2041 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2042 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2043 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2044 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2045 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2046 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2047 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2048 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2049 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2050 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2051 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |
| 2052 | 0,00 | 3,595.72 | 0,00 | 0,00 | 0,00 |

***Wood products carbon pool in the baseline***

To estimate the biomass carbon of the commercial volume extracted in the process of deforestation, the following equation was applied, which is applicable both for IPOÁ's APD and AUD components, according to "Option 2: Commercial inventory estimation", as recommended in the CP-W:

$$C_{XB,i} = C_{AB_{tree},i} * \frac{1}{BCEF} * Pcom_i$$

Where:

| | |
|---|---|
| $C_{XB,i}$ | Mean stock of extracted biomass carbon from stratum *i*; t $CO_2$-e $ha^{-1}$ |
| $C_{ABtree,i}$ | Mean aboveground biomass carbon stock in stratum *i*; t$CO_2$-e $ha^{-1}$ |
| $BCEF$ | iomass conversion and expansion factor (BCEF) for conversion of merchantable volume to total aboveground tree biomasss; dimesnionless |
| $i$ | 1,2,3, … *M* strata |

In order to calculate the proportion of biomass carbon extracted that remains sequestered in long-term wood products after 100 years, it was simply and conservatively assumed that all extracted biomass not retained in long-term wood products after 100 years is emitted in the year harvested, instead of tracking annual emissions through retirement, burning and decomposition. All factors are derived from Winjum et al. 1998[130].

[130] Winjum, J.K., Brown, S. and Schlamadinger, B. 1998. Forest harvests and wood products: sources and sinks of atmosph eric carbon dioxide. Forest Science 44: 272-284. https://academic.oup.com/forestscience/article/44/2/272/4626952

$$C_{WP,i} = \sum_{ty=s,w,oir,p,o} C_{XB,ty,i} * (1 - WW_{ty}) * (1 - SLF_{ty}) * (1 - OF_{ty})$$

Where:

| | |
|---|---|
| $C_{WP,i}$ | Carbon stock in long-term wood products pool (stock remaining in wood products after 100 years) from stratum i post deforestation; t $CO_2$-e $ha^{-1}$ |
| $C_{XB,ty,i}$ | Mean stock of extracted biomass carbon by class of wood product ty from stratum i; t $CO_2$-e ha-1 |
| $WW_{ty}$ | Wood waste. The fraction immediately emitted through mill inefficiency by class of wood product ty; dimensionless (0.24 for developing countries; Winjum et al. 1998 cited by CP-W) |
| $SLF_{ty}$ | Fraction of wood products that will be emitted to the atmosphere within 5 years of timber harvest by class of wood product ty; dimensionless (0.2 for sawnwood; Winjum et al. 1998 cited by CP-W) |
| $OF_{ty}$ | Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty; dimensionless (0.80 for sawnwood in tropical forests; Winjum et al. 1998 cited by CP-W) |
| Ty | ‘Wood product class – defined here as sawnwood (s) |
| *I* | 1,2, 3,… *M* strata |

The parameters used in calculation of wood products carbon pool in the baseline, as well as the results of estimates (sum of strata), are demonstrated in the table below, for the whole project period.

*Table 49. Summary of wood products in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| **Wood products carbon pool in baseline** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **Cumulative $tCO_2$** |
| 2022 | 19.46 | 19.46 | 63.98 | 63.98 |
| 2023 | 24.45 | 43.91 | 80.37 | 144.35 |
| 2024 | 24.39 | 68.30 | 80.18 | 224.53 |
| 2025 | 24.33 | 92.63 | 79.98 | 304.51 |
| 2026 | 24.27 | 116.90 | 79.79 | 384.30 |
| 2027 | 24.21 | 141.11 | 79.59 | 463.90 |
| 2028 | 24.15 | 165.27 | 79.40 | 543.30 |
| 2029 | 24.09 | 189.36 | 79.21 | 622.51 |
| 2030 | 24.04 | 213.40 | 79.02 | 701.52 |
| 2031 | 23.98 | 237.38 | 78.82 | 780.35 |
| 2032 | 23.92 | 261.30 | 78.63 | 858.98 |
| 2033 | 23.86 | 285.16 | 78.44 | 937.42 |
| 2034 | 23.80 | 308.96 | 78.25 | 1,015.67 |
| 2035 | 23.75 | 332.71 | 78.06 | 1,093.73 |
| 2036 | 23.69 | 356.39 | 77.87 | 1,171.60 |
| 2037 | 23.63 | 380.02 | 77.68 | 1,249.29 |
| 2038 | 23.57 | 403.60 | 77.49 | 1,326.78 |
| 2039 | 23.52 | 427.11 | 77.31 | 1,404.08 |
| 2040 | 23.46 | 450.57 | 77.12 | 1,481.20 |
| 2041 | 23.40 | 473.97 | 76.93 | 1,558.13 |
| 2042 | 23.34 | 497.32 | 76.74 | 1,634.87 |
| 2043 | 23.29 | 520.60 | 76.56 | 1,711.43 |
| 2044 | 23.23 | 543.84 | 76.37 | 1,787.80 |
| 2045 | 23.17 | 567.01 | 76.19 | 1,863.99 |
| 2046 | 32.14 | 599.16 | 105.67 | 1,969.66 |
| 2047 | 32.07 | 631.22 | 105.42 | 2,075.07 |
| 2048 | 22.96 | 654.19 | 75.49 | 2,150.56 |

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| Wood products carbon pool in baseline | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | Cumulative $tCO_2$ |
| 2049 | 22.91 | 677.09 | 75.30 | 2,225.87 |
| 2050 | 22.85 | 699.94 | 75.12 | 2,300.99 |
| 2051 | 22.80 | 722.74 | 74.94 | 2,375.93 |
| 2052 | 22.79 | 745.53 | 74.91 | 2,450.84 |

*Table 50. Summary of wood products in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense Tropical Lowland Rainforest | | | | |
|---|---|---|---|---|
| Wood products carbon pool in baseline | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | Cumulative $tCO_2$ |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense Tropical Lowland Rainforest | | | | |
|---|---|---|---|---|
| **Wood products carbon pool in baseline** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **Cumulative $tCO_2$** |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 51. Summary of wood products in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense Tropical Alluvial Rainforest | | | | |
|---|---|---|---|---|
| **Wood products carbon pool in baseline** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **Cumulative $tCO_2$** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense Tropical Alluvial Rainforest | | | | |
|---|---|---|---|---|
| Wood products carbon pool in baseline | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | Cumulative $tCO_2$ |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 52. Summary of wood products in the sum of strata, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Sum of strata | | | | |
|---|---|---|---|---|
| Wood products carbon pool in baseline | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | Cumulative $tCO_2$ |
| 2022 | 19.46 | 19.46 | 63.98 | 63.98 |
| 2023 | 24.45 | 43.91 | 80.37 | 144.35 |
| 2024 | 24.39 | 68.30 | 80.18 | 224.53 |

| Stratum: Sum of strata | | | | |
|---|---|---|---|---|
| Wood products carbon pool in baseline | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **Cumulative $tCO_2$** |
| 2025 | 24.33 | 92.63 | 79.98 | 304.51 |
| 2026 | 24.27 | 116.90 | 79.79 | 384.30 |
| 2027 | 24.21 | 141.11 | 79.59 | 463.90 |
| 2028 | 24.15 | 165.27 | 79.40 | 543.30 |
| 2029 | 24.09 | 189.36 | 79.21 | 622.51 |
| 2030 | 24.04 | 213.40 | 79.02 | 701.52 |
| 2031 | 23.98 | 237.38 | 78.82 | 780.35 |
| 2032 | 23.92 | 261.30 | 78.63 | 858.98 |
| 2033 | 23.86 | 285.16 | 78.44 | 937.42 |
| 2034 | 23.80 | 308.96 | 78.25 | 1,015.67 |
| 2035 | 23.75 | 332.71 | 78.06 | 1,093.73 |
| 2036 | 23.69 | 356.39 | 77.87 | 1,171.60 |
| 2037 | 23.63 | 380.02 | 77.68 | 1,249.29 |
| 2038 | 23.57 | 403.60 | 77.49 | 1,326.78 |
| 2039 | 23.52 | 427.11 | 77.31 | 1,404.08 |
| 2040 | 23.46 | 450.57 | 77.12 | 1,481.20 |
| 2041 | 23.40 | 473.97 | 76.93 | 1,558.13 |
| 2042 | 23.34 | 497.32 | 76.74 | 1,634.87 |
| 2043 | 23.29 | 520.60 | 76.56 | 1,711.43 |
| 2044 | 23.23 | 543.84 | 76.37 | 1,787.80 |
| 2045 | 23.17 | 567.01 | 76.19 | 1,863.99 |
| 2046 | 32.14 | 599.16 | 105.67 | 1,969.66 |
| 2047 | 32.07 | 631.22 | 105.42 | 2,075.07 |
| 2048 | 22.96 | 654.19 | 75.49 | 2,150.56 |
| 2049 | 22.91 | 677.09 | 75.30 | 2,225.87 |
| 2050 | 22.85 | 699.94 | 75.12 | 2,300.99 |
| 2051 | 22.80 | 722.74 | 74.94 | 2,375.93 |
| 2052 | 22.79 | 745.53 | 74.91 | 2,450.84 |

*Table 53. Summary of wood products in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| Wood products carbon pool in baseline | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | Cumulative $tCO_2$ |
| 2022 | 537.43 | 537.43 | 1,766.74 | 1,766.74 |
| 2023 | 940.00 | 1,477.43 | 3,090.15 | 4,856.89 |
| 2024 | 940.00 | 2,417.43 | 3,090.15 | 7,947.04 |
| 2025 | 940.00 | 3,357.43 | 3,090.15 | 11,037.19 |
| 2026 | 238.29 | 3,595.72 | 783.35 | 11,820.54 |
| 2027 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2028 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2029 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2030 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2031 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2032 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2033 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2034 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2035 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2036 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2037 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2038 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2039 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2040 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2041 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2042 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2043 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2044 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2045 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2046 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2047 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2048 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |

| | | | | |
|---|---|---|---|---|
| 2049 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2050 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2051 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2052 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |

*Table 54. Summary of wood products in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| **Stratum: Dense Tropical Lowland Rainforest** | | | | |
|---|---|---|---|---|
| **Wood products carbon pool in baseline** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **Cumulative $tCO_2$** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |

| | | | | |
|---|---|---|---|---|
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 55. Summary of wood products in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| **Stratum: Dense Tropical Alluvial Rainforest** | | | | |
|---|---|---|---|---|
| **Wood products carbon pool in baseline** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **Cumulative $tCO_2$** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |

| | | | | |
|---|---|---|---|---|
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 56. Summary of wood products in the sum of strata, that would occur within the Project Area in the baseline case – APD component.*

| **Stratum: Sum of strata** | | | | |
|---|---|---|---|---|
| **Wood products carbon pool in baseline** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **Cumulative $tCO_2$** |
| 2022 | 537.43 | 537.43 | 1,766.74 | 1,766.74 |
| 2023 | 940.00 | 1,477.43 | 3,090.15 | 4,856.89 |
| 2024 | 940.00 | 2,417.43 | 3,090.15 | 7,947.04 |
| 2025 | 940.00 | 3,357.43 | 3,090.15 | 11,037.19 |
| 2026 | 238.29 | 3,595.72 | 783.35 | 11,820.54 |
| 2027 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2028 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2029 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2030 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |

| | | | | |
|---|---|---|---|---|
| 2031 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2032 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2033 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2034 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2035 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2036 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2037 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2038 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2039 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2040 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2041 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2042 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2043 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2044 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2045 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2046 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2047 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2048 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2049 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2050 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2051 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |
| 2052 | 0.00 | 3,595.72 | 0.00 | 11,820.54 |

***Pasture carbon pools in the baseline***

Regarding calculation of the carbon pool remaining in pasturelands after deforestation, a conservative value of 27.8 $tCO_2$/ha ($CABnon{-}treepost$,i + $CBBnon{-}tree,post$,i) was applied[131] for both AUD and APD components. The proportion of baseline deforestation converted to pasture was defined as 100%.

[131] Source: http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf (Table 5, is the value of 7.57 tC/ha, for the Amazon biome, multiplied by the 44/12 conversion factor)

The table below summarizes the results obtained for pasture carbon pools in the baseline scenario, for 30 years of project:

*Table 57 - Summary of pasture carbon pool in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| **Baseline Pasture Carbon Pool** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 19.46 | 19.46 | 540.22 | 540.22 |
| 2023 | 24.45 | 43.91 | 678.62 | 1,218.83 |
| 2024 | 24.39 | 68.30 | 676.97 | 1,895.80 |
| 2025 | 24.33 | 92.63 | 675.32 | 2,571.12 |
| 2026 | 24.27 | 116.90 | 673.68 | 3,244.81 |
| 2027 | 24.21 | 141.11 | 672.05 | 3,916.85 |
| 2028 | 24.15 | 165.27 | 670.41 | 4,587.27 |
| 2029 | 24.09 | 189.36 | 668.79 | 5,256.05 |
| 2030 | 24.04 | 213.40 | 667.16 | 5,923.21 |
| 2031 | 23.98 | 237.38 | 665.54 | 6,588.75 |
| 2032 | 23.92 | 261.30 | 663.92 | 7,252.68 |
| 2033 | 23.86 | 285.16 | 662.31 | 7,914.99 |
| 2034 | 23.80 | 308.96 | 660.70 | 8,575.69 |
| 2035 | 23.75 | 332.71 | 659.10 | 9,234.79 |
| 2036 | 23.69 | 356.39 | 657.50 | 9,892.28 |
| 2037 | 23.63 | 380.02 | 655.90 | 10,548.18 |
| 2038 | 23.57 | 403.60 | 654.31 | 11,202.48 |
| 2039 | 23.52 | 427.11 | 652.72 | 11,855.20 |
| 2040 | 23.46 | 450.57 | 651.13 | 12,506.33 |
| 2041 | 23.40 | 473.97 | 649.55 | 13,155.88 |
| 2042 | 23.34 | 497.32 | 647.97 | 13,803.85 |
| 2043 | 23.29 | 520.60 | 646.40 | 14,450.25 |
| 2044 | 23.23 | 543.84 | 644.83 | 15,095.07 |
| 2045 | 23.17 | 567.01 | 643.26 | 15,738.33 |
| 2046 | 32.14 | 599.16 | 892.23 | 16,630.56 |
| 2047 | 32.07 | 631.22 | 890.06 | 17,520.63 |
| 2048 | 22.96 | 654.19 | 637.37 | 18,157.99 |
| 2049 | 22.91 | 677.09 | 635.82 | 18,793.81 |
| 2050 | 22.85 | 699.94 | 634.27 | 19,428.09 |
| 2051 | 22.80 | 722.74 | 632.73 | 20,060.82 |
| 2052 | 22.79 | 745.53 | 632.51 | 20,693.33 |

*Table 58 - Summary of pasture carbon pool in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| **Stratum: Dense tropical lowland rainforest** | | | | |
|---|---|---|---|---|
| **Baseline Pasture Carbon Pool** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 59 - Summary of pasture carbon pool in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case – AUD component.*

| **Stratum: Dense tropical aluvial rainforest** | | | | |
|---|---|---|---|---|
| **Baseline Pasture Carbon Pool** | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical aluvial rainforest | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 60 - Summary of pasture carbon pool in the sum of strata, that would occur within the Project Area in the baseline case – AUD component.*

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 19.46 | 19.46 | 540.22 | 540.22 |
| 2023 | 24.45 | 43.91 | 678.62 | 1,218.83 |
| 2024 | 24.39 | 68.30 | 676.97 | 1,895.80 |
| 2025 | 24.33 | 92.63 | 675.32 | 2,571.12 |
| 2026 | 24.27 | 116.90 | 673.68 | 3,244.81 |
| 2027 | 24.21 | 141.11 | 672.05 | 3,916.85 |

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| 2028 | 24.15 | 165.27 | 670.41 | 4,587.27 |
| 2029 | 24.09 | 189.36 | 668.79 | 5,256.05 |
| 2030 | 24.04 | 213.40 | 667.16 | 5,923.21 |
| 2031 | 23.98 | 237.38 | 665.54 | 6,588.75 |
| 2032 | 23.92 | 261.30 | 663.92 | 7,252.68 |
| 2033 | 23.86 | 285.16 | 662.31 | 7,914.99 |
| 2034 | 23.80 | 308.96 | 660.70 | 8,575.69 |
| 2035 | 23.75 | 332.71 | 659.10 | 9,234.79 |
| 2036 | 23.69 | 356.39 | 657.50 | 9,892.28 |
| 2037 | 23.63 | 380.02 | 655.90 | 10,548.18 |
| 2038 | 23.57 | 403.60 | 654.31 | 11,202.48 |
| 2039 | 23.52 | 427.11 | 652.72 | 11,855.20 |
| 2040 | 23.46 | 450.57 | 651.13 | 12,506.33 |
| 2041 | 23.40 | 473.97 | 649.55 | 13,155.88 |
| 2042 | 23.34 | 497.32 | 647.97 | 13,803.85 |
| 2043 | 23.29 | 520.60 | 646.40 | 14,450.25 |
| 2044 | 23.23 | 543.84 | 644.83 | 15,095.07 |
| 2045 | 23.17 | 567.01 | 643.26 | 15,738.33 |
| 2046 | 32.14 | 599.16 | 892.23 | 16,630.56 |
| 2047 | 32.07 | 631.22 | 890.06 | 17,520.63 |
| 2048 | 22.96 | 654.19 | 637.37 | 18,157.99 |
| 2049 | 22.91 | 677.09 | 635.82 | 18,793.81 |
| 2050 | 22.85 | 699.94 | 634.27 | 19,428.09 |
| 2051 | 22.80 | 722.74 | 632.73 | 20,060.82 |
| 2052 | 22.79 | 745.53 | 632.51 | 20,693.33 |

*Table 61 - Summary of pasture carbon pool in the Dense Tropical Submontane Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 537.43 | 537.43 | 14,917.27 | 14,917.27 |
| 2023 | 940.00 | 1,477.43 | 26,091.27 | 41,008.53 |
| 2024 | 940.00 | 2,417.43 | 26,091.27 | 67,099.80 |
| 2025 | 940.00 | 3,357.43 | 26,091.27 | 93,191.07 |
| 2026 | 238.29 | 3,595.72 | 6,614.14 | 99,805.20 |
| 2027 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2028 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2029 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2030 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2031 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2032 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2033 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2034 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2035 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2036 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2037 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2038 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2039 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2040 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2041 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2042 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2043 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2044 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2045 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2046 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2047 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2048 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2049 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2050 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2051 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2052 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |

*Table 62 - Summary of pasture carbon pool in the Dense Tropical Lowland Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical lowland rainforest | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 63 - Summary of pasture carbon pool in the Dense Tropical Alluvial Rainforest stratum, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Dense tropical aluvial rainforest | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical aluvial rainforest | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 64 - Summary of pasture carbon pool in the sum of strata, that would occur within the Project Area in the baseline case – APD component.*

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$ (cumulative)** |
| 2022 | 537.43 | 537.43 | 14,917.27 | 14,917.27 |
| 2023 | 940.00 | 1,477.43 | 26,091.27 | 41,008.53 |
| 2024 | 940.00 | 2,417.43 | 26,091.27 | 67,099.80 |
| 2025 | 940.00 | 3,357.43 | 26,091.27 | 93,191.07 |
| 2026 | 238.29 | 3,595.72 | 6,614.14 | 99,805.20 |
| 2027 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2028 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2029 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2030 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2031 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2032 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2033 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2034 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2035 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2036 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2037 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2038 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2039 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2040 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2041 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2042 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2043 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2044 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2045 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2046 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2047 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Baseline Pasture Carbon Pool | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$ (cumulative) |
| 2048 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2049 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2050 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2051 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |
| 2052 | 0.00 | 3,595.72 | 0.00 | 99,805.20 |

### 3.2.2 Project Emissions

The Project Area currently does not carry out any activities, so there are no project emissions to report at the time of audit.

*Table 65. Logging Gap in the project area.*

| Year | Total area exploited per year (ha/year) | Extracted wood ($m^3$/ha) | Total extracted wood (t/year) | Damage ($tCO_2$/year) | Gross emission ($tCO_2$/year) |
|---|---|---|---|---|---|
| 2022 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2023 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2024 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2025 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2026 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2027 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2028 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2029 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2030 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2031 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2032 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2033 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2034 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2035 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2036 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2037 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |

| Year | Total area exploited per year (ha/year) | Extracted wood ($m^3$/ha) | Total extracted wood (t/year) | Damage ($tCO_2$/year) | Gross emission ($tCO_2$/year) |
|---|---|---|---|---|---|
| 2038 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2039 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2040 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2041 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2042 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2043 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2044 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2045 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2046 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2047 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2048 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2049 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2050 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2051 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2052 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |

**Emissions arising in the logging gap**

In the project case, emissions occur as a direct result of the death of the timber tree and due to the death of trees killed when the timber tree is felled.

If there are where harvesting operations and management techniques vary in the project area, or where logging will take place in both broadleaf/mixed forests and stratified coniferous forests. The net in the case of the project is equal to the biomass of harvested wood plus the harvest damage factor multiplied by the volume harvested, summed across strata:

$$C_{LG,i,t} = \sum_{z=1}^{Z}(C_{EXT,z,i,t} + (LDF_{z,i} * V_{EXT,z,i,t} * \frac{44}{12}))$$

Where:

| | |
|---|---|
| $C_{LG,i,t}$ | Actual net project emissions arising in the logging gap , in stratum *i* in year t; t $CO_2$-e |
| $C_{EXT,\ z,i,t}$ | Biomass carbon stock of timber extracted within the project boundary for logging stratum *z*, in stratum *i* in year *t*; t $CO_2$-e |
| $LDF_{z,i}$ | Logging damage factor for logging stratum z, in stratum *i*; t C $m^{-3}$ |
| $V_{EXT,z,i,t}$ | Volume extracted from logging stratum *z*, in stratum *i* in year t; $m^3$ |
| *Z* | 1, 2, 3, …*Z* logging strata |
| *i* | 1, 2, 3, … *M* strata |
| *t* | 1, 2, 3, … *t* years elapsed since the start of the project activity |

Calculate the biomass of the total volume extracted from within each logging stratum:

$$C_{EXT,z,i,t} = \sum_{j=1}^{S} (V_{EXT,j,z,i,t} * D_j * CF_j * \frac{44}{12})$$

Where:

| | |
|---|---|
| $C_{EXT,z,i,t}$ | Biomass carbon stock of timber extracted within the project boundary for logging stratum *z*, in stratum *i* in year *t*; t $CO_2$-e |
| $V_{EXT,j,z,i,t}$ | The volume of timber extracted of species j for logging stratum *z*, in stratum *i* in year *t*; $m^3$ |
| $D_j$ | Basic wood density of species *j*; t d.m.$m^{-3}$ |
| $CF_j$ | Carbon fraction of biomass for tree species *j*; t C $t^{-1}$ d.m. |
| *z* | 1, 2, 3, …*Z* logging strata |
| *j* | 1, 2, 3, … $S_{PS}$ tree species |
| *t* | 1, 2, 3, … *t* years elapsed since the start of the project activity |

The logging damage factor (LDF) is a representation of the quantity of emissions that will ultimately arise per unit of extracted timber ($m^3$). These emissions arise from the noncommercial portion of the felled trees (the branched and stump) and trees incidentally killed during felling.

### 3.2.3 Leakage

***Market Leakage - LK-ME***

Similarly, to the Reference Area and Project Area, the Leakage Belt is also subject to risks of landgrabbing promoted by illegal organizations (i.e. family-scale land-grabber associations, landproperty documentation forgers), mostly supported by unscrupulous sawmills and political interests. Total GHG emissions due to market-effects leakage through decreased timber harvest is equal to the summed emissions from timber harvest in the baseline case potentially displaced plus a leakage factor:

$$LK_{MarketEffects,Timber} = \sum_{i=1}^{M} (LF_{Me} * AL_{T,i})$$

Where:

$LK_{MarketEffects,\,Timber}$ Total GHG emissions due to market- effects leakage through decreased timber harvest; t $CO_2$-e

$LF_{Me}$ Leakage factor for market-effects calculations; dimensionless

$AL_{T,\,i}$ Summed emissions from timber harvest in stratum *i* in the baseline case potentially displaced through implementation of carbon project; t $CO_2$-e

*i* 1,2, 3…*M* strata

The deduction factor (LFME) was adopted as 0.2, given that the biomass of total aboveground tree is considered to be 32,65% greater in the Amazon Biome than in the Project Area. According to the module VMD0011 (LK-ME), when the mean biomass is more than 15% greater than the biomass within the project boundaries, the LFME shall be considered 0.2. When evaluating the Leakage Belt deforestation factors and general forest characteristics (stratification, forest type, carbon stocks), the same parameters used for the Project Area are assumed to be equal in the Leakage Belt. This assumption is valid based on previous similarity analysis, which demonstrates that the Leakage Belt area and the Project Area present similarities in the following criteria: Politics and Legislation, Soil and Landscape, Climate and Vegetation, and Accessibility.

The total volume that would have been logged in the baseline in the Project Area, across strata and time periods, is estimated as follows:

$$AL_{T,i} = \sum_{t=1}^{t} \left( C_{BSL,XBT,i,t} \right)$$

Where:

| | |
|---|---|
| $AL_{T,\ i}$ | Summed emissions from timber harvest in stratum *i* in the baseline case potentially displaced through implementation of carbon project; t $CO_2$-e |
| $C_{BSL,\ XBT,\ i,t}$ | Carbon emission due to displaced timber harvests in the baseline scenario in stratum *i* in time *t*; t $CO_2$-e |
| *i* | 1,2, 3…*M* strata |
| *t* | 1, 2, 3, … *t* years elapsed since the projected start of the REDD+ project activity |

The carbon emission due to the displaced logging has two components: the biomass carbon of the extracted timber and the biomass carbon in the forest damaged in the process of timber extraction:

$$C_{BSL,XBT,I,T} = \left( \left[ V_{BSL,XE,I,t} * D_{mn} * CF \right] + \left[ V_{BSL,XE,I,t} * LDF \right] + \left[ V_{BSL,XE,I,t} * LIF \right] \right) * \frac{44}{12}$$

Where:

$C_{BSL,\ XBT,\ i,t}$ Carbon emission due to displaced timber harvests in the baseline scenario in stratum *i* in time *t*; t $CO_2$-e

$V_{BSL,\ XE,\ I,t}$ Volume of timber projected to be extracted from within the project boundary during the baseline in stratum *i* at time *t*; $m^3$

$D_{mn}$ Mean wood density of commercially harvested species; t d.m.$m^{-3}$

CF Carbon fraction of biomass for commercially harvested species j; t C td.$m^{-1}$

LDF Logging damage factor; t C $m^{-3}$ (default 0.53 t C $m^{-3}$ for broadleaf and mixed forests[132])

LIF Logging infrastructure factor; t C m-3 (default 0.29 t C m-3)

*i* 1,2, 3…*M* strata

*t* 1, 2, 3, …*t* years elapsed since the projected start of the REDD+ project activity

The table below summarizes the calculation steps and results of market leakage estimates for the IPOÁ REDD+ Project.

*Table 66 - Total leakage due to market effects per stratum (1)*

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$/year |
| 2022 | 19.46 | 19.46 | 2,747.33 | 549.5 |
| 2023 | 24.45 | 43.91 | 3,451.17 | 690.2 |
| 2024 | 24.39 | 68.30 | 3,442.79 | 688.6 |
| 2025 | 24.33 | 92.63 | 3,434.43 | 686.9 |
| 2026 | 24.27 | 116.90 | 3,426.08 | 685.2 |
| 2027 | 24.21 | 141.11 | 3,417.76 | 683.6 |
| 2028 | 24.15 | 165.27 | 3,409.46 | 681.9 |

[132] Source: https://verra.org/wp-content/uploads/2018/03/VMD0011-LK-ME-v1.0.pdf (pg 4)

| Stratum: Dense Tropical Submontane Rainforest | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$/year |
| 2029 | 24.09 | 189.36 | 3,401.18 | 680.2 |
| 2030 | 24.04 | 213.40 | 3,392.91 | 678.6 |
| 2031 | 23.98 | 237.38 | 3,384.67 | 676.9 |
| 2032 | 23.92 | 261.30 | 3,376.45 | 675.3 |
| 2033 | 23.86 | 285.16 | 3,368.25 | 673.6 |
| 2034 | 23.80 | 308.96 | 3,360.07 | 672.0 |
| 2035 | 23.75 | 332.71 | 3,351.90 | 670.4 |
| 2036 | 23.69 | 356.39 | 3,343.76 | 668.8 |
| 2037 | 23.63 | 380.02 | 3,335.64 | 667.1 |
| 2038 | 23.57 | 403.60 | 3,327.54 | 665.5 |
| 2039 | 23.52 | 427.11 | 3,319.45 | 663.9 |
| 2040 | 23.46 | 450.57 | 3,311.39 | 662.3 |
| 2041 | 23.40 | 473.97 | 3,303.34 | 660.7 |
| 2042 | 23.34 | 497.32 | 3,295.32 | 659.1 |
| 2043 | 23.29 | 520.60 | 3,287.31 | 657.5 |
| 2044 | 23.23 | 543.84 | 3,279.33 | 655.9 |
| 2045 | 23.17 | 567.01 | 3,271.36 | 654.3 |
| 2046 | 32.14 | 599.16 | 4,537.53 | 907.5 |
| 2047 | 32.07 | 631.22 | 4,526.51 | 905.3 |
| 2048 | 22.96 | 654.19 | 3,241.40 | 648.3 |
| 2049 | 22.91 | 677.09 | 3,233.52 | 646.7 |
| 2050 | 22.85 | 699.94 | 3,225.67 | 645.1 |
| 2051 | 22.80 | 722.74 | 3,217.83 | 643.6 |
| 2052 | 22.79 | 745.53 | 3,216.69 | 643.3 |

*Table 67. Total leakage due to market effects per stratum (2)*

| Stratum: Dense Tropical Lowland Rainforest | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$/year |
| 2022 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.0 |

| Stratum: Dense Tropical Lowland Rainforest | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2048 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.0 |

*Table 68. Total leakage due to market effects per stratum (3)*

| Stratum: Dense Tropical Alluvial Rainforest | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.0 |

| Stratum: Dense Tropical Alluvial Rainforest | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2040 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.0 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.0 |

*Table 69. Total leakage due to market effects: sum of strata*

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2022 | 19.46 | 19.46 | 2,747.33 | 549.5 |
| 2023 | 24.45 | 43.91 | 3,451.17 | 690.2 |
| 2024 | 24.39 | 68.30 | 3,442.79 | 688.6 |
| 2025 | 24.33 | 92.63 | 3,434.43 | 686.9 |
| 2026 | 24.27 | 116.90 | 3,426.08 | 685.2 |
| 2027 | 24.21 | 141.11 | 3,417.76 | 683.6 |
| 2028 | 24.15 | 165.27 | 3,409.46 | 681.9 |
| 2029 | 24.09 | 189.36 | 3,401.18 | 680.2 |
| 2030 | 24.04 | 213.40 | 3,392.91 | 678.6 |
| 2031 | 23.98 | 237.38 | 3,384.67 | 676.9 |

| Stratum: Sum of Strata | | | | |
|---|---|---|---|---|
| Market Leakage | | | | |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$/year |
| 2032 | 23.92 | 261.30 | 3,376.45 | 675.3 |
| 2033 | 23.86 | 285.16 | 3,368.25 | 673.6 |
| 2034 | 23.80 | 308.96 | 3,360.07 | 672.0 |
| 2035 | 23.75 | 332.71 | 3,351.90 | 670.4 |
| 2036 | 23.69 | 356.39 | 3,343.76 | 668.8 |
| 2037 | 23.63 | 380.02 | 3,335.64 | 667.1 |
| 2038 | 23.57 | 403.60 | 3,327.54 | 665.5 |
| 2039 | 23.52 | 427.11 | 3,319.45 | 663.9 |
| 2040 | 23.46 | 450.57 | 3,311.39 | 662.3 |
| 2041 | 23.40 | 473.97 | 3,303.34 | 660.7 |
| 2042 | 23.34 | 497.32 | 3,295.32 | 659.1 |
| 2043 | 23.29 | 520.60 | 3,287.31 | 657.5 |
| 2044 | 23.23 | 543.84 | 3,279.33 | 655.9 |
| 2045 | 23.17 | 567.01 | 3,271.36 | 654.3 |
| 2046 | 32.14 | 599.16 | 4,537.53 | 907.5 |
| 2047 | 32.07 | 631.22 | 4,526.51 | 905.3 |
| 2048 | 22.96 | 654.19 | 3,241.40 | 648.3 |
| 2049 | 22.91 | 677.09 | 3,233.52 | 646.7 |
| 2050 | 22.85 | 699.94 | 3,225.67 | 645.1 |
| 2051 | 22.80 | 722.74 | 3,217.83 | 643.6 |
| 2052 | 22.79 | 745.53 | 3,216.69 | 643.3 |

***Leakage outside the Leakage Belt (Step 4 - LK-ASU)***

Immigrants prevented from migrating into and deforesting the project area are conservatively assumed to migrate to an alternative forest area and to cause deforestation in the alternative area. The alternative forest area could be within the Leakage Belt or it could be elsewhere in the country. The proportion migrating to the Leakage Belt is calculated as the area of the Leakage Belt as a proportion of the total available forest area nationally (AVFOR). AVFOR was estimated as follows:

$$AVFOR = TOTFOR - PROTFOR - MANFOR$$

Where:

| | |
|---|---|
| *AVFOR* | Total available national forest area for unplanned deforestation; ha |
| *TOTFOR* | Total available national forest area; ha |
| *PROTFOR* | Total area of fully protected forests nationally; ha |
| *MANFOR* | Total area of forests under active management nationally; ha |

As the country has a great variety of forest biomes in all its extension, TOTFOR considered only the Amazon Rainforest biome. This is a conservative approach. Thus, as a representation of the total area of Amazon Rainforest in Brazilian Territory, TOTFOR consisted of multiplying size of the Brazilian Amazon by 97%, which represents its preserved area, according to SEMA[133]. As a result, TOTFOR represents 486,461,572 ha.

According to the Embrapa[134], PROTFOR is equivalent to 10.4% of the Brazilian territory, calculated as 88,566,400 ha. MANFOR is estimated in 1,400,000, according to IBAMA[135]. In this context, AVFOR is estimated as 396,495,172 ha.

The proportion of Leakage Belt area related to the total available national forest area (PROPLB) is calculated by dividing Leakage Belt area (LBFOR; 120,089.87 ha) by AVFOR. This procedure results in $PROP_{LB}$ equal to 0.000302879 (dimensionless).

The average carbon stock across the Leakage Belt ($C_{LB}$; 389.2 $tCO_2$/ha; based on similarity analysis, data from the Project Area was applied to Leakage Belt area) and the average carbon stock for all available forest area outside the Leakage Belt ($C_{OLB}$; 578,1 $tCO_2$/ha) were taken for calculation of the proportional difference in carbon stocks between areas of forest available for unplanned deforestation

[133] Source: http://meioambiente.am.gov.br/unidade-de-conservacao/
[134] Source: https://www.embrapa.br/car/sintese
[135] Source: Manejo sustentável autorizado pelo Ibama em 2019 totalizou 39 mil hectares

both inside and outside the Leakage Belt (PROPCS). PROPCS is calculated by dividing the stock outside the Leakage Belt ($C_{OLB}$) by the stock inside the Leakage Belt ($C_{LB}$), which results in a value of 1.4854.

The proportion of baseline deforestation caused by immigrating population ($PROP_{IMM}$) was estimated for a period from 2012 to 2020. For calculating $PROP_{IMM}$, local data were used for births, deaths and population. It is then assumed that the total annual population growth in a given municipality is attributed to: i) births and ii) immigration. Thus, by subtracting the number of annual births from the total annual population growth, it is possible to infer the number of immigrants. According to the number of immigrants, we have inferred the proportion of deforestation attributed to immigrant agents ($PROP_{IMM}$) as 1.33%.

The proportional leakage for areas with immigrating populations ($LK_{PROP}$) was then equal to the immigrating proportion multiplied by the proportion of available national forest area outside the Leakage Belt multiplied by the proportional difference in stocks between forests inside and outside the Leakage Belt.

$$LK_{PROP} = PROP_{IMM} * (1 - PROP_{LB}) * PROP_{CS}$$

Where:

$LK_{PROP}$ Proportional leakage for areas with immigrating populations; proportion

$PROP_{IMM}$ Estimated proportion of baseline deforestation caused by immigrating population, proportion

$PROP_{LB}$ rea of forest available for unplanned deforestation as a proportion of the total national forest area available for unplanned deforestation; proportion

$PROP_{CS}$ Proportional difference in stocks between areas of forest available for unplanned deforestation both inside and outside the Leakage Belt; proportion

$LK^{PROP}$ was estimated in 0.020

Leakage due to the proportion of the baseline deforestation actors who are displaced to areas outside the Leakage Belt was therefore equal to the change in stocks in the baseline scenario minus the change in stocks in the project scenario multiplied by the proportional leakage factor for areas with immigrating populations:

$$\Delta C_{LK-ASU,OLB} = (\Delta C_{BSL,LK,unplanned} - \Delta C_{P,LB,}) * LK_{PROP}$$

Where:

| | |
|---|---|
| $\Delta C_{LK-ASU,OLB}$ | Net $CO_2$ emissions due to unplanned deforestation displaced outside the Leakage Belt; t $CO_2$-e |
| $\Delta C_{BSL,LK,unplanned}$ | Net $CO_2$ equivalent emissions in the baseline from unplanned deforestation in the leakage belt; t $CO_2$-e |
| $\Delta C_{P,LB},$ | Net $CO_2$ equivalent emissions within the leakage belt in the project case; t $CO_2$-e |
| $LK_{PROP}$ | Proportional leakage for areas with immigrating populations; proportion |

For the difference in emissions from unplanned deforestation within the Leakage Belt in the baseline and in the project case, a 10% factor was considered (i.e. $\Delta C_{P,LB}$ was considered to be 10% higher than $\Delta C_{BSL,LK,unplanned}$). It is assumed that this factor is valid for the IPOÁ REDD+ Project, given that the project proponent will adopt a series of activities for leakage mitigation, as previously mentioned in this CCB-PD.

The tables below summarize the results obtained for calculation of Leakage outside the Leakage Belt.

*Table 70. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, per stratum (1)*

| **Stratum: Dense tropical submontane rainforest** | | | | | |
|---|---|---|---|---|---|
| **Leakage Outside** | | | | | |
| | | | **$\Delta C_{BSL,LK,unplanned}$** | **$\Delta C_{P,LB}$** | **$\Delta C_{LK-ASU,OLB}$** |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2022 | 169.37 | 169.37 | 122,227.14 | 134,449.86 | -240.99 |
| 2023 | 212.76 | 382.14 | 153,540.70 | 168,894.77 | -302.74 |
| 2024 | 212.25 | 594.38 | 153,167.72 | 168,484.49 | -302.00 |

| Stratum: Dense tropical submontane rainforest | | | | | |
|---|---|---|---|---|---|
| Leakage Outside | | | | | |
| | | | $\Delta C_{BSL,LK,unplanned}$ | $\Delta C_{P,LB}$ | $\Delta C_{LK\text{-}ASU,OLB}$ |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2025 | 211.73 | 806.11 | 152,795.64 | 168,075.21 | -301.27 |
| 2026 | 211.22 | 1,017.33 | 152,424.47 | 167,666.92 | -300.53 |
| 2027 | 210.70 | 1,228.03 | 152,054.20 | 167,259.62 | -299.80 |
| 2028 | 210.19 | 1,438.22 | 151,684.83 | 166,853.31 | -299.08 |
| 2029 | 209.68 | 1,647.91 | 151,316.36 | 166,447.99 | -298.35 |
| 2030 | 209.17 | 1,857.08 | 150,948.78 | 166,043.66 | -297.62 |
| 2031 | 208.66 | 2,065.74 | 150,582.10 | 165,640.31 | -296.90 |
| 2032 | 208.16 | 2,273.90 | 150,216.30 | 165,237.93 | -296.18 |
| 2033 | 207.65 | 2,481.55 | 149,851.40 | 164,836.54 | -295.46 |
| 2034 | 207.15 | 2,688.70 | 149,487.38 | 164,436.12 | -294.74 |
| 2035 | 206.64 | 2,895.34 | 149,124.24 | 164,036.67 | -294.03 |
| 2036 | 206.14 | 3,101.48 | 148,761.99 | 163,638.19 | -293.31 |
| 2037 | 205.64 | 3,307.12 | 148,400.62 | 163,240.68 | -292.60 |
| 2038 | 205.14 | 3,512.26 | 148,040.12 | 162,844.14 | -291.89 |
| 2039 | 204.64 | 3,716.90 | 147,680.50 | 162,448.55 | -291.18 |
| 2040 | 204.15 | 3,921.05 | 147,321.76 | 162,053.93 | -290.47 |
| 2041 | 203.65 | 4,124.70 | 146,963.88 | 161,660.27 | -289.77 |
| 2042 | 203.16 | 4,327.86 | 146,606.88 | 161,267.57 | -289.06 |
| 2043 | 202.66 | 4,530.52 | 146,250.74 | 160,875.82 | -288.36 |
| 2044 | 202.17 | 4,732.69 | 145,895.47 | 160,485.02 | -287.66 |
| 2045 | 201.68 | 4,934.36 | 145,541.06 | 160,095.17 | -286.96 |
| 2046 | 279.74 | 5,214.10 | 201,872.19 | 222,059.41 | -398.03 |
| 2047 | 279.06 | 5,493.16 | 201,381.80 | 221,519.98 | -397.06 |
| 2048 | 199.83 | 5,692.99 | 144,207.93 | 158,628.72 | -284.33 |
| 2049 | 199.35 | 5,892.34 | 143,857.62 | 158,243.38 | -283.64 |
| 2050 | 198.86 | 6,091.20 | 143,508.16 | 157,858.98 | -282.95 |
| 2051 | 198.38 | 6,289.57 | 143,159.55 | 157,475.50 | -282.27 |
| 2052 | 198.31 | 6,487.88 | 143,108.70 | 157,419.57 | -282.17 |

*Table 71. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, per stratum (2)*

| Stratum: Dense tropical lowland rainforest | | | | | |
|---|---|---|---|---|---|
| **Leakage Outside** | | | | | |
| | | | $\Delta C_{BSL,LK,unplanned}$ | $\Delta C_{P,LB}$ | $\Delta C_{LK-ASU,OLB}$ |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

| Stratum: Dense tropical lowland rainforest | | | | | |
|---|---|---|---|---|---|
| Leakage Outside | | | | | |
| | | | $\Delta C_{BSL,LK,unplanned}$ | $\Delta C_{P,LB}$ | $\Delta C_{LK\text{-}ASU,OLB}$ |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 72. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, per stratum (3)*

| Stratum: Dense tropical alluvial rainforest | | | | | |
|---|---|---|---|---|---|
| Leakage Outside | | | | | |
| | | | $\Delta C_{BSL,LK,unplanned}$ | $\Delta C_{P,LB}$ | $\Delta C_{LK\text{-}ASU,OLB}$ |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2022 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2023 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2024 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2025 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2028 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2029 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2030 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2031 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2032 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2033 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2034 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2035 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

| | | | | | |
|---|---|---|---|---|---|
| 2036 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2037 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2038 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2039 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2040 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2041 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2042 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2043 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2044 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2045 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2046 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2047 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2048 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2049 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2050 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2051 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2052 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

*Table 73. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, Sum of Strata*

| **Stratum: Sum of strata** | | | | | |
|---|---|---|---|---|---|
| **Leakage Outside** | | | | | |
| | | | **$\Delta C_{BSL,LK,unplanned}$** | **$\Delta C_{P,LB}$** | **$\Delta C_{LK\text{-}ASU,OLB}$** |
| **Year** | **ha/year** | **ha (cumulative)** | **$tCO_2$/year** | **$tCO_2$/year** | **$tCO_2$/year** |
| 2022 | 169.37 | 169.37 | 122,227.14 | 134,449.86 | -240.99 |
| 2023 | 212.76 | 382.14 | 153,540.70 | 168,894.77 | -302.74 |
| 2024 | 212.25 | 594.38 | 153,167.72 | 168,484.49 | -302.00 |
| 2025 | 211.73 | 806.11 | 152,795.64 | 168,075.21 | -301.27 |
| 2026 | 211.22 | 1,017.33 | 152,424.47 | 167,666.92 | -300.53 |
| 2027 | 210.70 | 1,228.03 | 152,054.20 | 167,259.62 | -299.80 |
| 2028 | 210.19 | 1,438.22 | 151,684.83 | 166,853.31 | -299.08 |

| | | | | | |
|---|---|---|---|---|---|
| 2029 | 209.68 | 1,647.91 | 151,316.36 | 166,447.99 | -298.35 |
| 2030 | 209.17 | 1,857.08 | 150,948.78 | 166,043.66 | -297.62 |
| 2031 | 208.66 | 2,065.74 | 150,582.10 | 165,640.31 | -296.90 |
| 2032 | 208.16 | 2,273.90 | 150,216.30 | 165,237.93 | -296.18 |
| 2033 | 207.65 | 2,481.55 | 149,851.40 | 164,836.54 | -295.46 |
| 2034 | 207.15 | 2,688.70 | 149,487.38 | 164,436.12 | -294.74 |
| 2035 | 206.64 | 2,895.34 | 149,124.24 | 164,036.67 | -294.03 |
| 2036 | 206.14 | 3,101.48 | 148,761.99 | 163,638.19 | -293.31 |
| 2037 | 205.64 | 3,307.12 | 148,400.62 | 163,240.68 | -292.60 |
| 2038 | 205.14 | 3,512.26 | 148,040.12 | 162,844.14 | -291.89 |
| 2039 | 204.64 | 3,716.90 | 147,680.50 | 162,448.55 | -291.18 |
| 2040 | 204.15 | 3,921.05 | 147,321.76 | 162,053.93 | -290.47 |
| 2041 | 203.65 | 4,124.70 | 146,963.88 | 161,660.27 | -289.77 |
| 2042 | 203.16 | 4,327.86 | 146,606.88 | 161,267.57 | -289.06 |
| 2043 | 202.66 | 4,530.52 | 146,250.74 | 160,875.82 | -288.36 |
| 2044 | 202.17 | 4,732.69 | 145,895.47 | 160,485.02 | -287.66 |
| 2045 | 201.68 | 4,934.36 | 145,541.06 | 160,095.17 | -286.96 |
| 2046 | 279.74 | 5,214.10 | 201,872.19 | 222,059.41 | -398.03 |
| 2047 | 279.06 | 5,493.16 | 201,381.80 | 221,519.98 | -397.06 |
| 2048 | 199.83 | 5,692.99 | 144,207.93 | 158,628.72 | -284.33 |
| 2049 | 199.35 | 5,892.34 | 143,857.62 | 158,243.38 | -283.64 |
| 2050 | 198.86 | 6,091.20 | 143,508.16 | 157,858.98 | -282.95 |
| 2051 | 198.38 | 6,289.57 | 143,159.55 | 157,475.50 | -282.27 |
| 2052 | 198.31 | 6,487.88 | 143,108.70 | 157,419.57 | -282.17 |

***Leakage Factor (final result)***

Based on the expected effectiveness of the proposed REDD+ project activities, the carbon stock changes and greenhouse gas emissions in the Leakage Belt that are expected to occur due to the implementation of the REDD+ project activity (that would not occur in the baseline case) were conservatively estimated as being 10% higher in the project case. It is assumed that this factor is valid for the IPOÁ REDD+ Project, given that the project proponent will adopt a series of activities for leakage mitigation, as previously mentioned in this CCB-PD.

Thus, the final results of leakage correspond to 10% of the sum of previously estimated leakage components (i.e. Market Leakage + Leakage outside the Leakage Belt), whose results are presented in the table below.

*Table 74. Total Leakage.*

| Stratum: Sum of strata | | | | | |
|---|---|---|---|---|---|
| Total Leakage | | | | | |
| | | | Leakage Outside | Market Leakage | Shifting Leakage |
| Year | ha/year | ha (cumulative) | $tCO_2$/year | $tCO_2$/year | $tCO_2$/year |
| 2022 | 19.46 | 19.46 | 240.99 | 549.47 | 79.0 |
| 2023 | 24.45 | 43.91 | 302.74 | 690.23 | 99.3 |
| 2024 | 24.39 | 68.30 | 302.00 | 688.56 | 99.1 |
| 2025 | 24.33 | 92.63 | 301.27 | 686.89 | 98.8 |
| 2026 | 24.27 | 116.90 | 300.53 | 685.22 | 98.6 |
| 2027 | 24.21 | 141.11 | 299.80 | 683.55 | 98.3 |
| 2028 | 24.15 | 165.27 | 299.08 | 681.89 | 98.1 |
| 2029 | 24.09 | 189.36 | 298.35 | 680.24 | 97.9 |
| 2030 | 24.04 | 213.40 | 297.62 | 678.58 | 97.6 |
| 2031 | 23.98 | 237.38 | 296.90 | 676.93 | 97.4 |
| 2032 | 23.92 | 261.30 | 296.18 | 675.29 | 97.1 |
| 2033 | 23.86 | 285.16 | 295.46 | 673.65 | 96.9 |
| 2034 | 23.80 | 308.96 | 294.74 | 672.01 | 96.7 |
| 2035 | 23.75 | 332.71 | 294.03 | 670.38 | 96.4 |
| 2036 | 23.69 | 356.39 | 293.31 | 668.75 | 96.2 |
| 2037 | 23.63 | 380.02 | 292.60 | 667.13 | 96.0 |
| 2038 | 23.57 | 403.60 | 291.89 | 665.51 | 95.7 |
| 2039 | 23.52 | 427.11 | 291.18 | 663.89 | 95.5 |
| 2040 | 23.46 | 450.57 | 290.47 | 662.28 | 95.3 |

| | | | | | |
|---|---|---|---|---|---|
| 2041 | 23.40 | 473.97 | 289.77 | 660.67 | 95.0 |
| 2042 | 23.34 | 497.32 | 289.06 | 659.06 | 94.8 |
| 2043 | 23.29 | 520.60 | 288.36 | 657.46 | 94.6 |
| 2044 | 23.23 | 543.84 | 287.66 | 655.87 | 94.4 |
| 2045 | 23.17 | 567.01 | 286.96 | 654.27 | 94.1 |
| 2046 | 32.14 | 599.16 | 398.03 | 907.51 | 130.6 |
| 2047 | 32.07 | 631.22 | 397.06 | 905.30 | 130.2 |
| 2048 | 22.96 | 654.19 | 284.33 | 648.28 | 93.3 |
| 2049 | 22.91 | 677.09 | 283.64 | 646.70 | 93.0 |
| 2050 | 22.85 | 699.94 | 282.95 | 645.13 | 92.8 |
| 2051 | 22.80 | 722.74 | 282.27 | 643.57 | 92.6 |
| 2052 | 22.79 | 745.53 | 282.17 | 643.34 | 92.6 |

### 3.2.4 Net GHG Emission Reductions and Removals

Net GHG Emission Reductions and Removals can be summarized as the “Estimated baseline emissions” minus the “Estimated project emissions” minus the “Estimated leakage emissions”, whose components are presented below.

***Estimate baseline emissions***

*= Baseline emissions from unplanned deforestation*

*+ Emissions from biomass burning in the baseline*

*- Sum of wood products and pasture carbon pools in the baseline case*

***Estimate project emissions***

*= Emissions arising from logging gap in Project Area*

*+ Emissions from constructing logging infrastructure for removal of timber in Project Area*

*- Wood Products Carbon Pool in Project Area*

***Estimate leakage emissions***

*= (Sum of Market Leakage and Leakage outside the Leakage Belt)*

** Leakage Factor*

*Table 75. Net GHG emission reductions from the Ipoá REDD project, detailing: baseline; project emissions; and leakage.*

| Year | Estimated baseline emissions or removals ($tCO_2e$) | Estimated project emissions or removals ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated net GHG emission reductions |
|---|---|---|---|---|
| **2022** | 414,352 | 0 | 629 | 413,723 |
| **2023** | 717,847 | 0 | 790 | 717,058 |
| **2024** | 702,888 | 0 | 788 | 702,100 |
| **2025** | 691,671 | 0 | 786 | 690,885 |
| **2026** | 168,846 | 0 | 784 | 168,062 |
| **2027** | 17,366 | 0 | 782 | 16,584 |
| **2028** | 17,324 | 0 | 780 | 16,544 |
| **2029** | 17,282 | 0 | 778 | 16,504 |
| **2030** | 17,240 | 0 | 776 | 16,464 |
| **2031** | 17,198 | 0 | 774 | 16,424 |
| **2032** | 17,156 | 0 | 772 | 16,384 |
| **2033** | 17,115 | 0 | 771 | 16,344 |
| **2034** | 17,073 | 0 | 769 | 16,304 |
| **2035** | 17,032 | 0 | 767 | 16,265 |
| **2036** | 16,990 | 0 | 765 | 16,225 |
| **2037** | 16,949 | 0 | 763 | 16,186 |
| **2038** | 16,908 | 0 | 761 | 16,147 |
| **2039** | 16,867 | 0 | 759 | 16,107 |
| **2040** | 16,826 | 0 | 758 | 16,068 |
| **2041** | 16,785 | 0 | 756 | 16,029 |
| **2042** | 16,744 | 0 | 754 | 15,990 |
| **2043** | 16,703 | 0 | 752 | 15,951 |
| **2044** | 16,663 | 0 | 750 | 15,913 |
| **2045** | 16,622 | 0 | 748 | 15,874 |
| **2046** | 23,056 | 0 | 1,038 | 22,018 |
| **2047** | 23,000 | 0 | 1,036 | 21,964 |
| **2048** | 16,470 | 0 | 742 | 15,729 |
| **2049** | 16,430 | 0 | 740 | 15,690 |

| Year | Estimated baseline emissions or removals ($tCO_2e$) | Estimated project emissions or removals ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated net GHG emission reductions |
|---|---|---|---|---|
| **2050** | 16,390 | 0 | 738 | 15,652 |
| **2051** | 16,350 | 0 | 736 | 15,614 |
| **2052** | 16,345 | 0 | 736 | 15,609 |
| **Total** | **3,146,488** | **0** | **24,076** | **3,122,412** |

The total net greenhouse gas emissions reductions corresponding to the AUD and APD components of the baseline are presented in the table below.

Emissions from both APD and AUD presented in Table below include: Carbon stock changes from initial land-use (forest) to post-deforestation land-use (pasture), and biomass burning.

*Table 76. Net GHG emission reductions from the IPOÁ REDD+ project, detailing: AUD and APD components.*

| Year | AUD emissions ($tCO_2$) | APD emissions ($tCO_2$) |
|---|---|---|
| 2022 | 13,960 | 400,392 |
| 2023 | 17,536 | 700,311 |
| 2024 | 17,493 | 685,394 |
| 2025 | 17,451 | 674,220 |
| 2026 | 17,409 | 151,438 |
| 2027 | 17,366 | 0 |
| 2028 | 17,324 | 0 |
| 2029 | 17,282 | 0 |
| 2030 | 17,240 | 0 |
| 2031 | 17,198 | 0 |
| 2032 | 17,156 | 0 |
| 2033 | 17,115 | 0 |
| 2034 | 17,073 | 0 |

| Year | AUD emissions ($tCO_2$) | APD emissions ($tCO_2$) |
|---|---|---|
| 2035 | 17,032 | 0 |
| 2036 | 16,990 | 0 |
| 2037 | 16,949 | 0 |
| 2038 | 16,908 | 0 |
| 2039 | 16,867 | 0 |
| 2040 | 16,826 | 0 |
| 2041 | 16,785 | 0 |
| 2042 | 16,744 | 0 |
| 2043 | 16,703 | 0 |
| 2044 | 16,663 | 0 |
| 2045 | 16,622 | 0 |
| 2046 | 23,056 | 0 |
| 2047 | 23,000 | 0 |
| 2048 | 16,470 | 0 |
| 2049 | 16,430 | 0 |
| 2050 | 16,390 | 0 |
| 2051 | 16,350 | 0 |
| 2052 | 16,345 | 0 |
| **Total** | **534,733** | **2,611,755** |
| **Total AUD + APD** | **3,146,488** | |

## 3.3 Monitoring

### 3.3.1 Data and Parameters Available at Validation

| Data / Parameter | **CF** |
|---|---|
| Data unit | tCt/td.m-1 |
| Description | Carbon fraction of dry matter in t Ct-1 d.m |

| | |
|---|---|
| Source of data | Values from the literature (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Table 4.3) shall be used if available, otherwise default value of 0.47 t C t-1 d.m. can be used |
| Value applied | 0.47 |
| Justification of choice of data or description of measurement methods and procedures applied | The default value was used for conservativeness purposes. |
| Purpose of data | Calculation of baseline emissions<br>Calculation of project emissions<br>Calculation of leakage |
| Comments | If new and more accurate carbon fraction data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period. |

| | |
|---|---|
| Data / Parameter | **44/12** |
| Data unit | Dimensionless |
| Description | Carbon mass to $CO_2e$ mass conversion factor |
| Source of data | From scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU |
| Value applied | 44/12 |
| Justification of choice of data or description of measurement methods and procedures applied | Conversion from C to $CO_2$ based on molecular weights |
| Purpose of data | Calculation of baseline emissions<br>Calculation of project emissions<br>Calculation of leakage |
| Comments | IPCC standard value |

| | |
|---|---|
| Data / Parameter | **R** |
| Data unit | t root d.m.t-1 shoot d.m. |

| | |
|---|---|
| Description | Root to shoot ratio appropriate to species or forest type / biome; note that as defined here, root to shoot ratio is applied as belowground biomass per unit area: aboveground biomass per unit area (not on a per stem basis) |
| Source of data | According to CP-AB - page 17; “Tropical rainforest”; “>125 t.ha-1”. |
| Value applied | 0.37 |
| Justification of choice of data or description of measurement methods and procedures applied | Local values are not known, and the value proposed in CP-AB is conservative. |
| Purpose of data | Calculation of baseline emissions<br>Calculation of project emissions<br>Calculation of leakage |
| Comments | N/A. |

| Data / Parameter | **WWty** |
|---|---|
| Data unit | Dimensionless |
| Description | Fraction of extracted biomass effectively emitted to the atmosphere during production by class of wood product ty |
| Source of data | Default value for developing countries: CP-W - page 14. |
| Value applied | 0.24 |
| Justification of choice of data or description of measurement methods and procedures applied | As per CP-W. |
| Purpose of data | Calculation of baseline emissions |
| Comments | Parameter values to be updated if new empirically based peer-reviewed findings become available. |

| Data / Parameter | **SLFty** |
|---|---|
| Data unit | Dimensionless |

| | |
|---|---|
| Description | Fraction of wood products that will be emitted to the atmosphere within 5 years of production by class of wood product ty |
| Source of data | As per CP-W: Sawnwood, page 13. |
| Value applied | 0.20 |
| Justification of choice of data or description of measurement methods and procedures applied | Default conservative value prescribed by CP-W. |
| Purpose of data | Calculation of baseline emissions |
| Comments | Parameter values to be updated if new empirically based peer-reviewed findings become available |

| | |
|---|---|
| Data / Parameter | **OFty** |
| Data unit | Dimensionless |
| Description | Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty |
| Source of data | The OFty is the complementary number of SLFty: the sum of both parameters must equal 1 (i.e., 100%). |
| Value applied | 0.80 |
| Justification of choice of data or description of measurement methods and procedures applied | As per CP-W. |
| Purpose of data | Calculation of baseline emissions |
| Comments | Parameter values to be updated if new empirically based peer-reviewed findings become available. |

| | |
|---|---|
| Data / Parameter | **PCOMi** |
| Data unit | Dimensionless |

| | |
|---|---|
| Description | Commercial volume as a percent of total aboveground volume in stratum i. |
| Source of data | For calculation of this parameter, the volume of merchantable wood in exploitation (in $m^3$/ha) has been based on "Florestal Santa Maria Project" (35.08 $m^3$/ha), whose Project Area is located about 120 km South of this Project Area and has the same two forest classes represented within it (According to Figure 22 of FSM VCS-PD[136]): Submontane dense tropical forest ("Ds") ", in the IBGE classification, which are the same as those predominating in this project.<br><br>Age classes are assumed to be similar, as both sites had not been deforested or managed until the date of their forest inventories (i.e., both were primary forests).<br><br>In this context, it is assumed that the composition of species and tree sizes are the same, given the identical IBGE classification for both typologies, given that both areas held primary forests at the inventory date. Importantly, based on literature consulted, both As and Ds typologies have very close aboveground biomasses (As: 254 t/ha; Ds: 251 t/ha): therefore, it is assumed that tree sizes are also similar in primary forest sites. In the "Florestal Santa Maria Project" aboveground biomass ranges from 224 to 268 t/ha, while the average aboveground biomass in this project is estimated as 253 t/ha.<br><br>Furthermore it is noted that the climate type is the same for both projects – specifically, Tropical monsoon climate area (type ("Am") of the Köppen climate classification system). And the predominant soil type in both project is the same, Red-Yellow Argisol (PVe), as shown under Figure 20 of the FSM PD.<br><br>Thus, as prescribed in CP-W, a forest inventory from a proxy area in the same region, representing the same forest type and age class was chosen for calculation of this parameter. |
| Value applied | Submontane Dense Tropical Rainforest: 0,07814<br>Lowland Dense Tropical Rainforest: 0,07791<br>Alluvial Dense Tropical Raiforest: 0,06863 |

[136]https://registry.verra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashoufifmln902309ksdflku098l71896923 (last visited in 10/09/2021)

| | |
|---|---|
| Justification of choice of data or description of measurement methods and procedures applied | As prescribed in CP-W, a forest inventory from a proxy area in the same region, representing the same forest type and age class was chosen for calculation of this parameter: Submontane Dense Tropical Rainforest (Ds), Lowland Dense Tropical Rainforest (Db) and Alluvial Dense Tropical Raiforest (Da) |
| Purpose of data | Calculation of baseline emissions |
| Comments | Updated at the time of baseline revision (at least every 10 years). Application of the commercial percentage of total volume introduces the simplifying assumption (and conservative, as it is only used in the ex-ante baseline calculations) that all commercial stocks are extracted (i.e., perfect efficiency). |

| | |
|---|---|
| Data / Parameter | **CWP,i** |
| Data unit | t $CO_2$e ha-1 |
| Description | Mean carbon stock entering the wood products pool from stratum I |
| Source of data | VCS Module VMD0005 REDD+ Methodological Module: Estimation of Carbon Stocks in The Long-Term Wood Products Pool (CP- W)131. This parameter has been calculated using default values prescribed in CP-W. |
| Value applied | 3.29 t $CO2_e$ $ha^{-1}$ for all three strata: Dense Tropical Submontane Rainforest; Dense Tropical Lowland Rainforest; and Dense Tropical Alluvial Rainforest. |
| Justification of choice of data or description of measurement methods and procedures applied | As per CP-W. |
| Purpose of data | Calculation of baseline emissions |
| Comments | N/A. |

| | |
|---|---|
| Data / Parameter | **Cab_tree** |

| | |
|---|---|
| Data unit | t $CO_2$-e ha-1 |
| Description | Mean aboveground biomass carbon stock in stratum i |
| Source of data | The value is the result of the division of the total aboveground carbon pool per stratum by the area, as indicated in Table 77. |
| Value applied | Submontane Dense Tropical Rainforest: 457<br>Lowland Dense Tropical Rainforest: 458<br>Alluvial Dense Tropical Rainforest: 520 |
| Justification of choice of data or description of measurement methods and procedures applied | As indicated in module CP-AB. |
| Purpose of data | Calculation of baseline emissions |
| Comments | N/A |

| Data / Parameter | **Cbb_tree** |
|---|---|
| Data unit | t $CO_2$-e ha-1 |
| Description | Mean belowground biomass carbon stock in stratum i |
| Source of data | The value is the result of the division of the total aboveground carbon pool per stratum by the area, as indicated in Table 78. |
| Value applied | Submontane Dense Tropical Rainforest: 268<br>Lowland Dense Tropical Rainforest: 269<br>Alluvial Dense Tropical Raiforest: 305 |
| Justification of choice of data or description of measurement methods and procedures applied | As indicated in module CP-AB. |
| Purpose of data | Calculation of baseline emissions |
| Comments | N/A |

| Data / Parameter | **CXB** |
|---|---|
| Data unit | t $CO_2$-e ha-1 |

| | |
|---|---|
| Description | Mean stock of extracted biomass carbon from stratum i |
| Source of data | 3.2.1 Baseline section, Wood products sub-section |
| Value applied | 27.03 |
| Justification of choice of data or description of measurement methods and procedures applied | Calculated according to Module CP-W |
| Purpose of data | Calculation of baseline emissions |
| Comments | N/A. |

| Data / Parameter | **Pasture carbon pool** |
|---|---|
| Data unit | t $CO_2e$ |
| Description | Pasture carbon pool in the baseline scenario |
| Source of data | Terceiro Inventário Brasileiro de Emissões e Remoções Antrópicas de Gases de Efeito Estufa Relatórios de Referência Setor Uso da Terra, Mudança do Uso da Terra e Florestas (Third Brazilian Inventory of Greenhouse Gas Emissions and Removals Reference Reports Land Use, Land Use Change and Forestry Sector). Ministry of Science, Technology and Innovation, 2015132. Deforestation measured through data from Mapbiomas project. |
| Value applied | 27.8 |
| Justification of choice of data or description of measurement methods and procedures applied | The post-deforestation biomass (pasture) according to the National GHG Inventory has been multiplied by the deforestation measured through data from Mapbiomas. |
| Purpose of data | Calculation of baseline emissions |
| Comments | Calculation based on country-specific values. |

| Data / Parameter | **BCEF** |
|---|---|
| Data unit | Dimensionless |

| | |
|---|---|
| Description | Biomass conversion and expansion factor for conversion of commercial wood volume per unit area to total aboveground tree biomass per unit area |
| Source of data | According to CP-AB - page 14, being the average of the three proposed factors. |
| Value applied | 1.32 |
| Justification of choice of data or description of measurement methods and procedures applied | BCEF was applied for conversion of merchantable volume to total aboveground tree biomass |
| Purpose of data | Calculation of baseline emissions<br>Calculation of project emissions<br>Calculation of leakage |
| Comments | N/A. |

| Data / Parameter | **$Cab_{icl}$** |
|---|---|
| Data unit | $tCO_2$/ha |
| Description | Average carbon stock per hectare in the above-ground biomass carbon pool of initial forest class icl |
| Source of data | National Inventory data: Ministério da Ciência, Tecnologia e Inovação, 2015). |
| Value applied | Submontane Dense Tropical Rainforest: 457<br>Lowland Dense Tropical Rainforest: 458<br>Alluvial Dense Tropical Raiforest: 520 |
| Justification of choice of data or description of measurement methods and procedures applied | 1.32 Biomass Expansion Factor (BCEF) was applied for conversion of merchantable volume to total aboveground tree biomass, according to CP-AB - page 14, being the average of the three factors presented in the module, for conservativeness.<br>0.589 t/m3 wood density was applied, according to Nogueira (2008). |
| Purpose of data | Calculation of baseline emissions |

| | |
|---|---|
| Comments | Literature studies used for this assessment, as well as respective calculations, are available for consultation by the audit team. |

| | |
|---|---|
| Data / Parameter | **$Cbb_{icl}$** |
| Data unit | $tCO_2$/ha |
| Description | Average carbon stock per hectare in the below-ground biomass carbon pool of initial forest class icl |
| Source of data | National Inventory data: Ministério da Ciência, Tecnologia e Inovação, 2015). |
| Value applied | Dense Tropical Submontane Rainforest: 268<br>Dense Tropical Lowland Rainforest: 269<br>Dense Tropical Alluvial Rainforest: 305 |
| Justification of choice of data or description of measurement methods and procedures applied | 0.37 root-shoot ratio (R) was applied, according to "2006 IPCC Guidelines for National Greenhouse Gas Inventories", V. 4, Ch. 4, AFOLU, pg. 4.49, Table 4-4. |
| Purpose of data | Calculation of baseline emissions |
| Comments | Literature studies used for this assessment, as well as respective calculations, are available for consultation by the audit team. |

| | |
|---|---|
| Data / Parameter | $CABnon{-}treepost$,i; $CBBnon{-}tree,post$,i |
| Data unit | $tCO_2$/ha |
| Description | Post-deforestation carbon stock in aboveground non-tree vegetation in stratum i; t $CO_2$-e ha-1; Post-deforestation carbon stock in belowground non-tree biomass in stratum i; t $CO_2$-e ha-1 |
| Source of data | Terceiro Inventário Brasileiro de Emissões e Remoções Antrópicas de Gases de Efeito Estufa Relatórios de Referência Setor Uso da Terra, Mudança do Uso da Terra e Florestas (Third Brazilian Inventory of Greenhouse Gas Emissions and Removals Reference Reports Land Use, Land Use Change and Forestry Sector). Ministry of Science, Technology and Innovation, 2015133. |

| | |
|---|---|
| Value applied | 27.8 (sum of above and belowground biomass) |
| Justification of choice of data or description of measurement methods and procedures applied | Value used in the National GHG inventory. |
| Purpose of data | Calculation of baseline emissions |
| Comments | Country-specific value provided for above and belowground biomass of pasture. |

| | |
|---|---|
| Data / Parameter | **$Cab_{fcl}$** |
| Data unit | $tCO_2/ha$ |
| Description | Average carbon stock per hectare in the above-ground biomass carbon pool of final post-deforestation class fcl |
| Source of data | Weighted average (by area obtained in Terra Class database): 2006 IPCC Guidelines for National Greenhouse Gas Inventories, V. 4, Chapter 6: Grassland, pg. 6.27, Table 6.4 (for Pasture: 76.1% of area) 2006 IPCC Guidelines for National Greenhouse Gas Inventories, V. 4, Chapter 4: Forest Land, pg. 4.63, Table 4.12 (for Pasture with regeneration: 23.9% of area) Value applied: 61.1 |
| Value applied | 27.8 |
| Justification of choice of data or description of measurement methods and procedures applied | Conservative default value from IPCC, to estimate post-deforestation land use carbon stock. |
| Purpose of data | Calculation of baseline emissions<br>Calculation of project emissions<br>Calculation of leakage |
| Comments | Conservative average to be used in calculations, based on uncertainties in source values. |

| | |
|---|---|
| Data / Parameter | **COMF** |
| Data unit | Dimensionless |
| Description | Combustion factor for stratum i (vegetation type) |
| Source of data | E-BPB refers to Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, “Primary tropical moist forest” 134 |
| Value applied | 0.50 |
| Justification of choice of data or description of measurement methods and procedures applied | Value was applied according to E-BPB: Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, “Primary tropical moist forest” |
| Purpose of data | Calculation of baseline emissions Calculation of project emissions |
| Comments | N/A. |

| | |
|---|---|
| Data / Parameter | **Ggi** |
| Data unit | g kg-1 dry matter burnt |
| Description | Emission factor for stratum i for gas g |
| Source of data | Defaults can be found in Volume 4, Chapter 2, of the IPCC 2006 Inventory Guidelines in table 2.5 (see Annex 2: emission factors for various types of burning for $CH_4$ and $N_2O$). |
| Value applied | $GCH_4$ = 4.8<br>$GN_20$ = 0.2 |
| Justification of choice of data or description of measurement methods and procedures applied | Conservative default values from IPCC 2006. |
| Purpose of data | Calculation of baseline emissions<br>Calculation of project emissions<br>Calculation of leakage emissions |
| Comments | N/A. |

| Data / Parameter | **Deforestation** |
|---|---|
| Data unit | ha |
| Description | Maps of forest cover areas converted into non-forest areas |
| Source of data | Measured through data from Mapbiomas project |
| Value applied | Yearly variable: deforestation values are presented for the Reference Region, Leakage Belt and Project Area (projections) in this VCS-PD |
| Justification of choice of data or description of measurement methods and procedures applied | The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for the monitoring period, other sources will be consulted such as PRODES, or a classification of imagery (Landsat8) will be carried out to measure deforested area:<br>Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September.<br>Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images.<br>For analysis of areas with cloud cover, visual interpretation of radar image would be performed.<br>Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very high. |

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|---|---|
| Purpose of data | Calculation of baseline emissions |
| Comments | N/A |

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|---|---|
| Data / Parameter | **Dmm** |
| Data unit | t d.m.m-3 |
| Description | Mean wood density of commercially harvested species |
| Source of data | Source: Brown, S., A. J. R. Gillespie, and A. E. Lugo, 1989. Biomass estimation methods for tropical forests with applications to forest inventory data. Forest Science, 35:881- 902. See pg. 890, Table 4, Moist |
| Value applied | 0.59 |
| Justification of choice of data or description of measurement methods and procedures applied | Country-specific data obtained from the same biome. |
| Purpose of data | Calculation of baseline emissions<br>Calculation of project emissions<br>Calculation of leakage emissions |
| Comments | N/A. |

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|---|---|
| Data / Parameter | **LDF** |
| Data unit | t C m-3 |
| Description | Factor for calculating the biomass of dead wood created during logging operations per cubic meter extracted (i.e. Logging Damage Factor) |
| Source of data | VMD0001 LK-ME135 |
| Value applied | 0.53 |
| Justification of choice of data or description of measurement methods and procedures applied | It is the default value for broadleaf and mixed forests, according to module VMD0001 LK-ME. Default value for broadleaf and mixed forests of 0.53 t C m-3 from 774 logging gaps measured by Winrock International in Bolivia, Belize, the Republic of Congo, Brazil and Indonesia may be used for tropical broadleaf forests |

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|---|---|
| Purpose of data | Calculation of leakage emissions<br>Calculation of project emissions |
| Comments | N/A. |

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|---|---|
| Data / Parameter | **LIF** |
| Data unit | t C m-3 |
| Description | Factor for calculating the emissions arising from the creation of logging infrastructure (roads, skid trails and decks) during logging operations per cubic meter extracted |
| Source of data | LK-ME, pg 8 |
| Value applied | 0.29 |
| Justification of choice of data or description of measurement methods and procedures applied | Conservative default value of 0.29 t $CO_2$-e m-3 calculated from 1,839 hectares of logging concessions analysed by Winrock International in the Republic of Congo and Brazil, may be used for tropical broadleaf forests. |
| Purpose of data | Calculation of leakage emissions<br>Calculation of project emissions |
| Comments | N/A. |

| | |
|---|---|
| Data / Parameter | **LFme** |
| Data unit | dimensionless |
| Description | Leakage factor for market effects calculations |
| Source of data | VMD0011 (LK-ME) |
| Value applied | 0.2 |
| Justification of choice of data or description of measurement methods and procedures applied | When the mean biomass is more than 15% greater than the biomass within the project boundaries, the LFME shall be considered 0.2. In the IPOÁ REDD+ Project, the biomass of total aboveground tree is considered to be 32.65% greater in the Amazon Biome than in the Project Area. |
| Purpose of data | Calculation of leakage |
| Comments | N/A. |

| Data / Parameter | **CLB** |
|---|---|
| Data unit | $tCO_2e$ ha-1 |
| Description | Area-weighted average aboveground tree carbon stock for forests available for unplanned deforestation inside the leakage belt |
| Source of data | Project Area data |
| Value applied | 389.2 |
| Justification of choice of data or description of measurement methods and procedures applied | Based on similarity analysis, data from the Project Area was applied to Leakage Belt area. It was taken a weighted average of aboveground biomass inside the Project Area. |
| Purpose of data | Calculation of leakage emissions |
| Comments | N/A. |

| Data / Parameter | **COLB** |
|---|---|
| Data unit | t $CO_2e$ ha-1 |
| Description | Area-weighted average aboveground tree carbon stock for forests available for unplanned deforestation outside the leakage belt |
| Source of data | Saatchi, R.A. Houghton, R.C. dos Santos Alvalá, J.V. Soares, and Yifan Yu. Distribution of Aboveground Live Biomass in the Amazon Basin. 2007.<br>As per LK-ASU, the number is derived from peer-reviewed literature that is appropriate, as it considers the same biome. |
| Value applied | 578.1 |
| Justification of choice of data or description of measurement methods and procedures applied | The value of 578,1 $tCO_2$/ha is the total mean biomass for floodplain forest in the Amazon basin of 157.66 tC/ha (according to Table 7 the study mentioned before), multiplied by 44/12, the conversion factor from tC to $tCO_2$. |
| Purpose of data | Calculation of leakage emissions |
| Comments | N/A. |

| Data / Parameter | ***DLF*** |
|---|---|
| Data unit | % |
| Description | Displacement Leakage Factor |
| Source of data | Local assessment |
| Value applied | 10 |
| Justification of choice of data or description of measurement methods and procedures applied | If deforestation agents do not participate in leakage prevention activities and project activities, the Displacement Factor shall be 100%. Where leakage prevention activities are implemented, the factor shall be equal to the proportion of the baseline agents estimated to be given the opportunity to participate in leakage prevention activities and project activities. The project design team estimates that 100% of potential deforestation agents in the Reference Region will be given the opportunity to participate in leakage prevention activities. Given that the PP is divulging the project activity and recruiting new project instances, it can be stated that most of neighbors are being given opportunity to participate in leakage prevention activities. Thus, the “Displacement Leakage Factor” (DLF) was conservatively defined as 10%. |
| Purpose of data | Calculation of leakage |
| Comments | This value is an ex-ante estimate. Accurate and actual values will be monitored and reported in verification periods |

### 3.3.2 Data and Parameters Monitored

| Data / Parameter | ***Aburn,i,t*** |
|---|---|
| Data unit | *ha* |
| Description | Area burnt in stratum *i* at time *t* |
| Source of data | Remote sensing data |
| Description of measurement methods and procedures to be applied | It is considered that burning is a common practice in the region, and that all deforested area undergoes burning in a given moment124. The Amazon Biome comprises a great amount of biomass, and post- deforestation residuals are very abundant. There is no |

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|---|---|
| | technology known and economically feasible to allow mechanical skidding or disposal of deforestation residual biomass. Selling of these residuals is not feasible, as transportation is more expensive than the residual material itself. In this context, the only feasible alternative for cattle ranchers for land clearing is setting fire in those residues, getting rid of them thoroughly, quickly, and cheaply. According to an article in the "Globo" newspaper: "Basically, whenever you deforest, you burn it. This is a viable way to get rid of all the weeds. Burning also helps prepare the soil for planting. It serves to reduce soil acidity, a problem in the Amazon, and deposits nutrients" - Philip Fearnside, American biologist and scientist125. |
| Frequency of monitoring/recording | Areas burnt will be monitored yearly, examination will occur prior to any verification event. |
| Value applied | This value varies annually, as a function of deforested area. |
| Monitoring equipment | Remote sensing |
| QA/QC procedures to be applied | Best practices in remote sensing<br><br>Land use / land change map for monitoring period;<br><br>Overlay land use / land change map with fire alerts location data from INPE-BDQUEIMADAS<br><br>(http://www.inpe.br/queimadas/abasFogo.php) into the period; Quantify pixels of deforested areas over fire alerts.<br><br>Monitor burned forest areas |
| Purpose of data | N/A |
| Calculation method | As burning of biomass is common practice in the region, it was considered that all the deforested areas were burnt – deforestation cycle<br>includes burning. |
| Comments | Remote sensing |

| | |
|---|---|
| Data / Parameter | $A_{DistPA}$ |
| Data unit | ha |
| Description | Area impacted by natural disturbance in the project stratum i converted to natural disturbance stratum q in year t; ha |
| Source of data | Remote Sensing imagery combined with ground verification or GPS coordinates |
| Description of measurement methods and procedures to be applied | Minimum monitoring unit shall be equal to a minimum of 11 Landsat pixels or one hectare. |
| Frequency of monitoring/recording | Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event |
| Value applied | 0 |
| Monitoring equipment | Remote sensing and GIS tools |
| QA/QC procedures to be applied | The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other sources will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area:<br>Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September. |

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|---|---|
| | Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images.<br>For analysis of areas with cloud cover, visual interpretation of radar image would be performed.<br>Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the mapping classification should be over 90%, considered very high. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Ex ante, estimations of emissions from natural disturbances shall be based on historic incidence of such event in the Project region |
| Comments | N/A |

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|---|---|
| Data / Parameter | **ABSLPAt** |
| Data unit | ha |
| Description | Annual area of baseline deforestation in the project area at year t |
| Source of data | Remote sensing data and GIS |
| Description of measurement methods and procedures to be applied | Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the project area provided by Mapbiomas project. |
| Frequency of monitoring/recording | Annually |
| Value applied | The value will be calculated ex-post, before every verification period. |
| Monitoring equipment | Remote sensing and GIS |
| QA/QC procedures to be applied | The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, |

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| | according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other souces will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area:<br>Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images.<br>For analysis of areas with cloud cover, visual interpretation of radar image would be performed.<br>Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very high1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September. |
| Purpose of data | • Calculation of project emissions<br>• Calculation of baseline emissions |
| Calculation method | Analysis of satellite images and maps |
| Comments | N/A |

| Data / Parameter | $V_{EXT\ z\ i\ t}$ |
|---|---|
| Data unit | $m^3$ |
| Description | Volume extracted from logging stratum z in stratum i in year t |
| Source of data | Records and reports documenting amount of wood extracted within project boundary. Official data of timber harvest reports from logging in sustainable forest management. Local field inventory. |

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|---|---|
| Description of measurement methods and procedures to be applied | All logged trees are previously inventoried. |
| Frequency of monitoring/recording | To be determined, according to the decision listed in the 'Comments' section. |
| Value applied | The value will be calculated ex-post. |
| Monitoring equipment | The same equipment applied in forest inventory. |
| QA/QC procedures to be applied | Forest inventory data can be crosschecked with official data of timber harvest reports from logging in sustainable forest management. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Official data of timber harvest reports from logging in sustainable forest management and timber inventory. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| Data / Parameter | $C_{EXT\ z\ i\ t}$ |
|---|---|
| Data unit | $tCO_2$-e |
| Description | Biomass carbon stock of timber extracted within the project boundary for logging stratum z, in stratum i in year t; |
| Source of data | Official data of timber volume reports from logging in sustainable forest management |
| Description of measurement methods and procedures to be applied | Official data of timber volume reports from logging in sustainable forest management will be converted to $tCO_2$-e, considering wood basic density and carbon fraction, multiplied by 44/12. All logged trees are previously inventoried. |

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|---|---|
| Frequency of monitoring/recording | To be determined, according to the decision listed in the 'Comments' section. |
| Value applied | The value will be calculated ex-post. |
| Monitoring equipment | The logged trees are previously inventoried. |
| QA/QC procedures to be applied | Forest inventory data can be crosschecked with official data of timber volume reports from logging in sustainable forest management. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Official data of timber volume reports from logging in sustainable forest management ($V_{EXT\ z\ i\ t}$) will be converted to $tCO_2$-e, considering wood basic density and carbon fraction, multiplied by 44/12. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| | |
|---|---|
| Data / Parameter | $\mathbf{C_{LG\ i\ t}}$ |
| Data unit | $tCO_2$-e |
| Description | Actual net project emissions arising in the logging gap, in stratum i in year t |
| Source of data | Official data of timber volume reports from logging in sustainable forest management |
| Description of measurement methods and procedures to be applied | Biomass carbon stock of timber extracted within the project boundary ($C_{EXT\ z\ i\ t}$) is summed to volume of logs in sustainable forest management ($V_{EXT\ z\ i\ t}$) multiplied by LDF (Logging damage factor), to estimate both amounts of emissions of timber extracted and emissions caused for logging damage. |

| | |
|---|---|
| Frequency of monitoring/recording | To be determined, according to the decision listed in the 'Comments' section. |
| Value applied | The value will be calculated ex-post. |
| Monitoring equipment | The logged trees are previously inventoried. |
| QA/QC procedures to be applied | Forest inventory data can be crosschecked with official data of timber volume reports from logging in sustainable forest management. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Biomass carbon stock of timber extracted within the project boundary ($C_{EXT\ z\ i\ t}$) is summed to volume of logs in sustainable forest management ($V_{EXT\ z\ i\ t}$) multiplied by LDF (Logging damage factor), to estimate both amounts of emissions of timber extracted and emissions caused for logging damage. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| Data / Parameter | $A_{DECKS,I,t}$ |
|---|---|
| Data unit | ha |
| Description | Area of logging decks in stratum i in year t |
| Source of data | Field measurements or reported measurements such as postharvest assessment reports and post-harvest maps that are based on field measurements. Remote sensing data and GIS. |
| Description of measurement methods and procedures to be applied | Systematic sampling must take place to ensure all decks within the area logged are identified and a conservative estimate of area produced. |

| | |
|---|---|
| Frequency of monitoring/recording | To be determined, according to the decision listed in the 'Comments' section. |
| Value applied | The value will be calculated ex-post |
| Monitoring equipment | According to the Annual Operational Plans of the Sustainable Management Plan. |
| QA/QC procedures to be applied | The measured area of logging decks in current logging gaps will be compared with those of previous logging gaps. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Data obtained from annual forest management. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| Data / Parameter | $A_{ROAD,i,t}$ |
|---|---|
| Data unit | ha |
| Description | Area of roads in stratum i in year t |
| Source of data | Reported measurements such as post-harvest assessment reports, post-harvest maps that are based on field measurements, or Annual Operational Plans of the Sustainable Management Plan. Remote sensing data and GIS. |
| Description of measurement methods and procedures to be applied | The area of roads created may be based on the length of roads multiplied by the average width of roads. The length of all roads created during selective logging within a stratum i in year t must be measured through systematically sampling the entire area logged to produce a conservative estimate of the length of roads created. The average width of the roads may be estimated through systematically sampling road width. Sufficient number of measurements of road width shall be measured to achieve a precision of equal or less than 15% of the mean at the 95% |

| | |
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| | confidence interval. Where different categories of roads exist, different average road widths must be used. |
| Frequency of monitoring/recording | To be determined, according to the decision listed in the 'Comments' section. |
| Value applied | The value will be calculated ex-post |
| Monitoring equipment | According to the Annual Operational Plans of the Sustainable Management Plan. |
| QA/QC procedures to be applied | The measured area of roads will be crosschecked with GIS analysis. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Estimations of emissions from road creation shall be based on logging management plans or average width of roads and length of roads produced for logging in the farm. GIS analysis will support final conclusions. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| Data / Parameter | $L_{sk}$ |
|---|---|
| Data unit | m |
| Description | Length of skid trail sk. |
| Source of data | Reported measurements such as post-harvest assessment reports, post-harvest maps that are based on field measurements, or Annual Operational Plans of the Sustainable Management Plan. |
| Description of measurement methods and procedures to be applied | The length of all skid trails within a stratum i in year t may be estimated through using systematic sampling with a random start of the entire area logged or within a sampled known logged area within the project boundary to produce a conservative |

| | |
|---|---|
| | estimate of the length of skid trails created. Where sampling is used, the total length of all skid trails can then be equal to the mean length of skid trails per unit area multiplied by the total area logged |
| Frequency of monitoring/recording | To be determined, according to the decision listed in the 'Comments' section. |
| Value applied | The value will be calculated ex-post |
| Monitoring equipment | According to the Annual Operational Plans of the Sustainable Management Plan. |
| QA/QC procedures to be applied | The measured length of skid trails in current logging gaps will be compared with those of previous logging gaps. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Estimations of emissions from skid trail creation shall be based on logging management plans or average length and number of skid trails produced due to logging in the farm. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| Data / Parameter | **CAB_tree, dest** |
|---|---|
| Data unit | $tCO_2$-e ha-1 |
| Description | Carbon stock in aboveground tree biomass assumed to be killed per unit area resulting from the creation of the skid trail in stratum i. |
| Source of data | Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data. |
| Description of measurement methods and procedures to be applied | As indicated in module CP-AB. |

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|---|---|
| Frequency of monitoring/recording | The maximum size tree assumed to be killed during skid trail creation must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event. |
| Value applied | Currently N/A, to be calculated ex-post, after SFMP implementation. |
| Monitoring equipment | On-the-ground measurement, data according to the Annual Operational Plans of the Sustainable Management Plan, and scientific data review. |
| QA/QC procedures to be applied | In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs.<br><br>Furthermore, the measured area of skid trails will be compared with those of previous skid trails. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Multiplication of skid trail dimensions by carbon stock in accordance with parameter CAB_tree, described in section 4.1 |
| Comments | N/A to this project during the current monitoring period as there is currently no logging in the project. |

| Data / Parameter | **CBB_tree,dest** |
|---|---|
| Data unit | $tCO_2$-e ha-1 |
| Description | Carbon stock in belowground tree biomass assumed to be killed per unit area resulting from the creation of the skid trail in stratum i |
| Source of data | Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data. |
| Description of measurement methods and procedures to be applied | As indicated in module CP-AB. |

| | |
|---|---|
| Frequency of monitoring/recording | Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event |
| Value applied | Currently N/A, to be calculated ex-post, after SFMP implementation. |
| Monitoring equipment | On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study. |
| QA/QC procedures to be applied | In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs. Furthermore, the measured area of skid trails will be compared with those of previous skid trails. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Multiplication of skid trail dimensions by carbon stock in accordance with parameter CBB_tree, described in section 4.1 |
| Comments | N/A to this project during the current monitoring period as there is currently no logging in the project. |

| | |
|---|---|
| Data / Parameter | **CS/U,ig** |
| Data unit | t C |
| Description | Carbon stock of trees snapped, leaning, or uprooted in logging gap lg |
| Source of data | Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data. |
| Description of measurement methods and procedures to be applied | If the forest management team data is used: Measurements (e.g. DBH, H) shall be taken on all trees (>10 cm DBH) that have been significantly damaged and are assumed to be killed by the felling of the timber tree. This shall include all trees where the bark of the stem has been damaged, where the stem has been snapped off, where the stem has been pushed over, and those trees that have been uprooted. The above and belowground carbon stock of each tree must be estimated using the same allometric regression equation and root to shoot ratio used in the module for estimating the carbon pool in trees (CP-AB) in the baseline scenario. |

| | |
|---|---|
| Frequency of monitoring/recording | Measurement shall take place one time prior to any verification event. |
| Value applied | Currently N/A, to be calculated ex-post, after SFMP implementation. |
| Monitoring equipment | On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study.<br>If peer-reviewed study data is used, the value of carbon stock of trees snapped, leaning, or uprooted due to logging will be applied in the logging area, that will be measured ex-post, when applicable, using remote sensing and GIS. |
| QA/QC procedures to be applied | In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs. |
| Purpose of data | Calculation of project emission |
| Calculation method | Literature data from reputed sources will be used and cross- checked with field data from forest management team, if available. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| Data / Parameter | **CB,ig** |
|---|---|
| Data unit | t C |
| Description | Carbon stock of branches knocked to the ground as a direct consequence of tree felling in logging gap lg |
| Source of data | Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data. |
| Description of measurement methods and procedures to be applied | As indicated in VM0007's M-REDD module. |

| | |
|---|---|
| Frequency of monitoring/recording | Measurement shall take place prior to every verification event. |
| Value applied | Currently N/A, to be calculated ex-post, after SFMP implementation. |
| Monitoring equipment | On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study. |
| QA/QC procedures to be applied | In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Literature data from reputed sources will be used and cross- checked with field data from forest management team, if available. |
| Comments | The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution. |

| | |
|---|---|
| Data / Parameter | **Wski**d |
| Data unit | m |
| Description | Mean width of skid trails |
| Source of data | Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data |
| Description of measurement methods and procedures to be applied | The average width of skid trails created within a stratum i can be based on reported widths; a conservative estimate based on machinery used; or additional field measurements.<br>Reported width: the average width of field measured skid trails within FSC Post Harvest Assessment reports from logging that has taken place within the project area;<br>Conservative estimate: Width edge of tires on largest skidder type * 140% can only be used where skidder type(s) is known and used to create all skid trails. (skidder defined as heavy vehicle used in logging operation to create path and pull cut trees to logging road);<br>Field measurements: average of field sampling based on systematic sampling with a random start. The width of the skid trail |

| | |
|---|---|
| | shall be equal to the distance between trees undamaged by skid trail creation on one side to trees undamaged by skid trail creation on the other side. Sufficient number of measurements of road width shall be measured to achieve a precision of equal or less than 15% of the mean at the 95% confidence interval. |
| Frequency of monitoring/recording | Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event |
| Value applied | Currently N/A, to be calculated ex-post, after SFMP implementation. |
| Monitoring equipment | On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study. |
| QA/QC procedures to be applied | In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs. |
| Purpose of data | Calculation of project emissions |
| Calculation method | Literature data from reputed sources will be used and cross- checked with field data from forest management team, if available. |
| Comments | N/A to this project during the current monitoring period as there is currently no logging in the project. |

| Data / Parameter | **Leakage outside** |
|---|---|
| Data unit | $tCO_2$/year |
| Description | Net $CO_2$ emissions due to unplanned deforestation displaced outside the leakage belt up to year t* |
| Source of data | Remote sensing data and GIS. Secondary data. |
| Description of measurement methods and procedures to be applied | The unplanned activities in leakage management areas that result in carbon stock decrease will be subject to monitoring, when significant. |

| | |
|---|---|
| Frequency of monitoring/recording | Annually |
| Value applied | Values are described in table 79. |
| Monitoring equipment | Remote sensing and GIS. Secondary data. |
| QA/QC procedures to be applied | N/A. |
| Purpose of data | Calculation of leakage |
| Calculation method | As per LK-ASU, Step 4. |
| Comments | N/A. |

| Data / Parameter | **MANFOR** |
|---|---|
| Data unit | ha |
| Description | Total area of forests under active management nationally |
| Source of data | Official country-specific data from IBAMA[126]. |
| Description of measurement methods and procedures to be applied | As per LK-ASU, a demonstration is required that areas will be protected against deforestation. Such a demonstration must include the existence of forest guards in sufficient numbers to prevent illegal colonization and an active management plan detailing harvest plans and return intervals, and/or evidence that the concession owner has previously evicted illegal colonists/squatters from the forest areas. Ex-ante it can be assumed that MANFOR must remain constant. |
| Frequency of monitoring/recording | Annually |
| Value applied | 1.400,000 |
| Monitoring equipment | N/A. |
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |

| | |
|---|---|
| Purpose of data | Calculation of leakage emissions |
| Calculation method | N/A. |
| Comments | N/A. |

| Data / Parameter | **PROTFOR** |
|---|---|
| Data unit | ha |
| Description | Total area of fully protected forests nationally |
| Source of data | Official country-specific data from ISA[137]. |
| Description of measurement methods and procedures to be applied | As per LK-ASU, a demonstration is required that areas will be protected against deforestation. Such a demonstration is made by government mechanisms and national polices.<br><br>Ex-ante it can be assumed that PROTFOR must remain constant. |
| Frequency of monitoring/recording | Annually |
| Value applied | 88.566,400 |
| Monitoring equipment | N/A. |
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |
| Purpose of data | Calculation of leakage emissions |
| Calculation method | N/A. |
| Comments | N/A. |

| Data / Parameter | **TOTFOR** |
|---|---|

[137]https://qv.icmbio.gov.br/QvAJAXZfc/opendoc2.htm?document=painel_corporativo_6476.qvw&host=Local&anonymous=true

| | |
|---|---|
| Data unit | ha |
| Description | Total available national forest area |
| Source of data | Official country-specific data. As the country has a great variety of forest biomes in all its extension, TOTFOR considered only the Amazon Rainforest biome. This is a conservative approach. Thus, as a representation of the total area of Amazon Rainforest in Brazilian Territory, TOTFOR consisted of multiplying size of the Brazilian Amazon by 97%, which represents its preserved area, according to SEMA[138]. |
| Description of measurement methods and procedures to be applied | As per LK-ASU, forest areas suitable for conversion to livestock. Ex ante it can be conservatively assumed that TOTFOR must remain constant for the baseline period. |
| Frequency of monitoring/recording | Annually |
| Value applied | 486.461,572 |
| Monitoring equipment | N/A. |
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |
| Purpose of data | Calculation of leakage emissions |
| Calculation method | N/A. |
| Comments | N/A. |

| Data / Parameter | **ΔCP,LB** |
|---|---|
| Data unit | $tCO_2e$ |
| Description | Net greenhouse gas emissions within the leakage belt in the project case |
| Source of data | As per Module M-REDD |

[138] https://meioambiente.am.gov.br/unidade-de-conservacao/

| | |
|---|---|
| Description of measurement methods and procedures to be applied | As per Module M-REDD. |
| Frequency of monitoring/recording | Annually |
| Value applied | To be measured ex-post. |
| Monitoring equipment | N/A. |
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |
| Purpose of data | Calculation of leakage emissions |
| Calculation method | N/A. |
| Comments | N/A. |

| Data / Parameter | **$PROP_{IMM}$** |
|---|---|
| Data unit | Proportion |
| Description | Estimated proportion of baseline deforestation caused by immigrating population |
| Source of data | Official country-specific (government) data. The proportion of baseline deforestation caused by immigrating population ($PROP_{IMM}$) was estimated for a period from 2012 to 2020. For calculating $PROP_{IMM}$, it was used local data[129] for births, deaths and population. It is then assumed that the total annual population growth in a given municipality is attributed to: i) births and ii) immigration. Thus, by subtracting the number of annual births from the total annual population growth, it is possible to infer the number of immigrants. |
| Description of measurement methods and procedures to be applied | Estimated as proportion of the area deforested in the past 5 years by population that migrated into the leakage belt and project area in |

| | |
|---|---|
| | the past 5 years (all areas within 2 km of the boundaries of the project area and the leakage belt must be considered here). |
| Frequency of monitoring/recording | Annually |
| Value applied | 1.33 |
| Monitoring equipment | N/A. |
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |
| Purpose of data | Calculation of leakage emissions |
| Calculation method | N/A. |
| Comments | N/A. |

| | |
|---|---|
| Data / Parameter | $A_{DefLB,i,t}$ |
| Data unit | ha |
| Description | Area of recorded deforestation in the leakage belt in the project case in stratum i in year t |
| Source of data | As per Module M-REDD. Satellite imagery. |
| Description of measurement methods and procedures to be applied | As per Module M-REDD. Satellite imagery analysis. |
| Frequency of monitoring/recording | Annually |
| Value applied | To be measured ex-post. |
| Monitoring equipment | N/A. |

| | |
|---|---|
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |
| Purpose of data | Calculation of leakage emissions |
| Calculation method | N/A. |
| Comments | N/A. |

| | |
|---|---|
| Data / Parameter | $\mathbf{A_{DefPA,I,u,t}}$ |
| Data unit | ha |
| Description | Area of recorded deforestation in the project area in the project case in stratum i converted to land use *u* in year *t* |
| Source of data | As per Module M-REDD. Satellite imagery. |
| Description of measurement methods and procedures to be applied | As per Module M-REDD. Satellite imagery analysis. |
| Frequency of monitoring/recording | Annually |
| Value applied | To be measured ex-post. |
| Monitoring equipment | N/A. |
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |
| Purpose of data | Calculation of leakage emissions |
| Calculation method | N/A. |
| Comments | N/A. |

| | |
|---|---|
| Data / Parameter | **RFt** |
| Data unit | % |
| Description | Risk factor used to calculate VCS buffer credits |

| | |
|---|---|
| Source of data | • VCS Non-Permanence Risk Report (v3.1),<br>• Remote sensing data and GIS,<br>• Supervisor report.<br>• Literature data. |
| Description of measurement methods and procedures to be applied | All sources of data from the VCS Non-Permanence Risk Report will be used to measure the various risk factors. |
| Frequency of monitoring/recording | Annually |
| Value applied | 10 |
| Monitoring equipment | VCS-approved AFOLU Non-Permanence Risk Tool |
| QA/QC procedures to be applied | Literature data from reputed sources will be used and critically checked. When possible, the average of two or more sources will be used. |
| Purpose of data | Calculation of project emissions |
| Calculation method | All the risk factors described in the VCS Risk Report were assessed. |
| Comments | N/A |

| Data / Parameter | **Project Forest Cover Monitoring Map** |
|---|---|
| Data unit | *ha* |
| Description | Map showing the location of forest land within the project area at the beginning of each monitoring period. If within the Project Area some forest land is cleared, the benchmark map must show the deforested areas at each monitoring event |
| Source of data | Remote sensing in combination with GPS data collected during ground truthing |
| Description of measurement methods and procedures to be applied | The minimum map accuracy must be 90% for the classification of forest/non-forest in the remote sensing imagery. If the classification accuracy is less than 90% then the map is not acceptable for further analysis. More remote sensing data and ground truthing data will be needed to produce a product that |

| | |
|---|---|
| | reaches the 90% minimum mapping accuracy |
| Frequency of monitoring/recording | Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event |
| Value applied | To be calculated ex-post |
| Monitoring equipment | Remote sensing and GIS |
| QA/QC procedures to be applied | The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other souces will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area:<br>Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September.<br>Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images.<br>For analysis of areas with cloud cover, visual interpretation of radar image would be performed.<br>Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very good. |
| Purpose of data | Project Emission |
| Calculation method | Remote sensing and GIS |
| Comments | *N/A* |

| Data / Parameter | **Leakage Belt Forest Cover Monitoring Map** |
|---|---|
| Data unit | ha |
| Description | Map showing the location of forest land within the leakage belt area at the beginning of each monitoring period. Only applicable where leakage is to be monitored in a leakage belt |
| Source of data | Remote sensing in combination with GPS data collected during ground truthing |
| Description of measurement methods and procedures to be applied | The minimum map accuracy must be 90% for the classification of forest/non-forest in the remote sensing imagery. If the classification accuracy is less than 90% then the map is not acceptable for further analysis. More remote sensing data and ground truthing data will be needed to produce a product that reaches the 90% minimum mapping accuracy. |
| Frequency of monitoring/recording | Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event |
| Value applied | To be calculated ex-post |
| Monitoring equipment | Remote sensing and GIS |
| QA/QC procedures to be applied | The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other souces will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area:<br>Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is |

| | |
|---|---|
| | performed during the period of low incidence of clouds and rainfall in the region, within July and September.<br>Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images.<br>For analysis of areas with cloud cover, visual interpretation of radar image would be performed.<br>Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very good |
| Purpose of data | *Leakage emissions* |
| Calculation method | Remote sensing and GIS |
| Comments | N/A |

| | |
|---|---|
| Data / Parameter | **ADefLB,i,u,t** |
| Data unit | ha |
| Description | Area of recorded deforestation in the leakage belt in stratum i converted to land use u in year t |
| Source of data | As per Module M-REDD v2.2. Remote sensing imagery. |
| Description of measurement methods and procedures to be applied | As per Module M-REDD. Satellite imagery analysis. |
| Frequency of monitoring/recording | Annually |
| Value applied | To be measured ex-post. |
| Monitoring equipment | N/A |
| QA/QC procedures to be applied | As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module. |
| Purpose of data | Calculation of leakage emissions |

| | |
|---|---|
| Calculation method | N/A. |
| Comments | N/A. |

### 3.3.3 Monitoring Plan

This Monitoring Plan was developed according to Methodology VM0007 Version 1.6. The methodology encompasses fourmain monitoring tasks:

i) Monitoring of project implementation;

ii) Monitoring of actual carbon stock changes and GHG emissions within the project area;

iii) Monitoring of leakage; and

iv) Ex post calculation of net anthropogenic GHG emissions reductions.

This Monitoring Plan describes how these tasks will be implemented. For each task, the monitoring plan includes the following aspects:

a) Technical description of the monitoring tasks;

b) Data to be collected;

c) Overview of data collection procedures;

d) Quality control and quality assurance procedures;

e) Data archiving; and

f) Organization and responsibilities of the parties involved in all the above.

***General informations***

Carbonext will perform the monitoring of carbon stocks from deforestation rates and greenhouse gas emissions produced or avoided during the REDD+ project, so it has a specialized and experienced REDD+ team. This includes social experts, who have the necessary skills to involve the community and stakeholders, ensuring community participation, activity design and socio-economic indicator monitoring, as described in *2.4.3 Management Team Experience.*

The main community stakeholders in this project are the communities Vila Batista, Vale da Benção and Nova Esperança. The communication channel between the stakeholders and the proponent is between the secretary and spokesperson Valdenir Lemos Ribeiro, hired by the Project Proponent.

Given the above, the organizational structure below is very simple, where the Project Proponents (Silvio and Carina) and the Project Designer (Carbonext) work together and exist a communication channel direct between site spokesperson Valdenir Lemos Ribeiro, and the technical developer team. The technical team is responsible for planning the activities and providing the guidance, while the site spokesperson is responsible for executing the project's actions and collective their respective items of evidence.

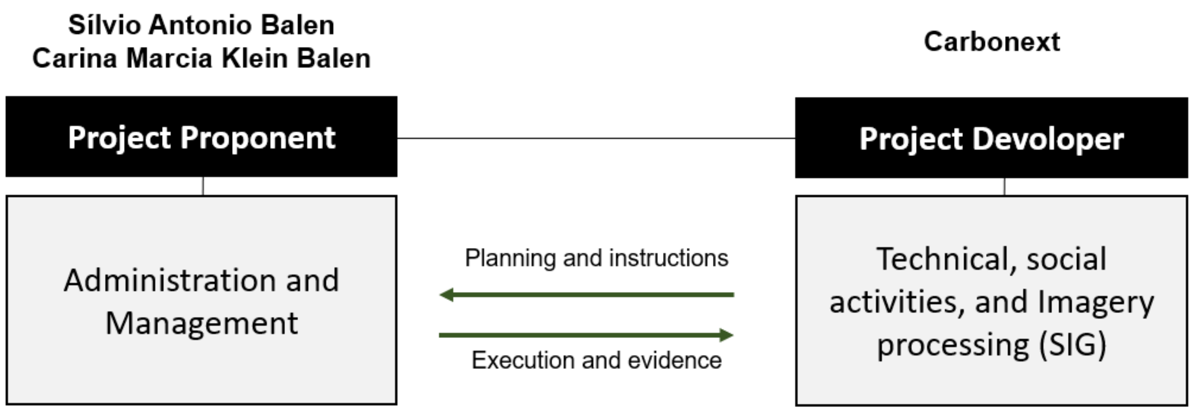


*Figure 39 - General overview of parties involved in monitoring activities.*

For the IPOÁ REDD+ Project, the PD activities (historical analysis of land use change) were carried out using GIS tools by Carbonext's team of experts using data from the PRODES project[139].Thus, we foresee that the other MRs will follow this strategy in order to guarantee methodological consistency. In case of unavailability of data from the PRODES system, we will consult the Mapbiomas base[140]. In case of unavailability of data from Mapbiomas, we will perform a supervised classification of optical images from Landsat or Sentinel 2 orbital sensors. If there is a high incidence of clouds in the mentioned images, the strategy will be to classify radar (SAR) images available from the Sentinel 1 sensor and thus measure all changes in land use within the period defined for the MR according to the requirements demanded by the methodology.

[139] http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes
[140] https://mapbiomas.org/

As a complementary strategy for monitoring deforestation, Carbonext has implemented a deforestation detection and warning system called MonitoraCarbon. Since the area has difficult access, the adoption of this system increases the chances of detecting possible changes in land use (deforestation), because it is programmed to perform continuous analysis as soon as an optical or radar image of the project area is available. Because it is an alert system, its process is designed to be sensitive enough to detect any change in the forest, undergo a rapid internal validation process, and issue alerts for monitoring agents to confirm and take appropriate action to contain the problem as efficiently as possible. The system is available for the IPOA project since 06/2022.

***Monitoring of Actual Carbon Stock Changes and GHG Emissions within the Project Area***

a) Technical description of the monitoring tasks.

Project activities implemented within the project area will be monitored to be consistent with the management plans of the project area and the PD.

b) Data to be collected.

Monitoring of deforestation-avoidance activities will be performed by means of evaluations andusing satellite imagery to continuously inspect the forest condition within the Project Area, as described in *3.3.2. Data and Parameters Monitored.* All images, maps and records generated during project implementation should be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

Monitoring of social and biodiversity parameters of project implementation will be based on data required in section *4.4.2. Monitoring Plan Dissemination* and *5.4.1 Biodiversity Monitoring Plan*, which attest those foreseen activities are being effectively implemented. A system will be implemented to record data on costs and investments, based on invoices, receipts, and contracts related to the implementation of activities, as well as photos, filming, reports, signed attendance lists, or other evidence that proves the activities were carried out.

c) Overview of data collection procedures.

All invoices, receipts and contracts related to activities implemented in this REDD+ project shall be conserved in printed version. Whenever possible, documentation should be kept in electronic format.

d) Quality control and quality assurance procedures.

The project proponent will train personnel to collect and keep all documentation in a sure place. All electronic documentation should also be sent to CARBONEXT to further security and checking.

e) Data archiving

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification. Backup copies of files should be available in the project proponent facilities, as well as in CARBONEXT facilities.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

The project proponent is responsible to implement this monitoring item. CARBONEXT is coresponsible in archiving, and quality control and quality assurance procedures, as described in *3.3.2. Data and Parameters Monitored.*

**Monitoring of land-use and land-cover change within the project area**

a) Technical description of the monitoring tasks.

Monitoring of land-use and land-cover change within the Project Area will be performed annually by analyzing satellite imagery of the Project Area.

Imagery analysis of land use will be based on secondary information from PRODES project, referring mainly to the classes "Forest Land" and "Non-Forest Land"

b) Data to be collected.

The categories of change that will be subject to MRV-A (monitoring, reporting, verification and accounting) are “Area of forest land converted to non-forest land”, “Area of forest land undergoing carbon stock decrease” and “Area of forest land undergoing carbon stock increase”. In this context, data to be collected consists of annual land use cover processed by PRODES project for the entire land coverage of Project Area and other such as Leakage belt and reference region.

The validation will be performed by using satellite images from Landsat 8 TM (orbit 230/64) to cover the total geographic scope of the Project Area and Reference Region. Other optical sensors could be use for accuracy assessment in case of cloud cover such as Sentinel 2 and CBERS.

c) Overview of data collection procedures.

The PRODES project contributes to understand the deforestation dynamics in Brazil. The data generated by this program is used in this project. PRODES data are applicable for use in this project, according to the criteria listed below (Methodology VM0007):

i) PRODES data cover the entire project area, leakage belt and reference region.
ii) PRODES data cover the entire reference period (beginning, middle and end) of the fixed baseline period.
iii) PRODES monitors conversion of forest land to non-forest land.
iv) Monitoring occurred during the entire fixed baseline period.

d) Quality control and quality assurance procedures.

The validation of land-use data used for modeling of land use will be performed by using the confusion matrix, in order to calculate the overall index of success by period and by class. Three specific classes will be used: forest, accumulated deforestation and other (hydrography, not forest, clouds, roads, residues, unclassified objects, and others).

With the help of the "Create Random Points" tool in ArcGIS Pro, at least 100 random points will be generated for each class / year as samples for evaluation, using satellite images as reference, making it possible to generate a confusion matrix for calculation of the accuracy indexes, and the Kappa index (indicators for validation of mapping accuracy). Land use classes must have higher values than 90% for the accuracy, as required in VMD0007 *3.3 methodology*.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT SIG expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

**Monitoring of carbon stock changes and non-$CO_2$ emissions from forest fires**

a) Technical description of the monitoring tasks.

Monitoring of carbon stocks is mandatory in the following cases:

Within the project area:

a) Areas subject to significant carbon stock decrease in the project scenario according to the ex-ante assessment. These will be areas subject to controlled deforestation and planned harvest activities, such as logging, fuel wood collection and charcoal production. In these areas, carbon stock changes must be estimated at least once after each harvest event.
b) Areas subject to unplanned and significant carbon stock decrease, e.g., due to uncontrolled forest fires and other catastrophic events. In these areas, carbon stock losses must be estimated as soon as possible after the catastrophic event.

Within leakage management areas:

a) Areas subject to planned and significant carbon stock decrease in the project scenario according to the ex-ante assessment. In these areas, carbon stocks must be estimated at least once after the planned event that caused the carbon stock decrease.

b) Data to be collected.

The results of monitoring activity data and carbon stocks must be reported using the same formats and tables used for the ex-ante assessment, according to Methodology VM0007 (the applicability of each table must be evaluated ex post, in the Monitoring Report).

Non-$CO_2$ emissions from forest fires are subject to monitoring and accounting, when significant.

Decreases in carbon stocks and increases in GHG emissions (e.g., in case of forest fires) due to natural disturbances (such as hurricanes, earthquakes, volcanic eruptions, tsunamis, flooding, drought, fires, tornados or winter storms) or man-made events, including those over which the project proponent has no control (such as acts of terrorism or war), are subject to monitoring and must be accounted under the project scenario, when significant.

If the area (or a sub-set of it) is affected by natural disturbances or man-made events generated VCUs in past verifications, the total net change in carbon stocks and GHG emissions in the area(s) that generated VCUs must be estimated, and an equivalent amount of VCUs must be cancelled from the VCS buffer.

c) Overview of data collection procedures

The PRODES project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. PRODES data are applicable for use in this project, according to the criteria listed below (Methodology VM0007):

- PRODES data cover the entire project area, leakage belt and reference region.
- PRODES data cover the entire reference period (beginning, middle and end) of the fixed baseline period.
- PRODES monitors conversion of forest land to non-forest land.
- Monitoring occurred during the entire fixed baseline period.

d) Quality control and quality assurance procedures

The validation of land-use data used for modeling of land use will be performed by using the confusion matrix, in order to calculate the overall index of success by period and by class. Three specific classes will be used: forest, accumulated deforestation and other (hydrography, not forest, clouds, roads, residues, unclassified objects, and others).

The validation will be performed by using satellite images from Landsat 8 TM (orbit 230/64) to cover the total geographic scope of the Project Area and Reference Region. Other optical sensors could be use for accuracy assessment in case of cloud cover such as Sentinel 2 and CBERS.

With the help of the "Create Random Points" tool in ArcGIS PRO, 100 random points will be generated for each class / year as samples for evaluation, using satellite images as reference, making it possible to generate a confusion matrix for calculation of the accuracy indexes, and the Kappa index (indicators for validation of mapping accuracy).

Land use classes must have higher values than 90% accuracy for the accuracy and Kappa index, as required in VM0007 1.6 methodology.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT SIG expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

***Monitoring of Leakage***

a) Technical description of the monitoring tasks

The sources of leakage identified as significant in the ex-ante assessment are subject to monitoring. Two sources of leakage are potentially subject to monitoring:

I. Decrease in carbon stocks and increase in GHG emissions associated with leakage prevention activities;

**II.** Decrease in carbon stocks and increase in GHG emissions in due to activity displacement leakage.

This Project Activity does not involve decrease in carbon stocks and increase in GHG emissions associated with leakage prevention activities. In this project, leakage prevention activities do not involve any carbon stock reduction due to deforestation or additional emissions caused by increased grazing activities. In this case, only the decrease in carbon stocks and increase in GHG emissions in due to activity displacement leakage will be monitored.

b) Data to be collected.

Deforestation above the baseline in the leakage belt area will be considered activity displacement leakage.

The result of the ex-post estimations of carbon stock changes must be reported using the same table formats used in the ex-ante assessment of baseline carbon stock changes in the leakage belt.

Where strong evidence can be collected that deforestation in the leakage belt is attributable to deforestation agents that are not linked to the Project Area, the detected deforestation will not be attributed to the Project Activity and considered leakage. The operational entity verifying the monitoring data shall determine whether the documentation provided by the project proponent represents sufficient evidence to consider the detected deforestation as not attributable to the Project Activity and therefore not leakage.

To estimate the increased GHG emissions due to forest fires in the leakage belt area the assumption is made that forest clearing is done by burning the forest. The parameter values used to estimate emissions shall be the same used for estimating forest fires in the baseline, except for the initial carbon stocks (Cab, Cdw) which shall be those of the initial forest classes burned in the leakage belt area.

Report the result of the estimations using the same table formats used in the ex-ante assessment of baseline GHG emissions from forest fires in the project:

- Parameters used to calculate emissions from forest fires in the leakage belt area

- Ex post estimated non-$CO_2$ emissions from forest fires in the leakage belt area.

c) Overview of data collection procedures.

The PRODES project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. PRODES data are applicable for use in this project, according to the criteria listed below (Methodology VM0007):

i) PRODES data cover the entire project area, leakage belt and reference region.
ii) PRODES data cover the entire reference period (beginning, middle and end) of the fixed baseline period.
iii) PRODES monitors conversion of forest land to non-forest land.
iv) Monitoring occurred during the entire fixed baseline period.

d) Quality control and quality assurance procedures.

The validation of land-use data used for modeling of land use will be performed by using the confusion matrix, in order to calculate the overall index of success by period and by class. Three specific classes will be used: forest, accumulated deforestation and other (hydrography, not forest, clouds, roads, residues, unclassified objects, and others).

The validation will be performed by using satellite images from Landsat 8 TM (orbit 230/64) to cover the total geographic scope of the Project Area and Reference Region. Other optical sensors could be use for accuracy assessment in case of cloud cover such as Sentinel 2 and CBERS.

With the help of the "Create Random Points" tool in ArcGIS 10.0, at least 100 random points will be generated for each class / year as samples for evaluation, using satellite images as reference, making it possible to generate a confusion matrix for calculation of the accuracy indexes, and the Kappa index (indicators for validation of mapping accuracy). Land use classes must have higher values than 90% for the accuracy and Kappa index, as required in VM0007 1.6 methodology.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT SIG expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

***Ex Post Calculation of Net Anthropogenic GHG Emission Reduction***

a) Technical description of the monitoring tasks.

The calculation of ex post net anthropogenic GHG emission reductions is similar to the ex-ante calculation with the only difference that ex post estimated carbon stock changes and GHG emissions must be used in the case of the project scenario and leakage.

b) Data to be collected.

Report the ex post estimated net anthropogenic GHG emissions and calculation of Verified Carbon Units (VCUt, and VBCt) using the same table format used for the ex-ante assessment:

- Ex post estimated net anthropogenic GHG emission reductions and VCUs

c) Overview of data collection procedures.

Data collection procedures are the same as described in previous steps. This step involves compilation of data from previous procedures to calculate ex post net anthropogenic GHG emission reduction.

d) Quality control and quality assurance procedures.

A map showing Cumulative Areas Credited within the project area shall be updated and presented to VCS verifiers at each verification event. The cumulative area cannot generate additional VCUs in future periods.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT SIG expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

**Revisiting the Baseline Projections for Future Fixed Baseline Period**

According to VM0007, the baseline will be revisited every 10 years. Thus, the first revision of the baseline is scheduled for 2032. For this purpose, the following tasks will be carried out:

- Updating information on agents, drivers and underlying causes of deforestation, which involves the following sub-tasks:

- Collecting information that is relevant to understand deforestation agents, drivers and underlying causes;

- Redoing step 3 of the ex-ante methodology, as specified in the methodology;

- Recalibrating the model for projection of future deforestation, using new “Factor Maps” for the subsequent fixed baseline period.

• Adjusting the land-use and land-cover change component of the baseline, which involves reassessing the following components of the baseline projections:

- The annual areas of baseline deforestation.

- The location of baseline deforestation.

- Adjustment of the annual areas of baseline deforestation.

- Adjustment of the location of the projected baseline deforestation.

Adjusting, as needed, the carbon component of the baseline (this task will only be carried out if more accurate methods for carbon stocks estimates are available in the occasion of baseline revision).

### 3.3.4 Dissemination of Monitoring Plan and Results (CL4.2)

The dissemination of the results of the monitoring and verification will be published mainly on the internet through the Carbonext's website, and also on the Verra platform page as usual. Considering that some local communities still don´t have access to internet, during the onsite visits print versions of the summary results will be made available to community members and provided to any individual or institution, under request. The established frequency is annually or before any new verification, or visitation periods, to enhance the importance and effectiveness of the project, an Operating Procedure on Annual Activities Report is being implemented. Other dissemination ways can be also adopted to give total transparency to the project, such as social networks.

### 3.3.5 Regional Climate Change Scenarios (GL1.1)

Identify likely regional or sub-national climate change and climate variability scenarios and impacts and identify potential changes in the local land use scenario due to these climate change scenarios in the absence of the project.

### 3.3.6 Climate Change Impacts (GL1.2)

Describe how current or anticipated climate changes are having or are likely to have an impact on the following in the project zone and surrounding regions:

Community well-being.

Biodiversity conservation status.

### 3.3.7 Measures Needed and Designed for Adaptation (GL1.3)

Based on the causal model described in response to G1.8, describe measures needed and designed to assist communities and biodiversity to adapt to the probable impacts of climate change.

# 4 COMMUNITY

## 4.1 Without-Project Community Scenario

### 4.1.1 Descriptions of Communities at Project Start (CM1.1)

***Social Aspects:***

Near the Project Area are the ribeirinhas communities[141] of Vila Batista, with a population of about 35 people and exists for more than 20 years and the Leader of the community is Carlinhos (Moacir Batista da Silva). Vila Batista, unlike other communities, is not centralized, that is, the houses in this community are very far apart, as can be seen in Figure 40. Basically, Vila Batista is composed by three points: the first is where the school, the church, and where Carlinhos, the community leader, lives; the second and the third point are the other houses.

Another community outside the project area, however nearby, is the riverside community of Nova Esperança where the leader is Magno Batista da Silva, Carlinhos' brother. The community has existed for three years and is located about 5 km from Fazenda do Maanaim with about 40 people (about seven families), the houses are centralized, and the community is growing because it is relatively new.

The riverside community Vale da Benção is led by Bráulio Pontes Machado and is located approximately 9 km from Fazenda do Maanaim, it used to be a larger community, but over time people moved to nearby towns, and now there are only three families, about 20 people. None of the communities have a formalized community association or cooperatives, but all the residents recognize the leaders of each one.

[141] Ribeirinhas populations: people who live on the banks of the rivers in the Amazon region.

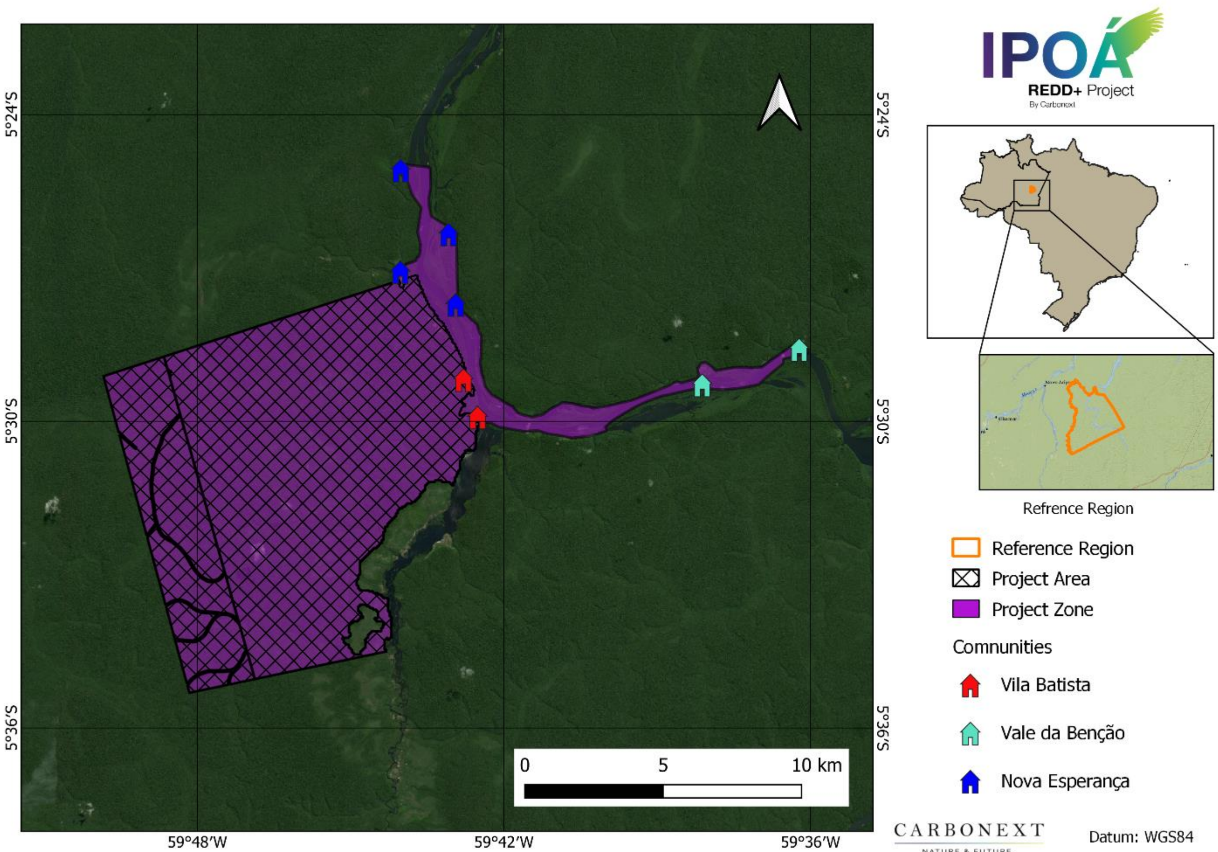


*Figure 40. Residents' homes in each community in the Project Zone.*

The communities have little or no infrastructure, which is the set of fundamental services for the socioeconomic development of a region such as sanitation, transportation, energy, and telecommunications. The communities have no conventional electricity, only in Magno's house in Nova Esperança, which has two LED reflectors that are charged from small solar plates and have the capacity to do artificial lighting. Some residents of the communities have cell phones, but as there is no telephone coverage few have carrier chips, and those who have them usually have telephone signal or internet access when they are in the municipalities of Borba and Novo Aripuanã, or even in the community of Arara where internet access is sold (R$ 3.00 per hour of use). The municipality of Novo Aripuanã in March 2022 registered 10,045 mobile phone accesses and has the density of service of

38.5 accesses per 100 inhabitants, a value 57% lower than the density in the state of Amazonas[142]. There is no police department, bank branch, communication antenna or health center operating in the communities. All community members are native speakers of Portuguese.

As far as education is concerned, according to the socioeconomic questionnaires applied to the residents, most people in the communities have studied up to Fundamental I (1st to 5th grade). Of the 44 people interviewed, 5% did higher education, 2% finished high school, 43% finished Fundamental 1, and 25% are illiterate. In Vila Batista there is a school, which is deteriorated and, according to reports, the students do not have classes during the whole school year. The Vale da Benção community used to have a school, but it no longer exists. Nova Esperança is articulating with the municipality of Borba to obtain support for a resident of the community to get authorization to be the teacher, and after getting this authorization they would build the school in the community.

The main leisure activity in the communities are the soccer games played among the men of the communities. In terms of religion, all are evangelical Christians from the Assembly of God, we will describe more in the item below about religion. To access other places, the communities use canoes, small wooden boats with engines known as rabetas[143] or small boats.

It is worth mentioning that there are also no communities or community groups within Fazenda de Campos, nor quilombos or indigenous groups within this area.

During the visit to the communities conducted in March 2022, the main economic activities of the communities were studied, which are based on the extraction of copaiba oil and nuts, cassava planting for flour production, and vegetable production. In addition, there were reports from residents of wood harvesting for the community's own use for building houses and other objects, but they do not have authorization or a cutting permit from the responsible bodies. Through third parties, we discovered that there is the sale of lumber, which is taken to the District of Canumã, Borba, and Novo Aripuanã to be sold. Another source of income is the payment generated by the Brazil Assistance Program[144].

The wild-harvesting activities described above are mainly carried out by men, while work within the household is the main responsibility of women. This unequal participation in work that generates

[142] Anatel - Mobile Telephony Accesses

[143] A Rabeta is a small boat, considered a canoe that uses a rabeta engine, or rabeta, as it is also called, which is attached to the hull of the boat, taking up less space.

[144] Auxílio Brasil — Português (Brasil) (www.gov.br)

financial income is the basis for a gender pay gap which, as found in field studies (as further discussed below), which is also reflected in literature on certain traditional amazon communities[145]. While both genders carry out the work for production of manioc flour, a job carried out collectively by the whole family, this is not remunerated work.

The communities have no electricity from the municipal grid, only in Magno's house is there a small solar panel able to provide artificial lighting. Many of the residents have cell phones, but there is no cell phone coverage, and because of this they can only access the internet when they go to the city or to a community that provides wi-fi.

***Diversity within the community:***

To diagnose the communities, we applied socioeconomic questionnaires in the three communities mentioned above: Vila Batista, Nova Esperança and Vale da Benção. Unfortunately, at the time, part of the communities is in the municipalities of Borba and Novo Aripuanã, and some men from the community were engaged in timber extraction near the Camaiu River. Because of this, we did not apply the questionnaires to all the communities, but we will consider the data obtained from these questionnaires to quantitatively describe some aspects of the communities.

Considering the total number of residents in the three communities obtained during the community visits in March 2022, we have a total of 105 in habitants: 35 habitants in Vila Batista, 40 habitants in Nova Esperança, and 30 habitants in Vale da Benção. Considering that the questionnaires were applied to six families, we have a number of 44 habitants evaluated, which corresponds to 42% of the total number of habitants of the communities. Thus, we have a sample of interviewees inside the Project Area that corresponds to 37% of the total population of Vila Batista, and outside the project area we have a sample of interviewees that corresponds to 44% of the total of residents that are outside the project area.

Regarding gender diversity in the communities, with the questionnaires applied, we observed that 30% of the people evaluated are women, 43% are men, and 27% are children under 12 years old,

[145]https://go.gale.com/ps/i.do?id=GALE%7CA565057147&sid=googleScholar&v=2.1&it=r&linkaccess=abs&issn=15168182&p=AONE&sw=w&userGroupName=nysl_me_jfkens

the age group defined by the Law of the Statute of the Child and Adolescent[146]. Extrapolating this ercentage to the total number of residents, we would have in absolute numbers: 31 women, 45 men and 29 children.

In relation to education inside and outside the project area, of the total number of people evaluated: 43% have completed elementary school, 25% are illiterate, 5% have a university education, and 2% have completed high school. Based on this data, we evaluated the number of men and women in each level of schooling presented in Table 16. We consider it extremely important to understand the level of schooling of these women in order to plan and develop activities that could expand the actions of women in the community from the most diverse perspectives.

*Table 80. Percentage of men and women by education level.*

| **Illiterate** | | **Elementary 1** | | **Elementary 2** | | **High School** | | **Higher Education** | |
|---|---|---|---|---|---|---|---|---|---|
| Women | Men | Women | Men | Women | Men | Women | Men | Women | Men |
| 45% | 55% | 32% | 68% | 0% | 0% | 100% | 0% | 50% | 50% |

Regarding the income of the families of the evaluated communities, it is noteworthy that in our evaluation women do not work in the forest, a fact consistent with studies by Rodrigues et al. 2015 who cite that in certain communities men contribute more to the financial income, and that women

[146] L8069 (planalto.gov.br).

hold mainly knowledge linked to home, and the production of cassava flour, which does not fit a logic "exclusionary of capitalism", but of common processing.

That is, they do not have paid work and do not contribute to the family financial income, but rather to their subsistence. In addition, as described in the economic aspects below, the families' income is not very diversified and highly dependent on the extraction of the Copaiba oil and the Brazil nuts.

***Economic aspects:***

During the 2022 visits to the communities that are in the project's zone of influence, socioeconomic surveys were conducted to diagnose these communities. In the questionnaires applied, we understood the economic dynamics, and the main sources of income of the communities.

The main economic activities of the communities are based on the extraction of copaiba oil and forest nuts; the planting of cassava for flour production and the extraction of wood on demand, and the communities do not have licenses or authorization to harvest wood. Besides the sale of the products, the residents of the communities also use these products for the community's own consumption.

The extraction of copaiba oil and nuts is done only by men, who enter the forests to search for areas to do the extraction. The communities of Vila Batista, Nova Esperança and Vale da Benção sell almost 70% of the copaíba oil in the community of Arara, which is closer to the city of Novo Aripuanã, and the rest is sold in the municipality of Borba or sells to boat dealers. The price of the oil is given by the quantity in kilograms of product, and the price of a kilo of copaíba oil, according to reports, varies between R$38.00 to R$50.00 per kg. The nuts, on the other hand, are sold in 20-liter cans and the price per can in Novo Aripuanã is as high as R$140.00.

According to Santana et al.[147], the Brazil nut tree (Bertholletia excelsa) has represented, since 1950, the main non-timber forest product of the Amazon region and its extraction generates work and income for more than 25 thousand families, besides contributing to the dynamics and diversity of the flora and fauna, as well as to climate regulation.

[147] Santana, A. C. de; Santana, Á. L; Santana, Á. L.; Martins, C. M. Valoração e sustentabilidade da castanha-do-brasil na Amazônia. Revista de Ciências Agrárias, 60, 77-89, 2017. Disponível em: http://periodicos.ufra.edu.br/index. php/ajaes/article/view/2690.

The price of nuts becomes strongly determined by demand, since supply varies very little due to the productive capacity of Brazil nut groves that are at their limit. Thus, market price adjustments respond to the variation in consumer income (SANTANA et al., 2017).

In the production of cassava flour, women help men in production and in the fields. Besides the flour they sell in the cities or also in the merchant boats that navigate the Sucunduri River. The residents produce other by-products derived from cassava, such as tapioca starch for personal consumption, as can be seen in the Figure 41 below.


*Figure 41. Vila Batista resident making Tapioca Gum.*

The harvesting of wood usually happens after the men of the community receive an invitation to do this work or some kind of order. During one of the field visits to the area in April 2022, some of the men from Nova Esperança and Vila Batista had moved from their communities to carry out this illegal timber extraction work near the Camaiú River. As in some cases they are commissioned to extract the wood, the value and destination of the wood are agreed upon before the cut, when it is not commissioned, the sale of wood normally occurs in the cities, such as in the District of Canumã, Borba, and Novo Aripuanã.

Some wood is also extracted for the manufacture of canoes, for which the use of Itauba wood is common. The woods that, according to reports, are the ones they normally extract for sale are: Ipê amarelo, Angelim pedra, Envira pente de macaco, Louro rosa, among others. Many of them with occurrence registered in the Juma RDS Management Plan[148].

It is worth mentioning that they have already harvested wood from the Manaaim farm years ago, but the last harvests for commercialization were outside the project area.


*Figure 42. Wood Cutting for Canoe Production.*

[148] https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf.

Another source of income for some families in the communities is the payment generated by the Brazil Aid Program[149] of the Federal Government of Brazil. Generally, the benefited families wait two months for the payment to travel to the municipalities of Borba and Novo Aripuanã to receive the benefit, since the travel expenses are significant.

As described, the scenario without a project presents socioeconomic indicators, which characterize a region with low income and welfare conditions due to the limitation of productive activities, lack of better production techniques, difficulties in offtake, as well as access to the consumer market. According to data from IBGE[150] in the municipality of Novo Aripuanã 52% of the population has monthly income of up to half the minimum wage per person. In the socioeconomic questionnaire applied to families in the communities during the community visit in April 2022, we found that none of the evaluated families have a monthly income per capita of half a minimum wage[151].

***Medical Aspects:***

There are no health care centers or health agents in the communities. For medical attention, the residents go to the municipalities of Borba or Novo Aripuanã. The municipality of Borba is generally more accessible because it is possible to get to the town by Rabeta, but according to reports from the residents themselves, this trip takes hours, practically the whole day. The communities do not have a faster means of transportation, such as the water ambulance (*Ambulancha*) to get to the nearest healthcare facility.

Once a year the communities receive medical care by the "Doctors of the Waters"[152] project which began in 2011 in order to bring medical and dental care and hygiene practices to the riverside populations of the Amazon Basin. In addition, residents told us that the *Doutores das Águas* "Doctors of the Waters" also donate clothes, hygiene kits, etc. However, since the pandemic, the number of donations has decreased.

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[149] Auxílio Brasil — Português (Brasil) (www.gov.br).

[150] IBGE | Cidades@ | Amazonas | Novo Aripuanã | Panorama.

[151] Considering the 2022 Minimum Wage of R$ 1,212.00. Source: Novo salário mínimo 2022: veja como registrar o reajuste no eSocial Doméstico — Português (Brasil) (www.gov.br).

[152] O Projeto – Doutores das Águas (doutoresdasaguas.org.br).

The diseases that most affect the region are malaria and influenza. In the present pandemic period (2020 – 2022), some residents have shown symptoms, but none have taken tests to identify whether or not there was an infection caused by the SARS-CoV-2 virus (coronavirus - COVID-19). All the residents have had the COVID-19 vaccine, but some adults have not had the 3rd or 4th dose.

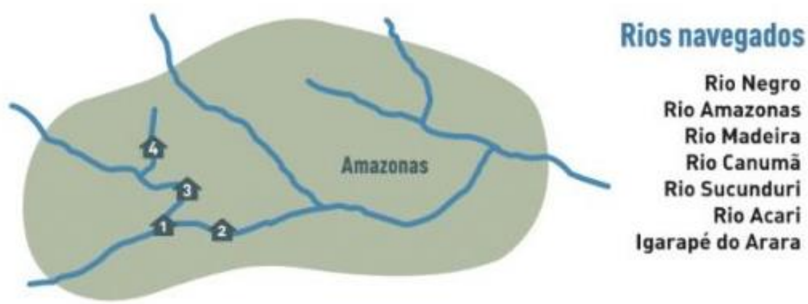


*Figure 43. Rivers navigated by the Doutores das Águas project.*

***Psychological Aspects:***

In the interviews conducted in April 2022, the residents of the three communities evaluated, reported liking to live in the communities because of the security, tranquillity, abundance, invoice of food, and because of the forest and the river. From the workshop with the women, it was possible to observe that the younger women between 14 and 20 years old, would like to live in the city mainly because of better health and education conditions.

In the interviews 83.4% of the interviewees reported having a good relationship with the other communities and only 16.6% reported that the relationship between the communities is regular.

***Environmental aspects:***

The Vila Batista community has existed for more 15 years and the Leader is Moarcir Batista da Silva (nickname Carlinhos). The Nova Esperança community has existed for three years, and the leader is Magno Batista da Silva, brother of Carlinhos. The Vale da Benção community has existed for more 10 years and is led by Braulio Bentes Machado. The communities do not have an association and do not have community cooperatives, but Magno is considered to be the leader of Nova Esperança and is trying to create an association for the community. At the time of the visit, there was no document indicating the leadership of these people. All members of the communities are native Portuguese speakers.

As described under “Economic Aspects” above, the community is not supplied with electricity from the grid.

None of the above-mentioned communities have potable water supply systems; sewage collection and treatment systems, nor the collection of solid waste generated. The main source of water supply for these communities is the Sucunduri river, through which the health agents of the Unified Health System - SUS perform the distribution of hypochlorite for domestic water treatment. The solid residues generated are burned and the human waste is thrown into an individual black well.

In the Nova Esperança and Vila Baptista communities, there were reports of dangerous situations while carrying out their economic activities, such as:

- Encounters with snakes while working in the forest
- Falling branches while extracting Copaíba oil
- Injuries due to the use of the chainsaw or the terracing

The community has not yet reported any risk of fires or landslides. However, the community of Nova Esperança has part of the community flooded by the rising level of the Sucunduri riverbed.

The main forest species[153] observed in the field were Ipê, Itaúba, Louro Rosa, Castanheira and Copaíba. The most sighted terrestrial animals are the white-lipped peccary, monkeys, capuchin monkeys, paca, armadillo, and agouti. The birds mentioned were parrot, Macaw, Curassow, Cujubim and Hoatzin. The main small watercourses are the Capinarana and das Pedras streams.

[153] Popular names of forest and animal species.

***Spiritual Aspects:***

All communities are Protestant Christian, the community is located in the Vila Batista community. Their custom is to attend this church on at least a weekly basis to its annual calendar of events.

The residents of all three communities declare themselves to be Protestant Christians, members of the Assembly of God.

The Vale da Benção community has a church where services are held on Tuesdays and Thursdays. On Sundays the services are held in the church of Vila Batista, which is outside the project area. However, the inhabitants of Vila Batista are building a new church inside the project area. The residents of Nova Esperança attend services in other communities, but they are interested in having a church within their own community.


*Figure 44. Church in the Vale da Benção community.*


*Figure 45. Building the new church in Vila Batista.*

The minister responsible for these churches is the Minister Paulo, who is in the Arara community, and the residents themselves conduct the services in his absence. Every year they hold the church's anniversary party.

### 4.1.2 Interactions between Communities and Community Groups (CM1.1)

In contact with the communities, we noticed in the reports of the residents of Vila Batista and Nova Esperança, that some years ago the leaders of both communities had a conflict that resulted in the separation of the community of Vila Batista, and years later from this separation there was the creation of the community of Nova Esperança. As the leaders of the current Vila Batista and Nova Esperança are brothers, and their wives are also sisters, the communities have contact with each other and a good relationship, but they prefer to treat each other as separate communities.

In questionnaires applied in the three riverside communities of this project, we asked them how they classified their relationship among the communities, and we obtained the following results: 83.4% of the interviewees reported having a good relationship with the other communities, and only 16.6% reported that the relationship among the communities is regular.

It is common in these contexts for neighbors or relatives to form strong bonds of mutual respect, mutual aid, reciprocal provision of services, and solidarity. However, this solid base of local social relations has not been sufficient to induce robust collective action capable of expanding the supply of collective public goods. In addition, knowledge related to the place where they live, including fauna, flora, important places to the community, climate, and dynamics of change, have been passed on from generation to generation, orally.

### 4.1.3 High Conservation Values (CM1.2)

The main objectives of the project are in line with the Resource Network's Guide to Identifying High Conservation Values (HCVs) described in the boxes below:

| | |
|---|---|
| High Conservation Value | HCV 4: Water bodies, Sucunduri River and its tributaries, water quality, fish resources, climate regulation. |
| Qualifying Attribute | *Climate regulation*: The Sucunduri River is an important ecosystem and provides ecosystem services related to the project activities and has an impact on the local climate.<br><br>Combining not only climate regulation but also other services (such as water, wildlife and nutrient cycling), the REDD+ project will be an important project for the conservation of this river, all aquatic life, and ecosystem services. In the region with the absence of the project activities, the well-being of the communities and other parts of the reference region would worsen.<br><br>*Water quality:* conservation to increase water quality and quantity is crucial for the well-being of the people who live there, and for the species that are directly and indirectly related to the river and waterways. Otherwise, this will reflect in a decrease in quality of life and biodiversity.<br><br>*Fishery Resources:* The conservation of the services that these ecosystems provide to the community and the entire Sucunduri River basin region are extremely important for the health and quantity of the fishery resources; for the aquatic and terrestrial fauna that feed on these fishery resources; for nutrient cycling; and for community nutrition. |
| Focal Area | Project Zone. |

| | |
|---|---|
| High Conservation Value | HCV 5: Water bodies, Sucunduri River and its tributaries, water quality, fish resources and mobility of communities. |
| Qualifying Attribute | According to the HCV Resource Network's Guide to Identifying High Conservation Values (HCV Resource Network's Guide to Identifying High Conservation Values), an ecosystem service must be presented in a critical situation, which can be the product of one of the following factors:<br><br>1. Their disruption would have severe negative impacts on the well-being, health, or survival of the communities that depend on them.<br><br>2. There are no viable, available, or affordable alternatives that could provide support in the absence of the resource.<br><br>All community members depend on the rivers for basic needs such as subsistence fishing, water use, and for locomotion and transportation.<br><br>Therefore, maintaining the quality of the rivers through conservation of the project area is essential for the rivers that are a source of food and water for the communities; transport for wild-harvesting products, for sale in the larger cities of the region. A deterioration of water quality and quantity would therefore result in an unprecedented negative impact on the well-being of the communities, and no alternative is available in the absence of this HCV.<br><br>The main welfare impacts from the project area conservation in this ecosystem are noted below:<br><br>Water quality: all communities are supplied with water and<br><br>for other uses from the rivers. There is no treated water available to the communities living in the project area.<br><br>Conservation to increase water quality and quantity is crucial for the well-being of the people living there. Otherwise, this will reflect in a decrease in quality of life and no alternative resource is available.<br><br>*Fishery resources*: the health and quantity of fishery resources depend not only on the conservation of forests, but also on the protection of riparian vegetation along the riverbanks. Respecting restrictions on fishing at certain times of the year is important to maintain the fish resources at a healthy level and ensure a level over the years. This resource, so important to the communities, would hardly have a substitute. In this sense, the REDD+ project is intended to mitigate the reduction of fish resources and prevent the deterioration of welfare.<br><br>*Community Mobility:* All the communities mentioned in this document use the Sucunduri River as their main transportation route. The river is the only way for them to access health care, education, and even to reach larger urban centers to sell their |

| | |
|---|---|
| | products. Besides this, it is in the cities that they can access the internet and telephone signal.<br><br>The Sucunduri River also provides access between the communities, for example the community of Nova Esperança needs to travel by boat to Vila Batista to attend their religious services, as described in section *4.1.1 Social Dimensions*. |
| Focal Area | Project Zone |

| | |
|---|---|
| High Conservation Value | HCV 6: Areas of cultivation and extraction of Brazil nuts and Copaíba oil. |
| Qualifying Attribute | Areas of cultivation and extraction: Within the project area, we have areas of ecological and economic importance, mainly for the community of Vila Batista, since it is in these areas that they extract Brazil nuts and Copaiba oil for sale and for their own use. Besides this, they also plant crops in the PA, which also qualifies as an area of high conservation value, since they plant for subsistence, and the cities and markets are far away. |
| Focal Area | Project Zone. |

### 4.1.4 Without-Project Scenario: Community (CM1.3)

In the without-Project Scenario, the community socio-economic indicators would remain low, due to current income sources, lack of improved production techniques, difficulties in production flow, as well as access to the consumer market, and poor welfare conditions due to limited productive activities. In this scenario, there would be no improvement in the communities' welfare conditions.

With the project we propose socioeconomic indicators to monitor the communities and thus evaluate and quantify these indicators. In the table above, we present the socioeconomic indicators analysed.

*Table 81. Monitored Socioeconomic Parameters.*

| **Activities depending on carbon credit sales** | | | |
|---|---|---|---|
| **Scope** | **Frequency** | **Geographic Area** | **Indicator** |
| **Income** | Annual | Project Zone | Per capita income per family |
| | Annual | Project Zone | Number of families that receive benefits and/or other assistance from the Government |
| | Annual | Project Zone | Number of people working with wild-collection |
| | Annual | Project Zone | Number of people working in non-extraction related jobs. |
| | Annual | Project Zone | Quantity of products from the wild-collection commercialized. |
| **Education** | Annual | Project Zone | Number of enrollments in schools. |

| | | | |
|---|---|---|---|
| | Annual | Project Zone | Student attendance |
| | Annual | Project Zone | Illiteracy rate |
| **Health** | Annual | Project Zone | Life Expectancy: Age of each member of the communities and accompaniment of the elderly. |
| | Annual | Project Zone | Natality Rate |
| **Public Service Offering** | Annual | Project Zone | Number of services offered by the public power. |

### 4.2 Net Positive Community Impacts

#### 4.2.1 Expected Community Impacts (CM2.1)

In the table below we describe the expected impacts that will result from the project activities, considering the scenario with the project:

| | |
|---|---|
| Community Group | *Overall community* |
| Impact(s) | *Provide tools for the resolution of socio-environmental conflicts promoting forest protection.* |
| Type of Benefit/Cost/Risk | *One of the strategic actions of the project is Communication*<br>*Project Communication and its related activities include the creation of communication and information dissemination channels between the project proponents and the communities in and around the project area, to raise awareness in the communities about the care of natural resources.*<br>*Another important benefit is Socio-educational. Here, community trainings on topics including REDD+ concepts, ecosystem management, health, among others. This will bring a great positive impact to the communities. The welfare benefit is indirect; however, it can lead to an increase in the quality and quantity of ecosystem services.*<br>*The impacts and activities described in this framework are ongoing but in their early stages. Therefore, more activities are being planned along these strategic lines for the future.* |
| Change in Well-being | *The activities will impact and improve well-being as:*<br>• *Ensuring the maintenance and enhancement of ecosystem services.*<br>• *Capacity building and training, providing community residents with more knowledge, tools, and methods for ecosystem conservation.*<br>• *The overall satisfaction of the community and improved quality of life.* |

| | |
|---|---|
| Community Group | *Overall community* |
| Impact(s) | *Workshops and dialogues with women in the communities, and the creation of a communication and denunciation channel.* |
| Type of Benefit/Cost/Risk | *With the women who live in the communities, we hold workshops and dialogues to listen to their main complaints and create a vehicle of trust with them. Not only do the conversations* |

| | |
|---|---|
| | *make it possible to plan actions that will be effective for them and for the whole community, but we also create a relationship of trust.*<br>*With the creation of a communication channel, already mentioned above, these women can report any type of aggression or harassment, ensuring the dignity and safety of these women. This is a real real benefit; however, it is still in its early stages, but with great potential to be great potential to be expanded.* |
| Change in Well-being | *The activities will impact and increase the well-being of these women, such as:*<br>• *The possibility of new specific projects for these women, training, and possibility of generating new forms of income for them.*<br>• *The creation of a communication and reporting channel to guarantee the physical, emotional, and psychological integrity of these women.*<br>• *The increase in the well-being and safety of the women in the communities.* |

### 4.2.2 Negative Community Impact Mitigation (CM2.2)

The project activities are only intended to increase positive impacts and reduce any possible negative impacts. Furthermore, in the public consultation processes, field visits, and in interviews, the community groups consulted had very positive feedback. They did not identify the project activities as a potential risk to them; rather, they perceived that the project activities would only help develop the region while protecting the local ecosystems.

The measures that will be designed to mitigate any negative impacts on the well-being of community groups and the maintenance or enhancement of HCV attributes related to community well-being will be planned and described in accordance with the precautionary principle. As we are in the process of field visits and data collection, this section will have further increments.

Constant communication between the project team and feedback from community members to the project team will ensure that any complaints about negative impact on communities will be identified and mitigated. And will ensure that the project and activities will bring positive impacts to the community.

The project also has a management process in place to identify any risks or negative impacts to the community and mitigate them.

### 4.2.3 Net Positive Community Well-Being (CM2.3, GL1.4)

The project activities are only designed to bring positive impacts, such as increased awareness communities to the conservation and care of natural resources, training, and environmental education with the same goal: to improve the quality and quantity of ecosystem services that are the basis of community well-being.

### 4.2.4 High Conservation Values Protected (CM2.4)

The HCVs related to community well-being will not be negatively affected by project activities. On the contrary, the project activities aim at protecting the ecosystem and important sites to the communities living in the project area. In the event that any potential negative impact is identified during the life of the project through the complaint mechanism, a mitigation plan will be created to reduce and ideally stop that impact.

## 4.3 Other Stakeholder Impacts

### 4.3.1 Impacts on Other Stakeholders (CM3.1)

The Project is designed to generate only positive impacts on the well-being of other stakeholders, and it won't generate negative impacts during the lifetime. No other stakeholders have been identified to use or depend on the resources in the Project Area or Reference Region. The positive impacts that are projected to occur are:

- Increase of awareness on environmental conservation and REDD+ projects.
- Partnership with universities in research projects.
- Strengthen ties between the communities and other stakeholders.

### 4.3.2 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

As mentioned in the previous item, there are not expected negative offsite impacts, thus, no mitigation strategies are required.

#### 4.3.3 Net Impacts on Other Stakeholders (CM3.3)

The project activities are not expected to cause any negative impacts on other stakeholders, only positive impacts. Thus, there is no need for mitigation.

## 4.4 Community Impact Monitoring

### 4.4.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The monitoring plan below identifies, in accordance with PD template guidelines, the communities, community groups and other stakeholders to be monitored, variables to be monitored, types of measurements and sampling methods, and the frequency of monitoring and reporting for each type and method.

*Table 82. Social parameters to be monitored.*

| Scope | Activity | When | Frequency | Target Community Groups/Unit | Goal |
|---|---|---|---|---|---|
| **Health** | Distribution of oral hygiene kits | Short Term | Occasional | - | Sufficient number for the entire population of the three communities. |
| | Distribution of soaps and ecological sanitary pads for women | Short term | Occasional | - | Sufficient number for the entire population of the three communities. |
| | Medical care by the NGO "Doutor das águas" | Long term | Occasional | N° of users, by gender and age | Attendance two times a year. |
| | Health transport programme | Long term | Annual | N° of users and coverage area by gender and age. | In development |

| Scope | Activity | When | Frequency | Target Community Groups/Unit | Goal |
|---|---|---|---|---|---|
| | Training: Universal access to sexual and reproductive health and reproductive rights | Medium Term | Biannual | Women between the ages of 15 and 49 who use contraceptives and take care of their reproductive health | Providing all women in the Project Zone with access to sexual and reproductive health and ensuring reproductive rights. |
| **Occupational safety** | Training and workshop: Occupational health and safety | Medium Term | Biannual | Fatal and non-fatal occupational injury frequency rates by sex and migration status | Train and equip all extractivists in the Project Zone |
| | Occupational safety PPE kits | Medium Term | Annual | | |
| **Communication and Access** | Installation, maintenance and/or facilitation of internet access | Short term | Permanent | Proportion of individuals in the project area by gender who use the internet provided | In development |
| | | | | Proportion of individuals in the project zone who have a telephone by gender. | In development |

| Scope | Activity | When | Frequency | Target Community Groups/Unit | Goal |
|---|---|---|---|---|---|
| **Educational programme** | Refurbishment of school establishment and provision of, and continued support for, certified teacher. | Medium Term | Annual | N° of students who passed from each grade level by gender | In development |
| | | | Biannual | N° of facility offerings at school and maintenance at school annually | In development |
| | Environmental Education | Medium Term | Biannual | - | In development |
| **Community training** | Dissemination of forest conservation practices | Medium Term | Annual | In development | In development |
| | Sustainable Extractivism | Medium Term | Annual | - | In development |
| | Survey on Hight Conservation Values areas | Medium Term | Every 5 years | - | Proportion of protection of survey on Hight Conservation Values |

The following forms will be applied after any of the trainings or at the meetings for data collection:

4. **Attendance List:** Contains the participant's name, type of affiliation with the project, and signature. and signature.
5. **Reaction Evaluation:** participants will be able to register their perceptions regarding the content presented, making it possible to evaluate the content and make suggestions for future training. This evaluation can be done anonymously and is not mandatory.

All communication with workers and families will be registered in meeting reports, which may be signed by all parties involved by all parties involved. The reports should be produced according to the monitoring frequency adopted. reports must clearly inform the results of the variables involved in the monitoring, by means of photos, interviews, and documents.

All records generated during project implementation will be retained and made available to the VCS verifiers for verification, demonstrating that the IPOÁ project activity was actually implemented. All documents and records will be kept securely for at least two years after the end of the project.

The Project Proponents are responsible for the financing, implementation and monitoring of the social aspects within the project credit period.

### 4.4.2 Monitoring Plan Dissemination (CM4.3)

The monitoring plan will be made available online within this Monitoring Report document for the voluntary carbon market audience on the IPOÁ REDD+ Project website. In addition, a copy of the Monitoring Report will be emailed to all identified stakeholders.

For the residents of the communities, the method of dissemination and communication is extremely important so that all residents of the Project Area have clarity about the stages and progress of the project. Therefore, the monitoring plan will be made available online within the present MR document, to the public of the voluntary carbon market, on the VERRA page of the IPOÁ REDD+ Project. To the demographics concerned in situ, the relevant method of disseminating the monitoring plan will be as below. communicated to the communities in Portuguese, in accessible language, with visual prompts, by a Carbonext field agent during a field visit planned before the beginning of MR1 and the start of MR2.

For the residents of the communities, the method of dissemination and communication is extremely important so that all residents of the Project Area have clarity about the stages and progress of the project. Therefore, the monitoring plan will be communicated to the communities in Portuguese, in accessible language, with visual prompts, by a Carbonext field agent during a field visit planned between MR1 and MR2.

## 4.5 Optional Criterion: Exceptional Community Benefits

### 4.5.1 Exceptional Community Criteria (GL2.1)

Not applicable.

### 4.5.2 Short-term and Long-term Community Benefits (GL2.2)

Not applicable.

### 4.5.3 Community Participation Risks (GL2.3)

Not applicable.

### 4.5.4 Marginalized and/or Vulnerable Community Groups (GL2.4)

Not applicable.

### 4.5.5 Net Impacts on Women (GL2.5)

Not applicable.

### 4.5.6 Benefit Sharing Mechanisms (GL2.6)

Not applicable.

### 4.5.7 Benefits, Costs, and Risks Communication (GL2.7)

Not applicable.

### 4.5.8 Governance and Implementation Structures (GL2.8)

Not applicable.

### 4.5.9 Smallholders/Community Members Capacity Development (GL2.9)

Not applicable.

# 5 BIODIVERSITY

## 5.1 Without-Project Biodiversity Scenario

### 5.1.1 Existing Conditions (B1.1)

Efforts were made to characterize the existing biodiversity conditions of the project area and its surroundings unit's management plans near of the project as a first analysis. After the first verification of the project, Carbonext will perform the Fauna and Flora Inventory - foreseen in item *5.4.1. Biodiversity Monitoring Plan* - within the IPOÁ Project Area. The main sources for this survey was Management Plan for The Juma Sustainable Development Reserve[154], which is located 13km from the Project Area, which is on the eastern (right) margin of the RDS. To compose the list of threatened species, we also used data from the Fauna Survey and Inventory Report for the Licensing Activity for Vegetable Supression Purposes carried out (*in loco*) for Maanaim and Campos Farms in March 2022. The methodology used to complete the document for the faunal and floristic biodiversity survey, from the JUMA RDS data, is described below.

***Fauna biodiversity***

To conduct the fauna survey, the Juma Reserve Management Plan considered secondary data from studies of the main terrestrial vertebrate groups: non-flying mammals, birds, reptiles, and amphibians. The results showed that species diversity in the RDS is exceptionally high, even by Amazonian standards. A total of 70 species of nonflying mammals (55 genera, 28 families, and 10 orders) were recorded. As for avifauna, 398 were recorded within the UC boundaries and 214 were recorded in the Reserve's surroundings, a total value (612) that represents half of the avifauna of the Brazilian Amazon, one third of the avifauna of Brazil as a whole, and among the richest avifauna of any protected area in the world. The herpetofauna survey resulted in 43 species of reptiles and 27 of amphibians.

[154] https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf

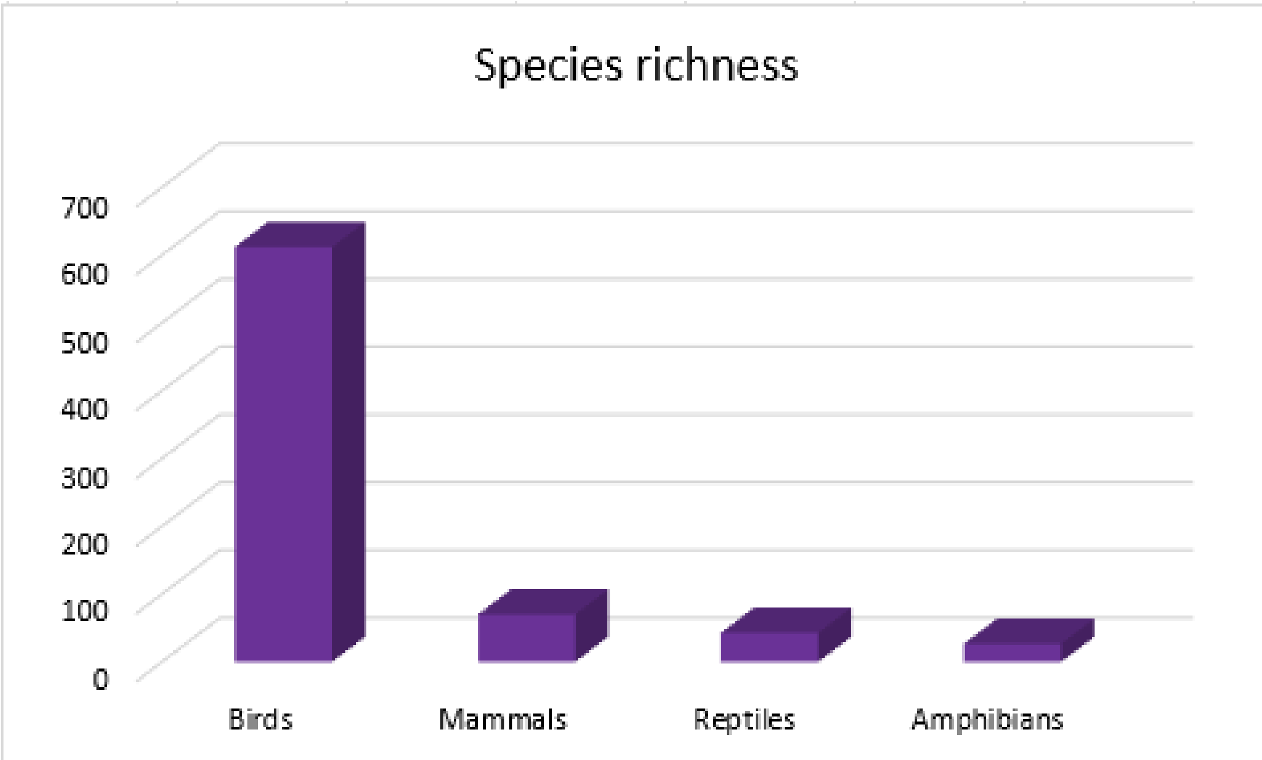


*Figure 46. Species richness of fauna likely to occur in the project area in in initial survey.*

According to Juma Reserve Management Plan, beyond that new species for science from all groups of fauna are being described from this region at a rate above normal, suggesting that the region has not yet been adequately studied and that there is still more diversity to be discovered. This highlights the importance of Carbonext carrying out the biodiversity inventory after the first verification, as well as the implementation of the IPOÁ REDD+ Project, in order to enable more specific studies for the region and preserve the habitat for this faunal diversity.

The JUMA RDS contains 11 species classified at some level of threat of extinction, of which 8 are mammal species and three chelonian species, as indicated by the following table. There are no bird species listed as endangered recorded in the RDS. Of the mammals, four are large species with a wide distribution in Brazil: *Tapirus terrestres* (tapir), *Trichechus inunguis* (manatee), *Priodontes maximus* (giant armadillo), *Pteronura brasiliensis* (giant river otter) which according to the JUMA RDS Management Plan are likely to have healthy populations at natural levels within the RDS. The largest monkey species (*Chiropotes* *albinasus*, *Ateles* sp., *Lagothrix* sp.) are considered "endangered" and the tiny *Callibella humilis* is "vulnerable." and the others are primates with restricted distributions. Furthermore, the fauna survey for the Maanaim and Campos Farms recorded, in addition to the *Tapirus terrestres* species other threatened species namely *Alouatta guariba clamitans*. The total number of species in some degree of threat of occurrence registered and possible occurrence for the area is described below:

*Table 83 - List of threatened and/or endemic birds, amphibians and reptiles in the region of the Juma RDS and Fazenda Maanaim and Campos Fauna Report.*

| Taxon | Popular name | Threat (IUCN) | Endemism | Source |
|---|---|---|---|---|
| **Podocnemidae** | | | | |
| *Podocnemis unifilis* | Yellow-spotted river turtle | VU | - | Vogt et al 2007 |
| *Podocnemis sextuberculata* | Six-tubercled turtle | VU | - | Vogt et al 2007 |
| **Testudinidae** | | | | |
| *Chelonoidis denticulata* | Yellow-footed tortoise | VU | - | Vogt et al 2007 |
| **Tapiridae** | | | | |
| *Tapirus terrestres* | South American tapir | VU | - | Röhe 2007; 4 |
| **Trichechidae** | | | | |
| *Trichechus inunguis* | Amazonian manatee | VU | - | Röhe 2007 |
| **Dasypodidae** | | | | |
| *Priodontes maximus* | Giant armadillo | VU | - | Röhe 2007 |
| **Mustelidae** | | | | |
| *Pteronura brasiliensis* | Giant otter | EN | - | Röhe 2007 |
| **Pitheciidae** | | | | |
| *Chiropotes albinasus* | White-nosed saki | EN | END | Röhe 2007 |
| **Atelidae** | | | | |
| *Ateles sp.,* | Monkey | EN* | - | Röhe 2007 |
| *Lagothrix sp.* | Monkey | EN | - | Röhe 2007 |
| *Alouatta guariba clamitans* | Southern brown howler | VU | - | 4 |
| **Callitrichidae** | | | | |
| *Callibella humilis* | Monkey | VU | EN | Röhe 2007 |
| **Psittacidae** | | | | |
| *Pyrrhura perlata* | - | VU | MD M | 1,2,3 |
| **Thamnophilidae** | | | | |
| *Herpsilochmus sp* | - | - | M/AR | 1,3 |
| *Rhegmatorhina hoffmannsi* | White-breasted antbird | LC | M/T | 1,2,3 |
| *Skutchia borbae* | Pale-faced bare-eye | LC | M/T | 1,2,3 |
| **Conopophagidae** | | | | |
| *Conopophaga melanogaster* | Black-bellied gnateater | LC | MD M | 1,2,3 |
| **Dendrocolaptidae** | | | | |
| *Dendrocolaptes hoffmannsi* | Hoffmanns's woodcreeper | VU | M/T | 1,2,3 |
| **Tyrannidae** | | | | |

| Taxon | Popular name | Threat (IUCN) | Endemism | Source |
|---|---|---|---|---|
| *Poecilotriccus senex* | Buff-cheeked tody-flycatcher | LC | M/T | 1,2,3 |
| **Troglodytidae** | | | | |
| *Odontorchilus cinereus* | Tooth-billed wren | NT | MD M | 1,3 |
| **Hylidae** | | | | |
| *Hypsiboas sp.* | - | NE | IND | Vogt et al 2007 |
| *Osteochephalus sp* | - | NE | IND | Vogt et al 2007 |
| **Leptodactylidae** | | | | |
| *Leptodactylus sp. 1* | - | NE | IND | Vogt et al 2007 |
| *Leptodactylus sp. 2* | - | NE | IND | Vogt et al 2007 |
| **Microhylidae** | | | | |
| *Chiasmocleis sp* | - | NE | IND | Vogt et al 2007 |
| **Strabomantidae** | | | | |
| *Pristimantis sp.* | - | NE | IND | Vogt et al 2007 |
| **Dipsadidae** | | | | |
| *Atractus sp* | - | NE | IND | Vogt et al 2007 |
| *Oxyrhopus sp* | - | NE | IND | Vogt et al 2007 |

**Threat level IUCN:** VU – Vulnerable; EN – Endangered; NE – Not Evaluated; LC – Least Concern
**Endemism:** MD M - endemic to the right margin of the Madeira river; M/T - endemic to the Madeira and Tapajós rivers interfluve; M/AR - endemic to the Maderia and Aripuanã river interfluves; IND - Endemism analysis to be identified in future research (possible new and endemic species).
**Sources:** 1 – Cohn-Haft et al. 2007 2 – Banco de dados não publicados de M. Cohn-Haft 3 – Whittaker 2009. 4 – Fazenda Maanaim and Campos Fauna Report

The Juma RDS can be considered one of the most important state protected areas in the state of Amazonas from the point of view of terrestrial fauna[155] due to the presence of species such as *Callibella humilis*, where approximately half of the species' global distribution is contained within the RDS. An important case is Mico acariensis, whose entire distribution fits in the area surrounding the Reserve. Besides these, some other examples include Mico manicorensis (Manicoré - Aripuanã), Mico chrysoleucus (Aripuanã - Canumã/Acarí), and a new species of bird (not yet described) Herpsilochmus sp. (Machado/Jiparaná - Aripuanã).

Most of the endemic species of the Aripuanã have no protection in other protected areas within their range[156]. Considering the distance between the RDS and PA and the possible presence of these taxa within the project area, we emphasize the importance of the existence of protected areas for the maintenance of habitats of this biodiversity.

[155] https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf
[156] https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf


*Figure 47. Tapirus terrestres (IUCN, 2022)*


*Figure 48. Priodontes maximus (IUCN, 2022)*

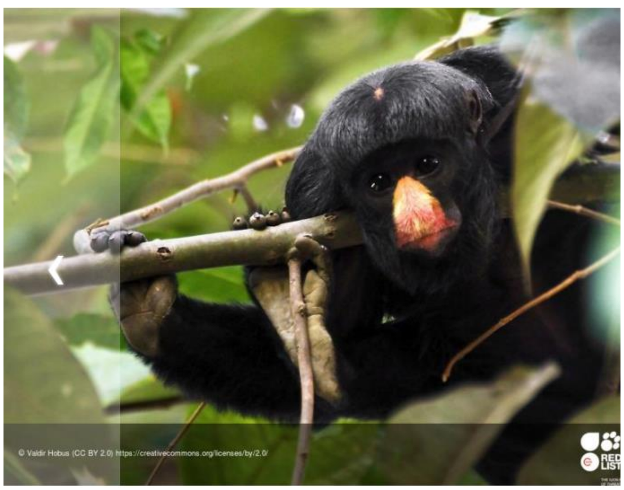


*Figure **49**. Chiropotes albinasus (IUCN, 2022)*

*Figure 50. Conopophaga melanogaster (Cláudio Dias Timm; Avibase)*[157]

***Floristic biodiversity***

The flora inventory in the JUMA RDS was conducted in October 2019 in the region of the AM 174 highway (Higuchi et al., 2010). The study found 358 tree species, with the taxon *Protium* sp. (red rosewood) -

[157] https://avibase.bsc-eoc.org/species.jsp?avibaseid=C0F59C9F8A109FC9

Burseraceae being the most frequent followed by *Eschweilera* sp. (yellow matamata) - Lecythidaceae, a pattern commonly found throughout the Amazon region (Higuchi et al., 2010). The family with the highest number of species was Mimosoideae (29), followed by Sapotaceae (21), Lauraceae and Papillionoidaeae (17), as observed by the following histogram:

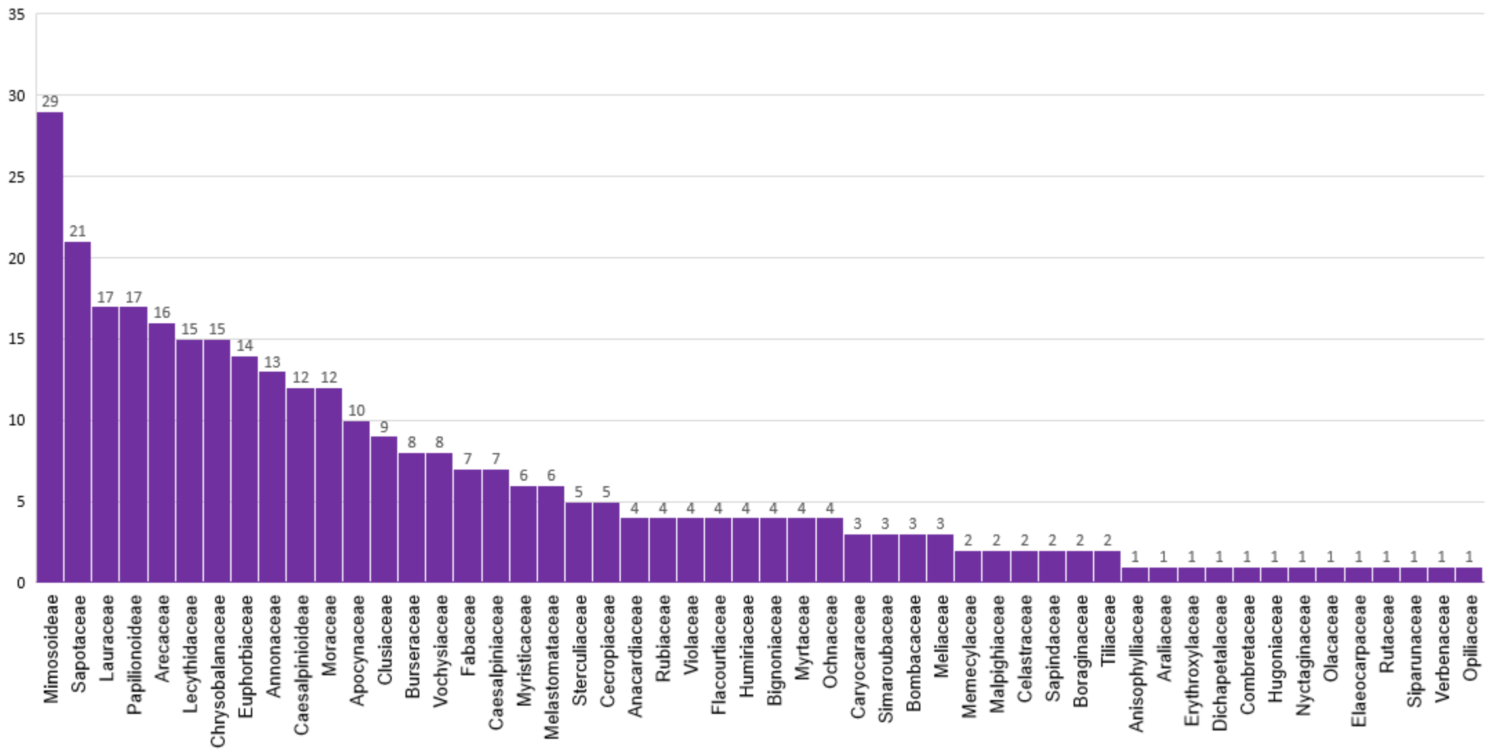


*Figure* **51***. Distribution of number of species by inventoried botanical families of probable occurrence for the project area.*

As for the occurrence of species, the presence of *Euterpe precatoria* (açai) and *Bertholletia excelsa* (Brazil nut tree) stands out in this study, both recorded in the literature as having significant participation as a source of employment and income generation for thousands of rural and urban workers in the state of Amazonas.[158]. The species *Bertholletia excelsa* (Brazil nut tree), moreover, is classified as "Vulnerable" by both the IUCN and CNCFlora, as indicated by table above. As described in item 4. Community, the Brazilian

[158] https://www.scielo.br/j/cerne/a/bcpmyb9fdlgfsgslqvnhnfs/?format=pdf&lang=pt

nut tree are possibly the same species reported by members of the communities, which use it for food and economic purposes. Thus, the importance of protecting these species and their natural habitats through the creation of preservation projects is highlighted.

From the list of inventoried species, available in the JUMA RDS Management Plan, we classified the taxa according to the level of extinction, based on the IUCN RED LIST (IUCN, 2018)[159] and the National Center for the Conservation of Flora (CNCFlora)[160]. According to the flora inventoried for the JUMA RDS, below are the species classified at some level of threat of extinction.

*Table 84. List of threatened and/or endemic flora in the region of the Juma RDS.*

| Taxon | Popular name | Threat (IUCN) | CNCFlora |
|---|---|---|---|
| **Bignoniaceae** | | | |
| *Tabebuia incana* | Yellow Arched Wood | VU | - |
| **Bombacaceae** | | | |
| Catostema albuquerquei | Mamãorana | EN | - |
| **Caesalpiniaceae** | | | |
| *Aldina heterophylla* | Macucu de paca | VU | VU |
| *Vouacapoua americana* | Wacapou | CR | EN |
| **Chrysobalanaceae** | | | |
| *Couepia longipendula* | Egg nut | EN | |
| *Licania adolphoduckei* | Macucu fluffy | EN | |
| **Lauraceae** | | | |
| *Mezilaurus itauba* | Itaúba | VU | VU |
| *Aniba ferrea* | Louro chumbo | EN | EN |
| *Aniba cf. permollis* | Louro rosa | EN | |
| **Lecythidaceae** | | | |
| *Bertholletia excelsa* | Brazilian nuts | VU | VU |
| *Lecythis prancei* | Jarana nuts | EN | |
| **Meliaceae** | | | |
| *Guarea convergens* | White gitó | VU | |
| *Trichilia areolata* | Red gitó | VU | |
| *Sorocea guilleminiana* | Wild jackfruit | VU | |
| **Papilionoideae** | | | |
| *Aldina heterophylla* | Macucu | VU | VU |
| **Rutaceae** | | | |
| *Hortia superba* | Pooper | VU | VU |
| **Sapotaceae** | | | |
| *Pouteria petiolata* | Abiurana cutiti | VU | |

[159] https://www.iucnredlist.org/
[160] http://cncflora.jbrj.gov.br/portal/

| Taxon | Popular name | Threat (IUCN) | CNCFlora |
|---|---|---|---|
| *Pouteria freitasii* | Abiurana sabiá | VU | VU |

**Threat level IUCN and CNCFlora :** VU – Vulnerable; EN – Endangered; CR - Critically Endangered

*Figure 52. Bertholletia excelsa*[161]

*Figure 53. Euterpe precatoria* [162]

***Importance of IPOÁ REDD+ for biodiversity***

It is noteworthy that the creation of the IPOÁ REDD+ Project directly impacts the preservation of fragments located 13km (Fazenda Campos) and 15km (Fazenda Maanaim) from the Juma RDS, thus it is possible to consider that several species recorded in the RDS have potential occurrence in the Project Area. Moreover, it is emphasized that other possible new mammal species are being studied by the researcher in the region, including jaguars, deer, tapirs, manatees, anteaters and primates[163]. The biodiversity inventory study to be conducted by Carbonext after the first monitoring period has the potential to contribute to science with fauna and flora studies of the Aripuanã region. If the occurrence of new endemic and threatened species is recorded, new conservation decision strategies can be planned for the region.

In addition, the Juma Reserve Management Plan points out the low occurrence of protected areas in the surrounding region of the RDS. Thus, for some species, if they depended entirely on the RDS to guarantee their persistence, they would qualify as threatened and possibly would not maintain sustainable populations.

[161] https://hablemosdeflores.com/bertholletia-excelsa/
[162] https://puppy-party.info/euterpe-precatoria-54/
[163] https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf

This also calls attention to the importance of pressures outside the RDS as well as the creation of protected areas. Noting that the Project Area of the IPOÁ REDD+ Project is located on the right bank (east) of the JUMA RDS, the conservation of these areas should contribute to the conservation of species mainly in the Juma RDS located on the right bank of the Aripuanã River, which apparently do not present physical barriers to the use of the conserved habitats of the Maanaim and Campos farms. The creation of the IPOÁ REDD+ Project, especially the Campos farm, may contribute, in addition, as a fauna refuge for species located in the Acari PARNA.

Finally, it is noteworthy that with the presence or absence of protected areas in the surroundings, the creation of the IPOÁ REDD+ project protects forest mosaics with important Amazon flora species, besides serving as habitat, ecological springboards, and refuges for different fauna species such as mammals, reptiles, and local avifauna.

### 5.1.2 High Conservation Values (B1.2)

For the Survey on High Conservation Areas, High Conservation Values (HCVs) were defined according to the "General Guide to Identifying High Conservation Values" relating to the region's biodiversity and communities. The definitions of HCVs consider a classified biological, ecological, social or cultural value of exceptional or critical importance, which can be classified into different categories. In the project zone, the following beneficial changes to biodiversity are expected:

| | |
|---|---|
| High Conservation Value | HCV 1: Species diversity - possible occurrence of threatened and endemic species in the Project zone |
| Qualifying Attribute | The Amazon Forest ecosystems comprise the largest and richest forest in the world in terms of biodiversity[164]. In the Amazon biome, about 183 species are threatened with extinction and among these 122 species are endemic[165]. According to studies conducted in the vicinity of the project area, endemic and endangered species are likely to occur in these regions, as noted in the tables under *5.1.1 Existing Conditions*. |

164 https://www.scielo.br/j/ea/a/yf6MQVWvKWGD33jN3jqw47n/?format=pdf&lang=pt
165 https://www.researchgate.net/profile/Maria-Fabiola-Barros/publication/350531320_Reflexoes_em_Biologia_da_Conservacao_-_Volume_2/links/60650c1292851c91b194596e/Reflexoes-em-Biologia-da-Conservacao-Volume-2.pdf#page=255

| | |
|---|---|
| Focal Area | It is necessary to conserve and protect in totality the current vegetation in the project area, maintaining biodiversity monitoring and carrying out educational activities for create awareness on the importance of biodiversity and conservation. |

| | |
|---|---|
| High Conservation Value | HCV 2: Ecosystems and mosaics at landscape level, and Intact Forest Landscapes - proximity of PA to protected areas |
| Qualifying Attribute | The PA of the REDD+ IPOÁ Project is near protect areas, as Juma Sustainable Development Reserve (13km) and Acari National Park (25km), Aripuanã National Forest (50km) and Indigenous Land like Coata-Laranjal (30km) and Setemã (48km)..The presence of protected areas in the vicinity, especially for the JUMA RDS, underscore the importance of protecting surrounding habitats. It is likely that these environments share similar species diversity, and perhaps animal populations that use the forest fragments as ecological stepping stones and other ecosystem services. Protecting areas near protected areas intensifies actions against deforestation in the Amazon |
| Focal Area | The IPOÁ REDD+ Project allows for the protection of the PA area, which serves as habitat for several species. Some species may belong to common populations between protected areas near the PA. In this way, conservation of the PA also contributes to the surrounding biodiversity. |

| | |
|---|---|
| High Conservation Value | HCV 5: Community needs - Brazil nut and copaíba extraction in the project zone and fishing in the Sucunduri River |
| Qualifying Attribute | The forest and aquatic ecosystems of the Amazon consist of a diverse source of food, nutritional and economic resources of importance to traditional communities. The main economic activities of the Nova Esperança community, located in the project zone, are the extraction of copaiba oil and Brazil nuts, cassava plantations for flour production, and vegetable production, the first two being directly dependent on forest resources. In addition, people from other communities have come to the Sucunduri River to engage in fishing activities. |

| | |
|---|---|
| Focal Area | The protection of the project area as well as the development of educational activities together with the surrounding communities impacts the quality of life and the basic needs of local communities in terms of livelihood, health, nutrition, water, among others. |

### 5.1.3 Without-project Scenario: Biodiversity (B1.3)

In the project reference region and near the project area, the main causes for the deterioration of the environment are the modification of land use for pasture as shown in the section 2*.1.6 Social Parameters*. Despite the fact that mining is the main economic activity in Novo Aripuanã, there are no apparent mining centers in the region.

The replacement of natural forest systems by pasture ecosystems and other anthropogenic uses is directly related to significant biodiversity loss in the Amazon. It is known that forests are considered the main ecosystems for terrestrial animals and the Amazon forest has the largest biodiversity in the world[166]. The removal of vegetation cover, degradation, and deforestation is related to direct and indirect losses of thousands of species across the planet. In addition to resulting in the direct death of arboreal individuals and other taxa of the plant kingdom, the loss of forest habitat forces the migration of populations of several animal species, when not in direct death, putting species at risk of extinction.

The IPOÀ REDD Project prevents 4,563.75 hectares from being deforested. In a scenario without the project, it is very likely the forest clearing for pasture conversion would reach the project area, negatively impacting biodiversity loss in the region. In addition, without the social educational actions foreseen in the item *4.4.2 Monitoring plan Dissemination* it is not possible to expect an environmental awareness in the communities of the region, which can be considered one of the main agents of change in environmental decision making in the region where they live.

[166] https://www.scielo.br/j/ea/a/yf6MQVWvKWGD33jN3jqw47n/?format=pdf&lang=pt

## 5.2 Net Positive Biodiversity Impacts

### 5.2.1 Expected Biodiversity Changes (B2.1)

| | |
|---|---|
| Biodiversity Element | Flora |
| Estimated Change | Positive - Biodiversity Conservation |
| Justification of Change | The REDD+ IPOÁ Project consists of preventing deforestation and degradation of Amazonian Forest ecosystems. Only in 2022, 190.39 km² of deforestation were registered in the state of Amazonas [167]. The presence of the project prevents vegetation removal and deforestation by pasture and other activities of degradation and deforestation, avoiding the threat to plant species and ensuring suitable habitat conditions for local biodiversity. |

| | |
|---|---|
| Biodiversity Element | Fauna |
| Estimated Change | Positive - Biodiversity Conservation |
| Justification of Change | The activities of the Project aim at the reduction of deforestation and forest degradation, based on the practices of mentioned in the items 5.*4 Biodversity Monitoring Plan*. The felling of the forest produces the extinction of species, with unpredictable systemic consequences for the environment. The maintenance of the vegetation cover will allows adequate living conditions for the countless Amazonian species of birds, mammals, amphibians, and reptiles in different stages of threat and endemism. |

[167] https://www.cnnbrasil.com.br/nacional/amazonia-legal-tem-recorde-de-alertas-de-desmatamento-no-1o-trimestre-de-2022/

### 5.2.2 Mitigation Measures (B2.3)

This project will have no negative impacts on biodiversity. The areas will be dedicated exclusively to the preservation of forest vegetation cover.

In the region, the pressures on biodiversity are related to external factors and the presence of REDD+ IPOÁ aims to mitigate these impacts. Based on the study of the local socioeconomic scenario and the reality of the communities observed in the field so far, we identified two main current external factors of risk to biodiversity, which are the a) Deforestation pressure for pastureland/land use change; b) Hunting habits in the communities

To mitigate negative impacts to biodiversity related to deforestation and forest degradation, the IPOÁ REDD+ Project will develop surveillance and monitoring activities as described in item *3.3 Monitoring*. In case deforestation alerts are observed via satellite or in field monitoring, mediated measures will be taken such as the registration of an occurrence bulletin and specific plans for landscape recovery, if necessary.

In general, the environmental education work developed with the community, as described in *5.4.1 Biodiversity Monitoring Plan*, promotes awareness about the protection of local biodiversity, as well as sustainable practices that avoid damage to the environment, such as raising awareness about the impacts caused by hunting and the importance of biodiversity

The following table highlights the activities to be implemented to conserve the identified HCVs and avoid negative impacts on biodiversity.

*Table* 85 - *Main biodiversity threats for the project region, mitigating measures, and related HCVs.*

| HCV Related | External threat to biodiversity | Possible impact agent | Mitigation Measure |
|---|---|---|---|
| HCV 1: Species diversity | Deforestation pressure for pastureland/land use change | Farmers; large landowners; local residents. | Constant surveillance and monitoring;<br><br>Encouraging farmers in the region to start RED+ projects. |
| | Hunting | Community | Development plan for Environmental education and awareness activities. |

In addition, the survey and monitoring of faunal and floristic biodiversity allows us to observe the stability of species and to demonstrate any possible changes in biodiversity. During the course of the project, as new agents are observed, new mitigation actions will be planned and developed.

### 5.2.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

As outlined in Section *5.1 Without-Project Biodiversity Scenario*, the baseline scenario will continue to compromise the integrity of Amazonian ecosystems with the advance of use for pasture, in addition to cattle and soy, as shown under the region, bringing negative damages, and putting biodiversity at risk. In addition, the activities to be developed described in the items *5.2.2 Mitigation Measures* and *5.4 Biodiversity Impact Monitoring* ensure the presence of vegetation cover, biodiversity integrity e and local ecosystem balance. As described in Section *5.2.1 Expected Biodiversity Changes*, the project will have positive gains for biodiversity in the project area, including through:

- *Inventory, followed by biodiversity monitoring every 5 years.*
- *Environmental education*
- *Workshops, activities related to environmental awareness, the importance of preserving species, and sustainable development.*
- *Sustainable extractivism training.*

### 5.2.4 High Conservation Values Protected (B2.4)

The project activities aim to avoid deforestation and deforestation and maintain forest cover No environmental impact activities will be carried out, so no negative effects on the HCV are anticipated.

### 5.2.5 Species Used (B2.5)

This item is not applicable as no restoration activities are planned in the area.

### 5.2.6 Invasive Species (B2.5)

This item is not applicable as no restoration activities are planned in the area.

### 5.2.7 Impacts of Non-native Species (B2.6)

This item is not applicable as no restoration activities are planned in the area.

### 5.2.8 GMO Exclusion (B2.7)

This item is not applicable as no restoration activities are planned in the area.

### 5.2.9 Inputs Justification (B2.8)

This item is not applicable as no restoration activities are planned in the area.

### 5.2.10 Waste Products (B2.9)

Most of the activities proposed by the project will be educational, such as trainings and lectures, which means that the residual materials can consist of paper. For the coffee-breaks, the team is prioritizing the use of reusable items in order to avoid the production of plastic waste. In addition, donations of basic hygiene kits, such as reusable ecological sanitary pads (no waste gration) and oral hygiene kits (contains plastic) are also planned. The waste generated (plastic and paper) are classified as class two B - non-hazardous and inert according to the ABNT NBR 10004 standard.

If not disposed of properly, waste material can harm the local aquatic and terrestrial biodiversity. Especially the populations of the 3 communities have the habit of burning all the garbage produced. Unfortunately, sending it for recycling ends up being an unviable alternative for the region's populations, due to the distance and logistics. Despite preventing this material from going into the forest or river, the burning of the material releases toxic substances that not only contribute to air pollution but also have carcinogenic potential.

Therefore, the project activities are being conducted in the most environmentally friendly way possible in order to avoid the generation of waste. As new activities are selected to compose the project, the identification, classification and management of its waste products will be properly conducted. Offsite Biodiversity Impacts

## 5.3 Offsite Biodiversity Impacts

### 5.3.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Measures (B3.2)

No negative offsite biodiversity impacts resulting from the implementation of the project's activities are expected. In addition, the proposed activities were designed to mitigate any possible leakages and negative impacts derived from the subsistence agriculture production practiced by local communities of the Reference Region. These activities will enhance sustainable practices and promote environmental awareness, thus providing more harmony among man in the field and the forest. Due to that, no mitigation measures were necessary.

### 5.3.2 Net Offsite Biodiversity Benefits (B3.3)

The net offsite biodiversity impact of the Project is positive, since there are no mapped negative offsite biodiversity impacts resulting from the project's activities and there are substantial benefits on fauna and flora conservation in the PA and on creating awareness on local communities.

## 5.4 Biodiversity Impact Monitoring

### 5.4.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

*Table 86. Monitoring Plan Biodiversity.*

| Variables | Area | Aspect | Unity | Monitoring Frequency | Goal |
|---|---|---|---|---|---|
| **Fauna and flora inventory** | Project Area | Endangered or endemic flora species | Nº | Every 5 years | Maintenance of total and number of endangered or endemic flora species or it decrease |
| | | Total flora species | Nº | | |
| | | Endangered or endemic fauna species | Nº | | Maintenance of total and number of endangered or endemic fauna species or it decrease |
| | | Total fauna species | Nº | | |
| **Dissemination of forest conservation practices** | Regional | Attendence | Attendes per year | 5 months | Disseminate, through workshops, educational practices on forest conservation and the importance of species. |
| **Sustainable Extractivism** | Project Zone | Attendence | Attendes per year | Anual | Forming partnerships with institutions to foster the local economy sustainable extractivism |

| Variables | Area | Aspect | Unity | Monitoring Frequency | Goal |
|---|---|---|---|---|---|
| **Environmental Education** | Project Zone | Attendence | Attendes per year | 5 months | Raising awareness about the impacts of hunting on biodiversity and others activities |
| **Survey on Hight Conservation Values areas** | Project Zone | Attributes to be defined by Survey | % of protection of survey on Hight Conservati on Values | Every 5 years | Constantly identify HCV species, especially after the Fauna and Flora inventory and greater interaction with the communitys |

### 5.4.2 Biodiversity Monitoring Plan Dissemination (B4.3)

The results of the monitoring carried out will be made public on the Internet and through the Carbonext website. All documents and information about the monitoring and verification results of this project will be published on the VERRA platforms. In addition, the results obtained from the monitored project will be shared with local stakeholders at the most convenient times, such as training, verification, or visit periods, to increase the relevance and effectiveness of the project.

## 5.5 Optional Criterion: Exceptional Biodiversity Benefits

### 5.5.1 High Biodiversity Conservation Priority Status (GL3.1)

This item does not apply to the present project at this time. This study will be possible after the Fauna and Flora inventory foreseen in item *5.4.1 Biodiversity Monitoring Plan*, which will be carried out every five years, the first one after the first one after receiving the credits.

To date, as described in *5.1.1 Existing Conditions*, it is expected that more threatened and endemic species will be found in the regions of the Campos and Maanaim farms, considering the surveys already conducted for the region and the stage of conservation of the forests belonging to the farms.

### 5.5.2 Trigger Species Population Trends (GL3.2, GL3.3)

This item does not apply to the present project at this time.

# 6 APPENDICES

The following appendices may be used if appropriate. Delete the instruction and heading if not used.

### Appendix 1: Actions of the IPOÁ REDD+ project

Through actions developed in collaboration with the communities involved, the activities and programs that will be developed throughout the project's lifetime were designed based on the main demands found and are presented in the table below.

*Table 87. Actions of the IPOÁ REDD+ project.*

| Scope | Activity | When | Frequency | SDG | Performance Indicator | SDG Indicador | Target Community Groups/Unit | Goal |
|---|---|---|---|---|---|---|---|---|
| Health | Distribution of oral hygiene kits | Short Term | Occasional | Health and well-being 3.8 | N° of hygiene kits distributed | - | - | Sufficient number for the entire population of the three communities. |
| | Distribution of soaps and ecological pads for women | Short Term | Occasional | Gender Equality 5.6.1 | N° of soaps and ecological pads for women distributed | - | - | Sufficient number for the entire population of the three communities. |
| | Medical care by the NGO "Doutor das águas" | Long Term | Occasional | Health and well-being 3.8 | N° of people attended | 3.8.1 | N° of users, by gender and age | Attendance two times a year. |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Health trasport programme | Long Term | Annual | Health and well-being 3.8 | N° of trips made for health care | 3.8.1 | N° of users and coverage area by gender and age. | In development |
| | Training: Universal access to sexual and reproductive health and reproductive rights | Medium Term | Biannual | Gender Equality 5.6 | In development | 5.6.1 | Women between the ages of 15 and 49 who use contraceptives and take care of their reproductive health | Providing all women in the Project Zone with access to sexual and reproductive health and ensuring reproductive rights. |
| **Occupational safety** | Training and workshop: Occupational health and safety | Medium Term | Biannual | Decent work and economic growth 8.8 | N° of training and lectures held | 8.8.1 8.8.2 | Fatal and non-fatal occupational injury frequency rates by sex and migration status | Train and equip all extractivists in the Project Zone |
| | Occupational safety PPE kits | Medium Term | Annual | | N° of kits distributed | | | |
| **Communication and Access** | Installation, maintenance and/or facilitation of internet access | Short Term | Permanent | Partnerships and means of implementation 17.8 | N° of people with internet access | 17.8.1 | Proportion of individuals in the project area by gender who use the internet provided | In development |
| | | | | Gender Equality 5.b | N° of people with mobile phone by gender | 5.b.1 | Proportion of individuals in the project zone who have a telephone by gender. | In development |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Education programme** | Refurbishment of school establishment and provision of, and continued support for, certified teacher. | Medium Term | Annual | Quality Education 4.1 | In development | 4.1.2 | N° of students who passed from each grade level by gender | In development |
| | | | Biannual | Quality Education 4.a e 4.1 | | 4.a.1 | N° of facility offerings at school and maintenance at school annually | In development |
| | Environmental Education | Medium Term | Biannual | - | In development | - | In development | Raising awareness about the impacts of hunting on biodiversity and other activities |
| **Community training** | Dissemination of forest conservation practices | Medium Term | Annual | Terrestrial Life 15.5 | Nº of people trained in copaiba oil production training events in partnership. | 15.5.1 | In development | In development |
| | Sustainable Extractivism | Medium Term | Annual | - | In development | - | In development | Forming partnerships with institutions to foster the local economy sustainable extractivism |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Survey on Hight Conservation Values areas | Medium Term | Every 5 years | - | % of protection of survey on Hight Conservation Values | - | In development | Constantly identify HCV species, especially after the Fauna and Flora inventory and greater interaction with the communitys |
| **Fauna inventory** | Research on endangered or endemic fauna species and total fauna species | Medium Term | Every 5 years | Terrestrial Life 15.5 | In development | 15.5.1 | Maintenance of total and number of endangered or endemic fauna species or it decrease | In development |
| **Flora inventory** | Research on number Endangered or endemic flora species and total flora species | Medium Term | Every 5 years | Terrestrial Life 15.5 | In development | 15.5.1 | Maintenance of total and number of endangered or endemic flora species or it decrease | In development |